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- Unit-wise Concise Notes
- Diagrams & Tables
- Key Terms & Definitions
- Important Points
- Quick Revision & Exam Focused

DR. AUQIB HAMID LONE

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JKSSB Computer Awareness

Concise Notes

Key definitions, comparisons, and examples for the JKSSB Computer Awareness syllabus.

Dr. Auqib Hamid Lone

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Prepared for the Computer Awareness component of JKSSB recruitment examinations and the SSC computer-knowledge syllabus.

Source of truth. These concise revision notes are derived from the course lessons. The lesson materials remain authoritative.

Provenance of practice items. Any question presented as a past year question is labelled `[PYQ …]` with its paper and key; adapted items are labelled `[PYQ-adapted …]`; generated items are labelled `[Original]`. Generated practice is never represented as an authentic exam question.

Companion volumes. *MCQ Bank with Answer Keys* (practice sets, unit-wise) and the video course on *Concepts That Click*.

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Welcome

Unit-wise revision notes for JKSSB Computer Awareness — from computer fundamentals and hardware to office applications, networking, and cybersecurity.

This volume gathers the complete JKSSB Computer Awareness course into a single, distributable revision companion. Each lesson distils its key definitions, comparisons, tables, and selected worked examples into one scannable pass, kept 1:1 with the course slides.

What this volume contains

- **Eleven units:** Fundamentals & data representation, CPU & hardware, memory & storage, I/O & peripherals, software & operating systems, programming, office applications, the internet, networking, cybersecurity, and e-governance.
- **One lesson per topic:** Definitions boxed for quick recall, alongside comparison tables and worked examples.
- **A matching MCQ set:** Practise recall, comparison, and one-step application with the companion *MCQ Bank with Answer Keys*.

How to revise

Read a lesson, cover the boxed definitions, then recall them from memory. Use the companion MCQ set to test the same lesson under exam conditions.

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About the Author

Dr. Auqib Hamid Lone — Assistant Professor & Blockchain Researcher

Dr. Auqib Hamid Lone is an **Assistant Professor of Computer Science and Engineering** at GCET Kashmir, teaching core undergraduate courses to B.Tech CSE students. His PhD, at NIT Srinagar, explored blockchain technology beyond cryptocurrencies — harnessing it to remove problems of trust and security in real applications.

Before academia he was a **Research Scientist in blockchain and cryptography at Parity Financial (London)**, where he worked on custody infrastructure, transaction construction, and zero-knowledge privacy layers. He publishes his teaching materials openly so students and fellow instructors can reuse, adapt, and improve them.

At a glance

GCET Kashmir | PhD, NIT Srinagar | Blockchain & Crypto | GATE 98.5%ile | AIR 31 UGC NET-JRF | 6+ Publications | 3 yrs Industry R&D, London

Field notes

- **Role:** Assistant Professor, Computer Science & Engineering
- **Institution:** Government College of Engineering & Technology (GCET), Kashmir
- **Focus:** Blockchain, applied cryptography, and teaching computing from first principles
- **Education:** PhD, NIT Srinagar
- **Office hours:** Mon–Sat, 2–3 PM

"Security is the balanced mix of feeling and reality."
— Dr. Auqib Hamid Lone

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Contents

Welcome . . . iii

Unit 1 — Fundamentals & Data Representation . . . **1**

Lesson 01 — What Is a Computer? . . . 2
- 01.1 A computer is a data-processing system . . . 2
- 01.2 Input, processing, output, and storage . . . 3
- 01.3 Characteristics and limitations . . . 3
- 01.4 Applications and role classification . . . 4

Lesson 02 — Types of Computers . . . 6
- 02.1 Choose the classification axis first . . . 6
- 02.2 Classification by signal representation . . . 6
- 02.3 Traditional size and capability classes . . . 7
- 02.4 Classification by purpose and form . . . 8
- 02.5 A reliable matching procedure . . . 9

Lesson 03 — Bits, Bytes & Units . . . 11
- 03.1 The basic units of digital data . . . 11
- 03.2 Bits and bytes in names and data rates . . . 12
- 03.3 Capacity units and the 1024 convention . . . 12
- 03.4 Keep capacity separate from rate . . . 13

Lesson 49 — Computer Generations & History . . . 15
- 49.1 What a computer generation represents . . . 15
- 49.2 The first four technology associations . . . 16
- 49.3 The fifth-generation association . . . 16
- 49.4 India's PARAM 8000 milestone . . . 16
- 49.5 Exam retrieval and common confusions . . . 17

Lesson 50 — Data Representation, Number Systems & Logic . . . 19
- 50.1 Bases and positional notation . . . 19
- 50.2 Converting decimal and binary . . . 20
- 50.3 Binary grouping for octal and hexadecimal . . . 20
- 50.4 Character codes, Unicode, and decimal digits . . . 21
- 50.5 RGB colour representation . . . 21
- 50.6 Logic gates and truth tables . . . 22

Unit 2 — CPU, Hardware & Architecture . . . **24**

Lesson 04 — Organization of a Computer . . . 25
- 04.1 The functional units of a computer . . . 25
- 04.2 The CPU: ALU, control unit and registers . . . 25
- 04.3 Input and output units . . . 26
- 04.4 Memory (primary storage) versus secondary storage . . . 26
- 04.5 The system bus: address, data and control . . . 26
- 04.6 The motherboard, expansion slots and interfaces . . . 27

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04.7 The stored-program (von Neumann) idea . . . 27
04.8 The instruction cycle . . . 27

Lesson 05 — CPU, ALU, Control Unit & Registers . . . 29
05.1 The CPU and its functional units . . . 29
05.2 The Arithmetic Logic Unit (ALU) . . . 29
05.3 The Control Unit (CU) . . . 29
05.4 Registers . . . 30
05.5 Clock speed, cycles, cores, and threads . . . 30
05.6 Exam retrieval and common confusions . . . 30

Lesson 06 — CPU vs GPU & Microprocessor . . . 32
06.1 Serial and parallel processing . . . 32
06.2 CPU versus GPU . . . 32
06.3 Cores, threads, and hyper-threading . . . 33
06.4 Cache depth: L1, L2, and L3 . . . 33
06.5 Clock speed and multi-core . . . 33
06.6 Microprocessor versus microcontroller . . . 33
06.7 Diagnosing a “slow” computer (EX-01) . . . 34
06.8 Exam retrieval and common confusions . . . 34

Lesson 59 — Current CPU Architecture, Data Representation & RISC/CISC . . . 36
59.1 von Neumann architecture and the bottleneck . . . 36
59.2 Addressing modes: immediate addressing . . . 36
59.3 Cache and the cache-hit ratio . . . 36
59.4 Data hazards even with register addressing . . . 37
59.5 Virtual memory versus cache . . . 37
59.6 Instruction cycle: Program Counter and Instruction Register . . . 37
59.7 Interrupts are processed at instruction boundaries . . . 37
59.8 Instruction timing: clock cycles vary . . . 37
59.9 Two’s complement and overflow detection . . . 38
59.10 Zero-extension versus sign-extension . . . 38
59.11 BCD: binary-coded decimal . . . 38
59.12 Byte-addressability and effective address vs instruction length . . . 39
59.13 RISC versus CISC . . . 39

Unit 3 — Memory, Storage & Backup . . . 41

Lesson 07 — Computer Memory Hierarchy . . . 42
07.1 Why a memory hierarchy exists . . . 42
07.2 The ordered hierarchy . . . 42
07.3 Registers and cache . . . 43
07.4 Main memory (RAM) and ROM . . . 43
07.5 Primary memory vs secondary storage . . . 44
07.6 Secondary and tertiary storage . . . 44
07.7 Locality of reference . . . 44
07.8 Exam retrieval and common confusions . . . 44

Lesson 08 — RAM, ROM & Cache . . . 46
08.1 Memory, volatility, and why the distinction matters . . . 46
08.2 RAM: volatile main memory . . . 46
08.3 DRAM versus SRAM . . . 46
08.4 ROM and the ROM family . . . 47
08.5 Cache memory and its levels . . . 47
08.6 Cache hits, misses, and hit ratio . . . 48
08.7 Exam retrieval and common confusions . . . 48

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Lesson 09 — Virtual Memory . . . 50
09.1 What virtual memory is and why it exists . . . 50
09.2 Virtual addresses, physical addresses, and the MMU . . . 50
09.3 Paging: fixed-size pages and frames . . . 51
09.4 Segmentation: variable-size logical segments . . . 51
09.5 Paging versus segmentation at a glance . . . 52
09.6 Page faults and demand paging . . . 52
09.7 Swapping and the disk-backed hierarchy . . . 52
09.8 Thrashing . . . 53
09.9 Exam retrieval and common confusions . . . 53

Lesson 10 — Storage Devices . . . 55
10.1 The storage hierarchy . . . 55
10.2 Magnetic storage . . . 55
10.2.1 Hard disk drive . . . 55
10.2.2 Magnetic tape . . . 56
10.3 Optical storage . . . 56
10.4 Flash and solid-state storage . . . 56
10.5 Cloud storage . . . 57
10.6 Hard disk versus solid-state drive . . . 57
10.7 Access methods: random and sequential . . . 57
10.8 Non-volatility of storage . . . 58
10.9 Exam retrieval and common confusions . . . 58

Lesson 11 — Backup Types . . . 60
11.1 Backup, storage and archive . . . 60
11.2 Full, incremental and differential backups . . . 60
11.3 Restore scenarios . . . 61
11.4 Image and file-level backups . . . 61
11.5 The 3-2-1 rule, off-site and offline copies . . . 62
11.6 Why RAID is not a backup . . . 62
11.7 Backup rotation . . . 62

Unit 4 — I/O, Peripherals & Multimedia . . . 64

Lesson 12 — Input Devices . . . 65
12.1 What is an input device . . . 65
12.2 Classifying input devices . . . 65
12.3 Direct vs indirect input . . . 66
12.4 The keyboard: text input . . . 66
12.5 Pointing devices . . . 66
12.5.1 The mouse . . . 67
12.5.2 The trackball . . . 67
12.5.3 The trackpad or touchpad . . . 67
12.5.4 The joystick . . . 67
12.5.5 The light pen . . . 67
12.6 Stylus and digitizer tablet . . . 67
12.7 Touchscreen . . . 67
12.8 Scanners, cameras, and webcams . . . 68
12.9 Barcode, QR code, and RFID . . . 68
12.10 Microphone and biometric input . . . 68
12.11 Summary of input devices . . . 68
12.12 Points of confusion between similar devices . . . 69
12.13 Exam retrieval and common confusions . . . 69

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Lesson 13 — OCR, OMR, MICR . . . 71
13.1 Reading machines and why they matter . . . 71
13.2 Terminology at a glance . . . 71
13.3 OCR — Optical Character Recognition . . . 72
13.4 OMR — Optical Mark Recognition . . . 72
13.5 MICR — Magnetic Ink Character Recognition . . . 72
13.6 The crucial trio: OCR vs OMR vs MICR . . . 73
13.7 Barcodes and QR codes . . . 73
13.8 RFID — Radio Frequency Identification . . . 74
13.9 Biometric input . . . 74
13.10 Full-form ladder and memory hooks . . . 74
13.11 Applications at a glance . . . 74
13.12 Exam retrieval and common confusions . . . 75

Lesson 14 — Output Devices . . . 77
14.1 What is an output device . . . 77
14.2 Classifying output devices . . . 77
14.3 Soft copy vs hard copy . . . 77
14.3.1 Uses and limits of each form . . . 78
14.4 Monitors . . . 78
14.4.1 CRT displays . . . 78
14.4.2 Flat-panel displays: LCD, LED, and OLED . . . 78
14.4.3 Display size, resolution, and refresh . . . 79
14.5 Projector . . . 79
14.6 Audio output: speakers and headphones . . . 79
14.7 Printers: overview . . . 79
14.8 Plotter . . . 80
14.9 Multifunction printer (MFP) . . . 80
14.10 Summary of output devices . . . 80
14.11 Points of confusion between similar devices . . . 80
14.12 Other output devices on the exam . . . 81
14.13 Exam retrieval and common confusions . . . 81

Lesson 15 — Printers and Displays . . . 83
15.1 Printers and hard-copy output . . . 83
15.2 Impact and non-impact printers . . . 83
15.3 Impact printers in detail . . . 84
15.4 Inkjet printers . . . 84
15.5 Laser printers . . . 84
15.6 Thermal printers . . . 85
15.7 3D printers . . . 85
15.8 Plotters . . . 85
15.9 Display terminology . . . 85
15.10 Exam retrieval and common confusions . . . 86

Lesson 16 — Computer Ports . . . 88
16.1 What a port is . . . 88
16.2 The two meanings of “port” . . . 88
16.3 USB: the Universal Serial Bus . . . 89
16.4 Video ports: HDMI, DisplayPort, and VGA . . . 90
16.5 Ethernet and the RJ-45 connector . . . 90
16.6 Audio jacks . . . 90
16.7 Storage interfaces: SATA and eSATA . . . 91
16.8 Classifying a port: four questions to ask . . . 91

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16.9 Exam retrieval and common confusions . . . 91

Lesson 65 — Current Multimedia, GUI and Media Formats . . . 94
65.1 What multimedia means . . . 94
65.2 Digitising media: sampling and quantization . . . 94
65.3 Compression: lossy and lossless . . . 95
65.4 Synchronization . . . 95
65.5 The graphical user interface and its elements . . . 95
65.6 WYSIWYG . . . 96
65.7 Image formats: PNG, JPEG, and GIF . . . 96
65.8 Audio and video formats: MP3 and MP4 . . . 96
65.9 Frame rate, resolution, and bit rate . . . 97
65.10 Plotter . . . 97
65.11 Impact versus non-impact printers (revisit) . . . 97
65.12 Multimedia hardware and applications . . . 97
65.13 Exam retrieval and common confusions . . . 98

Unit 5 — Software, Operating Systems, Windows & Utilities . . . 100

Lesson 17 — Hardware, Software and Firmware . . . 101
17.1 Hardware . . . 101
17.2 Software . . . 101
17.3 Hardware vs software . . . 102
17.4 System software and application software . . . 102
17.5 Firmware . . . 102
17.6 Device drivers . . . 103
17.7 Humanware . . . 103
17.8 The layered computing stack . . . 103
17.9 Why the stack matters . . . 104
17.10 Exam retrieval and common confusions . . . 104

Lesson 18 — Software Categories . . . 106
18.1 What is software? . . . 106
18.2 System software . . . 106
18.3 Application software . . . 107
18.4 Utility software . . . 107
18.5 Programming software and language translators . . . 108
18.6 Embedded software . . . 108
18.7 Middleware . . . 108
18.8 Software licensing models . . . 109
18.9 System software vs application software . . . 110
18.10 Exam retrieval and common confusions . . . 110

Lesson 19 — Operating System . . . 112
19.1 What an operating system is . . . 112
19.2 The OS as interface and resource manager . . . 112
19.3 Functions of an operating system . . . 113
19.4 The kernel . . . 113
19.5 User interfaces: GUI and CLI . . . 113
19.6 Booting the computer . . . 113
19.7 Process management . . . 114
19.8 Memory management . . . 114
19.9 File management . . . 114
19.10 Device management . . . 114
19.11 Multitasking and time-sharing . . . 114

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19.12 Security, user accounts and updates . . . 114
19.13 Examples of operating systems . . . 114
19.14 Exam retrieval and common confusions . . . 114

Lesson 20 — Windows Explorer . . . 116
20.1 What Windows Explorer is . . . 116
20.2 Drives, folders, subfolders, and files . . . 116
20.3 The file path . . . 116
20.4 File names and extensions . . . 117
20.5 Hidden files and attributes . . . 117
20.6 Properties . . . 117
20.7 Selecting files . . . 118
20.8 Copy . . . 118
20.9 Move . . . 118
20.10 Rename . . . 118
20.11 Delete and the Recycle Bin . . . 118
20.12 Recycle Bin vs permanent deletion . . . 119
20.13 Keyboard shortcuts and drag-and-drop . . . 119
20.14 Exam retrieval and common confusions . . . 119

Lesson 21 — Keyboard Shortcuts . . . 121
21.1 How to read a keyboard shortcut . . . 121
21.2 Copy, cut, and paste . . . 121
21.3 Undo, redo, and select all . . . 122
21.4 Save, print, find, and replace . . . 122
21.5 Alt-key combinations . . . 122
21.6 Function keys . . . 123
21.7 Windows-key shortcuts . . . 123
21.8 Other high-yield keys . . . 124
21.9 Special keys that stand alone . . . 124
21.10 Shortcuts in the office desktop workflow . . . 124
21.11 Exam retrieval and common confusions . . . 125
21.12 Quick-reference sheet . . . 125

Lesson 45 — Open Source . . . 127
45.1 Source code and what it means . . . 127
45.2 Open source versus proprietary software . . . 127
45.3 The open-source idea: view, modify, share . . . 127
45.4 Freeware, free software, public domain, commercial, shareware . . . 128
45.5 Free software and its four freedoms . . . 128
45.6 The Open Source Initiative . . . 128
45.7 Benefits and limitations of open source . . . 129
45.8 Full forms and key names to remember . . . 129
45.9 Repositories and community development . . . 129
45.10 Basic licensing concepts . . . 130
45.11 Copyright, licences, and piracy . . . 130
45.12 Common licences at a glance . . . 130
45.13 Well-known open-source and proprietary software . . . 131
45.14 Exam retrieval and common confusions . . . 131

Lesson 46 — Firmware, BIOS, UEFI and Drivers . . . 133
46.1 What firmware is . . . 133
46.2 Where firmware is stored . . . 133
46.3 Hardware, software, and firmware . . . 134
46.4 Firmware on everyday devices . . . 134

Concepts That Click
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46.5 System firmware: the BIOS . . . 134
46.6 The power-on boot sequence . . . 134
46.7 Boot devices and boot order . . . 135
46.8 POST: the power-on self-test . . . 135
46.9 BIOS setup and CMOS . . . 135
46.10 UEFI: the modern firmware . . . 136
46.11 BIOS versus UEFI . . . 136
46.12 Device drivers . . . 136
46.13 Firmware, driver, OS, and kernel . . . 137
46.14 Firmware updates and flashing . . . 137
46.15 Exam retrieval and common confusions . . . 137

Lesson 52 — OS PYQ Traps and System Utilities . . . 139
52.1 Why this lesson is a trap drill . . . 139
52.2 The trap catalogue for this lesson . . . 139
52.3 The operating system: role and classification . . . 140
52.4 System, application, and utility software . . . 140
52.5 Device drivers are software . . . 140
52.6 DMA: transferring data without the CPU . . . 141
52.7 Real-time operating systems . . . 141
52.8 DOS, UNIX, and Linux distributions . . . 141
52.9 Internal and external hardware . . . 142
52.10 The Windows Run box and the `winword` command . . . 142
52.11 Running example: the office desktop . . . 142
52.12 Defective options and how to read them . . . 143
52.13 Terminology ladder for this lesson . . . 143
52.14 Exam retrieval and common confusions . . . 143

Lesson 60 — Current OS, Processes, Linux and Utilities . . . 145
60.1 Deadlock and its four necessary conditions . . . 145
60.2 Deadlock recovery by resource preemption . . . 145
60.3 Processes and the scheduler . . . 145
60.4 Priority inversion versus priority inheritance . . . 146
60.5 Starvation and aging . . . 146
60.6 Preemptive priority scheduling . . . 147
60.7 Real-time operating systems (RTOS) . . . 147
60.8 The Linux kernel . . . 147
60.9 `/root` versus `/home` . . . 148
60.10 File systems . . . 148
60.11 Windows utilities: Store, WordPad/Notepad, Disk Cleanup . . . 148
60.12 Defragmenter and the SSD caution . . . 149
60.13 Imaging Fax and the `xcopy /s` command . . . 149
60.14 Exam retrieval and common confusions . . . 149

Unit 6 — Programming & Data Structures . . . 151

Lesson 51 — Programming Foundations, Compiler and Data Structures . . . 152
51.1 Programming languages and their levels . . . 152
51.2 Machine language . . . 152
51.3 Assembly language . . . 152
51.4 High-level languages . . . 152
51.5 Generations of programming languages . . . 153
51.6 Language translators: assembler, compiler, interpreter . . . 153
51.7 Source code, object code, and machine code . . . 153

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51.8 Algorithm . . . 154
51.9 Flowchart . . . 154
51.10 Debugging and types of errors . . . 155
51.11 Object-oriented programming basics . . . 155
51.12 Stack and LIFO . . . 156
51.13 Basic data structures . . . 156
51.14 Exam retrieval and common confusions . . . 156

Unit 7 — Office Applications & File Formats . . . 158

Lesson 22 — MS Word Fundamentals . . . 159
22.1 What MS Word is . . . 159
22.2 The Ribbon and its tabs . . . 159
22.3 The Quick Access Toolbar . . . 160
22.4 Creating, opening and saving documents . . . 160
22.5 Entering and editing text . . . 160
22.6 Font formatting . . . 160
22.7 Paragraph alignment . . . 161
22.8 Indentation . . . 161
22.9 Line and paragraph spacing . . . 161
22.10 Bullets and numbering . . . 161
22.11 Tables . . . 161
22.12 Images and objects . . . 162
22.13 Headers and footers . . . 162
22.14 Page setup . . . 162
22.15 Page break vs section break . . . 162
22.16 Printing a document . . . 162
22.17 Exam retrieval and common confusions . . . 162

Lesson 23 — MS Word High-Yield . . . 164
23.1 Why these Word tools are high-yield . . . 164
23.2 Finding and replacing text . . . 164
23.3 Spelling and grammar tools . . . 164
23.4 Hyperlinks and bookmarks . . . 165
23.5 Styles . . . 165
23.6 Comments and Track Changes . . . 165
23.7 Page Break versus Section Break . . . 165
23.8 Mail merge . . . 166
23.9 Saving, printing and exporting to PDF . . . 166
23.10 Exam retrieval and common confusions . . . 166

Lesson 24 — Excel Structure . . . 168
24.1 What Excel is and what a workbook is . . . 168
24.2 Workbook vs worksheet . . . 168
24.3 The worksheet grid: rows and columns . . . 168
24.4 The cell and the active cell . . . 169
24.5 Cell addresses and references . . . 169
24.6 Ranges . . . 169
24.7 The formula bar . . . 170
24.8 The name box . . . 170
24.9 Basic data types . . . 170
24.10 Exam retrieval and common confusions . . . 171

Lesson 25 — Excel Formulas . . . 173
25.1 What a formula is . . . 173

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25.2 The = rule and formula syntax . . . 173
25.3 Formula, function, and constant . . . 173
25.4 Operators . . . 174
25.5 Order of evaluation (precedence) . . . 174
25.6 Cell references . . . 174
25.7 Relative references . . . 175
25.8 Absolute references . . . 175
25.9 Mixed references . . . 175
25.10 Copying and filling formulas . . . 175
25.11 Worked examples . . . 176
25.12 Common formula errors . . . 176
25.13 Exam retrieval and common confusions . . . 177

Lesson 26 — Excel Functions . . . 178
26.1 From formulas to functions . . . 178
26.2 Function syntax and arguments . . . 178
26.3 SUM and AVERAGE . . . 179
26.4 COUNT and COUNTA: the near-neighbour pair . . . 179
26.5 MAX and MIN . . . 179
26.6 The IF function and nested IF . . . 179
26.7 A worked sheet, function by function . . . 180
26.8 Other basic functions worth knowing . . . 180
26.9 AutoSum . . . 181
26.10 Combining functions in one formula . . . 181
26.11 Error values . . . 181
26.12 Exam retrieval and common confusions . . . 182

Lesson 27 — Excel Data Tools . . . 184
27.1 What Excel data tools do . . . 184
27.2 Sorting . . . 184
27.3 Filtering . . . 185
27.4 Excel tables . . . 185
27.5 Charts . . . 186
27.6 PivotTables and PivotCharts . . . 186
27.7 Printing worksheets . . . 187
27.8 Protecting worksheets and workbooks . . . 187
27.9 What-if analysis . . . 187
27.10 Near-neighbour contrast: COUNTIFS vs SUMPRODUCT . . . 188
27.11 Exam retrieval and common confusions . . . 188

Lesson 28 — Access Database Basics . . . 190
28.1 What a database is . . . 190
28.2 Database vs DBMS . . . 190
28.3 What a DBMS does . . . 190
28.4 Microsoft Access in the Office suite . . . 190
28.5 The objects in an Access database . . . 191
28.6 Tables, records, and fields . . . 191
28.7 Data types in Access . . . 191
28.8 Field properties and validation . . . 192
28.9 Primary key and index . . . 192
28.10 Exam retrieval and common confusions . . . 192

Lesson 29 — Access Keys and Relationships . . . 194
29.1 Why keys and relationships matter . . . 194
29.2 The primary key . . . 194

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29.3 Candidate keys, alternate keys and composite keys . . . 195
29.4 The foreign key . . . 195
29.5 Types of relationships . . . 196
29.5.1 One-to-one . . . 196
29.5.2 One-to-many . . . 196
29.5.3 Many-to-many and the junction table . . . 196
29.6 Creating a relationship in Access . . . 197
29.7 Referential integrity . . . 197
29.8 Access file formats: `.mdb` vs `.accdb` . . . 197
29.9 Exam retrieval and common confusions . . . 198

Lesson 30 — Access Queries, Forms and Reports . . . 200
30.1 The Access objects and the object sequence . . . 200
30.2 The table is the source of data . . . 200
30.3 What a query is . . . 200
30.4 The Select query: retrieving on a condition . . . 200
30.5 Building a query: Design view and criteria . . . 201
30.6 Action queries: Update, Append and Delete . . . 201
30.7 Select versus action queries . . . 201
30.8 Lookup fields . . . 201
30.9 Basic SQL awareness . . . 202
30.10 Forms: entering and viewing data . . . 202
30.11 Reports: printable output . . . 202
30.12 Forms versus reports . . . 202
30.13 Exam retrieval and common confusions . . . 202

Lesson 31 — PowerPoint Basics . . . 204
31.1 What PowerPoint is . . . 204
31.2 Full forms and file extensions . . . 204
31.3 Slides, placeholders, and layouts . . . 205
31.4 Themes, templates, and the Slide Master . . . 205
31.5 Views: Normal, Outline, Sorter, Reading, Show, Notes Page . . . 205
31.6 Notes, handouts, and printing . . . 206
31.7 Running and navigating the slide show . . . 206
31.8 Exam retrieval and common confusions . . . 206

Lesson 32 — PowerPoint Animation and Transition . . . 208
32.1 Animation and transition: the core distinction . . . 208
32.2 The four animation effects . . . 208
32.3 Transitions and Morph . . . 209
32.4 Starting and controlling the slide show . . . 209
32.5 Speaker notes, views and handouts . . . 209
32.6 SmartArt conversion and its limits . . . 210
32.7 Exam retrieval and common confusions . . . 210

Lesson 33 — PDF and File Formats . . . 212
33.1 What a file format is . . . 212
33.2 Extension versus format (the core trap) . . . 212
33.3 Common document formats . . . 212
33.4 Common image extensions . . . 213
33.5 PDF: the Portable Document Format . . . 213
33.6 Text and scanned PDFs . . . 213
33.7 OCR: Optical Character Recognition . . . 214
33.8 Encryption and password protection . . . 214
33.9 Digital signature . . . 214

Concepts That Click
YouTube · @concepts.thatclick

33.10 Redaction and PDF/A . . . 215
33.11 Exam retrieval and common confusions . . . 215

Lesson 54 — Office Shortcut and Product Trap Lab . . . 217
54.1 Why this lesson is a trap drill . . . 217
54.2 MS Word: exact shortcut keys . . . 217
54.3 Word views and review tools . . . 217
54.4 Word: Section Break versus Page Break . . . 218
54.5 Word feature trap: Mail Merge . . . 218
54.6 PowerPoint: the show keys . . . 218
54.7 Handouts, Slide Master, and orientation . . . 218
54.8 PowerPoint: converting to SmartArt . . . 218
54.9 Excel: the formula palette . . . 218
54.10 Excel: cell references and the $ sign . . . 219
54.11 Product-category traps: Word, Excel, Access, MS-DOS . . . 219
54.12 Exam retrieval and common confusions . . . 219

Lesson 55 — File Formats and Named Technology Associations . . . 221
55.1 Extension and format are two different things . . . 221
55.2 Image formats . . . 221
55.3 PDF: the Portable Document Format . . . 221
55.4 The Office document family . . . 222
55.5 Plain-data formats: CSV and TXT . . . 222
55.6 Microsoft Access databases: `.mdb` and `.accdb` . . . 222
55.7 Named technology associations . . . 222
55.8 Software roles: browser, PDF reader, and remote support . . . 223
55.9 Exam retrieval and common confusions . . . 223

Lesson 61 — Current Office, Email and Statement Evaluation . . . 225
61.1 Word: structured content and content controls . . . 225
61.2 First-line and hanging indents . . . 225
61.3 The Navigation Pane . . . 225
61.4 Track Changes and revision metadata . . . 226
61.5 Section Break versus Page Break . . . 226
61.6 Excel: nested IF, blank and non-numeric input . . . 227
61.7 COUNTIFS and SUMPRODUCT . . . 227
61.8 Email: filters, signatures, POP3 and privacy . . . 228
61.9 PowerPoint SmartArt conversion . . . 228
61.10 Exam retrieval and common confusions . . . 229

Unit 8 — Internet, Web & Email . . . 231

Lesson 34 — Internet, WWW and Browser . . . 232
34.1 The Internet: the network itself . . . 232
34.2 Internet vs WWW . . . 232
34.3 Website, webpage, and homepage . . . 233
34.4 Browser . . . 233
34.5 Search engine . . . 233
34.6 Client and server . . . 233
34.7 ISP: the connection provider . . . 234
34.8 Hyperlink and URL . . . 234
34.9 Full-form ladder of this lesson . . . 234
34.10 Exam retrieval and common confusions . . . 234

Lesson 35 — URL, DNS, HTTP and HTTPS . . . 236
35.1 What a URL is . . . 236

Concepts That Click
@concepts.thatclick

35.2 Anatomy of a URL . . . 236
35.3 Ports, paths, and queries . . . 237
35.4 IP addresses: the real destination . . . 237
35.5 DNS: the phone book of the Internet . . . 237
35.6 HTTP: carrying web pages . . . 237
35.7 HTTPS and TLS: the secure form . . . 238
35.8 The full flow: from typed URL to secure page . . . 238
35.9 Exam retrieval and common confusions . . . 238

Lesson 36 — Web Browsing, Search, Upload and Download . . . 240
36.1 The browser, tabs, bookmarks, and history . . . 240
36.2 How a search engine works . . . 240
36.3 Sponsored results and organic results . . . 241
36.4 Cookies, cache, and history . . . 241
36.5 Downloading and uploading . . . 242
36.6 File conversion . . . 242
36.7 Exam retrieval and common confusions . . . 242

Lesson 37 — Email Basics . . . 244
37.1 What email is . . . 244
37.2 The email address . . . 244
37.3 Recipient fields: To, Cc and Bcc . . . 245
37.4 Subject, body and attachments . . . 245
37.5 Folders: Inbox, Sent, Drafts, Spam, Trash . . . 245
37.6 Reply, Reply All and Forward . . . 246
37.7 Webmail versus an email client . . . 246
37.8 Exam retrieval and common confusions . . . 246

Lesson 38 — SMTP, POP and IMAP . . . 248
38.1 Why email needs protocols . . . 248
38.2 SMTP: the sending protocol . . . 248
38.3 POP3: download to one machine . . . 249
38.4 IMAP: sync across devices . . . 249
38.5 POP3 vs IMAP and the full mail flow . . . 249
38.6 To, Cc, BCC, filters, and signatures . . . 250
38.7 Digital signature vs encryption, and secure transmission . . . 250
38.8 Spam, phishing, spoofing, and malicious content . . . 250
38.9 Exam retrieval and common confusions . . . 251

Lesson 39 — E-Banking . . . 252
39.1 What e-banking covers . . . 252
39.2 UPI in detail . . . 252
39.3 The four secret codes . . . 253
39.4 Authentication and multi-factor authentication . . . 253
39.5 QR codes and collect requests . . . 254
39.6 Safe-banking rules . . . 254
39.7 Exam retrieval and common confusions . . . 254

Lesson 53 — Networking Legacy Services and Web Standards . . . 256
53.1 How to read this lesson . . . 256
53.2 Packet switching vs circuit switching . . . 256
53.3 Telnet and FTP . . . 257
53.4 SMTP, POP3, and IMAP . . . 257
53.5 W3C and DHTML . . . 258
53.6 URL, DNS, and IP addresses . . . 258
53.7 IPv4 vs IPv6 . . . 258

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YouTube · @concepts.thatclick

53.8 The seven OSI layers . . . 258
53.9 Physical vs logical ports . . . 259
53.10 Exam retrieval and common confusions . . . 259

Lesson 63 — Current Web, HTML, JavaScript, VBScript and ASP.NET . . . 261
63.1 Static pages, markup, scripts, and the server . . . 261
63.2 HTML: a markup language in an ASCII-text document . . . 261
63.3 Tags, attributes, and validity . . . 262
63.4 The `h1` heading and the `marquee` tag . . . 262
63.5 DHTML: the dynamic combination . . . 262
63.6 XML: data, not display . . . 262
63.7 Java and JavaScript are different languages . . . 263
63.8 JavaScript string concatenation with `+` . . . 263
63.9 VBScript and the `Dim` statement . . . 263
63.10 ASP.NET and the `Session` object . . . 263
63.11 Meta-search engines: Dogpile . . . 263
63.12 Exam retrieval and common confusions . . . 263

Unit 9 — Networking . . . **265**

Lesson 40 — Networking Basics . . . 266
40.1 What a network is: nodes, hosts, clients, servers . . . 266
40.2 Networks by area: PAN, LAN, MAN, WAN . . . 266
40.3 WLAN, intranet, Internet, and ISP . . . 267
40.4 Topology: bus, star, ring, mesh, tree . . . 267
40.5 Transmission media: guided and unguided . . . 268
40.6 Networking devices and their roles . . . 269
40.7 Exam retrieval and common confusions . . . 269

Lesson 41 — Networking Protocols . . . 271
41.1 Why protocols and addresses matter . . . 271
41.2 TCP and UDP . . . 271
41.3 IPv4 and IPv6 . . . 272
41.4 ICMP: the control channel . . . 272
41.5 DNS and DHCP . . . 272
41.6 Application protocols: web, files, remote login, mail . . . 273
41.7 The seven OSI layers . . . 273
41.8 Port awareness . . . 274
41.9 Exam retrieval and common confusions . . . 274

Lesson 62 — Current Internet, Networking and Cloud IaaS . . . 275
62.1 TCP: ordered and reliable, but with no timing promise . . . 275
62.2 UDP and the TCP/UDP contrast . . . 276
62.3 ICMP: the network's own messenger . . . 276
62.4 BGP: routing between autonomous systems . . . 276
62.5 IPv6: address space, header, and no broadcast . . . 277
62.6 HTTP/3 and QUIC . . . 277
62.7 DNS and the .org / PIR / ICANN chain . . . 277
62.8 ARPANET and packet switching . . . 278
62.9 The router: joining networks by reading IP addresses . . . 278
62.10 IaaS: the provider manages the hardware . . . 278
62.11 Exam retrieval and common confusions . . . 279

Unit 10 — Cybersecurity & IT Law . . . **280**

Lesson 42 — Malware . . . 281

Concepts That Click
YouTube · @concepts.thatclick

42.1 What “malware” means . . . 281
42.2 Virus, worm, and Trojan . . . 281
42.3 Ransomware . . . 282
42.4 Spyware, adware, and the keylogger . . . 282
42.5 Rootkit and backdoor . . . 283
42.6 Bot and botnet, logic bomb, cryptojacking . . . 283
42.7 How malware reaches a machine, and how to stop it . . . 283
42.8 Exam retrieval and common confusions . . . 283

Lesson 43 — Hacking, Phishing and Spoofing . . . 285
43.1 Why attacks succeed . . . 285
43.2 Hacking . . . 285
43.3 Password attacks . . . 286
43.4 Social engineering . . . 286
43.5 Phishing . . . 286
43.6 Spoofing . . . 287
43.7 Phishing vs spoofing vs hacking . . . 287
43.8 Malicious links and attachments . . . 287
43.9 Safe habits and response after targeting . . . 288
43.10 Exam retrieval and common confusions . . . 288

Lesson 44 — Cybersecurity Prevention . . . 290
44.1 Prevention and the CIA triad . . . 290
44.2 Antivirus: signatures, scanning, and quarantine . . . 290
44.3 Firewall: the network gatekeeper . . . 291
44.4 Patching and updates . . . 291
44.5 Strong passwords and MFA . . . 291
44.6 Backup: the availability control . . . 292
44.7 Safe browsing . . . 292
44.8 Secure Wi-Fi . . . 292
44.9 Removable-media safety . . . 292
44.10 Which control does what . . . 292
44.11 Exam retrieval and common confusions . . . 293

Lesson 56 — Cybersecurity, IT Act and Security Controls . . . 295
56.1 Why security, and the CIA triad . . . 295
56.2 The Information Technology Act, 2000 . . . 295
56.3 Malware: the umbrella and its members . . . 296
56.4 Antivirus, quarantine, and patching . . . 296
56.5 Phishing and anti-phishing . . . 297
56.6 Encryption, digital signatures, and SSL/TLS . . . 297
56.7 Firewall, MFA, and the matching map . . . 297
56.8 Exam retrieval and common confusions . . . 298

Lesson 64 — Current Cybersecurity, Cryptography and Cyber Law . . . 300
64.1 Scope and study method . . . 300
64.2 Malware and the parasitic virus . . . 300
64.3 Phishing, ransomware, DDoS, and hacking . . . 301
64.4 Cryptography and DES . . . 301
64.5 Encryption vs digital signature . . . 301
64.6 Firewalls: three types . . . 301
64.7 Internet governance, privacy, and data protection . . . 302
64.8 Cyber law and copyright: awareness-level map . . . 302
64.9 Exam retrieval and common confusions . . . 302

Concepts That Click · @concepts.thatclick

Unit 11 — Governance & Integration **304**

Lesson 47 — Humanware and IT Governance 305
47.1 What humanware means 305
47.2 The people in a computer system 305
47.3 Hardware, software and humanware 305
47.4 The role of IT in governance 306
47.5 IT governance versus e-governance 306
47.6 E-governance basics 306
47.7 Digital public services 307
47.8 Electronic records 307
47.9 Authentication 307
47.10Exam retrieval and common confusions 307

Lesson 57 — Digital Platforms and E-Governance 309
57.1 What e-Governance means 309
57.2 The G2x classification 309
57.3 Digital India 310
57.4 NeGP, e-Kranti, and NeGD: three names, three things 310
57.5 e-NAM and agricultural marketing 311
57.6 Citizen portal, electronic records, and authentication 311
57.7 Exam retrieval and common confusions 311

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UNIT 1

Fundamentals & Data Representation

LESSON 01

What Is a Computer?

Video lesson: https://youtu.be/EdRZK57oqpo

01.1 A computer is a data-processing system

Start with an office task. A clerk enters marks into a spreadsheet, a formula adds the values and calculates an average, the result appears on screen, and the workbook can be saved. A computer is not merely the screen or the box on the desk. The task depends on data, instructions, processing, an output path, and possibly storage.

Definition — Computer

An electronic, programmable system that accepts data, processes it under instructions, produces information, and can store data or results.

Each part of the definition matters. *Electronic* describes the technology used by the system. *Programmable* means that instructions specify the operations to perform. The system accepts inputs, applies those instructions, and makes results available. It can also retain data or results for later use.

Definition — Data

Raw facts or symbols supplied to a system before they have been organized for the particular question being asked.

Definition — Information

Data that have been organized or processed so that they are useful for a particular purpose.

Definition — Instruction

A rule that tells the computer what operation to perform or what action to take.

Definition — Program

An organized set of instructions that directs a computer through a task.

Data and information are defined by their role in a task, not by whether the values are numbers or text. Individual marks are data when the question is "What are the three scores?" Their average is information when the question is "What was the student's average?" The average can itself become input data for a later task, so the same value may play different roles in different contexts.

Example 01.1. *A clerk enters* 68, 74, *and* 81. *These three marks are the input data. Adding them gives* 223*; dividing by three gives* $223/3 = 74.333\ldots$*, or approximately* 74.33 *to two decimal places. The total and average are processed results. They are useful information if the task is to summarize performance. Saving the workbook retains the records.*

Example 01.2. *Suppose the same three marks are each out of* 100. *The total possible score is therefore* 300. *The percentage is*

$$\frac{68+74+81}{300} \times 100 = \frac{223}{300} \times 100 = 74.333\ldots\%.$$

Rounded to two decimal places, the percentage is 74.33%. *The total, maximum possible score, formula, and rounding rule all matter; the computer cannot infer a missing maximum score from the marks alone.*

01.2 Input, processing, output, and storage

Definition — Input

Data and instructions that enter a computer system through a device, file, network, or software interface.

Definition — Processing

Operations carried out on input data according to instructions, usually by the processor and program involved in the task.

Definition — Output

The result made available by a computer system, for example on a display, in a printed document, as sound, or through another system.

Definition — Storage

The retention of data, instructions, or results for current or later use.

The basic three-stage model is IPO: Input, Processing, Output. Many introductory diagrams add Storage, giving IPOS. Storage is a distinct function, not another name for processing. The stages describe roles in a task; they do not require every computer job to follow a single rigid, one-direction path. A saved result can be read back as input to a later operation.

Stage	Question to ask	Marks-spreadsheet example
Input	What data or instructions enter?	The three marks and the average formula
Processing	What operation changes or organizes the data?	Addition, division, and percentage calculation
Output	What result is presented, and how?	The total or percentage shown on screen or printed
Storage	What is retained for later use?	The saved workbook and its records

The physical item does not determine the stage by itself. A monitor is normally an output device, but the classification question concerns the monitor's role in the described task. A saved file is stored data; opening that same file later makes its contents input to the new task. Likewise, information describes the usefulness of processed data, whereas output describes how the result is made available. A screen can present useful information, but the terms answer different questions.

Example 01.1. *In the marks spreadsheet, typing or importing the scores is input. The formula performs processing. The displayed total and average are output. Saving the workbook is storage. If the saved workbook is opened tomorrow, reading it is input to that later session.*

01.3 Characteristics and limitations

Definition — Speed

The ability to carry out many operations in a short time.

Definition — Accuracy

The ability to produce consistent, precise results when the data, instructions, and operation are correct.

Definition — Diligence

The ability to repeat work without tiring in the human sense, provided the system continues to receive power and function correctly.

Definition — Versatility

The ability to perform different kinds of tasks by running different programs.

Definition — Automation

The ability to continue a programmed operation after it has been started and the required inputs or conditions have been supplied.

Definition — Storage as a characteristic

The ability to retain data, instructions, and results for later access.

These characteristics describe capabilities, not a guarantee that every result is correct or useful. Speed concerns how quickly operations are performed; accuracy concerns whether the operation is correctly carried out on the supplied data. Storage concerns retaining material; it does not validate that material. Versatility comes from using different programs, while automation means a started procedure can continue according to its instructions and conditions.

Speed and accuracy must not be confused with judgement. A computer can calculate quickly and repeat the same operation consistently, but it cannot infer that a deliberately or accidentally incorrect mark is false merely because the value is stored in a spreadsheet. Accuracy is conditional on the quality of data and instructions.

Definition — GIGO

"Garbage In, Garbage Out": unsuitable or incorrect input can lead to unsuitable or incorrect output, even when the computer follows its instructions.

Example 01.1. *Suppose the correct marks are* 68*,* 74*, and* 81*, but a user enters* 18 *instead of* 81*. The formula may correctly compute* $68 + 74 + 18 = 160$ *and* $160/3 \approx 53.33$*. The arithmetic is consistent with the entered values, but the output is not a valid summary of the actual marks. The error entered at the input stage was not corrected by processing.*

The example illustrates the limitation often summarized by GIGO. If the input is wrong, a correct formula can still produce an incorrect result. If the input is correct but the formula is wrong, the result can also be wrong. The machine may execute either set of instructions consistently; it does not automatically decide that the source data are truthful, complete, or fair. Computers do not possess human judgement or common-sense understanding by default, so a person must interpret output in context and decide what action it justifies.

01.4 Applications and role classification

The same IPO/IPOS model applies across fields. What changes is the type of data and the useful result; the roles of input, processing, output, and storage remain identifiable.

Field	Input	Processing / output	Storage
Office	Text, figures, and instructions	Create a document or calculate a total	Document or workbook file
Banking	Account and transaction details	Validate and process a transaction; show a status	Account and transaction records
Education	Learner responses and assessment rules	Score work and present results or learning resources	Assessment and learner records
Healthcare	Patient details and appointment requests	Organize records, schedules, or diagnostic support	Patient and appointment records
Governance	Online form fields and supporting details	Process a request and provide a citizen service	Submitted forms and service records

For exam questions, classify the role within the described task. A keyboard is an input device; the formula is processing; a monitor presents output; a saved file is stored data. A physical object may support more than one stage, but the question usually asks for the role it is playing in that particular scenario.

Example 01.1. *An online expense form accepts entered amounts, checks and sums them using the programmed rules, displays the total, and stores the submitted record. If one amount is mistyped, the calculation may still be executed correctly on the wrong data. The user should correct the input before relying on the total.*

When a question gives a complete workflow, mark the stage of each action rather than assigning one label to the entire system. Entering a value is input; checking or summing it is processing; showing a result is output; keeping the transaction record is storage. If the stored record is opened and processed again, its role changes to input for that later operation.

Exercises

1. A user enters four prices, applies a tax formula, prints the invoice, and saves the file. Identify the input, processing, output, and storage.
2. Explain why a fast, consistent computer can still produce an incorrect result.
3. Three temperature readings are 18, 20, and 22 degrees. For a task asking for their average, identify the data and the resulting information.
4. Give one example from banking or governance and trace it through IPOS.

Solutions

1. The four prices are input; applying the tax formula is processing; the printed invoice is output; saving the invoice file is storage.
2. The computer follows its instructions and operates on the supplied data. Incorrect input or a wrong formula can therefore produce an incorrect result consistently; speed does not establish truth.
3. The three readings are input data. Their average is $(18 + 20 + 22)/3 = 20$ degrees and is information for the average-reading task.
4. Example: an online fee form accepts the user's details and amount (input), calculates the fee (processing), shows a receipt (output), and saves the transaction record (storage).

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LESSON 02

Types of Computers

Video lesson: https://youtu.be/d6zL3YS69kM

02.1 Choose the classification axis first

Computer labels are useful only when the basis of classification is clear. “Digital” describes how data are represented; “microcomputer” describes a traditional size/use class; “laptop” describes a form factor. These labels do not compete. One laptop can correctly be described using several axes at once.

Axis	Question	Clue / example
Signal representation	How are values represented?	Continuous signal: analog; distinct states: digital
Scale and workload	What kind of work or volume is handled?	Many transactions: mainframe; very large calculations: supercomputer
Purpose	How broad is the intended task range?	Many different programs: general-purpose; narrow task: special-purpose
Integration	Where is the computer installed?	Built into a larger product: embedded
Form factor	What physical form does it take?	Desktop, laptop, tablet, or smartphone

Definition — Classification axis

The feature used to group computers, such as signal representation, capability/workload, purpose, or physical form.

Before choosing an answer, identify the question being asked:

1. Is the stem about continuous or discrete signals?
2. Is it about personal use, transaction volume, or very large calculations?
3. Is it about a computer’s general purpose or its built-in role in a larger product?
4. Is it naming a shape, such as desktop, laptop, or tablet?

Selecting a familiar label without checking the axis is a common source of wrong answers.

02.2 Classification by signal representation

Definition — Analog computer

A computer that represents or processes quantities as continuously varying signals over a range.

Definition — Digital computer

A computer that represents or processes data as discrete values, commonly using binary digits 0 and 1.

Definition — Hybrid computer

A system combining analog measurement or input with digital processing, control, or display.

An analog quantity can take intermediate values within a range. For example, a moving pointer can indicate many positions along a speed range, not just a fixed list of allowed readings. A digital system represents information using distinct states; modern computers commonly use binary states, written as 0 and 1. The important contrast is continuous versus discrete representation, not old versus new or mechanical versus electronic.

Class	Representation	Recognition clue
Analog	Continuously varying quantity	A value can lie anywhere within a range
Digital	Separate symbols or states	A value is encoded using distinct states, commonly 0 and 1
Hybrid	Uses both forms at different stages	Measurement or input is analog; processing or display is digital

A hybrid arrangement can sense a physical signal and then use digital circuitry to process and present the measurement. Its classification describes the complete path, not just the display or one internal component.

Example 02.1. *A digital patient monitor may measure a changing physical signal through a sensor. The sensor/front end handles an analog quantity; conversion allows digital processing; the screen shows a numeric result. When a question asks about the complete measurement-and-processing arrangement, "hybrid" is the useful classification. The monitor's display alone does not determine the signal class of its entire system.*

02.3 Traditional size and capability classes

Definition — Microcomputer

A personal-use computer built around a microprocessor; common examples include desktop computers, laptops, tablets, and smartphones.

Definition — Minicomputer

A historical term for a mid-range computer class, often described as supporting multiple users; it is less common in current product marketing.

Definition — Mainframe

A computer class associated with reliable, high-volume input/output and large numbers of concurrent users or transactions.

Definition — Supercomputer

A high-performance computer designed for very large computational workloads, often using parallel processing.

Class	Primary clue	Example / caution
Microcomputer	Personal-use computer built around a microprocessor	Desktop, laptop, tablet, or smartphone; not defined only by its outer size
Minicomputer	Historical mid-range, multi-user class	A traditional category; less common in current product marketing
Mainframe	High transaction and input/output throughput for many users	Large banking or account-processing workload
Supercomputer	Very high computational performance, often parallel	Large climate or engineering simulation

Mainframes and supercomputers may both be powerful. The distinction in an exam question is usually the workload. A mainframe is associated with throughput, reliability, and many transactions or users. A supercomputer is associated with computationally intensive scientific or engineering problems. "Throughput" here means the amount of work completed over time; the mainframe clue is handling many transactions reliably, rather than solving one unusually large numerical problem. Parallel processing divides computational work so multiple processing units can perform parts of it at the same time. Physical size alone is not a dependable classification rule.

Example 02.1. *A bank processes a large number of account transactions for many customers. The emphasis is concurrent transaction throughput and service reliability, so the mainframe class is the best match. A research centre running a large-scale climate simulation emphasizes high-performance computation, so the supercomputer class is the better match.*

Example 02.2. *A question describes a "mid-range, multi-user computer" and asks for its traditional category. The intended answer is minicomputer. Do not infer that the label is common in modern consumer advertisements; it is a historical classification term.*

02.4 Classification by purpose and form

Definition — General-purpose computer

A computer capable of performing many different tasks by running different programs.

Definition — Special-purpose computer

A computer designed primarily for a narrower task or a limited set of related tasks.

Definition — Embedded computer

A computer built into a larger device to control or support that device's function.

Term	What it describes	Example
General-purpose	Broad range of tasks through different programs	Personal computer used for documents, browsing, and media
Special-purpose	Narrow task or small set of related tasks	Controller dedicated to an appliance function
Embedded	Integration into a larger product	Washing-machine controller
Form factor	Physical arrangement or shape	Desktop, laptop, tablet, smartphone

An embedded computer is commonly special-purpose, but these terms name different aspects: "embedded" describes where the computer is integrated; "special-purpose" describes the range of tasks it is intended to perform. A desktop computer that runs office, browsing, and media programs is a general-purpose system. A washing-machine controller is built into a larger appliance and handles a limited control task. A computer can therefore be both embedded and special-purpose; the terms are compatible, not competing answers. In contrast, "laptop" says how a computer is physically arranged and does not by itself tell whether the data representation is analog or digital.

Example 02.1. *A smartphone can be classified as digital by signal representation, a microcomputer by personal-use class, and a mobile device by form factor. These labels can all be true at the same time because they answer different questions.*

Example 02.2. *A washing machine's controller is embedded in the appliance and is special-purpose because it manages a limited set of appliance functions. Calling it a mainframe because it controls a whole machine confuses physical integration with workload class.*

02.5 A reliable matching procedure

When an exam stem describes a device, underline the clue that indicates the axis. "Continuous" points to analog; "discrete" or binary points to digital; "sensor plus digital processing" may point to hybrid. "Many concurrent transactions" suggests mainframe. "Large scientific calculations" suggests supercomputer. "Built into an appliance" points to embedded. "Desktop" or "laptop" identifies form factor, not signal type.

Example 02.1. *Classify a laptop used by an accounts clerk. By representation it is digital; by traditional size/use class it is a microcomputer; by form it is a laptop; and by purpose it is general-purpose if it runs many kinds of programs. None of these labels invalidates the others.*

Computer type and generation

A computer's type may describe its signal class, scale, purpose, or form. Generation instead names a historical technology grouping. Keep these axes separate: integrated circuits are associated with the third generation, while microprocessors are associated with the fourth.

This distinction prevents a common category error. A laptop can be called a digital microcomputer and can belong to a particular historical generation based on the technology used in its development. "Laptop" identifies its form, "microcomputer" its traditional user/capability class, "digital" its representation, and "generation" its place in a historical grouping. Each answer must match the axis named in the question.

Example 02.2. *For a washing-machine controller, "embedded" answers where the computer is installed. "Special-purpose" answers the range of work it is designed to perform. The controller can correctly have both labels at once. If a question asks how it represents data, that is a separate signal classification and must be answered from that clue.*

Exercises

1. Classify a smartphone on the signal, traditional class, and form axes.
2. A system is described as handling many simultaneous banking transactions reliably. Which class is the strongest match, and why?
3. Distinguish an embedded computer from a special-purpose computer.
4. Why is "laptop" not an answer to a question asking whether a computer is analog or digital?
5. A digital instrument measures a continuously changing physical quantity and processes the result digitally. Explain when "hybrid" is an appropriate label.

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Solutions

1. It is digital by representation, a microcomputer by personal-use class, and a smartphone by form factor.
2. Mainframe. The clues are many simultaneous transactions and reliability/throughput, rather than a single extremely large calculation.
3. Embedded describes integration into a larger product. Special-purpose describes a narrow intended task. An embedded controller is often special-purpose, but the terms describe different properties.
4. Laptop names physical form. Analog/digital names the representation of data, so the question asks about a different classification axis.
5. It is appropriate when the complete arrangement combines analog sensing or measurement with digital conversion and processing.

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LESSON 03

Bits, Bytes & Units

Video lesson: https://youtu.be/j_62CGCBElo

03.1 The basic units of digital data

Digital information is represented using bits. Groups of bits form larger units, but the terms do not all describe the same kind of quantity. A bit is a single binary digit; a word is an architecture-dependent processor unit; a character is an encoded symbol.

Definition — Bit

A binary digit that has one of two values, 0 or 1; the smallest unit of data in the usual computer-fundamentals description.

Definition — Nibble

A group of four bits.

Definition — Byte

A group of eight bits, equal to two nibbles.

Definition — Word

The natural data size a processor handles as a unit; its size depends on the processor architecture.

Definition — Character

A text symbol represented in digital storage by a character encoding; its storage size is not necessarily one byte.

Unit	Size / basis	Meaning
Bit	1 binary digit	One of the two values 0 or 1
Nibble	4 bits	Half of an 8-bit byte
Byte	8 bits	Two nibbles; common unit for file/storage amounts
Word	Architecture-dependent	Natural processor data size, such as 32 or 64 bits
Character	Encoding-dependent	A text symbol; its encoded storage may take more than one byte

The bit-to-byte relationships are fixed in this lesson:

$$1 \text{ nibble} = 4 \text{ bits}, \qquad 1 \text{ byte} = 8 \text{ bits} = 2 \text{ nibbles}.$$

The word size is not defined by this ladder. A processor may work with a natural word size such as 32 or 64 bits, but the term depends on the architecture and should not be treated as one universal number.

Example 03.1. *Write the byte `10110110` as two nibbles: `1011` and `0110`. Each group contains four bits; together the two groups still contain eight bits, so the original value is one byte.*

03.2 Bits and bytes in names and data rates

Capitalization changes the unit. Lowercase `b` denotes a bit, while uppercase `B` denotes a byte. A byte contains eight bits. Therefore a rate reported in megabits per second is not numerically equal to a rate in megabytes per second.

Definition — Transfer rate

The amount of data transferred in a stated time interval, commonly written in bits per second or bytes per second.

Definition — Capacity

The amount of data that a storage medium can hold.

The slash in a rate such as Mbps means "per second." It is not part of the data amount itself. Keep the numerical amount, whether it is bits or bytes, and the time unit visible while calculating. Under matching decimal prefixes,

$$1 \text{ byte} = 8 \text{ bits}, \qquad 1 \text{ MB} = 8 \text{ Mb}.$$

Thus dividing a rate in Mb/s by eight gives MB/s, while multiplying a file size in MB by eight gives Mb. These conversions change the unit, not the underlying amount of data.

To convert a bit rate to a byte rate, divide by eight, provided the prefixes match and the question is asking for the ideal conversion. Protocol overhead, congestion, and other factors can reduce actual network throughput, so a nominal rate is not a promise of application-level speed.

Example 03.1. *A link is rated at* 8 *Mbps. Using matching decimal prefixes and ignoring overhead, convert bits to bytes:*

$$8 \textit{ megabits/s} \div 8 = 1 \textit{ megabyte/s}.$$

This is a transfer rate, not a storage capacity.

Example 03.2. *Convert an ideal rate of* 24 *Mbps to megabytes per second. The prefixes already match, so only the bit-to-byte conversion is needed:*

$$24 \textit{ Mb/s} \div 8 = 3 \textit{ MB/s}.$$

The answer is a rate because the time denominator remains. If the question asked how long a file takes to transfer, the file amount would first need to be expressed in the same unit as the rate, then divided by that rate.

Example 03.3. *A* 4 *MB file contains* 32 *megabits under matching decimal prefixes because* $4 \textit{ MB} \times 8 = 32 \textit{ Mb}$*. If transferred over an ideal* 20 *Mbps link, the simple lower-bound calculation is* $32/20 = 1.6$ *seconds. Actual transfer can take longer because the ideal calculation ignores protocol overhead and other delays.*

03.3 Capacity units and the 1024 convention

For the conventional JKSSB-style ladder used in this lesson, each step is a factor of 1024:

$$1 \text{ KB} = 1024 \text{ bytes}, \quad 1 \text{ MB} = 1024 \text{ KB}, \quad 1 \text{ GB} = 1024 \text{ MB}, \quad 1 \text{ TB} = 1024 \text{ GB}.$$

Thus 1 MB under this convention is $1024 \times 1024 = 1{,}048{,}576$ bytes. The exam-style abbreviations KB, MB, GB, and TB are often used with this traditional binary ladder. Modern technical standards distinguish decimal SI units (powers of 1000) from binary IEC units such as KiB (powers of 1024). If a question specifies decimal units or the IEC prefixes, use those definitions instead. Do not silently mix conventions.

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Unit	Previous unit	Bytes under repeated 1024 convention
1 KB	1024 bytes	1,024 bytes
1 MB	1024 KB	1,048,576 bytes
1 GB	1024 MB	1,073,741,824 bytes
1 TB	1024 GB	1,099,511,627,776 bytes

To move from a larger unit to the next smaller unit, multiply by 1024. Repeat once for every step crossed. To move from a smaller unit to the next larger unit, divide by 1024. Writing the units beside the numbers helps prevent using the wrong direction.

Example 03.1. *Convert 3 KB to bytes under the 1024 convention:*

$$3\ KB \times 1024\ bytes/KB = 3072\ bytes.$$

The KB units cancel, leaving bytes.

Example 03.2. *Convert 2 MB to KB under the same convention:*

$$2\ MB \times 1024\ KB/MB = 2048\ KB.$$

This is one step down the capacity ladder, so multiply by 1024.

Example 03.3. *Convert 5 GB to MB using the same convention. GB is one step larger than MB, so multiply by 1024:*

$$5\ GB \times 1024\ MB/GB = 5120\ MB.$$

The units GB cancel, leaving MB. To convert 5120 MB back to GB, divide by 1024 and recover 5 GB.

03.4 Keep capacity separate from rate

Capacity answers "how much can be stored?" Transfer rate answers "how much is moved per unit of time?" MB is a byte-based amount; Mbps is a bit-based rate. The lowercase/uppercase distinction and the `/s` time component carry information. A drive with a larger capacity does not necessarily transfer data faster than a smaller drive.

Question	Capacity	Transfer rate
What does it measure?	Amount that can be held	Amount moved per unit of time
Common example	A 4 MB file or 1 TB drive	A 20 Mbps connection
Unit clue	No time denominator	Includes "per second" or `/s`

Example 03.1. *In the lesson's opening comparison, 4 MB describes the file's storage amount, while 20 Mbps describes the network's transfer rate. The first has no time denominator; the second is measured per second, so they are different kinds of quantities.*

Before calculating, perform a quick unit check: if the final question asks for bytes, convert bits to bytes; if it asks for seconds, divide an amount of data by a compatible rate; if it asks for a storage amount, do not attach a time unit. A numerically correct calculation with mismatched units is not a correct answer.

Exercises

1. How many bits are in 7 bytes? Show the conversion.
2. How many bytes are in 5 KB under the 1024 convention?

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3. Convert 24 Mbps to MB/s using matching decimal prefixes and ignoring overhead.
4. A device stores 4 GB. Is this a capacity or a transfer rate? What unit clue supports your answer?
5. A processor has a 64-bit word size. How many bytes are in one word?

Solutions

1. $7 \times 8 = 56$ bits.
2. $5 \times 1024 = 5120$ bytes.
3. $24/8 = 3$ MB/s before protocol overhead.
4. It is capacity: GB is a data amount, and there is no “per second” time component.
5. $64/8 = 8$ bytes per word.

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LESSON 49

Computer Generations & History

49.1 What a computer generation represents

Computer generations are conventional historical groupings based on the dominant hardware technology or design development associated with a period. They help organize the major changes in computer hardware, programming, and use. They are not exact universal year boundaries: a date range can differ between accounts, so the technology association is the safer cue for an objective question.

Definition — Computer generation

A conventional historical grouping of computers associated with a dominant technology or design development.

Term	Meaning in this lesson
Vacuum tube	An early electronic component used in first-generation computers; these systems were large and produced substantial heat.
Transistor	A smaller electronic component that replaced vacuum tubes in the conventional second-generation account.
Integrated circuit (IC)	A compact circuit incorporating multiple electronic components; the key third-generation association.
Microprocessor	A processor implemented as an integrated chip; the key fourth-generation association.
VLSI	Very Large Scale Integration: integrating a very large number of circuit components on a chip.
Machine language	Instructions expressed in the low-level codes directly used by the machine.
Assembly language	A symbolic, low-level notation for machine operations; it must be translated before execution.
High-level language	A programming language using more abstract statements than machine code or assembly.
Operating system	System software that manages the computer's operation and provides services to programs and users.
Multiprogramming	Keeping multiple programs available so the system can organize processor use while one program is waiting.

The language and system-software terms help explain what changed beyond the hardware labels. "Machine language" identifies the first generation's programming cue. Assembly and early high-level languages appear with the second-generation association. Operating systems and multiprogramming develop with the third-generation systems described here. These terms are not alternate names for the hardware generation; they are developments associated with it.

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49.2 The first four technology associations

Generation	Main association	Exam cue
First	Vacuum tubes	Large, heat-producing systems; machine-language programming.
Second	Transistors	Smaller and more reliable; assembly and early high-level languages.
Third	Integrated circuits (ICs)	Multiple electronic components integrated in compact circuits; smaller, more reliable systems; operating systems and multiprogramming develop.
Fourth	Microprocessors	VLSI integration; personal computers become widespread.

The progression is a change in how electronic components are built and combined. Vacuum tubes made early systems large and heat-producing. Transistors replaced those tubes and supported smaller, more reliable systems. Integrated circuits placed multiple components together in a compact circuit, continuing the trend toward reduced size and improved reliability. In the fourth-generation association, a microprocessor brings processor functions together on a chip; VLSI integration and the spread of personal computers are the accompanying cues.

The sequence is the important memory structure:

$$\text{vacuum tubes} \longrightarrow \text{transistors} \longrightarrow \text{integrated circuits} \longrightarrow \text{microprocessors.}$$

The most common near-neighbour error is to switch the third and fourth generations. An integrated circuit is the third-generation cue; the microprocessor is the fourth-generation cue.

Example 49.1. *An item asks which generation is associated with integrated circuits. Trace the sequence: vacuum tubes are first, transistors second, ICs third, and microprocessors fourth. Therefore the answer is the third generation. If the stem instead names a microprocessor, the conventional answer is fourth.*

49.3 The fifth-generation association

The conventional fifth-generation association commonly includes Artificial Intelligence (AI), parallel processing, and natural-language systems. This is a broad category, not a claim that every modern computer has one universally agreed start date or a single defining design. AI applications can run on systems that are not otherwise classified by one precise "fifth-generation" boundary. For an exam stem that explicitly uses this generation scheme, match the conventional association and avoid treating the label as a sharply dated technical standard.

Example 49.1. *If the stem asks which broad generation is commonly linked to AI and parallel processing in the conventional classification, select the fifth. If it asks for the microprocessor generation, select the fourth instead; "modern" alone is not a sufficient clue to erase the distinction.*

49.4 India's PARAM 8000 milestone

For the named national milestone, C-DAC's official high-performance computing history identifies: C-DAC, HPC Supercomputing Systems.

Definition — C-DAC

Centre for Development of Advanced Computing.

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Definition — PARAM 8000

An Indian supercomputer developed by C-DAC and unveiled in 1991; C-DAC identifies it as India's first supercomputer.

Keep the three parts of this fact separate. C-DAC is the organization; PARAM 8000 is the system; 1991 is the unveiling year. This milestone belongs to India's high-performance-computing history. It is not a generation label, and it should not be used to redefine the conventional technology sequence.

Example 49.1. *For a matching item with "Centre for Development of Advanced Computing," select C-DAC. For a prompt asking for India's first supercomputer, select PARAM 8000. If the prompt asks when the system was unveiled, use 1991. These answers refer to related but different parts of the same fact.*

49.5 Exam retrieval and common confusions

Use two retrieval directions. Given the technology, recall the generation; given the generation, recall the technology. Then isolate named milestones that sit outside the generation sequence.

- Vacuum tubes ↔ first generation.
- Transistors ↔ second generation.
- Integrated circuits ↔ third generation.
- Microprocessors ↔ fourth generation.
- AI and parallel processing ↔ common fifth-generation description.
- C-DAC / PARAM 8000 / 1991 ↔ separate Indian supercomputing milestone.

Three distinctions prevent most errors in this topic:

1. Do not swap the adjacent third- and fourth-generation cues: ICs are third; microprocessors are fourth.
2. Do not treat "fifth generation" as a single precisely dated hardware boundary. AI, parallel processing, and natural-language systems are the conventional association used in this lesson.
3. Do not mix a generation label with the PARAM 8000 fact. The latter identifies a named Indian supercomputer, its developer, and its unveiling year.

Exercises

1. Match vacuum tubes, transistors, integrated circuits, and microprocessors to the first four generations.
2. Explain why the presence of an IC inside a modern microprocessor does not change the conventional microprocessor/fourth-generation association.
3. Expand C-DAC and identify PARAM 8000's significance.
4. Why should fifth-generation date boundaries be treated cautiously?
5. Distinguish a generation milestone from the PARAM 8000 milestone.

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Solutions

1. First: vacuum tubes; second: transistors; third: integrated circuits; fourth: microprocessors.
2. The generation scheme uses the dominant historical milestone as its label. A later microprocessor can contain ICs without changing the conventional fourth-generation association.
3. C-DAC is Centre for Development of Advanced Computing. PARAM 8000 was developed by C-DAC and is identified as India's first supercomputer.
4. Generation boundaries are broad historical groupings and do not have one universally fixed transition date; technology associations are the more stable recall cue.
5. A generation milestone classifies a historical technology period. PARAM 8000 is a named Indian supercomputer and national HPC milestone, not one of the generation labels.

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LESSON 50

Data Representation, Number Systems & Logic

50.1 Bases and positional notation

Digital circuits commonly use two reliably distinguishable states, written as 0 and 1. A number system specifies a base (also called a radix) and the digits allowed in each position. The base determines the place values.

Definition — Number-system base

The number of distinct digit symbols in a positional number system, with place values given by powers of that base.

Name	Base	Allowed digits	Example
Binary	2	0–1	1011_2
Octal	8	0–7	27_8
Decimal	10	0–9	45_{10}
Hexadecimal	16	0–9, A–F	$2F_{16}$

The subscript identifies the base. In hexadecimal, A represents 10, B 11, C 12, D 13, E 14, and F 15. A digit that is not allowed in the stated base makes the numeral invalid: for example, digit 8 cannot appear in an octal number.

To test whether a numeral is valid, compare every digit with the allowed range for its base. Binary permits only 0 and 1; octal permits 0 through 7; decimal permits 0 through 9; hexadecimal permits 0 through 9 and A through F. A numeral can be converted only after it passes this check. For example, 128_8 is not a valid octal numeral because it contains the digit 8, even though the same characters could be interpreted under a different base.

Definition — Binary

A base-2 positional number system using digits 0 and 1.

Definition — Octal

A base-8 positional number system using digits 0 through 7.

Definition — Decimal

A base-10 positional number system using digits 0 through 9.

Definition — Hexadecimal

A base-16 positional number system using digits 0 through 9 and letters A through F, where A represents 10 and F represents 15.

Each position contributes its digit multiplied by a power of the base. For 1011_2, the rightmost position is 2^0, the next is 2^1, then 2^2, and then 2^3:

$$1011_2 = 1(2^3) + 0(2^2) + 1(2^1) + 1(2^0) = 8 + 0 + 2 + 1 = 11_{10}.$$

The rightmost position is the least-significant position because it has the smallest power.

Digit	1	0	1	1
Position from right	3	2	1	0
Place value	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
Contribution	8	0	2	1

The place-value method works in every base. Multiply each digit by the power assigned to its position, then add the contributions. The rightmost digit always has power zero; moving one place left increases the exponent by one. This is why leading zeros do not change a value: each added zero contributes zero at a new, higher place.

50.2 Converting decimal and binary

To convert a positive decimal integer to binary, divide repeatedly by two. At each step, keep the remainder, which must be either zero or one. Stop when the quotient becomes zero, then read the remainders from the last division upward. Reading them in the order produced gives the wrong direction.

Use this sequence every time:

1. Divide the current integer by 2.
2. Record the remainder, either 0 or 1.
3. Replace the current integer with the quotient and repeat until the quotient is 0.
4. Read the recorded remainders from last to first.
5. Verify the result by adding its binary place-value contributions.

Example 50.1. *Convert* 45_{10} *to binary:*

$$\begin{aligned} 45 &= 2(22) + 1 \\ 22 &= 2(11) + 0 \\ 11 &= 2(5) + 1 \\ 5 &= 2(2) + 1 \\ 2 &= 2(1) + 0 \\ 1 &= 2(0) + 1 \end{aligned}$$

The remainders from top to bottom are $1, 0, 1, 1, 0, 1$. *Reading bottom to top gives* 101101_2. *Check by place value:* $1(32) + 0(16) + 1(8) + 1(4) + 0(2) + 1(1) = 45$, *so* $45_{10} = 101101_2$.

Example 50.2. *Convert* 13_{10} *to binary. Repeated division gives* $13 = 2(6) + 1$, $6 = 2(3) + 0$, $3 = 2(1) + 1$, *and* $1 = 2(0) + 1$. *Reading the remainders upward gives* 1101_2. *Check:* $1(8) + 1(4) + 0(2) + 1(1) = 13$.

50.3 Binary grouping for octal and hexadecimal

Three binary bits encode one octal digit because $2^3 = 8$. Four binary bits encode one hexadecimal digit because $2^4 = 16$. To convert a binary numeral, start at the right and group bits into threes for octal or fours for hexadecimal. If the leftmost group is short, add leading zeros; leading zeros do not change the value.

Example 50.1. *Convert* 1111111_2 *to octal. Group from the right:* 001 111 111. *The groups are* $001_2 = 1_8$, $111_2 = 7_8$, *and* $111_2 = 7_8$, *so the answer is* 177_8. *Check:* $1(8^2)+7(8)+7 = 64+56+7 = 127$, *the same value as* 1111111_2.

Example 50.2. *Convert* 1111111_2 *to hexadecimal. Group from the right:* 0111 1111. *These groups are* 7 *and* 15, *and hexadecimal digit 15 is F. Therefore the answer is* $7F_{16}$. *Check:* $7(16) + 15 = 112 + 15 = 127$.

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Example 50.3. *Convert $2F_{16}$ to decimal. The hexadecimal digit F is 15:*

$$2F_{16} = 2(16^1) + 15(16^0) = 32 + 15 = 47_{10}.$$

50.4 Character codes, Unicode, and decimal digits

Definition — ASCII

A character-encoding standard whose original character set uses 7-bit codes.

Definition — Unicode

A standard that assigns code points to characters across many writing systems.

Definition — UTF-8

A variable-length encoding form for Unicode code points.

Definition — Binary-Coded Decimal (BCD)

A representation in which each decimal digit is encoded separately using a 4-bit code.

Representation	What it identifies or encodes	Key property
ASCII	Characters	Original ASCII uses 7-bit codes
Unicode	Characters from many writing systems	Assigns a code point to a character
UTF-8	Unicode code points as encoded bytes	Variable-length encoding
BCD	Decimal digits	Uses a separate 4-bit code for each digit

ASCII and Unicode concern character representation. Unicode code points identify characters; UTF-8 is one way to encode those code points as bytes. The code point and its byte encoding are different things: a character's Unicode identity does not by itself state how many bytes a particular encoding uses. UTF-8 is variable-length, so a character is not always stored using one byte. BCD has a different purpose: it represents each decimal digit separately using four bits. It is not the same as converting the entire multi-digit decimal value to binary.

Example 50.1. *Represent decimal 59 in BCD. Encode the digit 5 as 0101 and the digit 9 as 1001, giving 0101 1001. Ordinary binary for the numeric value 59 is 111011_2. These bit strings differ because BCD preserves the two decimal digits separately, while ordinary binary represents the value 59 as one base-2 numeral.*

The grouping explains why BCD and ordinary binary use different numbers of bits. BCD uses one 4-bit group for each decimal digit, so a two-digit decimal value uses two groups. Ordinary binary instead places the entire quantity in powers of 2. When an item asks for BCD, split the decimal number into its digits before encoding; do not convert the whole number directly to binary.

50.5 RGB colour representation

Definition — RGB

A colour model representing colour through red, green, and blue channel values.

In a common 24-bit RGB representation, each of the three channels uses 8 bits. Therefore the three channels require $8 + 8 + 8 = 24$ bits per pixel. Eight bits can encode $2^8 = 256$ possible values per channel, from 0 through 255. The total number of possible channel combinations in that representation is $256^3 = 16{,}777{,}216$. This does not mean every image format or display uses exactly 24-bit RGB; other bit depths, alpha channels, palettes, or colour spaces are possible.

The channel count and the bit count are separate quantities. RGB has three channels: red, green, and blue. In the common representation described above, each channel has 8 bits, so one pixel uses 24 bits total. A value within one channel ranges from 0 to 255 because an 8-bit field has $2^8 = 256$ possible patterns. To count all channel combinations, multiply the number of choices for each independent channel:

$$256 \times 256 \times 256 = 256^3 = 16{,}777{,}216.$$

This count follows only under the stated 8-bits-per-channel assumption.

Example 50.1. *In common 24-bit RGB, the red, green, and blue channels each use 8 bits. Adding the channel widths gives* $8 + 8 + 8 = 24$ *bits per pixel.*

50.6 Logic gates and truth tables

Definition — AND gate

A logic operation that outputs 1 only when every input is 1.

Definition — OR gate

A logic operation that outputs 1 when at least one input is 1.

Definition — XOR gate

A logic operation that outputs 1 when its two inputs differ.

Definition — NAND gate

The negation of AND: it outputs 0 only when every input is 1.

Definition — NOT gate

A one-input logic operation that reverses its input: 0 becomes 1 and 1 becomes 0.

A	B	AND	OR	XOR	NAND	NOT A
0	0	0	0	0	1	1
0	1	0	1	1	1	1
1	0	0	1	1	1	0
1	1	1	1	0	0	0

Read the truth table from the rule, not by memorizing an unlabelled row. AND requires both inputs to be 1. OR needs at least one 1. XOR checks whether the inputs differ. NAND first evaluates AND and then reverses that output. NOT has only one input and reverses it.

For two-input gates there are exactly four possible input pairs: 00, 01, 10, and 11. Evaluate every pair using the gate's rule:

- AND is 1 only on the 11 row.
- OR is 0 only on the 00 row.

- XOR is 1 on 01 and 10, where the inputs differ.
- NAND is the opposite of AND on every row.
- NOT does not use B; it reverses A in each row.

OR and XOR are easy to confuse. For input $A = 1$, $B = 1$, OR is 1 because at least one input is 1, but XOR is 0 because the two inputs are equal. NAND is checked only after evaluating AND and then taking the opposite result.

Example 50.1. *For $A = 1$ and $B = 0$, AND is 0 because both inputs are not 1; OR is 1 because one input is 1; XOR is 1 because the values differ; and NAND is 1 because it reverses the AND result. NOT(A) is 0.*

Example 50.2. *For $A = 1$ and $B = 1$, AND is 1 and OR is 1. XOR is 0 because the inputs do not differ. NAND is the opposite of AND, so it is 0. NOT(A) is 0.*

Exercises

1. Convert 29_{10} to binary and verify the result by place value.
2. Convert 101101_2 to octal and hexadecimal.
3. Explain the difference between BCD for decimal 25 and ordinary binary for the value 25.
4. In common 24-bit RGB, how many bits are assigned to each channel? Is that rule universal for all images?
5. For $A = 0$, $B = 1$ and for $A = 1$, $B = 1$, find AND, OR, XOR, NAND, and NOT(A).

Solutions

1. Repeated division gives $29 = 2(14) + 1$, $14 = 2(7) + 0$, $7 = 2(3) + 1$, $3 = 2(1) + 1$, and $1 = 2(0) + 1$. Reading upward gives 11101_2. Check: $16 + 8 + 4 + 0 + 1 = 29$.
2. Group from the right: $101\ 101$, so the octal result is 55_8. Pad and group by four: $0010\ 1101$, so the hexadecimal result is $2D_{16}$.
3. BCD for 25 is $0010\ 0101$, encoding 2 and 5 separately. Ordinary binary for the value 25 is 11001_2.
4. Eight bits per channel, since $8 + 8 + 8 = 24$. No; this describes a common 24-bit representation, not every image or display.
5. For $A = 0$, $B = 1$: AND=0, OR=1, XOR=1, NAND=1, and NOT(A)=1. For $A = 1$, $B = 1$: AND=1, OR=1, XOR=0, NAND=0, and NOT(A)=0.

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UNIT 2

CPU, Hardware & Architecture

LESSON 04

Organization of a Computer

Video lesson: https://youtu.be/92XQ_n1t_cc

04.1 The functional units of a computer

Start with the office desktop. A clerk types marks on the keyboard, the CPU processes them through a spreadsheet formula, the monitor and printer present the result, and the file is kept on the SSD for later use. This single machine contains every functional unit that the exam asks you to name.

Definition — Functional unit

A major part of a computer that performs one broad job — accepting input, producing output, holding data, performing operations, or controlling the machine.

Unit	Job in this lesson
Input unit	Accepts data and instructions from the outside world.
Output unit	Presents the result to the user.
Memory (primary storage)	Holds data and instructions currently in use.
Arithmetic Logic Unit (ALU)	Performs arithmetic and logical operations.
Control unit	Directs the sequence of operations and the flow of data.

The memory unit described here is *primary storage*, that is, main memory. Secondary storage such as a hard disk or an SSD is a separate idea and is treated in Section 4.

04.2 The CPU: ALU, control unit and registers

The **Central Processing Unit (CPU)** is the processing centre of the computer. It is built from three parts: the arithmetic logic unit, the control unit, and registers.

Definition — Central Processing Unit (CPU)

The unit that fetches, decodes and executes instructions; it is formed by the ALU, the control unit and the registers.

Definition — Arithmetic Logic Unit (ALU)

The part of the CPU that performs arithmetic operations (such as addition and subtraction) and logical operations (such as comparison) on binary data.

Definition — Control unit

The part of the CPU that fetches and decodes instructions, issues control signals, and coordinates the operation of all other units.

Definition — Register

A small, very fast storage location inside the CPU used to hold data, addresses or control information during processing.

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The ALU does the calculating, but it does not decide what to do. The control unit decides the sequence of operations and the flow of data, and then tells the ALU and the other units when to act. Registers are the CPU's own working storage; they are not the same as main memory or secondary storage.

Example 04.1. *In the marks spreadsheet, the formula that adds three marks is carried out by the ALU. The control unit has already fetched the formula's instruction from memory, decoded it, and signalled the ALU to act. The intermediate values are kept in registers while the operation runs. The final total is then written back to memory and later shown on the monitor.*

04.3 Input and output units

The input unit brings data and instructions into the system. The output unit presents the result.

Input devices	Output devices
Keyboard, mouse, scanner, microphone, light pen, digitizer	Monitor, printer, plotter, speaker, sound-card output

A device is classified by the role it plays in the described task, not by its shape. A monitor is normally an output device because it presents results. A scanner is an input device because it brings data in. When a question lists devices, mark each one as input or output before choosing the option.

04.4 Memory (primary storage) versus secondary storage

This is a high-value contrast.

Definition — Primary storage (main memory)

The memory the CPU accesses directly while processing; it holds the programs and data currently in use. Random Access Memory (RAM) is volatile; Read Only Memory (ROM) is non-volatile.

Definition — Secondary storage

Storage that keeps data and files permanently but is not accessed directly by the CPU during processing; examples are the hard disk, the Solid State Drive (SSD) and the pen drive.

The key rule for the exam is that the **CPU works with main memory, not with secondary storage**. When a program on the hard disk is run, it is first loaded into main memory; the CPU then works from main memory. Secondary storage is reached through main memory and the operating system.

Example 04.1. *The spreadsheet file is saved on the SSD. When the clerk opens it, the file is first loaded into RAM. All the additions and comparisons are then performed on data held in RAM and in the CPU registers. Writing the result back to the SSD happens later, through the operating system. At no point does the ALU process the file straight from the SSD.*

04.5 The system bus: address, data and control

The **system bus** is the set of wires that carries information between the CPU, memory and the other units. It has three parts, and their directions differ — this is a common trap.

Bus	Direction	Role
Address bus	CPU → memory	Carries the address of the memory location the CPU wants to use.
Data bus	Bidirectional	Carries the actual data and instructions between the CPU and memory.
Control bus	CPU → units	Carries control and timing signals, such as read and write.

Definition — System bus

The address bus, data bus and control bus taken together; they connect the CPU with memory and the input/output units.

Remember the three questions: the address bus answers *where*, the data bus answers *what*, and the control bus answers *when and how*. The address bus is one-way (from the CPU outwards) and carries the address, not the contents of the location. The data bus is two-way. The control bus carries signals, not data or addresses.

Example 04.1. *The CPU wants the value stored at memory location 200. It places the address 200 on the address bus and sends a read signal on the control bus. The memory puts the contents of location 200 onto the data bus, and the CPU receives it. The address travelled out on the address bus; the data travelled back on the data bus.*

04.6 The motherboard, expansion slots and interfaces

The **motherboard** is the main circuit board that connects and links the CPU, main memory and the buses. Expansion slots on the motherboard allow extra cards to be added, such as a graphics card or a network card. Interfaces and ports connect external devices — keyboard, printer, monitor — to the system. The system bus runs along the motherboard between the CPU and the other units, so the motherboard is the physical platform on which the units communicate.

04.7 The stored-program (von Neumann) idea

In early machines, changing the task often meant rewiring the machine. The **stored-program concept** changed this.

Definition — Stored-program concept (von Neumann architecture)

The design in which a program's instructions are stored in main memory together with the data, so the CPU can fetch and execute them without rewiring the machine.

Because instructions and data share the same memory, the CPU fetches an instruction from memory, executes it, and moves on to the next. This repeating pattern is the instruction cycle, described next.

04.8 The instruction cycle

The **instruction cycle** is the sequence the CPU follows for each instruction.

1. **Fetch**: the CPU obtains the next instruction from main memory.
2. **Decode**: the control unit works out what the instruction means.
3. **Execute**: the ALU or another unit performs the operation.
4. **Store** (when required): the result is written back to memory or to a register.

The order is fixed: fetch, then decode, then execute. The optional store step writes the result back. Data and instructions move between the units over the system bus, and the whole cycle repeats for the next instruction.

Example 04.1. *Consider an instruction "add the value in memory to a register". During fetch, the CPU gets the instruction from memory. During decode, the control unit identifies it as an addition and signals the ALU. During execute, the ALU performs the addition. The updated value is then stored back in the register or in memory, and the cycle begins again.*

Three distinctions prevent most errors in this topic:

1. The CPU accesses main memory directly, not secondary storage.
2. Address, data and control buses are not interchangeable: the address bus is one-way and carries addresses; the data bus is two-way and carries data and instructions; the control bus carries control signals.
3. Memory (primary) holds current work; storage (secondary) keeps files permanently.

Exercises

1. Name the five functional units of a computer and state one job of each.
2. What are the three parts of the CPU, and what does each do?
3. State the direction and role of the address bus, the data bus and the control bus.
4. A file is opened from a hard disk and edited. Explain why the CPU does not process the file directly from the hard disk.
5. List the steps of the instruction cycle in order.

Solutions

1. Input unit (accepts data and instructions); output unit (presents results); memory/primary storage (holds data and instructions in use); ALU (performs arithmetic and logical operations); control unit (directs operations and data flow).
2. The ALU, the control unit and the registers. The ALU computes; the control unit fetches, decodes and signals; the registers hold data and addresses during processing.
3. Address bus: one-way from CPU to memory, carrying the address of the location. Data bus: bidirectional, carrying data and instructions. Control bus: carries control and timing signals from the CPU to other units.
4. The CPU accesses only main memory directly. The file is first loaded into main memory, the CPU processes the copy in main memory and registers, and the operating system writes the changes back to the hard disk.
5. Fetch, decode, execute (and store when a result must be written back).

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LESSON 05

CPU, ALU, Control Unit & Registers

Video lesson: https://youtu.be/b6w8uT4GBYQ

05.1 The CPU and its functional units

The Central Processing Unit (CPU) is the part of the computer that executes instructions. It repeatedly fetches an instruction from main memory, decodes it, and then executes it. To do this work the CPU is organised into three functional units: the Arithmetic Logic Unit (ALU), the Control Unit (CU), and the registers. Data and control signals move between these units over internal buses. The CPU works directly with main memory (RAM); it does not reach into secondary storage such as a hard disk by itself.

Definition — CPU (Central Processing Unit)

The processor that fetches, decodes, and executes instructions, made up of the ALU, the Control Unit, and registers.

A simple way to keep the roles apart is this: the ALU computes, the Control Unit coordinates, and the registers hold the working values. Almost every objective question about this topic turns on not swapping those three roles.

05.2 The Arithmetic Logic Unit (ALU)

The ALU is the computing part of the CPU. It performs two broad families of operations. **Arithmetic** operations include addition, subtraction, increment, decrement, multiplication, and division. **Logical** operations include AND, OR, NOT, XOR, shifts, and comparisons. The ALU operates on data held in binary form, and its operations are carried out with logic-gate operations on those binary values. When the ALU finishes, it returns its result to a register or the accumulator; it does not keep a permanent copy.

Definition — ALU (Arithmetic Logic Unit)

The CPU unit that performs arithmetic and logical operations on binary data using logic gates.

Example 05.1. *For inputs* 1 *and* 1*, an addition produces the binary result for two, and a logical XOR produces* 0 *because the two bits are equal. Both results are produced by the same unit — the ALU — and handed back for further use.*

05.3 The Control Unit (CU)

The Control Unit directs and sequences the operations of the computer. It decodes each instruction and then issues the control signals that time and trigger the other units. It also controls the flow of data between the ALU, the registers, and memory. What the Control Unit does **not** do is compute: it does not add, subtract, or carry out logical operations. Those are the ALU's job. Students frequently write "the Control Unit performs the calculations," which is exactly the error to avoid.

Definition — Control Unit (CU)

The CPU unit that directs and sequences operations, decodes instructions, and issues control signals; it does not perform arithmetic or logic.

Example 05.1. *When an addition instruction is executed, the Control Unit fetches and decodes the instruction and signals the ALU to add the two values. The ALU performs the addition; the Control Unit only directs the sequence.*

05.4 Registers

Registers are the smallest and fastest storage in a computer. They are located inside the CPU, so they are faster than cache and far faster than main memory. Because they are very small, they hold only the values needed for the work currently in progress: instructions, addresses, data, and intermediate results. The following table lists the key registers and their purposes.

Register	Purpose
Program Counter (PC / IP)	Holds the address of the **next** instruction to be fetched.
Instruction Register (IR)	Holds the instruction currently being decoded or executed.
Memory Address Register (MAR)	Holds the address of the memory location to be accessed.
Memory Data Register (MDR)	Holds the data being transferred to or from memory.
Accumulator (ACC)	Holds intermediate arithmetic and logic results.
General-purpose registers	Hold temporary operands and results.
Status / flags register	Holds condition flags such as zero, carry, sign, and overflow.

The Program Counter is the register most often confused. It holds the address of the **next** instruction to fetch, not the address of the instruction currently being executed. The instruction currently being decoded or executed is held in the Instruction Register. Memory access is split between two registers: the MAR carries the address, and the MDR carries the data itself. The accumulator is the natural home for an ALU result that will be used again immediately.

Example 05.1. *Suppose the CPU is about to fetch the instruction at address 200. The Program Counter holds 200. Once the instruction is fetched, it is placed in the Instruction Register for decoding. If the instruction needs to read a value stored at address 900, the MAR holds 900 and the MDR holds the value brought back from memory.*

05.5 Clock speed, cycles, cores, and threads

The CPU's operations are paced by a clock. **Clock speed** is measured in hertz (Hz); one gigahertz (GHz) is one billion clock cycles per second. A **clock cycle** is a single tick of that clock. Do not assume every instruction takes exactly one cycle: different instructions can require different numbers of clock cycles. A **core** is an independent processing unit within the CPU; a quad-core chip contains four cores. A **thread** is a stream of instructions, and a system can run multiple threads across the available cores.

Definition — Clock cycle

One tick of the system clock; the basic unit of timing for CPU operations.

Example 05.1. *The office desktop in our running example is a quad-core machine running at 2.4 GHz. It has four independent cores, and each core can complete clock cycles at about 2.4 billion per second. Running a spreadsheet calculation and a browser at the same time lets different threads keep the cores busy.*

05.6 Exam retrieval and common confusions

Three distinctions prevent most errors on this topic.

1. Do not swap the ALU and the Control Unit. The ALU performs arithmetic and logic; the Control Unit only directs operations and issues control signals.
2. Do not say the Program Counter holds the current instruction. It holds the address of the **next** instruction; the current instruction is in the Instruction Register.

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3. Do not place registers below cache or RAM in speed. Registers are the fastest storage, and the accumulator is where intermediate arithmetic and logic results are kept.

Exercises

1. State the three main functional units of the CPU and one role of each.
2. Expand ALU, CU, PC, IR, MAR, and MDR.
3. Does the Control Unit perform arithmetic? Explain in one line.
4. Which register holds the address of the next instruction, and which register holds the instruction currently being executed?
5. Why are registers described as the fastest storage, and what does the accumulator hold?

Solutions

1. ALU — performs arithmetic and logical operations on binary data; Control Unit — directs and sequences operations and issues control signals; registers — hold instructions, addresses, data, and intermediate results.
2. ALU = Arithmetic Logic Unit; CU = Control Unit; PC = Program Counter; IR = Instruction Register; MAR = Memory Address Register; MDR = Memory Data Register.
3. No. The Control Unit directs and coordinates operations; the ALU performs the arithmetic and logical calculations.
4. The Program Counter (PC/IP) holds the address of the next instruction; the Instruction Register (IR) holds the instruction currently being decoded or executed.
5. Registers are located inside the CPU and are smaller and faster than cache and RAM. The accumulator (ACC) holds intermediate arithmetic and logic results.

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LESSON 06

CPU vs GPU & Microprocessor

Video lesson: https://youtu.be/Vaod5Ox7lzM

06.1 Serial and parallel processing

Serial processing means that one task is carried out at a time, in sequence: the first operation finishes before the next begins. **Parallel processing** means that several tasks or operations proceed at the same time. A serial workload rewards a processor with a few very fast units; a parallel workload rewards a processor with many units that can work simultaneously.

Definition — Serial processing

Carrying out one task at a time, in sequence, so that each step completes before the next starts.

Definition — Parallel processing

Carrying out several tasks or operations at the same time.

The difference matters because it explains why two processors can both be described as "fast" while being designed for opposite jobs. A processor that excels at finishing a single chain of instructions quickly is a latency-optimised design. A processor that excels at completing a very large number of small independent operations per second is a throughput-optimised design.

06.2 CPU versus GPU

The CPU (Central Processing Unit) is the general-purpose processor of a computer. It has **few, powerful cores**, and each core is capable of running a sequential instruction stream quickly. Its design goal is **low latency**: to complete one task with the least possible delay. In the office-desktop running example (EX-01), the CPU executes the operating system, MS Word, and spreadsheet logic step by step.

The GPU (Graphics Processing Unit) is a processor built around **many simpler cores**. Each core does a small amount of work, and the design goal is **high throughput**: a very large number of independent operations per second. This makes the GPU a natural fit for pixels and graphics, and also for scientific and machine-learning workloads that can be expressed as many parallel calculations.

Feature	CPU	GPU
Cores	Few, powerful	Many, simple
Design goal	Low latency	High throughput
Best suited to	Sequential, general-purpose work	Parallel work (pixels, ML)
Core type	General-purpose	Specialised, simpler
Example use	EX-01 desktop logic	Image and graphics operations

Example 06.1. *An item asks which processor is better suited to an image-processing task that repeats the same operation on millions of pixels. Because those pixel operations are independent and can run in parallel, the GPU — with its many simple cores — is the better fit. The CPU's few powerful cores would each handle the chain of pixels one after another.*

The most common near-neighbour error is to treat GPU cores as if they were interchangeable general-purpose CPU cores. They are not. GPU cores are simpler and optimised for parallel throughput, so they are not a full replacement for the CPU's general-purpose cores.

06.3 Cores, threads, and hyper-threading

A **core** is a physical processing unit. A quad-core CPU contains four such units. A **thread** is a stream of instructions that a core executes. **Hyper-threading** lets a single physical core present two logical threads, so the core can stay busy while one thread is waiting.

Two logical threads on one core are not two physical cores. A quad-core processor with hyper-threading may report eight logical threads, but it still has four physical cores. Clock speed and core count together, not thread count alone, determine raw capacity.

Definition — Hyper-threading

A technique in which one physical core presents two logical threads to the operating system, improving utilisation while a thread waits.

06.4 Cache depth: L1, L2, and L3

Cache is small, fast memory held close to the core. It is organised in levels. **L1** is the smallest and fastest level and sits nearest the core. **L2** is larger and a little slower than L1. **L3** is the largest and slowest of the three and is typically shared across cores. The order of speed and closeness is L1, then L2, then L3, then main memory.

Level	Relative size	Relative speed and location
L1	Smallest	Fastest; closest to the core
L2	Larger	Slower than L1
L3	Largest	Slowest of the three; often shared across cores

06.5 Clock speed and multi-core

Clock speed is measured in **GHz** (gigahertz), which counts billions of cycles per second. A higher clock speed generally means more work completed per second, all else being equal. A **multi-core** processor places several cores on one chip, which enables parallel work. However, clock speed alone never decides real performance: the number of cores, the cache size, the amount of memory, and the input/output speed all contribute.

06.6 Microprocessor versus microcontroller

A **microprocessor** is a central processing unit implemented on a single chip. It is a general-purpose processor, and software decides what it does. On its own it still needs external memory and input/output devices to form a complete working system. The EX-01 office desktop is built around a microprocessor.

A **microcontroller** integrates more on the same chip: a microprocessor **plus** memory **plus** input/output peripherals. It therefore forms a near-complete computer on one chip and is used for **embedded/special-purpose** control, for example in washing machines, remote controls, and appliance or vehicle controllers.

Definition — Microprocessor

A central processing unit implemented on a single chip; a general-purpose device that needs external memory and I/O.

Definition — Microcontroller

A microprocessor together with integrated memory and input/output peripherals on one chip, used in embedded and special-purpose systems.

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Feature	Microprocessor	Microcontroller
Contents	CPU on a chip	CPU + memory + I/O on a chip
Purpose	General-purpose	Embedded / special-purpose
System	Needs external parts	Self-contained on one chip
Example	EX-01 office desktop	Appliance controller

The key difference is the **level of integration**, not clock speed. A microcontroller is not "just a microprocessor": it is a more integrated device that already includes memory and I/O on the same chip.

06.7 Diagnosing a "slow" computer (EX-01)

The office desktop in this running example is a quad-core 2.4 GHz machine with 16 GB of RAM and an SSD. Suppose one particular program still runs slowly. A common but incorrect conclusion is that the CPU must be slow, because "more cores should mean faster". More cores do not help a single-threaded program, which uses only one core at a time. The real cause is usually a **bottleneck**: memory pressure, disk input/output, or software that is not multithreaded.

Example 06.1. *A question describes a quad-core 2.4 GHz desktop with 16 GB RAM and an SSD, and asks why one application is slow. The best answer is to diagnose a stated bottleneck — a single-threaded program, insufficient memory, or slow disk input/output — rather than to declare the CPU slow. A slow program is not automatically a "slow CPU."*

06.8 Exam retrieval and common confusions

Use two retrieval directions. Given the processor, recall its core profile; given the workload, recall the better processor.

- CPU ↔ few powerful general-purpose cores, low latency.
- GPU ↔ many simple cores, high throughput, parallel.
- Serial ↔ one task at a time in sequence.
- Parallel ↔ several tasks or operations at once.
- Microprocessor ↔ CPU on a single chip, general-purpose.
- Microcontroller ↔ microprocessor + memory + I/O on one chip, embedded.
- Cache order ↔ L1 (smallest, fastest) to L3 (largest, slowest).

Three distinctions prevent most errors in this topic:

1. Do not treat GPU cores as general-purpose CPU cores. They are simpler and optimised for parallel throughput.
2. Do not confuse serial with parallel. Serial is sequential, one task at a time; parallel is simultaneous.
3. Do not attribute a slow program on a capable machine automatically to a slow CPU. Check the multithreading, memory, and disk-I/O bottleneck; and do not confuse a microprocessor (CPU on a chip) with a microcontroller (CPU + memory + I/O on one chip).

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Exercises

1. Distinguish serial processing from parallel processing.
2. State one design goal of the CPU and one of the GPU, and give the core profile of each.
3. Explain the difference between a microprocessor and a microcontroller.
4. What does hyper-threading allow, and why is it not the same as adding a physical core?
5. A quad-core 2.4 GHz desktop with 16 GB RAM and an SSD runs one program slowly. Give the most appropriate diagnosis and say why.

Solutions

1. Serial processing performs one task at a time in sequence; parallel processing performs several tasks or operations at the same time.
2. The CPU is optimised for low latency and has few powerful general-purpose cores. The GPU is optimised for high throughput and has many simpler cores for parallel work.
3. A microprocessor is a CPU on a single chip and is general-purpose, needing external memory and I/O. A microcontroller adds memory and I/O peripherals on the same chip, forming an embedded, special-purpose system; the difference is the level of integration.
4. Hyper-threading lets one physical core present two logical threads, so it can stay busy while one thread waits. It does not add a physical core; a quad-core chip with hyper-threading still has four physical cores.
5. The program is likely limited by a bottleneck — a single-threaded workload, insufficient memory, or slow disk input/output — rather than by a slow CPU, since extra cores do not speed up a single-threaded program. A slow program is not automatically a “slow CPU.”

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LESSON 59

Current CPU Architecture, Data Representation & RISC/CISC

59.1 von Neumann architecture and the bottleneck

A **von Neumann** (stored-program) computer keeps both instructions and data in the same memory. The processor fetches an instruction, decodes it, and executes it; when the instruction needs a data operand, that operand is read from the same memory. Because a single shared path carries instructions and data, the processor frequently has to wait for memory. This limitation is called the **von Neumann bottleneck**: the rate at which work can proceed is capped by the rate at which instructions and data can be transferred between the CPU and memory over that shared path. Faster memory, a cache, and wider paths all reduce — but do not remove — the bottleneck.

Definition — von Neumann architecture

A stored-program design in which instructions and data are held together in the same memory and fetched over a shared path between the CPU and memory.

The important contrast is *same memory for instructions and data*. A system that keeps them physically separate is a different (Harvard-style) design and is not the von Neumann model.

59.2 Addressing modes: immediate addressing

An **addressing mode** describes how an instruction states where its operand is. **Immediate addressing** embeds the operand value directly inside the instruction. For example, an instruction "add 5 to the register" carries the value 5 in the instruction encoding itself. Because the value is already present, no extra memory access is required to obtain the operand.

The near-neighbour contrasts matter for exams:

- *Immediate*: the operand is in the instruction.
- *Register*: the operand is in a named register.
- *Direct*: the instruction names a memory address holding the operand.

The single line to remember is: immediate addressing means the operand is a constant carried with the instruction.

59.3 Cache and the cache-hit ratio

A **cache** is a small, fast memory placed between the CPU and main memory. It holds copies of recently used instructions and data. When the CPU asks for an item and the cache has it, that access is a **hit**; if the cache does not have it, the access is a **miss** and the item must come from main memory (and is usually placed in the cache for next time).

Definition — Cache-hit ratio

$$\text{hit ratio} = \frac{\text{number of hits}}{\text{total accesses}} = \frac{\text{hits}}{\text{hits} + \text{misses}}.$$

A **hit** is faster because cache access time is much lower than the access time of main memory. On a hit the CPU avoids the slow main-memory access; on a miss it pays that cost. This is why a higher hit ratio improves average performance.

Example 59.1. *If a program makes 900 cache hits and 100 cache misses, the total accesses are* $900 + 100 = 1000$ *and the hit ratio is* $900/1000 = 0.90$ *(90%).*

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59.4 Data hazards even with register addressing

A **data hazard** is a dependency that prevents an instruction from proceeding correctly in the cycle originally planned. A common case is a read-after-write dependency: an instruction needs the value of a register that an earlier instruction has not yet written. Pipelining is what exposes these dependencies, because several instructions are in progress at once.

Register addressing avoids an extra memory access and removes some hazards caused by memory operands, but it does not eliminate all hazards. Two instructions can still depend on the same register. This is the trap: a candidate may reason that "no memory operand means no hazard," which is false. Hazards arise from dependencies between instructions, whatever kind of operand is used.

59.5 Virtual memory versus cache

Cache and virtual memory are frequently confused, but they solve different problems.

Point	Cache	Virtual memory
Location	Small fast memory next to the CPU.	Uses main memory plus secondary storage as backing store.
Purpose	Reduce average memory access time by keeping recently used blocks close.	Make the memory available to programs appear larger than physical RAM.
Management	Largely hardware.	Operating system with hardware support.

Both involve small "fast" storage for larger "slow" storage, but the cache shortens access time while virtual memory expands the apparent size of memory. They are not the same mechanism and are not interchangeable.

59.6 Instruction cycle: Program Counter and Instruction Register

During the fetch stage the CPU reads an instruction from memory. The **Program Counter (PC)** holds the address of the **next** instruction to be fetched; once the current instruction has been fetched, the PC is advanced to point at the following one. The **Instruction Register (IR)** holds the instruction that is currently being fetched, decoded, and executed.

The exam trap is the PC. It is easy to say the PC holds "the address of the current instruction," but at the moment the instruction is executing, the PC already points at the *next* instruction. The correct statement is: the PC holds the address of the next instruction to be fetched.

59.7 Interrupts are processed at instruction boundaries

An **interrupt** is a signal that causes the CPU to suspend the current program temporarily and transfer control to a handler routine, after which execution usually resumes. Interrupts are recognised at **instruction boundaries** — between one instruction and the next — during the program's execution, not only when the whole program finishes.

The trap is the claim that "an interrupt is processed only after the entire program completes." That is false. Mid-program interrupts are the normal case: the processor finishes the instruction in progress, saves its state, and services the interrupt before continuing.

59.8 Instruction timing: clock cycles vary

Different machine instructions perform different amounts of work. A simple register-to-register operation may complete in a single clock cycle, while a division, a complex operation, or an instruction that must access slow memory can take many cycles. Instruction timing therefore depends on the instruction, its operands, and the memory system.

The trap is the claim that "all machine instructions take the same number of clock cycles." The count varies by instruction; averaging over a program gives an average cycles-per-instruction, not a constant per instruction.

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59.9 Two's complement and overflow detection

In an n-bit **two's-complement** representation the most significant bit is the **sign** bit: 0 for non-negative and 1 for negative values. A negative number is formed by inverting the bits of its magnitude and adding one. With n bits the representable range is -2^{n-1} through $+2^{n-1}-1$.

Overflow occurs when the arithmetic result cannot be represented in the available bits. The reliable test is the **sign** rule:

Definition — Overflow test

*Overflow occurs when two operands of the **same sign** are added and the result has the **opposite sign**. If the two operands have different signs, no overflow occurs.*

Overflow is not the same as a carry out of the most significant bit. A carry can be produced without overflow, and overflow can occur without the final carry telling the full story. Use the sign test, not the carry flag.

Example 59.1 (Overflow). *Work in 8-bit two's complement. Add $+7$ and $+1$:*

$$0111\ (+7)\ +\ 0001\ (+1)\ =\ 1000.$$

The operands are both positive, but the sum 1000 has sign bit 1 and represents -8. Two same-sign operands produced an opposite-sign result, so this is overflow.

Example 59.2 (Carry without overflow). *Again in 8 bits, add -1 and -1:*

$$1111\,1111\ (-1)\ +\ 1111\,1111\ (-1)\ =\ 1\,1111\,1110.$$

There is a carry out of the most significant bit, but the discarded 9th bit leaves $1111\,1110$, which is exactly -2. The result is correct, so there is no overflow. This shows that a carry out of the MSB is not the same as overflow.

59.10 Zero-extension versus sign-extension

Extension widens a value from fewer bits to more bits while preserving its meaning. The rule depends on whether the value is unsigned or signed: zero-extension fills the new leftmost bits with 0 and is used for **unsigned** values; sign-extension copies the sign bit into the new leftmost bits and is used for **signed** values. The common trap is to swap the two.

8-bit pattern	Unsigned	Signed	16-bit sign-extended
0000 0101	5	+5	0000 0000 0000 0101
1111 1011	251	−5	1111 1111 1111 1011

Zero-extending $1111\,1011$ would give $0000\,0000\,1111\,1011$ (value 251), which preserves the *unsigned* value. Sign-extending it gives $1111\,1111\,1111\,1011$ (value -5), which preserves the *signed* value. Using the wrong rule changes the value.

59.11 BCD: binary-coded decimal

BCD (Binary-Coded Decimal) represents each **decimal digit** with its own group of **4 bits**. The ten digit codes are:

0	1	2	3	4	5	6	7	8	9
0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

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Example 59.1 (BCD encoding). *Encode the decimal number* 59 *in BCD. Take each digit separately:* 5 *is* 0101 *and* 9 *is* 1001. *So* 59 *becomes*

$$0101\ 1001 \quad \textit{(BCD)}.$$

For contrast, the ordinary binary value of 59 *is* 111011_2. *BCD keeps one 4-bit code per decimal digit; it does not pack the whole number as plain binary.*

The trap is the claim that BCD is "just the binary of the whole number." It is not: each decimal digit is coded in 4 bits, and the codes 1010 through 1111 are not valid BCD digits.

59.12 Byte-addressability and effective address vs instruction length

Memory is **byte-addressable** when every individual byte has its own unique address. A word is a group of bytes at consecutive addresses. Because bytes are individually addressable, a word access begins at the address of its first byte.

The **effective address** is the actual location that an instruction uses for its operand. Depending on the addressing mode, it may be given directly or computed at run time from registers and constants. Instruction length is the number of bytes the instruction itself occupies. These are different quantities: a long instruction does not imply a distant operand. The address of the operand is set by the addressing mode, not by the instruction's size. Confusing the two is the trap; keep "how big is the instruction" separate from "where is the operand."

59.13 RISC versus CISC

The two broad instruction-set philosophies are **RISC** (Reduced Instruction Set Computer) and **CISC** (Complex Instruction Set Computer).

Aspect	RISC	CISC
Instruction set	Small, simple, regular.	Large and complex.
Instruction length	Fixed.	Variable.
Registers	Many general-purpose registers.	Fewer registers.
Memory access	Load/store architecture: only load and store touch memory.	Instructions may operate directly on memory operands.
Cycles per instruction	Typically one, suited to pipelining.	Often multiple cycles.
Control emphasis	Hardwired control; work shifted to the compiler.	Microcoded control; work done in hardware.
Typical examples	ARM, MIPS, RISC-V.	x86 and older mainframe designs.

RISC keeps instructions simple and fixed-length so they pipeline well, relying on many registers and a load/store discipline. CISC offers powerful, variable-length instructions that can do more in one step, at the cost of more complex decoding. The exam cue is the pair: *RISC = reduced, fixed, many registers, load/store*; *CISC = complex, variable-length, memory operands allowed.*

Exercises

1. State what is shared in von Neumann architecture and define the von Neumann bottleneck.
2. What is immediate addressing, and why does it need no extra memory access for the operand?
3. A program makes 850 cache hits and 150 misses. Compute the cache-hit ratio.
4. Explain why register addressing does not eliminate all data hazards.
5. In 8-bit two's complement, add $+7$ and $+1$ and state whether overflow occurs. Justify using the sign test.
6. Encode the decimal number 59 in BCD, and contrast it with the plain binary value.

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Solutions

1. Instructions and data share the same memory. The bottleneck is the limited rate at which instructions and data can be transferred between the CPU and memory over the shared path.
2. Immediate addressing carries the operand inside the instruction encoding, so the value is already available and no memory fetch is required for it.
3. Total accesses $= 850 + 150 = 1000$, so the hit ratio is $850/1000 = 0.85$ (85%).
4. Register operands avoid a memory access, but two instructions can still depend on the same register (for example a read-after-write dependency), so hazards remain.
5. $0111\ (+7) + 0001\ (+1) = 1000$, which has sign bit 1 and represents -8. Two same-sign operands produced an opposite-sign result, so overflow occurs.
6. BCD: $5 \to 0101$, $9 \to 1001$, giving $0101\ 1001$. Plain binary of 59 is 111011_2; BCD keeps one 4-bit code per decimal digit instead of packing the number as binary.

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UNIT 3

Memory, Storage & Backup

LESSON 07

Computer Memory Hierarchy

07.1 Why a memory hierarchy exists

A running computer never keeps all its data and instructions in one place. Instead it uses several stores arranged in a hierarchy. The reason is a three-way trade-off: the memory that is fastest is also the smallest and the most expensive per bit, while the memory that is largest and cheapest is the slowest. No single memory can be fast, large, and cheap at the same time. The hierarchy places the small, fast, costly memories close to the CPU and the large, slow, cheap memories further away.

Definition — Memory hierarchy

The ordered arrangement of storage levels in a computer, from the fastest and smallest (registers) through cache and main memory down to the slowest and largest (secondary and offline storage), chosen to balance speed, cost, and capacity.

Term	Meaning in this lesson
Register	The fastest and smallest storage, located inside the CPU.
Cache	A small, very fast memory that holds recently and frequently used data.
Main memory (RAM)	The primary memory that holds active programs and data; it is volatile.
ROM	Non-volatile read-only primary memory that holds firmware such as the BIOS.
Secondary storage	Non-volatile storage such as an SSD or HDD, used for files.
Tertiary / offline	Backup and archival storage such as magnetic tape or cloud backup.

The rest of this lesson uses these terms in one consistent sense. A **register** is the CPU's own storage. **Cache** is the small fast memory between the CPU and main memory. **Main memory** means RAM: the primary memory that holds what is running. **Secondary storage** means SSD and HDD, which keep files without power. **Tertiary or offline** storage is the slowest level, used for backup and archive.

07.2 The ordered hierarchy

The levels, read from fastest to slowest, are registers, cache, main memory (RAM), secondary storage, and then tertiary or offline storage.

Level	Speed	Size	Typical role
Registers	Fastest	Smallest	Values in use right now
Cache (L1/L2/L3)	Very fast	Small	Recently and frequently used data
Main memory (RAM)	Moderately fast	Larger	Active programs and data
Secondary (SSD/HDD)	Slower	Large	Permanent file storage
Tertiary/offline (tape, cloud)	Slowest	Largest	Backup and archive

Reading downward, capacity tends to grow and speed tends to fall. The anchor to remember is at the top: **registers are the fastest** storage in the computer. Below them cache is still very fast but slower than the registers. Main memory (RAM) is larger than cache but slower. Secondary storage (SSD or HDD) is larger again, cheaper per bit, and slower, but it keeps its contents without power. Tertiary or offline storage (magnetic tape, cloud backup) is the slowest and cheapest level and is used for backup and long-term archive.

07.3 Registers and cache

Registers are located inside the CPU and are the fastest and smallest storage in the machine. Because they are so small, they hold only the values needed at this instant: instructions, addresses, data, and intermediate results.

Cache is a small, very fast memory that sits between the CPU and main memory. It holds recently and frequently used data and instructions, and it is faster than main memory. Cache does not store new information of its own; it keeps *copies* of data that really live in RAM. It is therefore not a backup, and it loses its contents when power is removed, just as RAM does.

Definition — Cache

A small, very fast memory that holds recently and frequently used data and instructions so the CPU can avoid the slower trip to main memory.

Cache is usually described in levels. **L1** is the smallest and fastest and sits closest to the CPU core. **L2** is larger and a little slower than L1. **L3** is the largest and slowest of the three, is often shared across cores, and is still faster than main memory. For exam purposes, the key point is the ordering: L1 is fastest and smallest, and L3 is largest and slowest among the three, but all cache is faster than RAM.

Example 07.1. *While a student edits a document on the office desktop, the same lines and the nearby rows are read again and again. The cache keeps those recently used and nearby values ready, so the CPU does not have to wait for main memory on every access. The cache holds copies of those values; the master copy remains in RAM.*

07.4 Main memory (RAM) and ROM

Main memory is the computer's primary memory. The programs and data that are currently in use are held here while the machine runs, and the CPU works directly with main memory. **RAM** (Random Access Memory) is *volatile*: its contents are lost when the power is switched off. RAM is larger than cache but slower.

ROM (Read-Only Memory) is the other part of primary memory. It is *non-volatile*: it keeps its contents without power. In normal operation it can be read but not written; changing it needs a special process. ROM holds firmware such as the startup (BIOS) instructions.

Definition — Volatile memory

Memory that loses its contents when the power is removed. RAM is the common example.

Definition — Non-volatile memory

Memory that retains its contents without power. ROM, SSD, and HDD are common examples.

Example 07.1. *An item states, "RAM is non-volatile and will keep the file after shutdown." This is wrong. RAM is volatile; its contents are lost when the power goes off. It is **ROM** that is non-volatile. The wrong option has swapped the two primary-memory types.*

07.5 Primary memory vs secondary storage

Primary memory is directly accessible by the CPU and holds the active programs and data. It includes RAM, which is volatile, and ROM, which is non-volatile. **Secondary storage**, such as an SSD or HDD, is non-volatile and keeps files and programs between sessions. It is larger and cheaper per bit but slower than primary memory. The CPU does not access secondary storage directly; data moves through main memory under the control of the operating system.

07.6 Secondary and tertiary storage

Secondary storage comes in two common forms. A **solid-state drive (SSD)** uses flash memory and has no moving parts; a **hard disk drive (HDD)** uses rotating magnetic disks. Both are non-volatile and keep data without power. **Tertiary or offline** storage, such as magnetic tape and cloud backup, is the slowest and cheapest level and is used for backup and long-term archive.

07.7 Locality of reference

Programs do not access memory at random; they tend to reuse the same data and instructions. This behaviour is called **locality of reference**, and it comes in two forms. **Temporal locality** means the same item is accessed again soon after it was last used, such as a loop counter used on every iteration. **Spatial locality** means items stored at nearby addresses are accessed next, such as the consecutive elements of an array. This reuse is exactly why cache works: by keeping recently used and nearby data ready, cache reduces the average time the CPU waits for data.

Definition — Locality of reference

The tendency of a running program to reuse recently accessed data (temporal locality) and data at nearby addresses (spatial locality).

07.8 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. RAM is **volatile**; ROM is **non-volatile**. Do not swap them (TRAP-01).
2. Cache is **not** a backup. It holds copies of data that live in RAM.
3. RAM is main memory, **not** storage for files. Files belong in secondary storage.
4. Cache is **faster** than main memory; reversing this is a common wrong answer (PYQ-03).
5. The order of speed is registers, then cache, then RAM, then secondary storage. Registers are fastest.

Example 07.1. *An item asks which memory is fastest. Trace the hierarchy from the top: registers are fastest and smallest, then cache, then main memory (RAM), then secondary storage. The answer is registers. If the stem instead asks which level holds recently used data, the answer is cache, and if it asks which is volatile primary memory, the answer is RAM.*

Exercises

1. List the levels of the memory hierarchy from fastest to slowest.
2. Explain why the faster memories of the hierarchy are placed close to the CPU.
3. Distinguish volatile memory from non-volatile memory, giving one example of each.
4. Why is cache described as faster than main memory, and why is it not a backup?
5. Define temporal locality and spatial locality, and explain how they relate to cache.

Solutions

1. Registers, cache (L1, L2, L3), main memory (RAM), secondary storage (SSD/HDD), then tertiary/offline storage.
2. Faster memory is smaller and more expensive per bit. Placing it close to the CPU keeps the values the CPU needs soonest at the fastest level, so the average access time is low while cost and capacity are balanced.
3. Volatile memory loses its contents when power is removed — RAM is the example. Non-volatile memory retains its contents without power — ROM, or an SSD/HDD, is the example.
4. Cache sits between the CPU and main memory and is a smaller, faster technology, so it is faster than RAM. It only keeps copies of data that already exist in RAM, so if the cache is lost the data is still in RAM; it is therefore not a backup.
5. Temporal locality is the reuse of the same item soon after it was last accessed; spatial locality is the use of data at nearby addresses. Because programs show both patterns, keeping recently used and nearby data in the small fast cache pays off.

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LESSON 08

RAM, ROM & Cache

08.1 Memory, volatility, and why the distinction matters

Main memory (also called primary memory) holds the data and instructions that the CPU uses directly. Secondary storage such as a hard disk or an SSD is not accessed by the CPU directly; its contents are brought into main memory first. This lesson concentrates on the primary-memory components a computer-awareness paper actually asks about: RAM, ROM, and cache.

The single most tested distinction in this lesson is **volatility**.

Definition — Volatile memory

Memory whose stored contents are lost when the power supply is removed.

Definition — Non-volatile memory

Memory whose stored contents remain available after the power supply is removed.

RAM is volatile; ROM is non-volatile. A practice item may phrase this as a NOT/incorrect-statement question, so learn both directions: "RAM loses its contents on power-off," and "ROM retains its contents without power."

08.2 RAM: volatile main memory

RAM stands for **Random Access Memory**. It is the main memory in which running programs and their active data are held. The term "random access" means that any storage location can be reached in roughly the same amount of time; it is not a statement about volatility. RAM is **volatile**: when the office desktop (EX-01) is switched off, whatever was in RAM is lost unless it was saved to secondary storage.

Example 08.1. *A stem states that RAM keeps its data intact after the computer is switched off. This is false. RAM is volatile, so its contents are lost on power-off; it is ROM (and flash) that retains data without power. If the stem instead asks which memory holds the running programs, select RAM.*

08.3 DRAM versus SRAM

RAM comes in two main types, and the exam routinely contrasts them. Both are RAM and both are volatile; the difference lies in speed, refresh, cost, and use.

Feature	DRAM	SRAM
Full form	Dynamic Random Access Memory	Static Random Access Memory
Refresh	Needs periodic refresh	No refresh needed
Speed	Slower	Faster
Density and cost	Denser and cheaper per bit	Less dense and costlier
Typical use	Main memory (the computer's RAM)	Cache memory

DRAM stores each bit in a tiny capacitor that leaks charge, so it must be **refreshed** periodically. That refresh overhead is the price of its high density and low cost, which is why DRAM is the natural choice for large main memory. **SRAM** uses a different cell design that holds its state without refresh, making it faster but more expensive and less dense; it is therefore used for the small, fast **cache**.

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Example 08.1. *An item asks which RAM type is used to build cache memory. SRAM is faster and needs no refresh, so it is used for cache; DRAM is denser and cheaper and serves as main memory. If the stem asks which type needs periodic refreshing, the answer is DRAM.*

08.4 ROM and the ROM family

ROM stands for **Read-Only Memory**. It is **non-volatile** and is **read-only in normal operation**: the processor reads its contents, and changing them requires a special programming process rather than ordinary memory writes. ROM commonly stores **firmware**, such as the start-up instructions a computer runs before an operating system loads.

Definition — ROM

Read-Only Memory: non-volatile primary memory that is normally read but not written during ordinary operation.

Several ROM variants appear in exams. They differ mainly in *when* they are programmed and *how* they are erased.

Type	Key property
PROM	Programmable Read-Only Memory: can be programmed **once** by the user; it cannot be erased and reprogrammed.
EPROM	Erasable Programmable Read-Only Memory: erased by exposing the chip to **ultraviolet (UV)** light.
EEPROM	Electrically Erasable Programmable Read-Only Memory: erased and rewritten **electrically**, without removing the chip.
Flash	An **EEPROM-derived** non-volatile memory erased in **blocks**; widely used in USB drives and SSDs.

Example 08.1. *An item asks which memory can be erased by ultraviolet light. PROM is one-time programmable and cannot be erased; EEPROM is erased electrically; flash is erased in blocks. The UV-erasable type is EPROM. If the stem instead asks which type is programmable only once, select PROM.*

08.5 Cache memory and its levels

Cache is a small, very fast memory placed between the CPU and main memory. It holds frequently or recently used data and instructions so the CPU does not have to wait as often for slower main memory. Cache is **faster than main memory** — a fact the paper has tested directly (PYQ-03). It is temporary working memory, **not** a backup and not a substitute for storage.

Cache is organised in levels.

Definition — L1 cache

The level-1 cache: the smallest, fastest cache, located closest to the CPU core.

Definition — L2 cache

The level-2 cache: larger and slightly slower than L1.

Definition — L3 cache

The level-3 cache: the largest and slowest of the three levels, typically shared across the cores of the processor.

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As the level number rises, size grows and speed falls, and distance from the core increases: L1 < L2 < L3 in size, with the speed order reversed. Even L3, however, is faster than main memory.

08.6 Cache hits, misses, and hit ratio

Performance depends on how often the CPU finds what it needs already in the cache.

Definition — Cache hit

A memory access in which the required data is found in the cache.

Definition — Cache miss

A memory access in which the required data is not found in the cache, so it must be fetched from main memory.

The **hit ratio** measures how often accesses succeed in the cache:

$$\text{hit ratio} = \frac{\text{hits}}{\text{total accesses}}.$$

Example 08.1. *If the cache receives 100 memory accesses and 90 of them are found in the cache, then the hit ratio is* $90/100 = 0.90$. *The remaining 10 accesses are misses and must go to main memory. A higher hit ratio means the CPU waits less, so the system performs better.*

08.7 Exam retrieval and common confusions

Use the following distinctions to answer both recall and NOT/incorrect items reliably:

- RAM is **volatile**; ROM is **non-volatile** (TRAP-01).
- ROM is **read-only in normal operation**; writing needs a special process (TRAP-02).
- **Cache is faster** than main memory, not slower (PYQ-03).
- **SRAM** is used for cache; **DRAM** is used for main memory — and DRAM is the type that needs refresh.
- Cache is **not** a backup; it is temporary working memory.
- ROM family: PROM (once), EPROM (UV), EEPROM (electrical), flash (block erase).
- Hit ratio = hits ÷ total accesses.

Exercises

1. Explain why RAM is described as volatile and ROM as non-volatile.
2. Distinguish DRAM from SRAM in terms of speed, refresh, and typical use.
3. Expand PROM, EPROM, EEPROM, and flash, and state how each is erased (or whether it can be).
4. Order L1, L2, and L3 cache by size and by speed, and state which level is typically shared across cores.
5. A cache handles 200 memory accesses and 150 are hits. Compute the hit ratio.

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Solutions

1. RAM is volatile because it loses its contents when power is removed; ROM is non-volatile because it retains its contents without power.
2. DRAM needs periodic refresh, is slower but denser and cheaper, and is used for main memory. SRAM needs no refresh, is faster but costlier and less dense, and is used for cache.
3. PROM: Programmable Read-Only Memory, programmable once and not erasable. EPROM: Erasable Programmable Read-Only Memory, erased by ultraviolet light. EEPROM: Electrically Erasable Programmable Read-Only Memory, erased electrically. Flash: EEPROM-derived, erased in blocks.
4. By size: L1 $<$ L2 $<$ L3, so L3 is the largest. By speed: L1 is fastest and L3 the slowest of the three. L3 is typically shared across cores.
5. Hit ratio $= 150/200 = 0.75$.

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LESSON 09

Virtual Memory

09.1 What virtual memory is and why it exists

Physical RAM is limited. A modern application, its data, and the other programs open at the same time can together demand more memory than the machine physically has. **Virtual memory** is the technique that lets the system keep running anyway: it uses secondary storage (the disk) to hold parts of a program that are not currently needed in RAM. In this way the *addressable* memory seen by a program can be much larger than the physical memory actually installed.

Definition — Virtual memory

A memory-management technique that uses secondary storage to extend the addressable memory beyond physical RAM, giving each process the impression of a large, contiguous memory space.

The office-desktop running example (EX-01) makes this concrete. The desktop has 16 GB of RAM and runs MS Word, a spreadsheet, and a web browser at once. If the combined working set of those programs exceeds 16 GB, the operating system moves some of the least-used pages to disk and brings them back when they are referenced again. No single program is told that RAM ran out; each continues to use its own large address space.

Two ideas are worth separating at the start:

- Virtual memory extends **capacity**. It lets a program use more memory than the physical RAM provides.
- Virtual memory does **not** add speed. Because it relies on the disk, it is **much slower** than RAM.

09.2 Virtual addresses, physical addresses, and the MMU

A program does not work with the real location of its data in RAM. It works with **virtual addresses**, which form a large, uniform address space. The hardware translates each virtual address into the actual **physical address** in RAM when the instruction executes.

Definition — Virtual address

The address generated by a running program; it belongs to the program's large virtual address space and is not the real RAM location.

Definition — Physical address

The actual location of data in main memory (RAM), which the hardware uses after translation.

The translator is the **MMU**, the Memory Management Unit.

Definition — MMU (Memory Management Unit)

The hardware that translates virtual addresses into physical addresses at run time, using the page table maintained by the operating system.

The MMU is central to every memory access. It does *not* store files and it does not itself create storage; it maps one address form to another. Keeping this role straight avoids two common exam

errors: attributing file storage to the MMU, or describing translation as a software-only operation with no hardware component.

Example 09.1. *A question asks what the MMU does. The correct answer is that it translates virtual addresses into physical addresses. An option saying that the MMU stores files permanently is wrong: file storage is the job of secondary storage, not the address translator. An option saying the MMU holds the BIOS is also wrong; that describes firmware, not the unit that maps addresses.*

09.3 Paging: fixed-size pages and frames

Paging divides memory into equal, fixed-size blocks. The virtual address space of a process is split into **pages**, and physical memory is split into **frames** of exactly the same size. Any page can be placed in any free frame, and the page table records which page currently sits in which frame.

Definition — Page

A fixed-size block of the virtual address space in paging.

Definition — Frame

A fixed-size block of physical memory (RAM) into which a page is loaded; frames are the same size as pages.

Because every block is the same size, paging makes memory allocation simple and predictable. There is no awkward leftover space caused by blocks of many different sizes. The cost is a small amount of unused space in the last page of a program, but the mechanism itself is uniform.

Two related terms must not be swapped. **Pages** are the *virtual* blocks; **frames** are the *physical* blocks. They are the same size, but a page lives in the program's virtual view while a frame lives in RAM.

09.4 Segmentation: variable-size logical segments

Segmentation divides a program into **variable-size segments**, each corresponding to a logical unit of the program: the code, the data, the stack, a procedure, or a module. Because these logical units differ in size, segment sizes differ as well.

Definition — Segment

A variable-size block of a program that corresponds to a logical unit such as code, data, stack, or a module.

Segmentation reflects how a programmer thinks about a program: one segment per logical part. Paging, by contrast, reflects no logical unit at all — its pages are simply equal slices chosen for uniform allocation. This is the key near-neighbour distinction between the two techniques.

Example 09.1. *A stem asks which technique divides a program according to logical units such as code and data. The answer is segmentation, because those units are logical and of different sizes. If the stem instead describes equal fixed-size blocks and the terms "page" and "frame", the answer is paging.*

09.5 Paging versus segmentation at a glance

Feature	Paging	Segmentation
Block size	Fixed	Variable
Blocks	Pages (virtual) and frames (physical)	Logical segments
Sizes equal?	Yes, all equal	No, they differ
Logical view	Not visible to the programmer	Matches program structure
Unit	Equal slices	Code, data, stack, module

The single most common error is to call paging's blocks "segments" or to say that segmentation uses equal-sized blocks. Keep the pairing firm: *paging is fixed-size; segmentation is variable-size and logical.*

09.6 Page faults and demand paging

When a program references a page that is *not* currently in RAM, the hardware raises a **page fault**. The operating system then loads the required page from the disk into a free frame and lets the program continue.

Definition — Page fault

An event that occurs when a referenced page is not currently in RAM; the page must be loaded from secondary storage (disk) into a free frame.

Definition — Demand paging

A policy in which a page is loaded into memory only when it is actually referenced, rather than loading the whole program in advance.

A page fault is a normal, expected event — **not a hardware fault**. It simply means the needed page is on the disk for the moment. Demand paging is what makes a program able to start before every one of its pages is in RAM: the remaining pages are brought in on demand, as execution reaches them.

Example 09.1. *A question says a running program references a page that is not in RAM and asks what happens. The correct answer is a page fault followed by loading that page from the disk. An option describing a physical hardware failure is wrong, because a page fault is a routine memory-management event, not a broken component.*

09.7 Swapping and the disk-backed hierarchy

Swapping is the movement of pages or whole processes between RAM and secondary storage (the disk). Because virtual memory depends on this movement, it is **disk-backed**: the disk supplies the extra capacity that RAM does not have.

Definition — Swapping

Moving pages or processes between main memory (RAM) and secondary storage (disk) so that the memory actually needed at a moment can reside in RAM.

The memory hierarchy, from fastest and smallest to slowest and largest, runs from CPU registers, to cache, to RAM, to disk-backed virtual memory. Two facts follow directly:

1. Virtual memory is **much slower than RAM**, because the disk is far slower than main memory.

2. Virtual memory is **not additional physical RAM**. It borrows disk space; it does not add RAM chips.

This is also where virtual memory is most often confused with **cache**. They are opposite in purpose and location. Cache is small, fast memory sitting close to the CPU and it *speeds up* repeated access to hot data. Virtual memory is large, disk-backed, and it *extends capacity* using slower storage. Cache helps speed; virtual memory helps size.

Example 09.1. *A stem contrasts cache and virtual memory. The correct statement is that cache is small, fast memory close to the CPU, while virtual memory is slower and disk-backed. Options that call both "secondary storage", or that say virtual memory speeds up the CPU, reverse the roles and are wrong.*

09.8 Thrashing

Thrashing is *excessive paging*. When too many processes compete for too little RAM, the system spends most of its time moving pages between RAM and disk, and very little time doing useful work. The CPU stays busy, yet throughput collapses.

Definition — Thrashing

A condition of excessive paging in which the system spends most of its time swapping pages between RAM and disk, leaving little time for useful work.

Thrashing is a symptom of memory pressure, not a hardware failure. Adding physical RAM, or reducing the number of active processes, usually relieves it.

09.9 Exam retrieval and common confusions

Use two retrieval directions. Given the technique, recall its block rule; given a description, name the technique.

- Virtual memory ↔ extends memory using disk; slower than RAM; not extra physical RAM.
- MMU ↔ Memory Management Unit; translates virtual to physical addresses.
- Paging ↔ fixed-size pages (virtual) and frames (physical).
- Segmentation ↔ variable-size logical segments.
- Page fault ↔ referenced page not in RAM; load it from disk.
- Demand paging ↔ load a page only when it is referenced.
- Swapping ↔ move pages/processes between RAM and disk.
- Thrashing ↔ excessive paging.

Three distinctions prevent most errors in this topic:

1. Do not say virtual memory is **extra physical RAM**. It is disk-backed and **slower** than RAM; it adds capacity, not speed.
2. Do not confuse **paging** (fixed-size pages and frames) with **segmentation** (variable-size logical segments).
3. Do not call a **page fault** a hardware fault. It means the referenced page must be loaded from the disk; and do not confuse **cache** (small, fast, near the CPU) with virtual memory (large, disk-backed).

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Exercises

1. Explain why virtual memory is said to extend memory but not to add physical RAM.
2. Expand MMU and state its function in one sentence.
3. Distinguish paging from segmentation in terms of block size and the unit being divided.
4. What is a page fault, and what happens after one occurs?
5. Define thrashing and give one practical way to reduce it.

Solutions

1. Virtual memory extends the addressable memory by using secondary storage (disk) to hold parts of a program not currently in RAM. It does not add RAM chips, so it is not additional physical RAM; because it is disk-backed, it is much slower than RAM.
2. MMU is the Memory Management Unit. It translates virtual addresses generated by a program into physical addresses in RAM at run time.
3. Paging uses fixed-size, equal blocks: pages in the virtual space and frames in physical memory. Segmentation uses variable-size blocks that correspond to logical program units such as code, data, or a module.
4. A page fault occurs when a program references a page that is not currently in RAM. The operating system then loads that page from the disk into a free frame and resumes execution.
5. Thrashing is excessive paging, in which the system spends most of its time swapping pages between RAM and disk and little time on useful work. Adding physical RAM (or reducing the number of active processes) usually reduces it.

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LESSON 10

Storage Devices

10.1 The storage hierarchy

Storage is organized into levels according to how close it sits to the processor and what job it does. The CPU works directly with primary storage; secondary storage holds data permanently until it is needed and feeds it to main memory; tertiary storage is an archival level used for large, rarely accessed collections.

Definition — Primary storage

The directly CPU-accessible working memory of a computer, such as RAM, ROM, and cache; it holds the data and instructions currently in use.

Definition — Secondary storage

Non-volatile storage that retains data when the power is off and is not accessed directly by the CPU — for example a hard disk, SSD, optical disc, or USB flash drive.

Definition — Tertiary storage

An archival storage level for large volumes of infrequently used data, typically robotic tape libraries whose media are loaded on demand.

Level	Examples	Key property
Primary	RAM, ROM, cache	Directly accessible by the CPU
Secondary	HDD, SSD, optical disc, USB drive	Non-volatile; holds files
Tertiary	Robotic tape library	Archival; loaded on demand

The CPU does not read a file straight from secondary storage. The operating system brings the required data into main memory first, and the CPU then works with that copy. This is why the hierarchy matters for an exam scenario: a large program on disk is limited by the memory and storage that actually feed it, not only by processor speed.

10.2 Magnetic storage

Magnetic storage records data by magnetizing a surface, and it is read back by detecting those magnetic patterns. Hard disks and magnetic tape are the two common magnetic media, but they differ sharply in how they are reached.

10.2.1 Hard disk drive

A hard disk drive (HDD) stores data on *rotating platters* coated with a magnetic material. A *read/write head* mounted on an actuator arm floats just above the platter surface and reads or writes as the platter spins. The head moves inward and outward to reach different tracks. Because the platters spin and the arm moves, an HDD is a *mechanical* device with moving parts.

Definition — Hard disk drive (HDD)

Magnetic secondary storage that stores data on rotating platters read and written by a moving read/write head; it is mechanical and contains moving parts.

10.2.2 Magnetic tape

Magnetic tape stores data on a long, thin magnetic ribbon wound in a reel or cartridge. The tape is read or written in order, position by position, so a record far from the current position takes longer to reach. This makes tape a *sequential-access* medium. It is inexpensive and dense, which is why it is used for backup and long-term archival rather than for routine access.

Definition — Magnetic tape

Magnetic storage on a long tape that is read and written in order; it is a sequential-access medium used mainly for backup and archival.

10.3 Optical storage

Optical storage records data as *pits and lands* — tiny physical markings — on a disc and reads them with a *laser*. The reflected light pattern encodes the data. Optical discs are removable and non-volatile, so they are convenient for distributing and archiving files.

The three common families are the compact disc (CD), the digital versatile disc (DVD), and the Blu-ray disc. Blu-ray uses a shorter-wavelength laser and tighter tracks, allowing it to store more than a DVD, which in turn stores more than a CD.

Definition — Optical storage

Storage in which a laser reads data encoded as pits and lands on a disc, such as a CD, DVD, or Blu-ray disc.

Medium	Approximate capacity (single layer)	Reading method
CD	700 MB	Laser
DVD	4.7 GB	Laser (shorter wavelength)
Blu-ray	25 GB	Laser (shortest wavelength)

Order the capacities from smallest to largest as CD < DVD < Blu-ray. A common test error is to swap the middle entries or to reverse the order entirely; fix the sequence by remembering that CD is the oldest and smallest and Blu-ray is the newest and largest.

Example 10.1. *An item asks which optical disc holds the most data. Compare the standard single-layer figures: CD ≈ 700 MB, DVD ≈ 4.7 GB, and Blu-ray ≈ 25 GB. Since* $25\ GB > 4.7\ GB > 700\ MB$*, the Blu-ray disc holds the most.*

10.4 Flash and solid-state storage

Flash memory stores data in semiconductor cells that hold their state without power. There are no rotating platters and no moving heads, so flash devices are *solid-state*. Common examples are the solid-state drive (SSD), the USB flash drive, and the memory card.

Definition — Flash memory

Non-volatile semiconductor storage that keeps data in cells with no moving parts; the technology behind SSDs, USB flash drives, and memory cards.

Definition — Solid-state drive (SSD)

A flash-based storage device with no moving parts. It is faster and more shock-resistant than a mechanical hard disk and has a high transfer rate.

Because nothing spins and no head has to move, flash can start a transfer almost immediately and is far less sensitive to being bumped or dropped. Among a hard disk drive, an optical drive, magnetic tape, and an SSD, the **SSD** has the highest transfer rate.

10.5 Cloud storage

Cloud storage keeps data on remote servers in a data centre and makes it available over the Internet. The user sees a folder or an application, while the provider manages the underlying hardware. Even though the storage feels abstract, it ultimately rests on physical secondary storage — arrays of disks and solid-state drives — on the provider's machines.

Definition — Cloud storage

Storage in which data is held on remote servers and accessed over the Internet rather than on a locally attached device.

Cloud storage is not main memory and it is not the same as primary storage. It is remote secondary storage reached through a network connection, useful for backup, sharing, and access from several devices.

10.6 Hard disk versus solid-state drive

The comparison below is the most frequently tested near-neighbour pair in this topic. The hard disk is magnetic and mechanical; the SSD is flash and solid-state. Keep the technology and the consequences together.

Aspect	Hard disk drive (HDD)	Solid-state drive (SSD)
Technology	Magnetic	Flash memory
Moving parts	Yes — spinning platters, moving head	None
Speed	Usually slower transfer rate	Higher transfer rate (fastest here)
Shock resistance	More vulnerable to shock	More shock-resistant
Typical use	Bulk data at low cost per GB	Fast system and boot drive

Example 10.1. *A candidate reads "the fastest storage device is the hard disk." That is incorrect. The hard disk is magnetic and mechanical, with a spinning platter and a moving head. The SSD is flash-based with no moving parts and has the higher transfer rate. The correct choice is the SSD (KB PYQ-27).*

10.7 Access methods: random and sequential

An access method describes how quickly a device can reach a stored location.

Definition — Random (direct) access

An access method in which any location can be reached in approximately equal time; RAM and hard disks behave this way.

Definition — Sequential access

An access method in which locations are reached in order, so the time depends on how far the target is from the current position; magnetic tape works this way.

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Access method	Behavior	Examples
Random (direct)	Any location in about equal time	RAM, hard disk
Sequential	Reached in order	Magnetic tape

Memory and disks are approximately random access because the electronics or the head can be directed to a location without stepping through everything before it. Tape is sequential because the ribbon must physically pass over the read head in order.

10.8 Non-volatility of storage

Non-volatility is the property that stored contents survive when the power is switched off. Storage devices are designed to be non-volatile so that files remain available after shutdown. Primary storage such as RAM is *volatile*: its contents disappear when power is removed.

Definition — Non-volatile storage

Storage that retains its contents without power; hard disks, SSDs, optical discs, and USB drives are non-volatile.

The one-line rule to carry into the exam is: RAM is volatile; storage is non-volatile. This is exactly why unsaved work in RAM can be lost in a power failure while a file already saved to disk remains.

10.9 Exam retrieval and common confusions

Use a two-direction recall drill. Given a device, state its technology and its access method; given a technology or capacity, state the matching device.

- HDD ↔ magnetic, mechanical, random access.
- Magnetic tape ↔ magnetic, sequential access, backup.
- CD / DVD / Blu-ray ↔ optical, capacities 700 MB / 4.7 GB / 25 GB (single layer).
- SSD / USB drive / memory card ↔ flash, no moving parts; the SSD has the highest transfer rate.
- Cloud storage ↔ remote servers over the Internet.
- RAM volatile versus storage non-volatile.

Three distinctions prevent most errors in this topic:

1. Do not call the SSD magnetic or mechanical. The magnetic and mechanical device is the hard disk; the SSD is flash with no moving parts.
2. Do not claim that the hard disk has the highest transfer rate; the SSD does (KB PYQ-27).
3. Do not reverse the optical ordering. It is CD < DVD < Blu-ray for standard single-layer capacities, and do not confuse volatile RAM with non-volatile storage.

Example 10.1. *An item asks which device stores data without any moving parts. Eliminate the hard disk (spinning platters and a moving head) and magnetic tape (a moving ribbon). An optical drive also spins its disc. The flash-based SSD stores data in semiconductor cells with no moving parts, so it is the answer.*

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Exercises

1. Distinguish primary, secondary, and tertiary storage and give one example of each.
2. Explain why an SSD is described as having no moving parts, and why it is more shock-resistant than a hard disk.
3. Order CD, DVD, and Blu-ray by single-layer capacity from smallest to largest, with the approximate figures.
4. Classify RAM, a hard disk, and magnetic tape as random access or sequential access.
5. A student writes: "RAM is non-volatile, and the hard disk is the fastest storage device." Correct both errors.

Solutions

1. Primary storage is directly accessible by the CPU (for example RAM). Secondary storage is non-volatile and holds data when power is off (for example a hard disk). Tertiary storage is archival storage loaded on demand (for example a robotic tape library).
2. An SSD stores data in flash semiconductor cells, with no spinning platter and no moving read/write head. With no moving parts to be disturbed, it tolerates shock better than a hard disk and transfers data faster.
3. CD $\approx$ 700 MB, then DVD $\approx$ 4.7 GB, then Blu-ray $\approx$ 25 GB, for single-layer media.
4. RAM is random access; a hard disk is approximately random access; magnetic tape is sequential access.
5. RAM is *volatile*, not non-volatile: its contents are lost without power. The device with the highest transfer rate is the *SSD*, not the hard disk; the hard disk is magnetic and mechanical with moving parts.

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LESSON 11
Backup Types

11.1 Backup, storage and archive

These three words describe different jobs, and exam questions often test the distinction. **Storage** is where your working data lives day to day — the primary copy you actually open and edit. A **backup** is a separate copy kept so that data can be recovered after loss, damage, corruption or an attack. An **archive** is older data retained for the long term, usually moved out of routine use. A backup exists for recovery; an archive exists for retention. Neither is simply "another folder" on the same disk.

Definition — Backup

A separate copy of data kept for the purpose of recovery after loss, damage, corruption or attack.

Definition — Archive

Older data kept for long-term retention, normally not used in everyday work.

The office desktop of the running example (KB EX-01) has one primary copy of its documents. If the only copy is the one on the desktop's own drive, then the user has storage but no backup. The moment a second, independent copy exists, recovery becomes possible.

11.2 Full, incremental and differential backups

A backup type decides *how much* is copied on each run. The three basic types differ only in the answer to one question: since which earlier point are changes included?

Definition — Full backup

Copies every selected file and folder (or the entire volume) on every run. It is the recovery baseline for the other types.

Definition — Incremental backup

Copies only the files changed or added since the last backup of any type *— whether that was a full, an incremental or a differential backup.*

Definition — Differential backup

Copies everything changed or added since the last full backup*, not since the previous backup.*

The single phrase that separates incremental from differential is therefore the word *full*. An incremental looks only as far back as the immediately preceding backup; a differential always looks back to the last full backup, so it keeps growing until the next full backup resets it.

Type	What it copies	Restore requirement
Full	Everything selected, every run.	One complete set. Fastest single restore, but largest and slowest to create.
Incremental	Changes since the last backup of any type.	The last full backup plus every incremental backup in sequence.
Differential	Changes since the last full backup.	The last full backup plus only the most recent differential.

There is a consistent trade-off. Smaller and faster backups produce longer or more complex restores. A full backup is heavy to create but trivial to restore; an incremental is light to create but may need many files chained together at restore time; a differential sits in between.

Example 11.1. *On Monday the office desktop is given a full backup. On Tuesday, a single document changes. With an incremental strategy, Tuesday's backup contains only that one change. With a differential strategy, Tuesday's backup contains that change measured against Monday's full backup. If only that one file changed, the two results look identical on Tuesday — the difference appears on Wednesday, when the incremental again holds only Wednesday's change, while the differential holds Tuesday's change plus Wednesday's change.*

11.3 Restore scenarios

Restore behaviour is the part candidates most often miss, because it is not visible on the day of the backup — only when recovery is needed.

Example 11.1 (Restore from an incremental chain). *A full backup is taken Sunday. Incrementals are taken Monday, Tuesday and Wednesday. On Thursday the disk fails. To recover Wednesday's data you need the Sunday full backup and then* all three *incrementals — Monday, Tuesday and Wednesday — applied in order. Missing any one link loses the changes it carried.*

Example 11.2 (Restore from a differential chain). *A full backup is taken Sunday. Differentials are taken Monday, Tuesday and Wednesday. On Thursday the disk fails. To recover Wednesday's data you need only the Sunday full backup and the* most recent *differential — Wednesday's. The earlier differentials are not required, because Wednesday's already includes all changes since Sunday.*

Both scenarios produce the same recovered data. The difference is operational: the incremental restore is longer and depends on every tape or file in the chain; the differential restore is shorter and depends on only two sets.

11.4 Image and file-level backups

Beyond the three time-based types, two coverage-based terms appear in exams.

Definition — Image backup (system image)

A complete clone of a system — operating system, settings, applications and data — that can restore a whole machine.

Definition — File-level backup

A backup of selected files and folders rather than a whole system.

An image restores a working machine in one step, which makes it ideal after a drive failure or a corrupted operating system. A file-level backup is smaller and more flexible for everyday documents, but recovering a machine from it means reinstalling the operating system first and then restoring the files. The office desktop could use an image for disaster recovery and file-level backups for its working documents.

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11.5 The 3-2-1 rule, off-site and offline copies

The **3-2-1 rule** is the standard exam reminder for resilient backup:

1. **3** copies of the data — the original plus two backups.
2. On **2** different types of media (for example, an external disk and cloud storage).
3. With **1** copy kept **off-site**, at a different location.

The numbers map to copies, media and location, in that order. A common trap is to swap them, or to read the "2" as two off-site copies.

An **off-site** copy is kept at a different physical place, so a single fire, flood or theft cannot destroy every copy at once. An **offline** copy is disconnected from the network; it is especially useful against ransomware, which spreads through connected drives. A **cloud backup** is an off-site copy held on remote servers. These are not mutually exclusive: the office desktop's cloud backup is both off-site and, when not mounted, can serve as an offline copy as well.

Example 11.1. *The office desktop keeps its working files on the local SSD (copy 1), an external USB disk (copy 2, different medium, kept offline), and a cloud account (copy 3, off-site). This satisfies the 3-2-1 rule: three copies, two media types, one off-site.*

11.6 Why RAID is not a backup

RAID (**Redundant Array of Independent Disks**) combines several disks for performance or redundancy. If one disk fails, a mirrored or parity RAID keeps the system running. That protects *uptime*, not *recovery*. RAID is not a backup because it does not protect against the things a backup exists for: accidental deletion, corruption or ransomware. When a file is deleted, or an encrypted copy is written by ransomware, the change is written to every member of the RAID array at once. There is no earlier clean copy to go back to. Redundancy duplicates the current state — including the damage.

11.7 Backup rotation

Backup rotation means keeping several dated backup sets and reusing them on a fixed schedule, commonly the grandfather–father–son scheme (daily, weekly, monthly). Rotation matters because a problem such as silent corruption may not be noticed for days; if the single backup set has already been overwritten with the corrupted data, there is no clean copy left. Keeping multiple versions across time ensures that an older, clean copy remains available.

Exercises

1. Define full, incremental and differential backup in one line each.
2. A full backup runs Sunday; differentials run Monday to Wednesday. The disk fails on Thursday. Which sets are needed to restore Wednesday's data?
3. State the 3-2-1 rule and explain what each number refers to.
4. Why is RAID not a substitute for a backup? Give one concrete danger.
5. Explain the purpose of backup rotation.

Solutions

1. Full: copies everything selected each run. Incremental: copies changes since the last backup of any type. Differential: copies changes since the last full backup.
2. The Sunday full backup and the most recent (Wednesday) differential only. The earlier differentials are unnecessary because Wednesday's includes all changes since the full backup.
3. Three copies of the data, on two different types of media, with one copy kept off-site.

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4. RAID provides redundancy for uptime, not recovery. A deletion, corruption or ransomware write is applied to every RAID member at once, so no earlier clean copy survives.
5. Rotation retains several dated backup versions so that an older, clean copy is still available if corruption is discovered only after the newest backup has already been overwritten.

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UNIT 4

I/O, Peripherals & Multimedia

LESSON 12

Input Devices

12.1 What is an input device

A computer works in a cycle: input, process, and output. In the input step, data and instructions are fed into the machine. An **input device** is the hardware that accepts data or control signals from the user or from the environment and passes them to the computer, converting a human action or a physical signal into machine-readable form. Without an input device the computer would have nothing to process.

Definition — Input device

Hardware that accepts data or control signals from the user or the environment and supplies them to the CPU, converting them into a form the computer can process.

Common input devices include the keyboard, mouse, touchscreen, scanner, microphone, digital camera, barcode reader, and biometric sensor. The following table fixes the meaning of each term used in this lesson.

Term	Meaning in this lesson
Keyboard	Text input device that converts a keystroke into a character code.
Mouse	Pointing device moved on a surface to control the on-screen pointer.
Trackball	Pointing device whose ball is rotated in place; the device stays still.
Touchpad	Flat, touch-sensitive pad on a laptop; the pointer moves on the screen.
Joystick	Lever that moves on two or three axes, used for games and control.
Light pen	Pen-shaped input device with a light sensor, used on a CRT screen.
Stylus / digitizer	Pen plus tablet that converts strokes into digital coordinates.
Scanner	Converts a paper document or picture into a digital image.
Biometrics	Input from a physical or behavioural human trait (fingerprint, iris, face).

12.2 Classifying input devices

Input devices are grouped by the type of data they capture. This is the classification JKSSB question-setters use most often, so learn one example for each row.

Category	Typical devices
Text input	Keyboard
Pointing input	Mouse, trackball, touchpad, joystick, light pen, stylus, touch-screen
Audio input	Microphone
Image / video input	Scanner, digital camera, webcam
Biometric input	Fingerprint, iris, and face sensors
Code / tag input	Barcode, QR code, RFID

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A single device may belong to more than one row if it captures more than one kind of data, but for exam purposes use the primary role: a touchscreen is a pointing input device, a webcam is an image/video input device, and a microphone is an audio input device.

12.3 Direct vs indirect input

A second useful classification is whether the data is captured at its source or encoded by the user.

Direct input devices capture the data themselves. Examples are the touchscreen, light pen, scanner, barcode reader, biometric sensor, and microphone: the device reads the object, code, trait, or sound and turns it directly into computer data.

Indirect input devices require the user to act on the device to encode the data. Examples are the keyboard, mouse, trackball, touchpad, and joystick: the user translates an intention into keystrokes or pointer movements, which the device then sends to the computer.

Definition — Direct input

Input in which the device captures the data at its source (for example, a scanner reading a page or a biometric sensor reading a fingerprint) without the user encoding it by hand.

The distinction matters because it explains why some devices are fast and error-free while others depend on the user. A barcode reader or a biometric sensor captures the data exactly as it is, so there is no typing mistake; a keyboard, being indirect, is only as accurate as the person typing. In a question, if the stem stresses that the device *reads* or *captures* an object, code, trait, or sound, think of a direct input device; if it stresses that the *user types or moves* something to enter the data, think of an indirect input device.

12.4 The keyboard: text input

The keyboard is the most common text input device. Its standard letter layout is called **QWERTY**, after the first six letters of the top row. A keyboard has alphanumeric keys, function keys (F1 to F12), modifier keys such as Ctrl, Alt, and Shift, and usually a numeric keypad. Each time a key is pressed, the keyboard sends the corresponding character code to the computer. The keyboard is an indirect input device: the user types the data, and the device encodes it.

Keyboards differ in construction even though their role is the same. A **membrane keyboard** uses a flexible membrane under the keys and is cheap and quiet. A **mechanical keyboard** uses a separate physical switch under each key and gives a firmer, more definite feel. A **wireless keyboard** connects to the computer by Bluetooth or by a USB receiver instead of a cable. Whatever the construction, a keyboard is still a text input device: it sends character codes for the keys that are pressed.

12.5 Pointing devices

A pointing device controls the position of an on-screen pointer (cursor) and is used to point, click, and drag. The table below fixes each pointing device and its nearest neighbour, because JKSSB questions often ask candidates to tell two similar pointing devices apart.

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Device	How it works	Nearest neighbour
Mouse	The device slides on a surface and the pointer follows.	Trackball: the ball rotates while the device stays still.
Trackball	The ball is rotated in place; the device is stationary.	Mouse: the whole device moves.
Touchpad	A flat finger pad moves the pointer on the screen (indirect).	Touchscreen: the display itself is touched (direct).
Joystick	A lever moves on two or three axes for games and control.	Mouse: flat-surface pointing.
Light pen	A light sensor is pointed at a CRT screen.	Stylus: used on a digitizer surface.
Stylus + digitizer	A pen stroke is converted into digital coordinates.	Light pen: uses a light sensor, not a tablet.

12.5.1 The mouse

The **mouse** is a hand-held pointing device moved on a flat surface. It has one or more buttons and usually a scroll wheel. An **optical or laser mouse** tracks movement with light, which is the common modern design; an older **mechanical mouse** used a rolling ball. A mouse is an input device (PYQ-08).

12.5.2 The trackball

The **trackball** is, in effect, an upside-down mouse: the ball is on top and is rotated with the fingers while the device itself stays still. This saves desk space because the device does not have to be moved. The nearest-neighbour trap is that a trackball and a mouse both point, but with a mouse the *device* slides, while with a trackball the *ball* rotates in place.

12.5.3 The trackpad or touchpad

The **trackpad** (also called the touchpad) is a flat, touch-sensitive pad found on laptops. The user slides a finger across the pad to move the pointer on the screen and taps to click. It is an indirect device because the finger does not touch the object on the screen; the pad only controls the pointer.

12.5.4 The joystick

The **joystick** is a lever that can be moved on two or three axes. It is used mainly for games and for some industrial control tasks.

12.5.5 The light pen

The **light pen** is a pen-shaped input device with a light sensor at its tip. It is pointed at a CRT screen to select items or to draw. A light pen is an input device, not an output device (TRAP-12).

12.6 Stylus and digitizer tablet

A **stylus** is a pen-like tool used to write or draw on a tablet or a touch screen. A **digitizer** (graphics tablet) is a flat surface that converts the movement of the stylus into digital coordinates. Together they are used for freehand drawing, capturing signatures, computer-aided design (CAD), and handwriting recognition. Because a digitizer captures a drawing as data, it is an input device (TRAP-12). Do not confuse the stylus with the light pen: the light pen works with a light sensor, while the stylus works with a digitizer surface.

12.7 Touchscreen

A **touchscreen** is a display that also works as an input device. The user touches the screen directly to select and enter data. Because the item touched is the item selected, a touchscreen is a direct input device. The two common technologies are **resistive** and **capacitive** touchscreens. Touchscreens are used in smartphones, tablets, ATMs, and information kiosks.

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The nearest-neighbour trap is the touchscreen versus the touchpad. On a touchscreen the user touches the display itself, so the input is direct. On a touchpad the user touches a separate pad and the pointer moves on the screen, so the input is indirect. A touchpad is not a touchscreen.

12.8 Scanners, cameras, and webcams

A **scanner** converts a paper document or a picture into a digital image. Common types are the **flatbed** scanner (the page is placed on a glass surface), the sheet-fed scanner, the handheld scanner, and the drum scanner. Software such as OCR (Optical Character Recognition) or OMR (Optical Mark Recognition) can then read text or marks from the scanned image. A scanner is an input device (TRAP-12).

A **digital camera** captures photographs as digital images and stores them on a memory card. A **webcam** is a small digital camera connected to a computer and used mainly for live video, such as video calls and streaming. Both are image/video input devices.

12.9 Barcode, QR code, and RFID

A **barcode** is a one-dimensional pattern of parallel lines of varying width, read by a barcode scanner. It is used on product labels and on book ISBNs.

A **QR code** (Quick Response code) is a two-dimensional matrix of black and white squares. It can store more data than a barcode and can be read from any direction. The full form of QR is **Quick Response**.

RFID stands for **Radio Frequency Identification**. An RFID system has a tag and a reader; the two exchange radio waves, so no line of sight is needed, unlike a barcode. RFID is used in toll collection, access cards, and inventory tracking. A barcode, a QR code, and an RFID tag are all read by an input device that feeds the data into the computer.

12.10 Microphone and biometric input

A **microphone** is an audio input device. It converts sound, such as voice or music, into an electrical signal that the computer digitises. It is used for recording, voice calls, and speech recognition. The matching output device is the **speaker**: a microphone takes audio in, a speaker sends audio out.

Biometric input identifies a person from a physical or behavioural trait. Common traits are the **fingerprint**, the **iris**, the **face**, the voice, and the palm. A sensor captures the trait and feeds the measured data to the computer. Biometrics are used for attendance systems, device unlocking, and authentication. Because the sensor reads a human trait and passes the data to the system, biometric input is a direct input form.

Example 12.1. *A student wants to scan a printed page into the computer. The correct device is a scanner, which converts the page into a digital image. If the student instead wanted to type the same text by hand, the device would be a keyboard, which is an indirect text input device. If the goal were to capture a live video of the student, the device would be a webcam.*

12.11 Summary of input devices

The table below gathers the devices in this lesson for quick revision. Use it as a checklist: for each device, be ready to state its category and its one-line function, and to name the device it is most often confused with.

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Device	Category	Function
Keyboard	Text	Converts a keystroke into a character code.
Mouse	Pointing	Moves the on-screen pointer; point, click, drag.
Trackball	Pointing	Ball rotated in place; device stays still.
Touchpad	Pointing	Flat finger pad; pointer moves on the screen.
Joystick	Pointing	Lever on two or three axes for games/control.
Light pen	Pointing	Light sensor pointed at a CRT screen.
Stylus + digitizer	Pointing	Converts pen strokes into digital coordinates.
Touchscreen	Pointing (direct)	Touch the display to select and enter.
Scanner	Image	Converts a document or picture into a digital image.
Digital camera	Image/video	Captures photographs as digital images.
Webcam	Image/video	Small camera for live video on a computer.
Microphone	Audio	Converts sound into an electrical signal.
Biometric sensor	Biometric	Reads a fingerprint, iris, or face trait.
Barcode / QR code	Code/tag	Read optical codes that identify an item.
RFID	Code/tag	Reads a tag by radio waves; no line of sight.

12.12 Points of confusion between similar devices

Beyond classifying a device, JKSSB items often test a single distinction between two devices that look or sound alike. The table below collects the pairs used in this lesson.

Pair	The distinction to state
Mouse vs trackball	Mouse: the device slides. Trackball: the ball rotates, device still.
Touchscreen vs touchpad	Touchscreen: touch the display (direct). Touchpad: separate pad moves the pointer (indirect).
Light pen vs stylus	Light pen: light sensor on a CRT screen. Stylus: used with a digitizer tablet.
Scanner vs digital camera	Scanner: converts a flat document or picture into an image. Camera: captures a scene.
Barcode vs QR code	Barcode: 1D parallel lines. QR code: 2D matrix, holds more data.
Barcode/QR vs RFID	Barcode/QR are read optically and need line of sight; RFID uses radio waves.
Microphone vs speaker	Microphone: audio input. Speaker: audio output.
Data glove vs haptic glove	Data glove: captures hand movement (input). Haptic glove: gives force feedback (output).

12.13 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

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1. A light pen is an **input** device; a plotter is an **output** device. Do not swap them (TRAP-12).
2. A punch-card reader, a digitizer tablet, and a scanner are all **input** devices. A drum printer and a sound card are **output** devices (TRAP-12).
3. A **trackball** rotates its ball in place; a **mouse** slides over a surface.
4. A **touchscreen** is touched directly; a **touchpad** is a separate pad that moves the pointer indirectly.
5. **RFID** uses radio waves and needs no line of sight; a **barcode** and a **QR code** are read optically.
6. **Haptic gloves**, a **tactile display**, and a **Voice Response Unit (VRU)** give feedback to the user and are **output**; a **data glove** captures hand movement and is **input** (PYQ-09).

Example 12.1. *An item lists four devices — light pen, punch-card reader, plotter, and scanner — and asks which are input devices. Classify each one: the light pen is input, the punch-card reader is input, the plotter is output, and the scanner is input. The answer is the light pen, the punch-card reader, and the scanner. The plotter is the only output device in the list.*

Exercises

1. Define an input device and give three examples.
2. Classify these devices by the type of data they capture: keyboard, microphone, scanner, fingerprint sensor.
3. Distinguish direct input from indirect input, giving one example of each.
4. Explain the difference between a trackball and a mouse, and between a touchscreen and a touchpad.
5. Differentiate a barcode, a QR code, and RFID.
6. State whether each is input or output: light pen, plotter, digitizer tablet, drum printer, sound card.

Solutions

1. An input device accepts data or control signals from the user or the environment and passes them to the CPU in machine-readable form. Examples: keyboard, mouse, scanner (any three valid devices).
2. Keyboard — text input; microphone — audio input; scanner — image input; fingerprint sensor — biometric input.
3. Direct input captures the data at its source, such as a scanner or a biometric sensor. Indirect input needs the user to encode the data, such as a keyboard or a mouse.
4. With a mouse, the device slides over a surface; with a trackball, the ball is rotated in place and the device stays still. A touchscreen is touched directly on the display; a touchpad is a separate pad that moves the pointer indirectly.
5. A barcode is a 1D pattern of lines read optically; a QR code is a 2D matrix that holds more data; RFID uses radio waves between a tag and a reader and needs no line of sight.
6. Light pen — input; plotter — output; digitizer tablet — input; drum printer — output; sound card — output.

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LESSON 13

OCR, OMR, MICR

13.1 Reading machines and why they matter

A computer works only with data that has already been converted into machine-readable form. When the data exists on paper — as printed text, as a filled-in bubble on a form, or as a line of characters on a cheque — the machine needs a way to read it. A **reading machine** is the device or technology that captures this physical information and turns it into digital data the system can process. The physical reader (often a scanner, sometimes a dedicated reader) performs the capture; the *recognition method* decides what the captured pattern means. This lesson covers the three most examined recognition methods — OCR, OMR, and MICR — together with the code-and-tag devices (barcode, QR code, RFID) and biometric input, because questions frequently ask candidates to tell these near-neighbours apart.

The single most important idea is the *read-target* of each method. **OCR** reads *text*. **OMR** reads *marks*. **MICR** reads *characters printed in magnetic ink*. Almost every item on this topic can be solved by fixing this one distinction in memory.

All the devices in this lesson are *input* devices or reading *technologies*: they bring data into the system. Most of them are also *direct* input, because the device reads the object, mark, code, or trait itself, instead of asking the user to type it. This is why a reading machine is faster and less error-prone than a keyboard for a repetitive task such as reading a thousand answer sheets or a thousand cheques.

Definition — Reading machine

A device or technology that captures printed characters, marks, codes, or tags from a physical medium and converts them into machine-readable data for the computer to process.

13.2 Terminology at a glance

The table below fixes the meaning of every term used in this lesson. Read it down once; then, for each term, be ready to state the read-target and one place it is used.

Term	Meaning in this lesson
OCR	Optical Character Recognition — converts an image of printed or written text into editable text.
OMR	Optical Mark Recognition — detects the presence or absence of marks at fixed positions on a form.
MICR	Magnetic Ink Character Recognition — reads characters printed in magnetic ink, used on bank cheques.
Barcode	A one-dimensional pattern of parallel lines read optically to identify an item.
QR code	A two-dimensional matrix of squares (Quick Response) that holds more data than a barcode.
RFID	Radio Frequency Identification — a tag and reader exchange radio waves; no line of sight is needed.
Biometrics	Input that identifies a person from a physical or behavioural trait (fingerprint, iris, face).

13.3 OCR — Optical Character Recognition

OCR stands for **Optical Character Recognition**. It takes a scanned image of *printed or handwritten text* and converts it into machine-encoded, *editable text*. The key word is *character*: OCR recognises individual letters, digits, and symbols, so the output is text that a computer can search, copy, and edit. This is what makes OCR different from OMR, which recognises only whether a position has been marked and never identifies a letter.

OCR is used to digitise the printed word. A book, a bundle of invoices, a passport page, or an old form can be scanned into an image, and OCR software then extracts the text from that image. A useful way to remember the pairing is: the **scanner** is the hardware that creates the image, while **OCR** is the technology that turns the image of text into actual, selectable text.

Definition — OCR

A recognition technology that reads characters from a scanned image of printed or handwritten text and converts them into machine-encoded, editable text.

OCR works best on clean, clearly printed text in a known font; its accuracy drops when the page is skewed, smudged, or handwritten. When the input is handwritten rather than printed, the same idea is sometimes called *Intelligent Character Recognition* (ICR), but for exam purposes both count as OCR-style reading of text. OCR is *not* a storage device and *not* a printer: it only converts an existing image into text.

Example 13.1. *A student scans a page of a textbook into the computer. The scanner produces an* image*, so the words on the page cannot yet be copied or searched. Running OCR on that image produces a text file in which every word can be selected and edited. The scanner supplied the image; OCR supplied the text.*

13.4 OMR — Optical Mark Recognition

OMR stands for **Optical Mark Recognition**. It reads a form by detecting the *presence or absence of a mark* at fixed positions. Typically the candidate fills a small bubble or shades a box with a pencil or pen, and the OMR reader checks which positions have been darkened. Crucially, OMR reads *marks*, not characters: it does not know what a letter "says", only that a particular position has been filled.

Because it detects marks at fixed coordinates, OMR is ideal for forms where the answer is one of a small set of choices. Its classic use is the **MCQ answer sheet**: the machine reads which bubble has been shaded for each question. It is also used for questionnaires, surveys, census forms, and ballots. A very common exam trap is to confuse OMR with OCR; remember that OMR works on *marks*, while OCR works on *text*.

Definition — OMR

A recognition technology that detects the presence or absence of marks at fixed positions on a form, used mainly to read objective (MCQ) answer sheets and survey forms.

OMR is extremely fast because it does not have to interpret a shape as a character; it only has to decide whether a small region is dark or light. That speed is why OMR is the standard method for large-scale examinations where millions of answer sheets must be processed quickly. Its natural limit is that it can record only *which* option was marked, not any written answer: if the candidate writes a sentence instead of shading a bubble, OMR cannot read it.

13.5 MICR — Magnetic Ink Character Recognition

MICR stands for **Magnetic Ink Character Recognition**. It reads characters printed in a special *magnetic ink* that contains iron oxide. The characters are printed in a standard font along the bottom of a *bank cheque* as the **MICR line**, which normally encodes the cheque number, the

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bank and branch code, and the account number. A MICR reader magnetises the ink and detects the shapes of the characters as they pass the read head.

Banking is the defining application of MICR. Cheque processing must be very fast and highly secure, and MICR satisfies both needs: the reading is quick, and because the ink is magnetic it is difficult to forge or to alter without detection. When a question mentions a *bank cheque* or *magnetic ink*, the answer is almost always MICR.

Definition — MICR

A recognition technology that reads characters printed in magnetic ink, used mainly in banking to process the MICR line printed at the bottom of a cheque.

Unlike ordinary ink, magnetic ink can be read even if the cheque is slightly crumpled or overstamped, because the reader senses the magnetic field of the ink rather than a purely visual image. This ruggedness and the difficulty of counterfeiting magnetic ink are exactly the reasons banks adopted MICR for cheque clearing.

13.6 The crucial trio: OCR vs OMR vs MICR

These three names are the heart of the lesson, and JKSSB questions routinely swap their meanings in the options. The table below is the single most useful thing to memorise.

Method	What it reads	Where it is used
OCR	Printed or written characters	Digitising books, documents, and passports
OMR	Marks on a form	MCQ answer sheets, surveys, ballots
MICR	Characters in magnetic ink	Bank cheques and cheque clearing

The distinction to hold is: **MICR** uses *magnetic ink*; **OCR** reads *printed text*; **OMR** reads *marks*. This is the JKSSB trap TRAP-25, and the real paper item behind it (PYQ-30) asks simply that magnetic ink means MICR.

It also helps to picture *how* each one captures its data. OMR and OCR both read an optical (light) signal: the reader sees the paper. MICR is the odd one out, because its reader senses a *magnetic* signal from the ink. So a quick way to separate them in a hurry is: two are *optical*, one is *magnetic*.

Method	Signal read	Input pattern	Cannot do
OCR	Optical (light)	Text characters	Cannot read marks alone
OMR	Optical (light)	Marks / shaded bubbles	Cannot read written characters
MICR	Magnetic	Magnetic-ink characters	Not used for forms or plain text

13.7 Barcodes and QR codes

A **barcode** is a one-dimensional (1D) pattern of parallel lines of varying width and spacing. A barcode scanner shines a light beam across the lines and reads the reflected pattern, which represents a number or code that identifies a product. Because the scanner must see the lines, a barcode requires *line of sight*. Barcodes are used on product labels, on the ISBN of books, and on postal items.

A **QR code** — **Quick Response** code — is a *two-dimensional* (2D) matrix of black and white squares. It stores more data than a barcode and can be read from any direction. Because it can hold a URL, scanning a QR code can open a website or an app directly. Both a barcode and a QR code are read *optically*, so the nearest-neighbour trap is simply that a QR code is two-dimensional and carries more information, while a barcode is one-dimensional.

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Definition — QR code

A two-dimensional matrix of black and white squares (Quick Response code) that stores more data than a barcode and can be read from any direction.

Example 13.1. *A shop assistant passes a product over a laser at the billing counter: the scanner reads the 1D* barcode *on the packet and looks up its price. A customer points a phone camera at the square* QR code *on a poster: the phone reads the 2D code and opens the shop's website. The barcode carried a short number; the QR code carried a whole web address.*

13.8 RFID — Radio Frequency Identification

RFID stands for **Radio Frequency Identification**. An RFID system has two parts: a *tag* attached to the object and a *reader* that interrogates it. The two exchange *radio waves*, so no line of sight is required, unlike a barcode. A passive tag carries no battery and is powered by the reader's signal; an active tag has its own battery and can broadcast over a longer range.

RFID is used where the tag may be out of sight or where many tags must be read quickly: toll collection (for example FASTag), access-control cards, passports, library books, and inventory tracking. The exam contrast is always barcode/QR versus RFID: barcode and QR are read *optically* and need line of sight, while RFID uses *radio waves* and can read a tag without it being visible.

13.9 Biometric input

Biometrics identifies a person from a physical or behavioural trait. The common physical traits are the **fingerprint**, the **iris**, the **face**, and the palm; the voice is a common behavioural trait. A sensor captures the trait and feeds the measured data to the computer as *input*; the system then compares it with the stored template. Biometric input is a *direct* form of input because the sensor reads the trait itself rather than asking the user to type or encode it.

Biometrics are used for attendance systems, device unlocking, and authentication, and they appear in national identity and benefit systems. A useful checklist item for the exam is that a fingerprint reader, an iris scanner, and a face-recognition camera are all *input* devices; a device that merely prints or displays the result is *output*. Do not confuse biometric traits (fingerprint, iris, face) with the optical reading machines: a fingerprint sensor reads a part of the human body, not a paper code.

13.10 Full-form ladder and memory hooks

Direct-recall items on this topic usually ask for a full form, so learn the expansions exactly. The same expansions appear on the slides; repeat them until they can be produced without hesitation.

Short form	Full form
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
QR	Quick Response
RFID	Radio Frequency Identification

Three one-line hooks carry most of the content of this lesson. First, *OCR reads text, OMR reads marks, MICR reads magnetic ink.* Second, *a barcode is one-dimensional, a QR code is two-dimensional, and RFID uses radio waves.* Third, *a biometric sensor reads a human trait — fingerprint, iris, face, or voice.* If a question seems unfamiliar, reduce it to one of these three sentences and the answer usually becomes clear.

13.11 Applications at a glance

The table gathers the reading machines and code devices with their defining application, for quick revision.

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Technology	Defining application
OCR	Converting scanned printed text into editable text
OMR	Reading marks on objective (MCQ) answer sheets
MICR	Reading the magnetic-ink line at the bottom of a bank cheque
Barcode	Reading 1D product labels and book ISBNs
QR code	Reading 2D codes that can open a link or store more data
RFID	Reading tags by radio waves (tolls, access cards, inventory)
Biometrics	Identifying a person from a fingerprint, iris, face, or voice

13.12 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **MICR** uses **magnetic ink**; **OCR** reads **printed text**; **OMR** reads **marks**. Do not interchange them (TRAP-25, PYQ-30).
2. A real PYQ asks which technology uses magnetic ink on cheques: the answer is **MICR** (PYQ-30).
3. **OMR**, not OCR, is used for **MCQ answer sheets**, because it reads marks at fixed positions.
4. A **QR code** is two-dimensional and holds more data than a one-dimensional **barcode**; both are read optically.
5. **RFID** uses **radio waves** and needs no line of sight; a barcode or QR code must be *seen* by the reader.
6. A **fingerprint** or **iris** scanner is **input**; biometrics identify a person from a physical or behavioural trait.

Example 13.1. *An item states that a bank reads the characters at the bottom of a cheque using OCR. This is wrong. The cheque characters are printed in magnetic ink, so the technology is **MICR**. OCR would be used to read ordinary* printed text *from a scanned document, and OMR would be used to read* marks *on a form — neither of which matches a cheque's magnetic-ink line.*

Example 13.2. *An item lists four reading methods and asks which two are* optical. *OCR and OMR both use a light signal to read the paper, so they are the two optical methods. MICR is* magnetic, *not optical, and RFID is neither optical nor magnetic — it uses radio waves. Separating the methods by their signal (optical, magnetic, radio) is a fast way to answer such a question.*

Exercises

1. Expand OCR, OMR, and MICR, and state in one line what each reads.
2. Which technology would read an objective (MCQ) answer sheet, and why?
3. Explain why a bank uses MICR rather than OCR for cheque processing.
4. Distinguish a barcode from a QR code on two points.
5. State one way in which RFID differs from a barcode.
6. Give three examples of biometric traits and state the purpose for which biometric input is commonly used.
7. Classify these as optical, magnetic, or radio reading: OMR, MICR, RFID, OCR.

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Solutions

1. OCR = Optical Character Recognition, which reads printed or written characters; OMR = Optical Mark Recognition, which reads marks on a form; MICR = Magnetic Ink Character Recognition, which reads characters in magnetic ink.
2. OMR, because it detects the presence or absence of a mark at fixed positions, which is exactly how a candidate fills bubbles on an MCQ answer sheet.
3. The cheque characters are printed in magnetic ink, which MICR reads directly; the reading is fast, and the magnetic ink is difficult to forge or alter. OCR reads ordinary printed text, not magnetic-ink characters.
4. A barcode is a one-dimensional pattern of parallel lines and holds less data; a QR code is a two-dimensional matrix of squares, holds more data, and can be read from any direction.
5. RFID exchanges radio waves between a tag and a reader and needs no line of sight, whereas a barcode is read optically and must be visible to the scanner. RFID can also read many tags at once.
6. Fingerprint, iris, and face (voice and palm are also valid). Biometric input is commonly used for attendance, authentication, and device unlocking.
7. OMR and OCR are *optical*; MICR is *magnetic*; RFID is *radio*.

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LESSON 14

Output Devices

14.1 What is an output device

A computer works in a cycle: input, process, and output. In the output step, the result of processing is handed back to the user or to another device. An **output device** is the hardware that receives processed data from the CPU and presents it in a form a human can see, hear, or read. Without an output device the result of the computation would stay locked inside the machine.

Definition — Output device

Hardware that receives processed data from the CPU and presents it to the user or to another device in a readable form, such as on a screen, through a speaker, or on paper.

Common output devices include the monitor, projector, speaker, headphones, printer, and plotter. The following table fixes the meaning of each term used in this lesson.

Term	Meaning in this lesson
Monitor	The main visual output device; shows text, images, and video as soft copy.
Projector	Throws a large image onto a screen or wall as soft copy.
Speaker	Converts an electrical signal into sound for a group of listeners.
Headphones	Produce sound for a single listener; worn on or over the ears.
Printer	Produces hard copy on paper; impact or non-impact.
Plotter	Draws large, precise line drawings such as engineering designs.
MFP	Multifunction Printer; combines printing, scanning, copying, and often faxing.

14.2 Classifying output devices

Output devices are grouped by the form in which they present the result. This is the classification JKSSB question-setters use most often, so learn one example for each row.

Category	Typical devices
Visual / soft copy	Monitor, projector
Audio	Speaker, headphones
Printed / hard copy	Printer, plotter

A single device may belong to more than one row if it produces more than one kind of output, but for exam purposes use the primary role: a monitor is a visual output device, a speaker is an audio output device, and a printer is a printed (hard copy) output device.

14.3 Soft copy vs hard copy

Output comes in two forms. **Soft copy** is output that is displayed electronically, such as an image on a monitor or a projected slide; it is not printed and can usually still be edited. **Hard copy** is output that is printed on a physical medium such as paper, produced by a printer or a plotter; once printed it is fixed.

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Definition — Soft copy

Output that is shown on a screen or projected electronically and is not printed, such as the image on a monitor or projector.

Definition — Hard copy

Output that is printed on a physical medium such as paper, produced by a printer or a plotter.

The distinction matters because a single item can test it in either direction. If the stem mentions a screen, a display, or a projected image, the answer is soft copy; if it mentions paper, printing, or a physical printout, the answer is hard copy.

Point	Soft copy	Hard copy
Form	Electronic, on a screen	Physical, on paper
Examples	Monitor, projector	Printer, plotter
Editable	Easily edited	Fixed once printed

14.3.1 Uses and limits of each form

Soft copy and hard copy serve different needs, and an exam item may ask which is better for a stated purpose. Soft copy is convenient because it can be edited, searched, copied, and shared electronically, and it costs nothing to display; its limit is that it needs power and a screen, and it is easy to lose if it is not saved. Hard copy is convenient because it needs no power to read, is permanent, and can be signed or filed as a legal record; its limits are that it can be bulky, is fixed once printed, and consumes paper and ink. The rule of thumb is: to read, edit, or send, use soft copy; to sign, file, or archive, use hard copy.

14.4 Monitors

A **monitor** is the main visual output device of a computer. It displays text, images, and video as soft copy. Screen size is measured diagonally in inches, and picture sharpness is measured in pixels, which together give the resolution of the display. A monitor shows output to the user; it does not take input, with the single exception of a touchscreen, which senses touch in addition to displaying.

14.4.1 CRT displays

A **CRT** display uses a **Cathode Ray Tube**. An electron beam sweeps across a phosphor-coated screen inside a vacuum tube to form the image. CRT monitors are bulky and deep, and they are now largely replaced by flat panels. For exam purposes the key point is the full form: CRT = **Cathode Ray Tube**, and it is the older, bulky technology.

14.4.2 Flat-panel displays: LCD, LED, and OLED

A **flat-panel display** is thin, light, and flat. The common technologies are LCD, LED, and OLED.

Type	What it means	Backlight
LCD	Liquid Crystal Display; cells of liquid crystal control light	Needs a backlight
LED	Light Emitting Diode display; an LCD panel lit by LEDs	LED backlight
OLED	Organic Light Emitting Diode; each pixel emits its own light	No backlight (self-emitting)

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The nearest-neighbour trap is LCD versus LED versus OLED. An **LCD** needs a separate backlight to be visible. An **LED** display is really an LCD panel whose backlight is made of LEDs. An **OLED** display needs no backlight at all, because each pixel makes its own light.

14.4.3 Display size, resolution, and refresh

The size of a monitor is given as the diagonal length of the screen in inches. The picture is made of tiny dots called **pixels**; the number of pixels across and down the screen is the **resolution**, such as 1920 by 1080. A higher resolution shows more detail in the same screen area. The **refresh rate** is how many times per second the screen is redrawn, measured in hertz (Hz); a higher refresh rate gives smoother motion. These are the three numbers a display specification quotes, and they apply to monitors and to laptop screens alike.

14.5 Projector

A **projector** throws a large image onto a screen or a wall. It is used in classrooms, meetings, and cinemas to show the same computer output to many people at once. The projected image is soft copy, because it is shown on a surface and not printed. A projector is an output device; it does not capture anything, so it must not be confused with a scanner or a camera.

14.6 Audio output: speakers and headphones

A **speaker** converts an electrical signal into sound that a group of listeners can hear. **Headphones** produce sound for a single listener and are worn on or over the ears. Both are audio output devices.

The nearest-neighbour pair is the speaker against the microphone. A **microphone** takes sound in and is an audio input device; a **speaker** or a **headphone** sends sound out and is an audio output device.

Example 14.1. *A student plays a recorded lecture on the office desktop. The sound leaves the computer through the speakers, so the speakers are audio output devices. If the student then records a reply into the same computer, the microphone is used; that is an audio input device. The same machine performs both jobs through two different devices.*

14.7 Printers: overview

A **printer** produces hard copy on paper. Printers fall into two broad classes according to how they mark the paper. An **impact printer** strikes a ribbon against the paper; a **non-impact printer** marks the paper without striking it, using ink, toner, or heat. The common types are the dot-matrix printer (impact) and the inkjet and laser printers (non-impact). Laser printers are common in offices, while inkjet printers are common in homes. A printer is an output device (TRAP-12).

Point	Impact	Non-impact
Method	Strikes a ribbon and the paper	No striking; uses ink, toner, or heat
Examples	Dot-matrix	Inkjet, laser
Noise	Noisier	Quieter
Graphics quality	Lower	Higher

A **dot-matrix** printer is an impact printer and is the noisiest of the common types. An **inkjet** printer sprays tiny drops of ink onto the paper, and a **laser** printer forms the image with toner that is fused onto the paper by heat. Both are non-impact printers.

The choice follows the job. A dot-matrix printer is cheap to run and can print multi-part carbon forms, but its quality is low and it is noisy. An inkjet printer is quiet and inexpensive to buy and is suited to colour photographs and home use, though the ink can be costly. A laser printer is fast, quiet, and gives sharp text, which is why it is the usual office choice. Whatever the type, every printer is an output device producing hard copy.

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Example 14.1. *An office prints a hundred letters every morning. The best choice is a laser printer, because it is fast, quiet, and gives sharp text. If the same office had to print only the occasional colour photograph at home, an inkjet printer would be enough. If it needed multi-part carbon forms for a carbon copy, a dot-matrix printer would be the right impact printer.*

14.8 Plotter

A **plotter** draws large, precise line drawings, such as engineering and architectural designs, on paper larger than a normal printer's. It draws using a pen or a vector-based method, which makes it suitable for the continuous lines of technical drawings. A plotter is an output device: it produces hard copy but it does not read anything, so it must not be confused with a scanner (TRAP-12).

Example 14.1. *An item asks which output device is used to produce large engineering drawings. The answer is a **plotter**, because it is designed for large, precise line drawings. A normal printer is meant for ordinary pages, a monitor shows soft copy, and a scanner is an input device.*

14.9 Multifunction printer (MFP)

An **MFP** is a **Multifunction Printer**, also called an **all-in-one** device. It combines **printing, scanning, copying**, and often **faxing** in a single unit. Because it does several jobs, an MFP is the one common device that is both an input and an output device: its scanner part takes input, while its printer part gives output. MFPs are common in offices and small businesses.

14.10 Summary of output devices

The table below gathers the devices in this lesson for quick revision. Use it as a checklist: for each device, be ready to state its category (visual, audio, or printed) and its one-line function, and to name the device it is most often confused with.

Device	Category	Function
Monitor	Visual / soft copy	Shows text, images, and video on a screen.
Projector	Visual / soft copy	Throws a large image onto a screen or wall.
Speaker	Audio	Converts an electrical signal into sound for a group.
Headphones	Audio	Produces sound for a single listener.
Printer	Printed / hard copy	Prints text and graphics on paper; impact or non-impact.
Plotter	Printed / hard copy	Draws large, precise line drawings.
MFP	Hybrid	Combines printing, scanning, copying, and often faxing.

14.11 Points of confusion between similar devices

Beyond classifying a device, JKSSB items often test a single distinction between two devices that look or sound alike. The table below collects the pairs used in this lesson.

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Pair	The distinction to state
Printer vs plotter	Printer: standard pages of text and graphics. Plotter: large, precise line drawings.
Monitor vs projector	Monitor: image on its own screen. Projector: image thrown onto a screen or wall.
Speaker vs microphone	Speaker: audio output. Microphone: audio input.
Soft copy vs hard copy	Soft copy: shown on a screen. Hard copy: printed on paper.
LCD vs OLED	LCD: needs a backlight. OLED: self-emitting, no backlight.
Impact vs non-impact	Impact: strikes the paper (dot-matrix). Non-impact: no striking (inkjet, laser).

Example 14.1. *An office buys one machine that must print letters, scan signed forms, and photocopy pages. The right choice is a Multifunction Printer (MFP), because it does all three jobs in one unit. If the office instead needed very large engineering plans, the correct device would be a plotter, and if it only needed to show a presentation to a room, the correct device would be a projector. Matching the output device to the job is the practical form of this topic's classification.*

14.12 Other output devices on the exam

A few output devices appear in real JKSSB items without being part of the main list. A **sound card** produces audio output: it turns the digital sound data into a signal for the speakers. A **Voice Response Unit (VRU)** speaks responses to a caller and so gives audio output. **Haptic gloves** and a **tactile display** give touch or force feedback to the user and are therefore output devices. All of these feed information *back* to the user and must be told apart from input devices of similar names, such as a **data glove**, which captures hand movement as input (PYQ-09).

Definition — Sound card

An expansion card that produces audio output, converting digital sound data into a signal that speakers or headphones can play.

14.13 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. **Soft copy** is on a screen (monitor, projector); **hard copy** is printed on paper (printer, plotter). Do not swap them.
2. A **plotter** and a **printer** are **output** devices; a **scanner**, a **light pen**, and a **digitizer** are **input** devices (TRAP-12).
3. **CRT** = Cathode Ray Tube and is bulky; modern monitors are flat panels (LCD, LED, OLED).
4. An **LCD** needs a backlight, an **LED** display is an LCD with an LED backlight, and an **OLED** display is self-emitting with no backlight.
5. A **dot-matrix** printer is an **impact** printer; inkjet and laser printers are **non-impact**.
6. A **plotter** draws large engineering drawings (PYQ-29), while a normal printer is meant for ordinary pages.
7. A **sound card**, a **Voice Response Unit (VRU)**, a **haptic glove**, and a **tactile display** give output to the user; a **data glove** captures hand movement as input (PYQ-09).

Example 14.1. *An item lists four devices — scanner, plotter, drum printer, and light pen — and asks which are output devices. Classify each one: the scanner is input, the plotter is output, the drum printer is output, and the light pen is input. The answer is the plotter and the drum printer. This is exactly the input/output classification tested by TRAP-12.*

Exercises

1. Define an output device and give three examples.
2. Distinguish soft copy from hard copy, giving one example of each.
3. Expand CRT, LCD, and OLED, and state which of them needs a backlight.
4. Explain the difference between an impact printer and a non-impact printer, giving one example of each.
5. Which output device is used for large engineering drawings, and why?
6. State whether each is input or output: plotter, scanner, projector, digitizer tablet, speaker.

Solutions

1. An output device receives processed data from the CPU and presents it to the user in a readable form. Examples: monitor, printer, speaker (any three valid devices).
2. Soft copy is shown on a screen and is not printed, such as an image on a monitor or projector. Hard copy is printed on paper, such as a page from a printer or plotter.
3. CRT = Cathode Ray Tube; LCD = Liquid Crystal Display; OLED = Organic Light Emitting Diode. The LCD needs a backlight; the OLED does not, because each pixel emits its own light.
4. An impact printer strikes a ribbon against the paper, such as a dot-matrix printer. A non-impact printer marks the paper without striking it, such as an inkjet or a laser printer.
5. A plotter is used, because it is designed to draw large, precise line drawings such as engineering and architectural designs (PYQ-29).
6. Plotter — output; scanner — input; projector — output; digitizer tablet — input; speaker — output.

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LESSON 15

Printers and Displays

15.1 Printers and hard-copy output

A printer is an output device. It accepts data from the computer and records it as **hard copy** — output on a physical medium, normally paper. Output that stays on a screen is called **soft copy**; a printer is the device that turns soft copy into hard copy. Printers are described from three angles: by the **mechanism** they use (dot matrix, inkjet, laser, thermal, 3D), by whether they touch the paper (**impact** or **non-impact**), and by the **unit printed** (a character, a line, or a page).

Definition — Printer

An output device that records computer output as a permanent, human-readable copy on paper or another medium.

Basis	Types
By mechanism	Dot matrix, inkjet, laser, thermal, 3D printer
By contact	Impact (strikes the paper) vs non-impact (no contact)
By unit printed	Character printer, line printer, page printer

The same machine can carry several labels at once. A laser printer, for example, is a non-impact printer, a page printer, and a laser-mechanism printer. In an exam item, read which label is being asked for before choosing.

15.2 Impact and non-impact printers

An **impact printer** forms characters by physically striking an inked ribbon against the paper. The contact makes these printers relatively noisy, but because the type presses through the paper, they can produce **multiple carbon copies** of multi-part forms. Common impact printers are the dot matrix printer, the daisy wheel printer, and line printers such as the drum printer and chain printer.

A **non-impact printer** forms characters without striking the paper. These printers are quieter and generally give better print quality, but they cannot make carbon copies. Common non-impact printers are the inkjet printer, the laser printer, and the thermal printer.

Definition — Impact printer

A printer that strikes an inked ribbon against the paper to form a character.

Definition — Non-impact printer

A printer that forms characters without touching the paper with a striking mechanism.

Point	Impact	Non-impact
Contact with paper	Yes (strikes)	No
Noise	Louder	Quieter
Carbon copies	Possible	Not possible
Examples	Dot matrix, daisy wheel, drum/chain line printers	Inkjet, laser, thermal

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Example 15.1. *An item lists "inkjet, laser, thermal, dot matrix" and asks which is an impact printer. Trace the test: which device strikes an inked ribbon against the paper? Only the dot matrix printer does. The other three form the image without contact, so they are non-impact.*

15.3 Impact printers in detail

Dot matrix printer

The dot matrix printer is an impact printer. Its print head holds rows of tiny pins or hammers. The pins strike an inked ribbon to build each character as a pattern of dots inside a matrix. It is noisy and its print quality is low, but it is cheap to run and can print carbon copies and multi-part invoices and receipts. For this reason it survives in billing and point-of-sale work.

Daisy wheel printer

A daisy wheel printer is an impact printer that uses a spinning wheel with raised type characters, much like an electric typewriter. It produces good text quality but cannot print graphics or images well.

Line printers

A line printer prints an entire line of text in one operation instead of one character at a time. A **drum printer** uses a rotating drum that carries the full character set, with a hammer striking the paper wherever a character is needed. A **chain (band) printer** moves a chain or band of type characters past the hammers. Line printers are high-speed impact printers associated with older mainframes and large batch printing.

Definition — Line printer

A printer that prints one full line of characters at a time; the drum printer and chain printer are line printers.

Print units

Printers are often grouped by the unit they print. A **character printer** prints one character at a time (dot matrix, daisy wheel). A **line printer** prints one line at a time (drum, chain). A **page printer** prints one complete page at a time (laser and LED printers). The unit grows from character to line to page.

15.4 Inkjet printers

An inkjet printer is a non-impact printer. Its print head sprays tiny **droplets of liquid ink** through fine nozzles onto the page. Because there is no striking, it is quieter than an impact printer, and it prints good-quality full-colour images and photographs on many kinds of paper. Its drawbacks are that it is slower per page than a laser printer, that wet ink can **smudge**, and that its ink cartridges must be replaced when empty. Inkjet printers use liquid ink, not toner.

Definition — Inkjet printer

A non-impact printer that forms characters and images by spraying fine droplets of liquid ink onto the paper.

15.5 Laser printers

A laser printer is a non-impact printer that uses a **laser beam** and **toner**, a fine dry powder, in a xerographic (electrostatic) process. The laser draws the page image onto a charged drum; the toner sticks to the charged areas; the toner is then transferred to the paper and fused in place with heat and pressure. The result is fast, sharp, high-quality text, which is why laser printers are common in offices. A laser printer is a **page printer**: it prints a whole page at a time. Because it uses toner rather than liquid ink, its output does not smudge when it gets wet.

Definition — Laser printer

A non-impact page printer that uses a laser beam to draw an image on a charged drum and toner powder to form the printed page.

Point	Inkjet	Laser
Contact	Non-impact	Non-impact
Image formed by	Liquid ink droplets	Laser beam + toner
Unit printed	Character/line	Page printer
Speed	Slower	Faster
Strengths	Colour photos	Sharp text, offices

15.6 Thermal printers

A thermal printer is a non-impact printer that uses **heat**. It heats specially treated **thermal paper** (or a heat-sensitive ribbon) to form characters. In its simplest form it needs no ink or toner cartridge, and it operates silently. It is used for point-of-sale receipts, tickets, barcode labels, and older fax machines. It is usually limited to a single colour. Thermal, laser, and inkjet printers are all non-impact; dot matrix is impact.

15.7 3D printers

A 3D printer builds a solid three-dimensional object **layer by layer** from a digital model. This process is called **additive manufacturing**. It does not print ink onto paper; instead it adds material such as plastic, resin, or metal until the object is complete. 3D printers are used for prototypes, models, spare parts, and some medical implants. The word "printer" is used, but the output is a physical object rather than a document.

15.8 Plotters

A plotter is an output device that **draws** line graphics, usually with one or more pens that move over the paper. It works with **vector** drawings, so it is ideal for large engineering drawings, maps, and CAD plans, and it handles sheet sizes that ordinary printers cannot. A plotter draws lines; a printer lays down dots or characters. Both are output devices. A plotter must not be confused with a scanner or a digitizer, which are input devices.

Example 15.1. *An item says, "A plotter is an input device." This is wrong. A plotter produces output by drawing lines on paper, so it is an **output** device. The items that capture drawings as data — the scanner and the digitizer tablet — are the input devices.*

15.9 Display terminology

A monitor displays soft copy, and several exact terms describe it.

Definition — Pixel

The smallest addressable element of a display; short for "picture element". Every image on the screen is built from pixels.

Definition — Resolution

The number of pixels a display can show, written as columns × rows, for example 1920×1080. *More pixels mean a sharper image.*

The sharpness of a printed page is measured in **DPI**, which stands for **Dots Per Inch**.

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Definition — Refresh rate

*How many times per second a display redraws the image. It is measured in **hertz (Hz)**. A higher refresh rate (for example 120 Hz) gives smoother motion and less flicker.*

Definition — Response time

*How long a pixel takes to change from one state to another. It is measured in **milliseconds (ms)**. A lower response time means less motion blur or ghosting.*

Brightness is the amount of light a display emits, commonly measured in **nits** (candela per square metre, cd/m^2). **Contrast** is the difference between the brightest white and the darkest black the display can produce. The **contrast ratio** compares the two, for example $1000:1$; a higher ratio gives deeper blacks and more vivid images. In short: brightness is how much light, and contrast is the range between light and dark.

Display term	Meaning	Unit
Pixel	Smallest addressable element	—
Resolution	Number of pixels shown	pixels (or DPI for print)
Refresh rate	Screen redraws per second	hertz (Hz)
Response time	Time for a pixel to change	milliseconds (ms)
Brightness	Light emitted by the screen	nits (cd/m^2)
Contrast ratio	Brightest white : darkest black	ratio

15.10 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **Impact vs non-impact**. Only a printer that *strikes* an inked ribbon is impact. Dot matrix, daisy wheel, and drum/chain line printers are impact; inkjet, laser, and thermal are non-impact.
2. **Dot matrix is impact**. It is easy to group it with inkjet and laser by "modern-ness", but it strikes the ribbon, so it is impact.
3. **Inkjet vs laser**. Inkjet sprays *liquid ink*; laser uses a *laser beam and toner*. Do not swap the consumables.
4. **Page vs line printer**. A laser printer is a page printer (one page at a time); a drum or chain printer is a line printer (one line at a time).
5. **Printer vs plotter**. A plotter *draws* vector line graphics for engineering drawings; it is an output device, not a printer of characters and not an input device.
6. **Display units**. Refresh rate is in *Hz* (higher is better); response time is in *ms* (lower is better); brightness is in *nits*; resolution is a pixel count.

Example 15.1. *An item asks for the difference between an inkjet and a laser printer. The correct contrast is the image-forming method and consumable: the inkjet sprays liquid ink droplets, while the laser uses a laser beam and toner powder. Both are non-impact. A choice that calls the laser a character printer, or says the inkjet uses toner, is wrong.*

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Exercises

1. Define an impact printer and a non-impact printer, and give two examples of each.
2. Explain how a dot matrix printer forms a character, and state whether it is impact or non-impact.
3. Distinguish an inkjet printer from a laser printer in terms of the image-forming method and the consumable used.
4. What is a page printer? Name one example and contrast it with a line printer.
5. Define pixel, resolution, refresh rate, response time, brightness, and contrast, giving the unit of measurement for each where one applies.

Solutions

1. An impact printer forms characters by striking an inked ribbon against the paper (dot matrix, daisy wheel, drum/chain line printers). A non-impact printer forms characters without striking the paper (inkjet, laser, thermal).
2. The dot matrix print head has rows of pins that strike an inked ribbon to build each character as a pattern of dots in a matrix. Because it strikes the ribbon, it is an impact printer.
3. An inkjet printer sprays fine droplets of liquid ink through nozzles; a laser printer uses a laser beam to draw an image on a charged drum and toner powder to form the page. The inkjet consumable is liquid ink; the laser consumable is toner.
4. A page printer prints a whole page at a time; a laser printer is an example. A line printer prints a whole line at a time (drum or chain printer), a smaller unit than a page.
5. A pixel is the smallest addressable element of a display (no unit); resolution is the number of pixels shown, as columns $\times$ rows (pixel count, or DPI for print); refresh rate is the number of screen redraws per second (hertz, Hz); response time is the time for a pixel to change (milliseconds, ms); brightness is the light emitted by the screen (nits, cd/m^2); contrast is the difference between brightest white and darkest black, expressed as a contrast ratio.

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LESSON 16

Computer Ports

16.1 What a port is

Every device that connects to a computer does so through a **port**. A port is a socket or interface on the machine into which a cable or a device is plugged. It lets data and, in many cases, electric power pass between the computer and the attached device. Ports are not all alike: they differ in their shape, in the type of signal they carry (digital or analog), and in the job they are meant to do.

The common physical ports found on a personal computer are USB (including USB-C), HDMI, DisplayPort, VGA, Ethernet through an RJ-45 connector, the 3.5 mm audio jack, and SATA. Knowing which port suits which device is the central skill of this lesson.

Definition — Port

A socket or interface on a device where a cable or another device is connected, allowing data (and often power) to flow between the computer and the attached device.

Term	Meaning in this lesson
Port	The socket on the device (a USB port, an HDMI port).
Connector	The plug end of a cable that fits the port.
Cable	The wire that joins two connectors.
Physical port	A hardware socket on the machine.
Logical port	A software number that names a network service.

A first view of the ports covered in this lesson, with their signal type and their normal use, is given below. The rest of the lesson explains each one in turn.

Port	Signal	Typical use
USB / USB-C	Digital (data + power)	Keyboard, mouse, pen drive, phone
HDMI	Digital (video + audio)	TV, monitor, projector
DisplayPort	Digital (video + audio)	PC monitors, graphics cards
VGA	Analog (video only)	Older monitors and projectors
RJ-45 (Ethernet)	Digital (network)	Wired LAN cable, router, switch
3.5 mm audio jack	Analog (audio)	Headphones, speakers, microphone
SATA / eSATA	Digital (storage)	Internal and external drives

16.2 The two meanings of “port”

The word **port** has two entirely different meanings in computing, and the exam frequently tests the difference. A **physical port** is a piece of hardware — a socket such as a USB port, an HDMI port, or an RJ-45 network port. A **logical port**, also called a network port, is a software number that identifies a particular service running on a computer during network communication.

A web server, for example, listens on logical port 80 for ordinary web requests and on port 443 for secure web requests. The physical network cable, meanwhile, is plugged into an RJ-45 port on the network card. Thus one machine uses both meanings at the same time without any contradiction: the socket is hardware, and the port number is software.

Definition — Logical (network) port

A number, in the range 0 to 65535, that identifies a specific service or process on a computer for the purpose of network communication. It is not a physical socket.

Example 16.1. *An item states, "The monitor is connected to logical port 80." This is nonsense. Port 80 is a logical port used by web (HTTP) traffic, not a socket for a monitor. A monitor connects to a physical video port such as HDMI, DisplayPort, or VGA. The wrong statement has mixed the two meanings of the word* port.

A few well-known logical port numbers are worth remembering, because the exam sometimes asks which service uses a given port. The list below is illustrative; the concept being tested is that these are *numbers*, not sockets.

Service	Logical port	Full form
HTTP	80	HyperText Transfer Protocol
HTTPS	443	HTTP Secure
FTP	21	File Transfer Protocol
SMTP	25	Simple Mail Transfer Protocol
DNS	53	Domain Name System

16.3 USB: the Universal Serial Bus

USB stands for **Universal Serial Bus**. It is a standard interface for connecting a wide range of peripherals to a computer. Two of its best-known features are **plug and play** and **hot swapping**: a device can be attached or removed while the computer is running, without switching the machine off. A USB connection carries both **data** and **power**, which is why many small devices can be used or charged directly from a USB port.

Typical USB devices include the keyboard, the mouse, a printer, a pen drive (flash drive), an external hard disk, a webcam, and a smartphone. The connector at the end of a USB cable comes in several shapes:

- **USB Type-A**: the familiar rectangular connector found on most PCs.
- **USB Type-B**: often used for printers and scanners.
- **Mini-USB** and **Micro-USB**: smaller connectors used on older phones and devices.
- **USB Type-C**: a modern, small, reversible connector.

USB-C, or USB Type-C, deserves special attention. It is a *connector type*, not a separate bus. It is **reversible**, so it can be plugged in either way up, and a single USB-C connection can carry data, power, and video together. It is now common on smartphones, laptops, external SSDs, and docking stations. The key exam point is precision: USB is the standard, USB-C is one connector shape for that standard, and a USB port is the socket on the machine.

The USB standard itself has developed over time through versions such as USB 1.1, USB 2.0, USB 3.0 (and its later 3.1 and 3.2 refinements), and USB4. Each newer version generally offers a higher data-transfer rate, and the version is independent of the connector shape: a Type-A or Type-C socket can appear on a machine that supports different USB versions. For exam purposes, the safe points are the meaning of the abbreviation, the plug-and-play and hot-swap behaviour, the combined data-and-power carrying, and the reversibility of USB-C.

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16.4 Video ports: HDMI, DisplayPort, and VGA

Displays are attached through video ports. Three names dominate the syllabus.

HDMI stands for **High-Definition Multimedia Interface**. It carries **digital video and digital audio** over a single cable, which is why one HDMI cable can replace a separate video cable and a separate audio cable. It is very common on televisions, monitors, projectors, set-top boxes, and game consoles.

DisplayPort is also a **digital** audio/video interface, but it is designed mainly for computers and monitors. It appears on graphics cards, laptops, and desktop monitors, and it supports high refresh rates and multiple monitors through daisy-chaining. **Mini DisplayPort** is a smaller variant of the same interface.

Point	HDMI	DisplayPort
Signal	Digital video + audio	Digital video + audio
Main use	TV, home theatre, monitor	PC monitors, graphics cards
Note	Very common on TVs	Common on PCs; multi-monitor

Both HDMI and DisplayPort are digital. The right choice in practice usually depends on which sockets the devices offer, not on one being inherently better than the other.

VGA stands for **Video Graphics Array**. It is an **analog** video interface: it carries video only and no audio. It uses a 15-pin D-sub connector, often coloured blue and held in place by two screws. VGA is still seen on older monitors and projectors but has largely been replaced by HDMI and DisplayPort.

Definition — Digital vs analog video

Digital video ports (HDMI, DisplayPort) send the picture as discrete numbers and also carry audio; the analog port (VGA) sends a continuously varying signal and carries video only.

Example 16.1. *An item reads, "VGA carries both video and audio." This is false. VGA is an analog video-only interface with no audio channel. HDMI and DisplayPort, by contrast, carry digital video together with digital audio. The wrong option has transferred the audio capability of the digital ports onto VGA.*

16.5 Ethernet and the RJ-45 connector

Ethernet is the standard for **wired** computer networking, and it is commonly associated with the IEEE 802.3 family of standards. The physical socket used for a wired network connection is the **RJ-45** connector. RJ-45 stands for **Registered Jack 45**, and it has eight pins, used with twisted-pair cable such as Cat5e or Cat6.

Typical devices attached through an RJ-45 port are a router, a switch, a modem, and the network card of a desktop computer. A near-neighbour that causes confusion is **RJ-11**, the smaller telephone jack. RJ-45 is the network connector with eight pins; RJ-11 is the telephone connector and is smaller. The two look similar but serve different purposes.

16.6 Audio jacks

The common audio socket on a personal computer is the **3.5 mm audio jack**. It connects headphones, earphones, speakers, and microphones. The plug wiring has standard forms: **TRS** (Tip-Ring-Sleeve) carries stereo audio, while **TRRS** (Tip-Ring-Ring-Sleeve) adds a microphone channel, which is why a phone headset with a microphone uses a TRRS plug. On desktop PCs a colour convention is often followed: green for audio output, such as headphones or speakers, and pink for a microphone. Some equipment uses 2.5 mm or 6.35 mm jacks instead of the 3.5 mm size.

16.7 Storage interfaces: SATA and eSATA

SATA stands for **Serial Advanced Technology Attachment**. It is the interface that connects **internal storage** — a hard disk drive, a solid-state drive, or an optical drive — to the motherboard. The SATA data cable is thin and has seven pins, and a separate power connector feeds the drive. **eSATA** stands for **external SATA**, and it is used to connect an external drive. It is important to classify eSATA correctly: it is a *storage interface*, not a general-purpose peripheral port like USB. SATA is about attaching drives, whereas USB, HDMI, and the audio jack are about attaching other kinds of peripherals and displays.

16.8 Classifying a port: four questions to ask

When a question shows a port or names it, four questions settle almost every case. First, *is it a physical port or a logical one?* A socket is physical; a number such as 80 or 443 is logical. Second, *what kind of signal does it carry?* HDMI and DisplayPort are digital audio/video, VGA is analog video, the 3.5 mm jack is analog audio, and RJ-45 is a digital network link. Third, *what class of device does it attach?* Displays go to video ports, drives go to SATA, network equipment goes to RJ-45, and general peripherals go to USB. Fourth, *which near-neighbour is it often confused with?* USB-C is confused with USB generally; RJ-45 is confused with RJ-11; VGA is confused with HDMI and DisplayPort.

Definition — Port class

A way of grouping ports by what they attach: peripheral ports (USB), display ports (HDMI, DisplayPort, VGA), the network port (RJ-45), audio ports (3.5 mm jack), and storage interfaces (SATA, eSATA).

Two ports are easily confused with a near-neighbour. The table below keeps the distinctions in one place.

Port	Nearest confusion	The distinction
USB-C	USB (the standard)	USB-C is a connector shape; USB is the bus.
HDMI	DisplayPort	Both digital A/V; HDMI is common on TVs, DisplayPort on PCs.
VGA	HDMI / DisplayPort	VGA is analog, video-only; the others are digital with audio.
RJ-45	RJ-11	RJ-45 is the 8-pin network jack; RJ-11 is the smaller phone jack.
SATA	USB	SATA attaches drives (storage); USB is a general peripheral port.
Logical port	Physical port	Logical port is a number; physical port is a socket.

16.9 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. A **physical port** is a hardware socket; a **logical (network) port** is a software number, such as HTTP 80 or HTTPS 443. Never mix the two meanings of "port".
2. **VGA is analog** and video-only; **HDMI and DisplayPort are digital** and carry both video and audio.
3. **RJ-45** (network, eight pins) is not **RJ-11** (the smaller telephone jack).

4. **USB-C is a connector type** and is reversible; "USB" is the standard, and a USB port is the socket.
5. **SATA is a storage interface**; do not call it a display or network port.
6. Say the term precisely: a *port* is the socket, a *connector* is the plug, and a *cable* is the wire. This is the same discipline as file *format* vs file *extension* (TRM-09).

Example 16.1. *An item asks which port carries digital audio and digital video over one cable. Name the digital audio/video ports: HDMI and DisplayPort both qualify, whereas VGA is analog video-only, RJ-45 is networking, and SATA is storage. If the stem prefers the port common on television sets, choose HDMI; if it stresses PC monitors and multi-monitor setups, choose DisplayPort.*

The full forms below are the ones most likely to be asked directly, so they should be memorised exactly.

Abbreviation	Full form
USB	Universal Serial Bus
HDMI	High-Definition Multimedia Interface
VGA	Video Graphics Array
RJ-45	Registered Jack 45
SATA	Serial Advanced Technology Attachment
eSATA	external SATA
TRS	Tip-Ring-Sleeve
TRRS	Tip-Ring-Ring-Sleeve
HTTP	HyperText Transfer Protocol
HTTPS	HTTP Secure

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Exercises

1. Distinguish a physical port from a logical port, giving one example of each.
2. Expand USB, HDMI, VGA, RJ-45, and SATA.
3. Which video port is analog and video-only, and which two common video ports are digital and also carry audio?
4. What is the difference between RJ-45 and RJ-11?
5. What is the difference between SATA and eSATA, and why is SATA not a general peripheral port?
6. Which port is commonly used to connect a monitor and supports high refresh rates and multiple monitors, and which connector is used for a wired Ethernet connection?
7. Explain the difference between TRS and TRRS audio plugs.
8. Name two features of USB that make attaching devices convenient.

Solutions

1. A physical port is a hardware socket such as a USB port or an RJ-45 port. A logical port is a software number that identifies a network service, such as HTTP port 80.
2. USB = Universal Serial Bus; HDMI = High-Definition Multimedia Interface; VGA = Video Graphics Array; RJ-45 = Registered Jack 45; SATA = Serial Advanced Technology Attachment.

3. VGA is the analog, video-only port. HDMI and DisplayPort are digital and carry both video and audio.
4. RJ-45 is the eight-pin connector used for wired Ethernet networking; RJ-11 is the smaller telephone jack.
5. SATA connects internal drives (HDD, SSD, optical drive) to the motherboard. eSATA is external SATA, used for an external drive. SATA is a storage interface — it attaches drives — so it is not a general-purpose peripheral port such as USB.
6. DisplayPort is the monitor port for high refresh rates and multiple monitors; RJ-45 is the connector for a wired Ethernet connection.
7. TRS (Tip-Ring-Sleeve) has three contact areas and carries stereo audio; TRRS (Tip-Ring-Ring-Sleeve) has four and adds a microphone channel.
8. USB supports plug and play and hot swapping, and it carries both data and power over the same connection.

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LESSON 65

Current Multimedia, GUI and Media Formats

65.1 What multimedia means

Multimedia is the presentation of content that combines more than one medium in a single document or program. The media may be **text, graphics, audio, video**, and **animation**. Text and graphics are *static* media: they do not change with time. Audio, video, and animation are *time-based* media: their meaning depends on being played in the correct order and at the correct speed. Because a computer stores everything as binary data, every medium — sound, picture, or moving image — must first be captured and converted into numbers before it can be stored, compressed, and replayed.

Definition — Multimedia

Content that combines two or more media types — such as text, graphics, audio, video, and animation — in a single digital presentation.

Element	Meaning
Text	Readable characters; the base medium of almost every document.
Graphics	Still images, drawings, and photographs.
Audio	Sound, music, and speech.
Video	Moving pictures, usually accompanied by sound.
Animation	A sequence of images played fast enough to create motion.

65.2 Digitising media: sampling and quantization

An analogue signal, such as a sound wave or a scene's light, is *continuous*: it has a value at every instant. A computer, however, stores only discrete numbers. Converting a continuous signal into digital form is done by **analogue-to-digital conversion**, and it has two distinct steps, **sampling** and **quantization**.

Definition — Sampling

*The process of measuring a continuous signal at regular intervals and recording each measurement. The number of measurements taken per second is the **sampling rate**, measured in hertz (Hz).*

A **higher sampling rate** captures the signal more often, so the reproduction is more faithful, but the file becomes larger. A lower sampling rate saves space but loses detail. The sampling rate is the horizontal resolution of the digitised signal: how often we look at it.

Definition — Quantization

The process of mapping each measured sample to one of a fixed set of discrete levels, because a computer can store only a limited number of distinct values per sample.

The number of available levels is set by the **bit depth** of each sample. With a higher bit depth — more bits per sample — there are more levels, so each recorded value is closer to the true value; the rounding error, called *quantization noise*, is smaller, and quality is better, but again the file is larger.

Example 65.1. *Think of drawing a smooth curve on graph paper. **Sampling** decides how many points across the page you mark — more points follow the curve more closely. **Quantization***

decides how finely the vertical axis is ruled — more lines let you record each point's height more precisely. The two choices together determine how closely the digitised version matches the original, and how big the file becomes.

The common exam confusion is to swap the two. Remember the one-line distinction: **sampling is how often we measure; quantization is how precisely each measurement is stored.**

65.3 Compression: lossy and lossless

Raw multimedia files are large, so they are almost always **compressed** — stored in a form that takes less space. There are two kinds of compression, and the difference is whether the original can be recovered exactly.

Definition — Lossless compression

Compression that allows the original data to be reconstructed exactly. No information is discarded. PNG and GIF images and ZIP archives are lossless.

Definition — Lossy compression

Compression that discards some data to reduce the file size further, so the reconstruction is only an approximation of the original. JPEG images and MP3 audio are lossy.

Lossless compression works by removing redundancy — repeated or predictable data — while keeping every value. Lossy compression goes further: it removes detail that a human eye or ear is unlikely to notice. Lossy files are usually much smaller, which is why lossy formats dominate the internet for photographs, music, and video. When exactness matters — for a diagram, screenshot, or archive — a lossless format is used instead.

65.4 Synchronization

A multimedia file often carries more than one **stream** at once: a video stream and an audio stream, for instance. **Synchronization** is the process of keeping those streams aligned in time so they play together correctly. If the streams drift out of step, the sound no longer matches the picture — the narrator's voice is heard before or after the lips move, which is called a loss of *sync*. Synchronization is what makes a video play with the sound in the right place at the right moment; it is a timing requirement peculiar to time-based media.

Definition — Synchronization

The alignment in time of separate media streams — such as audio and video — so that they are presented together correctly during playback.

65.5 The graphical user interface and its elements

A **GUI** is a **Graphical User Interface**: it lets the user work with pictures, icons, and menus instead of typing commands. A **CLI** is a **Command-Line Interface**: it accepts typed text commands. The GUI is easier for beginners and is the standard on modern desktops; the CLI is often faster for an expert and is still used for administration.

Definition — GUI (Graphical User Interface)

A user interface in which the user interacts with the computer through visual elements such as windows, icons, menus, and a pointer, rather than by typing text commands.

The four classic GUI elements are remembered by the short form **WIMP**: **Windows, Icons, Menus, Pointers**.

Element	Meaning
Window	A rectangular on-screen area that holds a program or a document.
Icon	A small picture that represents a file, folder, or program.
Menu	A list of commands that the software offers.
Pointer	The on-screen arrow, moved by the mouse or touchpad, used to select.

Definition — Dialogue box

A small window that appears to ask the user for input or to confirm an action, such as Save As or Print. It usually contains text boxes, drop-down lists, check boxes, radio buttons, and OK/Cancel buttons.

65.6 WYSIWYG

WYSIWYG stands for **What You See Is What You Get**. It means that what is shown on the screen is exactly how the document will look when printed — the same layout, fonts, spacing, and page breaks. A word processor such as MS Word works in WYSIWYG mode, so the page visible on the monitor matches the printed page.

Example 65.1. *While editing a document in MS Word, the text, margins, and page breaks on screen match the printed output. This is WYSIWYG. When the user opens the Print dialogue box and chooses options, those settings are entered in a dialogue box, not typed on the command line. The two ideas are often asked together: WYSIWYG describes the on-screen view; a dialogue box is one GUI element that collects input.*

65.7 Image formats: PNG, JPEG, and GIF

Three image formats appear most often in competitive-exam questions.

Format	Full form	Nature and typical use
PNG	Portable Network Graphics	Lossless; supports transparency; used for logos and web graphics.
JPEG	Joint Photographic Experts Group	Lossy; best for photographs and complex pictures.
GIF	Graphics Interchange Format	Lossless but limited to 256 colours; used for simple graphics and short animation.

JPEG is chosen for photographs because its lossy compression makes them much smaller with little visible loss. **PNG** is chosen when exact colours or transparency are needed, because it is lossless. **GIF** is chosen for simple graphics and small animations, but it can show only 256 colours, so it is unsuitable for photographs. **BMP** (Bitmap) is an older, largely uncompressed image format that produces large files and is not used for the web.

65.8 Audio and video formats: MP3 and MP4

MP3 stands for **MPEG-1 Audio Layer III** (also written MPEG Audio Layer III). It is a **lossy** audio format: it removes sound detail that the human ear is unlikely to notice, which is why an MP3 file is far smaller than an uncompressed one. Its small size made MP3 the standard for portable music and downloads. By contrast, **WAV** is an uncompressed audio format.

MP4 stands for **MPEG-4 Part 14**. It is a **container** format: a single MP4 file can hold video, audio, and subtitles together. It is widely used for streaming, downloads, and mobile video. A container such as MP4 holds the streams; the *codec* inside the container is what actually compresses the video and audio. **AVI** (Audio Video Interleave) is an older video container.

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Definition — MPEG

MPEG stands for Moving Picture Experts Group — the group that defines the MP3 audio layer and the MP4 container, among other standards.

65.9 Frame rate, resolution, and bit rate

Video is a rapid sequence of still frames. The **frame rate** is the number of frames shown per second (fps); a higher frame rate produces smoother motion but a larger file. The **resolution** is the number of pixels in each frame (for example, 1920×1080); more pixels mean a sharper image and a larger file. The **bit rate** is the amount of data stored or sent per second; a higher bit rate keeps more detail but needs more space or bandwidth. These three settings, together with the choice of lossy or lossless compression, decide the final quality and size of a video or audio file.

Definition — Frame rate

The number of still frames displayed per second in a video. A higher frame rate gives smoother motion at the cost of a larger file.

Example 65.1. *A video at 30 frames per second with a high bit rate looks smooth and sharp but takes much more space than the same clip at a lower frame rate and bit rate. Streaming services lower the bit rate when the network is slow, trading quality for smooth playback — a daily example of the quality-versus-size trade-off that runs through the whole lesson.*

65.10 Plotter

A **plotter** is an **output** device that draws on paper. Unlike a printer, which lays down text and small images, a plotter produces **vector graphics**: lines and curves defined by coordinates rather than by a grid of dots. It is used for **large engineering drawings**, maps, blueprints, and banners, where continuous, accurate lines are needed. Early pen plotters moved a pen across the paper; modern inkjet plotters spray ink while the sheet moves. A plotter is always an output device, never an input device (PYQ-29, PYQ-08).

65.11 Impact versus non-impact printers (revisit)

The basic classification of printers reappears in later units, so keep it firmly in mind. An **impact printer** forms an image by striking the paper; a **non-impact printer** does so without any striking action.

Point	Impact	Non-impact
Action	Strikes the paper to print	Forms the image without striking
Examples	Dot-matrix, daisy-wheel, line printer	Inkjet, laser, thermal printer
Carbon copies	Possible (same strike hits several sheets)	Not possible
Noise	Comparatively noisy	Comparatively quiet

The quick test is the mechanism: if the device physically hits the paper, it is impact; if it does not, it is non-impact. Both kinds are output devices. A *dot-matrix* printer is impact; an *inkjet* and a *laser* printer are non-impact.

65.12 Multimedia hardware and applications

Playing and creating multimedia needs both hardware and software. On the hardware side, a **sound card** processes audio, **speakers** and **headphones** play it, and a **microphone** captures it. A **graphics card (GPU)** renders images and video, a **monitor or projector** displays them, and

a **camera** or **webcam** captures still and moving pictures. Because audio and video files are large, multimedia also places heavy demands on **memory and storage**.

On the software side, editors for images, audio, and video are used to capture, edit, compress, and synchronize the media streams. The main application areas of multimedia are education (e-learning and tutorials), entertainment (music, games, and streaming films), business (presentations and advertising), and medicine (medical imaging and simulation).

Short form	Full form
GUI	Graphical User Interface
CLI	Command-Line Interface
WYSIWYG	What You See Is What You Get
WIMP	Windows, Icons, Menus, Pointers
PNG	Portable Network Graphics
JPEG	Joint Photographic Experts Group
GIF	Graphics Interchange Format
MP3	MPEG-1 Audio Layer III
MP4	MPEG-4 Part 14
MPEG	Moving Picture Experts Group
AVI	Audio Video Interleave
BMP	Bitmap

65.13 Exam retrieval and common confusions

Several distinctions prevent most errors on this topic.

1. **Sampling** is how often a signal is measured; **quantization** is how precisely each sample is stored. Do not swap them.
2. **Lossless** formats restore the original exactly (PNG, GIF); **lossy** formats discard data (JPEG, MP3, MP4).
3. **JPEG** is lossy and best for photographs; **PNG** is lossless and supports transparency; **GIF** is lossless and supports simple animation.
4. **MP3** is audio (MPEG-1 Audio Layer III); **MP4** is video (MPEG-4 Part 14). Do not swap the full forms.
5. **Synchronization** keeps audio and video aligned in time; it is not compression and not sampling.
6. **GUI** is Graphical User Interface; its elements are **Windows, Icons, Menus, Pointers** (WIMP).
7. **WYSIWYG** means the screen matches the print; a **dialogue box** is a small window that collects input.
8. A **plotter** is an output device for large drawings; an **impact** printer strikes the paper and a **non-impact** printer does not.

Example 65.1. *An item asks, "Which format is lossless and supports transparency?" Trace the formats: JPEG is lossy, MP3 is lossy audio, MP4 is lossy video, and GIF is lossless but limited to 256 colours without transparency support. The answer is PNG, the lossless format that supports transparency.*

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Exercises

1. Define multimedia and name its five building blocks.
2. Distinguish sampling from quantization, and explain how each affects file size.
3. Differentiate lossy and lossless compression, giving one example of each.
4. What is synchronization in multimedia, and why is it needed?
5. Expand GUI, WYSIWYG, and WIMP.
6. State one suitable use each for PNG, JPEG, and GIF.
7. Expand MP3 and MP4, and state the medium each is used for.
8. Explain why a plotter is preferred over a printer for large engineering drawings.
9. Distinguish impact printers from non-impact printers with one example of each.

Solutions

1. Multimedia is content combining more than one medium in a single presentation. Its five building blocks are text, graphics, audio, video, and animation.
2. Sampling measures a continuous signal at regular intervals; a higher sampling rate captures more detail but enlarges the file. Quantization maps each sample to one of a fixed set of levels set by the bit depth; more levels improve quality but also enlarge the file.
3. Lossless compression restores the original exactly (for example PNG); lossy compression discards some data and is only an approximation (for example JPEG).
4. Synchronization is the alignment in time of separate media streams such as audio and video. It is needed so that sound and picture play together correctly rather than drifting apart.
5. GUI = Graphical User Interface; WYSIWYG = What You See Is What You Get; WIMP = Windows, Icons, Menus, Pointers.
6. PNG: logos and web graphics needing transparency or exact colours. JPEG: photographs and complex pictures. GIF: simple graphics and short animations.
7. MP3 = MPEG-1 Audio Layer III, used for audio (music and speech). MP4 = MPEG-4 Part 14, used for video (a container holding video, audio, and subtitles).
8. A plotter produces vector graphics — continuous lines defined by coordinates — on large sheets, which suits precise engineering drawings, maps, and blueprints better than a printer.
9. Impact printers strike the paper (for example dot-matrix); non-impact printers form the image without striking (for example inkjet or laser).

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UNIT 5

Software, Operating Systems, Windows & Utilities

LESSON 17

Hardware, Software and Firmware

17.1 Hardware

Hardware is the set of physical, tangible parts of a computer system — the parts you can see and touch. A computer's hardware includes the central processing unit (CPU), the motherboard, random access memory (RAM), the hard disk drive or solid-state drive, the keyboard, the mouse, the monitor, the printer, and every other physical component. Hardware carries out the actual work of the machine: it executes instructions, stores data, and moves signals.

Hardware alone does nothing useful. A processor with no instructions to run is only silicon and metal. The instructions come from software, and the two together make a working computer. For exam purposes, the test for hardware is simple: if it is a physical object that you can touch, it is hardware.

Definition — Hardware

The physical equipment of a computer system — the tangible components such as the CPU, memory, storage, input devices, output devices, and communication hardware.

Hardware is usually grouped by function. Input devices, such as the keyboard, mouse, and scanner, bring data into the computer. Output devices, such as the monitor, printer, and speaker, present results. Processing hardware, mainly the CPU and the graphics processing unit (GPU), performs computation. Memory hardware, such as RAM, ROM, and cache, holds data and instructions. Storage hardware, such as the hard disk drive (HDD) and solid-state drive (SSD), keeps files without power. Communication hardware, such as the network interface card and the motherboard, connects the parts together.

Hardware is also described as internal or external. **Internal hardware** is fixed inside the computer case — the motherboard, CPU, RAM, and internal drives. **External hardware**, also called a peripheral, is connected from outside, such as the keyboard, mouse, monitor, printer, and an external drive. A question may ask whether a named part is internal or external; the touch test still applies, because both kinds are physical and both are hardware.

A single concrete machine runs through this course: the **office desktop** — a personal computer with a CPU, memory, a disk, a keyboard, a mouse, a monitor, and a printer, running an operating system and MS Word. Every term in this lesson can be pointed at on that one machine, which makes the abstract layers easier to remember.

17.2 Software

Software is the set of programs and instructions that tell the hardware what to do. Unlike hardware, software is intangible: you cannot touch it. It is kept on secondary storage, such as a hard disk or solid-state drive, and it is loaded into main memory when it runs. Software is what turns a collection of physical parts into a usable computer.

Definition — Software

The collection of programs, procedures, and related data that direct the operation of a computer. Software is logical and intangible, and it instructs the hardware.

Software falls into two broad kinds. **System software** runs and manages the computer itself — the operating system, device drivers, and utility programs. **Application software** helps the user perform a task — a word processor such as MS Word, a web browser, a spreadsheet, or a game. Every program a candidate meets in JKSSB questions belongs to one of these two kinds, but the boundary matters: the operating system is system software, not application software.

Software is kept on secondary storage as files. When it is launched, the operating system copies it into main memory, and the CPU executes it from there. This is why a program must be *loaded* before it can run, and why data held only in RAM is lost when the power goes off. The permanent copy lives on disk; the running copy lives in memory.

17.3 Hardware vs software

The two are easy to confuse only when the wording is careless, so hold the contrasts in a table.

Point	Hardware	Software
Nature	Physical and tangible	Logical and intangible
Can be touched?	Yes	No
Role	Executes the work	Gives the instructions
Stored or fixed	Fixed as physical parts	Stored as files, loaded to run
Example	CPU, printer, keyboard	MS Word, a driver, an OS
How it fails	Wears out, breaks, burns out	Bugs, corruption, wrong logic

Neither can work alone. Hardware and software together form a computer system, and a question that names one and describes the other is usually testing this single distinction. A candidate should be able to sort a mixed list — a CPU, MS Word, a printer driver, the BIOS, and a user — into its correct class at a glance.

Example 17.1. *A question lists "CPU, MS Word, printer driver, BIOS, operator" and asks which are software. **MS Word** and the **printer driver** are software. The CPU is hardware; the BIOS is firmware (software stored in hardware); and the operator is a person, that is, humanware. Sorting is done by asking, for each item, whether it is physical, a program, embedded code, or a person.*

17.4 System software and application software

System software manages the machine and provides services to other programs. The operating system is the main example, together with device drivers and system utilities. **Application software** is written to let the user accomplish a task, and it depends on the system software beneath it.

For the exam, the stable rule is: the operating system is system software. This is the point tested by the real Junior Assistant 2026 concept recorded as PYQ-06 (device driver = software; OS = system software). A device driver, too, is system software — it is a program, not a physical part.

17.5 Firmware

Firmware is software that is embedded in hardware and stored in non-volatile memory. It is held in read-only memory (ROM), erasable programmable ROM (EPROM), or flash memory inside the device, so it is available the moment the device is powered on. Firmware provides the low-level control instructions a device needs and starts it up.

Definition — Firmware

Software permanently stored in a hardware device, usually in non-volatile memory such as ROM or flash. It is software by nature but is treated as part of the hardware because it is fixed to a specific device.

Common examples of firmware include the **BIOS** (Basic Input Output System) or **UEFI** on a personal computer, which initialises the hardware and begins loading the operating system; the firmware inside a printer, a router, a keyboard, or a solid-state drive. Firmware is usually described as sitting between hardware and software. It is a program, so it is software; it is stored on the physical device, so it behaves like part of the hardware.

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Firmware differs from ordinary software in how tightly it is bound to the device. Firmware is written for one specific piece of hardware, is stored in ROM or flash on that device, and changes rarely (it is updated by a process called flashing). Ordinary software is general-purpose, is stored on secondary storage, and is installed and updated frequently.

Firmware is the **first software to run** when a device is switched on. On a personal computer, the CPU executes the BIOS or UEFI firmware from ROM before any operating system is available; that firmware checks the hardware and then hands control to the operating system. This is exactly why firmware must be non-volatile: it has to be present even before anything can be loaded. A candidate should not confuse firmware with the operating system — the firmware runs first and *starts* the computer, while the operating system is what the machine is finally running.

17.6 Device drivers

A device driver is a small program that lets the operating system communicate with a piece of hardware. It acts as a translator: the operating system sends a general request such as "print this page", and the driver converts it into the exact signals that a specific printer understands. A printer driver, for example, lets the operating system print on one particular printer model.

Definition — Device driver

A program that enables the operating system to communicate with, and control, a specific hardware device by translating general commands into device-specific signals.

The single most important exam point about a driver is that it is **software**, even though it is tied to hardware and is often named after a device. A candidate may be tempted to call a driver hardware because it belongs to a printer or a keyboard, but that is wrong. The driver is a program, it is stored like software, and a device will not work properly without the right driver installed. This is TRAP-10, and it is also the heart of the real Junior Assistant 2026 concept PYQ-06.

Example 17.1. *A question states, "A device driver is a hardware component that connects the printer to the computer." This is wrong on two counts: the driver is a **program**, not a physical component, and it is software, not hardware. The physical cable and port are hardware; the driver that makes the printer work is software. The trap is that the driver carries the device's name.*

17.7 Humanware

Humanware, also called **peopleware**, is the human component of a computer system. It covers everyone involved: the end users who operate the machine, the programmers who write software, the system administrators who maintain it, and the operators who run it. A computer system is usually described as hardware, software, and humanware together.

Definition — Humanware

The people who design, operate, maintain, and use a computer system; the human component that hardware and software ultimately serve.

Humanware is not a device and not a program. In a classification question, if the named item is a person or a role — user, programmer, operator, administrator — it is humanware. It is the layer that makes decisions and uses the results, and without it the hardware and software would have no purpose.

17.8 The layered computing stack

The parts of a computer system can be arranged as a stack of layers, each one resting on the layer below. The order, from the bottom up, is:

Layer	Role
Hardware	The physical components at the base of the system
Firmware	Stored on the hardware; starts it and controls it at a low level
Device driver	Lets the operating system communicate with the hardware
Operating system	Manages all resources and provides services to programs
Application software	Lets the user perform a task

Reading the stack from the bottom, the physical parts come first. Firmware sits directly on them and brings the hardware to life. The device driver sits above the firmware and gives the operating system a clean way to talk to each device. The operating system manages memory, files, processes, and devices, and offers services to the programs above it. The application software at the top is what the user actually opens and works in. A command travels *down* the stack — from an application, through the operating system and the driver, to the hardware — and results travel back up.

This stack is the single mental picture for the whole lesson: hardware at the bottom, software above it, and firmware and drivers as the thin bridges that connect the two.

17.9 Why the stack matters

The layered stack is not only a way to sort terms; it explains the order in which work is done. When a user clicks Print in a word processor, the command begins at the application-software layer. The application asks the operating system to print; the operating system passes the job to the device driver for that printer; the driver translates the request into the signals the printer hardware understands; and the printer's own firmware controls the physical printing. The result then travels back up the stack to the user. Every layer depends on the one below it, and a fault in any single layer can stop the whole chain — which is why "reinstall the driver" is such a common fix.

Example 17.1. *A printer fails to print although the paper and ink are fine. Because the system has layers, the fault could lie in the application (a wrong page range), the operating system (a stuck print queue), the driver (missing or outdated), or the hardware (a loose cable). Testing from the bottom up checks each layer in turn. In exam terms, the layer that "lets the operating system talk to the printer" is the* ***device driver****.*

17.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. Hardware is **physical and tangible**; software is **logical and intangible**. The simple test is touch: if you can touch it, it is hardware.
2. A **device driver** is **software**, not hardware. Its name follows the device, but it is a program (TRAP-10; PYQ-06).
3. The **operating system** is **system software**, not application software (PYQ-06).
4. **Firmware** is software stored in non-volatile memory inside hardware, such as the BIOS; it is software by nature, hardware by storage.
5. **Humanware** (peopleware) is the **people** in the system, not a device or a program.

Example 17.1. *An item asks, "Which of the following is software?" with options CPU, a printer, a device driver, and a keyboard. The answer is the* ***device driver****: it is a program, whereas the CPU, the printer, and the keyboard are physical. If the options instead are an operating system, MS Word, a BIOS chip, and a game, the item is testing system software versus application software and firmware, and the operating system is the system software.*

Exercises

1. Define hardware and software, and give one example of each.
2. Distinguish system software from application software. To which kind does the operating system belong?
3. What is firmware, where is it stored, and why is it called “between hardware and software”?
4. Explain why a device driver is classified as software even though it is named after a hardware device.
5. What is humanware? Give two roles it includes.
6. List the layers of the computing stack from the bottom (hardware) to the top (application).
7. Give one example each of internal hardware and external hardware.
8. Using the print command as an example, explain how a request travels down the computing stack.

Solutions

1. Hardware is the physical, tangible equipment of a computer, such as the CPU or a printer. Software is the programs and instructions that tell the hardware what to do, such as MS Word or a device driver.
2. System software runs and manages the computer itself — the operating system, device drivers, and utilities. Application software lets the user do a task, such as a word processor or a game. The operating system is **system software**.
3. Firmware is software embedded in a hardware device and stored in non-volatile memory such as ROM or flash, for example the BIOS. It is software by nature but is fixed to a specific device, so it is described as sitting between hardware and software.
4. A device driver is a program, so it is software. It is named after the device it serves, which tempts a candidate to call it hardware, but it is stored and works like other software; it merely translates the operating system’s commands for one specific device.
5. Humanware, or peopleware, is the human component of a computer system. It includes roles such as end users and system administrators (also programmers or operators).
6. Hardware, then firmware, then device driver, then operating system, then application software.
7. Internal hardware: the motherboard (also the CPU, RAM, or an internal drive). External hardware: the keyboard (also the mouse, monitor, printer, or an external drive).
8. The user clicks Print in the application software. The application asks the operating system, which passes the job to the printer’s device driver. The driver translates the command into the signals the printer hardware understands, and the printer’s firmware controls the physical printing. The result returns up the stack to the user.

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LESSON 18

Software Categories

18.1 What is software?

A computer system has two parts: hardware and software. **Hardware** is the physical part that you can see and touch, such as the CPU, memory, and input/output devices. **Software** is the set of programs, that is, the instructions that tell the computer what to do. Programs are stored in memory and executed by the CPU, and without software the hardware cannot perform any useful task. Where hardware is the body of the computer, software is the set of instructions that gives it purpose.

Definition — Software

A set of programs or instructions that directs the computer to perform a task. Hardware is the physical equipment; software is the logical instructions that run on it.

Software does not exist as a physical object; it exists as information. A programmer writes the instructions in a programming language, and these instructions are then translated, stored in memory, and executed by the CPU. The same piece of hardware — a desktop computer, for example — can behave completely differently depending on which software is loaded onto it. This is why software is described as the logical part of a computer system: it supplies the behaviour that the physical parts only make possible.

Point	Hardware	Software
Nature	Physical; can be touched	Logical; a set of instructions
Made of	Electronic and mechanical components	Programs and data
Example	CPU, RAM, keyboard, monitor	Operating system, MS Word, a browser
Failure	Fails due to wear, damage, or age	Fails due to errors (bugs) or corruption

Software is commonly grouped by *what it is for* into six categories: system software, application software, utility software, programming software, embedded software, and middleware. These categories describe purpose and they can overlap in practice, but each answers a different question about a program's role.

Category	Role in one line
System software	Controls the hardware and provides a platform for other software.
Application software	Helps the user perform a specific task.
Utility software	Performs maintenance and housekeeping tasks.
Programming software	Used to write, translate, and test programs.
Embedded software	Built into a device to control that device.
Middleware	Connects different applications and systems.

18.2 System software

System software manages and controls the computer hardware and provides the *platform* on which application software runs. It works close to the machine, usually starts when the computer

is switched on, and is not written for one user's task. Common examples are the operating system, device drivers, firmware such as the BIOS, and utility programs.

The **operating system (OS)** is the most important system software. It acts as the interface between the user and the hardware and manages memory, files, processes, and input/output devices. Windows, Linux, macOS, and Android are familiar operating systems. Its main functions are process management (starting, scheduling, and stopping programs), memory management (allocating and freeing main memory), file management (creating, reading, and organising files), device management (controlling hardware through drivers), and providing a user interface. **Firmware** such as the BIOS or UEFI also belongs to the system-software layer: it initialises the hardware when the machine is switched on and then loads the operating system.

A **device driver** is also system software: it allows the operating system to communicate with a particular hardware device. It is easy to mistake a device driver for hardware, but it is a piece of software (TRAP-10).

Definition — System software

Software that controls and coordinates the hardware and provides the services and platform that other programs need in order to run.

18.3 Application software

Application software helps the user perform a specific task. It runs on top of the system software and is written for the user's work rather than to manage the machine. Word processors, spreadsheets, web browsers, media players, and accounting packages are all application software.

Application software is often divided into two kinds. **General-purpose** software is a ready-made package sold to many users, such as MS Word or MS Excel; it is also called an *application package*. **Specific-purpose** software, also called *tailor-made* or custom software, is built for one organisation or one job, such as a payroll system or a railway reservation system.

Application software covers many kinds of user work. Word processors prepare documents; spreadsheets handle calculations and tables; presentation tools build slide shows; database packages store and query records; graphics and multimedia packages edit images, audio, and video; browsers and email clients handle communication; and accounting packages keep business records. Most modern application software has a graphical user interface (GUI), so the user works with windows, icons, and menus instead of typing commands.

Definition — Application software

Software that is written to help the user perform a specific task or set of tasks, running on top of the system software rather than managing the machine.

Example 18.1. *The office desktop runs MS Windows and MS Office. Windows is system software because it manages the hardware; MS Word and MS Excel are application software because they help the user create documents and spreadsheets. The same machine shows all the main categories in use at once.*

18.4 Utility software

Utility software is a small program that performs maintenance or housekeeping tasks on the system. Antivirus programs, disk cleanup tools, disk defragmentation tools, backup utilities, and file-compression tools are common examples. Utility software keeps the system healthy, fast, and safe, and it belongs to the system software family.

A simple way to keep utilities and applications apart is to ask what the program is for. A utility *maintains the system* (for example, scanning for viruses or cleaning temporary files), whereas an application helps the user do *work* (for example, typing a letter or preparing a budget).

Definition — Utility software

A small program within the system software family that carries out maintenance and housekeeping tasks such as virus scanning, disk cleanup, defragmentation, backup, and compression.

Example 18.1. *A user notices that the office desktop has become slow and that the free disk space keeps shrinking. Running a disk-cleanup tool to remove temporary files and a disk-defragmentation tool to reorganise the hard disk restores speed and space. Both tools are utility software: they maintain the system rather than help the user produce a document.*

18.5 Programming software and language translators

Programming software is used to write, translate, and test computer programs. The core members are the **language translators**: a **compiler**, an **interpreter**, and an **assembler**. A compiler translates a whole program into machine code at once, while an interpreter translates and runs the program line by line. An assembler translates assembly language into machine code.

Translator	How it works
Compiler	Translates the whole program into machine code before it runs.
Interpreter	Translates and executes the program one line at a time.
Assembler	Translates assembly language (mnemonic codes) into machine code.

Because a compiler translates the whole program first, it reports errors together and the translated program generally runs faster; because an interpreter works line by line, it can start running immediately and is handy for testing small pieces of code.

Around the translators sit other supporting tools: the **linker**, which joins separate pieces of a program; the **loader**, which brings a program into memory to run; the **debugger**, which helps find errors; and the text editor used to write the code. An **IDE** (Integrated Development Environment) bundles an editor, a compiler or interpreter, and a debugger into one package, which makes programming more convenient.

18.6 Embedded software

Embedded software is built into a device in order to control that device. It is usually stored in read-only memory as **firmware**, runs automatically when the device is switched on, and is written for one dedicated job. Washing machines, microwave ovens, network routers, and the engine-control units in modern cars all contain embedded software. The user normally never sees this software directly; it simply makes the device behave correctly.

Definition — Embedded software

Software built into a device and stored as firmware to control the device's fixed, dedicated functions.

18.7 Middleware

Middleware is software that connects different applications and systems so that they can work together. It sits between the operating system and the application software and provides common services such as messaging, authentication, and data exchange. An application server that links a web front end to a back-end database is a typical example. Middleware allows programs that were built separately to communicate without being rewritten for each other.

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Definition — Middleware

Software that sits between the operating system and applications and connects different programs or systems, providing common services such as messaging, authentication, and data exchange.

Middleware is common in large or distributed systems. On a web application, for example, the browser (client), the web server, and the database are three different programs. Middleware passing requests and replies between them lets these separate programs behave as one service, without each program needing to know the internal details of the others.

18.8 Software licensing models

A software **licence** states how a program may be used, copied, modified, and distributed. The main models are summarised below.

Model	Key point
Proprietary	Source code is kept closed; use is governed by the vendor's licence.
Open-source	Source code is available to view, modify, and redistribute under its licence.
Free software	Gives the freedom to run, study, modify, and share the software.
Freeware	Distributed free of cost, but the source code may not be available.
Shareware	Offered as a limited trial; payment is needed for full or continued use.
Public domain	Has no copyright and may be used freely for any purpose.
Commercial	Sold for profit; usually proprietary.

Proprietary software keeps its source code secret and restricts copying and modification. MS Windows and MS Office are examples. **Open-source software** publishes its source code and its licence allows users to view, change, and share it; Linux, Mozilla Firefox, and LibreOffice are examples. Note that "open-source" refers to *source-code availability*, not to price. Common open-source licences include the **GNU General Public License (GPL)**, the **Apache License**, and the **MIT License**; each sets out the conditions under which the source code may be used and changed.

Proprietary software is usually supplied under an **EULA** (End User License Agreement), which the user accepts before installing the product and which forbids copying, modifying, or redistributing the program. Because the source code is closed, only the vendor can improve or fix the software.

Free software is a related but distinct idea: it refers to freedom, not to price. Free software gives the user the freedom to run the program, study and modify it, and redistribute copies. It is important to see that both free software and open-source software make the source code available; they differ mainly in the philosophy they express, with free software emphasising the user's freedom. **Freeware**, by contrast, simply means free of cost; its source code may remain closed. **Shareware** is a trial version distributed free for a limited period, after which the user must pay for continued or full use. **Public-domain software** carries no copyright at all and may be used freely. **Commercial software** is sold for profit; it is usually also proprietary, although a commercial product can occasionally be open-source.

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Definition — Open-source vs freeware vs shareware

Open-source means the source code *is available; freeware means the software is free of* cost *(source may be closed); shareware is a* trial *that must later be paid for. The three are not the same.*

Example 18.1. *An item states, "Freeware, open-source, and shareware all mean that the source code is freely available." This is wrong. Only open-source guarantees source-code availability. Freeware is free of cost but may be closed-source, and shareware is a trial that later requires payment. The wrong option has merged three different licensing models (TRAP-23).*

18.9 System software vs application software

The two broadest groups are worth comparing side by side, because many exam items turn on exactly this distinction.

Point	System software	Application software
Purpose	Runs and manages the machine	Helps the user do a task
Runs	Close to the hardware; starts at boot	On top of the system software
Needed for	The computer to work at all	Performing the user's work
Examples	Operating system, device driver, BIOS	MS Word, MS Excel, a web browser

The same physical computer can run one operating system (system software) and many applications at the same time. The operating system is what makes the hardware usable; the applications are what the user actually works with.

18.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. **System vs application**: system software manages the hardware; application software helps the user perform a task. The operating system is system software, while MS Word is application software.
2. **Utility vs application**: a utility *maintains* the system (antivirus, disk cleanup); an application helps the user do *work*.
3. **A device driver is software**, not hardware; the operating system is system software (TRAP-10).
4. **Open-source ≠ freeware ≠ shareware** (TRAP-23): open-source gives the source code, freeware is free of cost, and shareware is a paid trial.
5. **Free software is about freedom, not price**; it should not be confused with freeware, which is about price alone.

Example 18.1. *An item asks which of a compiler, an antivirus program, MS Excel, and the operating system is* not *system software. Trace the categories: a compiler is programming software (system family), an antivirus program is utility software (system family), and the operating system is system software. MS Excel is application software, so it is the odd one out.*

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Exercises

1. Define software and distinguish it from hardware.
2. List the six main categories of software and give one example of each.
3. Distinguish system software from application software, with one example of each.
4. Explain the difference between utility software and application software.
5. Distinguish open-source software, freeware, and shareware, and state which one guarantees access to the source code.
6. Classify each of the following as system, application, utility, programming, embedded, or middleware software: MS Word; an antivirus program; a compiler; the firmware in a router; an application server.

Solutions

1. Software is a set of programs (instructions) that tells the computer what to do; hardware is the physical equipment that runs those instructions.
2. System software (operating system), application software (MS Word), utility software (antivirus), programming software (compiler), embedded software (washing-machine firmware), and middleware (application server).
3. System software manages and controls the hardware and provides a platform; application software helps the user perform a specific task. The operating system is system software; MS Excel is application software.
4. Utility software performs maintenance and housekeeping tasks such as virus scanning or disk cleanup; application software helps the user carry out work such as writing a document or preparing a budget.
5. Open-source software publishes its source code and allows it to be viewed, modified, and shared; freeware is free of cost but may keep its source code closed; shareware is a trial that later requires payment. Only open-source software guarantees access to the source code.
6. MS Word is application software; an antivirus program is utility software; a compiler is programming software; the firmware in a router is embedded software; and an application server is middleware.

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LESSON 19

Operating System

19.1 What an operating system is

A computer's hardware cannot be used directly by a person or by ordinary application programs. Something must sit between the user and the machine, accept the user's instructions, and control the hardware on the user's behalf. That something is the **operating system**. An operating system (OS) is the main piece of **system software** that controls the whole computer. It starts the machine, runs other programs, and manages every resource. It makes the computer usable, because without an OS the hardware alone cannot run application software.

Definition — Operating system

The main system software that sits between the user and the computer hardware, provides a user-friendly interface, and manages the computer's resources such as the processor, memory, storage, and input/output devices.

Term	Meaning in this lesson
Kernel	The core part of the OS that interacts directly with the hardware.
Booting	Starting the computer and loading the OS into main memory.
GUI	Graphical User Interface — interaction through icons, windows, and clicks.
CLI	Command Line Interface — interaction by typing text commands.
Process	A program that is currently executing.
Multitasking	Running more than one program at a time.
Time-sharing	Sharing one computer among several users at the same time.

The rest of this lesson uses these terms in one consistent sense. The **kernel** is the privileged core of the OS. **Booting** is the startup sequence. A **GUI** is a graphical interface and a **CLI** is a text-command interface. A **process** is a running program. **Multitasking** and **time-sharing** are two ways of sharing the CPU.

19.2 The OS as interface and resource manager

The operating system has **two main jobs**. The first is to provide a **user-friendly interface**: the user gives instructions and receives results through the OS, without needing to understand the hardware. This is the definition of the OS as an interface layer, and it is a regularly tested idea (PYQ-21). The second job is to act as a **resource manager**: the OS shares the CPU, main memory, storage, and devices among all the programs that are running, so that each one gets a fair turn and none interferes with the others.

It is important to know **where** the operating system is. The OS is stored permanently on secondary storage, such as an SSD or HDD. At startup it is loaded into **primary storage**, that is, main memory. While the computer is running, the OS is *resident in primary storage* (PYQ-22): it stays in main memory for the whole session so that it can respond instantly whenever the hardware or an application needs it.

Example 19.1. *An item states that the OS is kept in secondary storage while the computer is running. This is wrong. The OS is stored on secondary storage when the machine is off, but once the computer starts it is copied into primary storage (main memory) and remains there while running. The wrong option confuses the place where the OS is stored with the place where it runs.*

19.3 Functions of an operating system

The work of an OS is usually grouped into a few large functions, each of which is a favourite exam topic:

1. **Process management**: creating processes, scheduling the CPU among them, and removing them when they finish.
2. **Memory management**: deciding how much main memory each program gets and keeping programs apart.
3. **File management**: organising data into files and folders and handling their creation, reading, writing, and deletion.
4. **Device management**: controlling input and output devices through device drivers.
5. **Security**: managing user accounts, logins, and permissions, and applying updates.
6. **User interface**: presenting either a GUI or a CLI to the user.

19.4 The kernel

The **kernel** is the **core** of the operating system. It is the part of the OS that interacts **directly with the hardware**, and it always stays in main memory while the computer runs. The kernel provides the most important services: process scheduling, memory handling, and device control. Other software, including the user interface, relies on the kernel to reach the hardware.

Definition — Kernel

The central, privileged part of the operating system that manages the hardware and provides the fundamental services on which all other software depends.

19.5 User interfaces: GUI and CLI

Two broad kinds of interface are used to work with a computer. A **Graphical User Interface (GUI)** lets the user interact through pictures, icons, windows, menus, and clicks; it is easier for beginners. A **Command Line Interface (CLI)** requires the user to type text commands; it gives precise control but demands that the commands be known. Windows, macOS, and Linux desktops are examples of GUIs, while the command prompt and the terminal are examples of CLIs.

Point	GUI	CLI
Full form	Graphical User Interface	Command Line Interface
Input method	Clicks, icons, windows	Typed text commands
Ease of use	Easier for beginners	Needs the commands to be known
Example	Windows desktop, icons	Command prompt, terminal

19.6 Booting the computer

Booting is the process of starting a computer and loading the operating system into main memory. A **cold boot** starts from the power-off state, while a **warm boot** restarts a computer that is already on. The startup firmware, **BIOS** (Basic Input/Output System) or its modern replacement **UEFI**, first checks the hardware and then finds and loads the operating system from secondary storage. BIOS/UEFI initialises the hardware and hands control to the OS; it is not the kernel and it is not a device driver.

19.7 Process management

A **process** is a program that is currently executing. A program sitting on disk is passive; once it is loaded and running it becomes an active process. The operating system creates processes, gives each a turn on the CPU through **CPU scheduling**, lets them communicate where necessary, and removes them when they finish. Scheduling decides which process runs next and for how long, which is what allows several programs to make progress.

19.8 Memory management

The operating system decides **how much main memory** each program receives. It keeps one program's memory **separate** from another's so that programs cannot overwrite each other's data. It frees memory when a program ends so that the space can be reused. Where needed, the OS uses **virtual memory** to give programs more space than the physical RAM provides.

19.9 File management

The OS organises data on storage devices into **files** and **folders**. It handles creating, reading, writing, renaming, and deleting files, and it keeps a **directory** that records where each file is stored. It also controls **permissions**: which accounts may open, change, or delete a given file. Together these tasks are called the **file system**, the way the OS names, stores, and finds files.

19.10 Device management

The operating system controls all **input/output devices**: the keyboard, mouse, printer, disk, and so on. Each device communicates with the OS through a **device driver**. A device driver is **software**, not hardware — a point that is regularly tested (TRAP-10). The OS also uses **spooling** to queue print jobs and feed the printer at the printer's own speed, so that the computer does not have to wait for a slow device to finish.

19.11 Multitasking and time-sharing

Multitasking means that the operating system runs **more than one program at a time** (PYQ-23). The CPU switches between programs so quickly that they appear to run together, even though a single CPU core does only one thing at any instant. **Time-sharing** lets **many users** share one computer at the same time, each receiving a slice of CPU time in turn. Both multitasking and time-sharing depend on the OS **scheduler**.

Example 19.1. *An item defines multitasking as using more than one monitor, or as storing more than one file on a disk. Both are wrong. Multitasking is about the* ***processor*** *running more than one program at a time. The number of monitors and the number of files stored are unrelated to multitasking.*

19.12 Security, user accounts and updates

The operating system provides **user accounts**, each with a username and password. **Authentication** (the login step) verifies who is using the computer. **Permissions** decide which files and settings each account may use, so that one user cannot disturb another's work. Finally, system and software **updates** should be applied regularly, because they fix security holes and add improvements. Keeping the OS up to date is one of the simplest ways to protect the machine.

19.13 Examples of operating systems

Common operating systems include **Microsoft Windows**, the open-source **Linux** family, and Apple's **macOS** on desktops and laptops, plus the mobile operating systems **Android** and **iOS**. Android is built on the Linux kernel. An application such as MS Word or a web browser is *not* an operating system; it runs on top of one.

19.14 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. The OS is **system software**, not application software. A device driver is **software**, not hardware (TRAP-10).
2. While running, the OS is resident in **primary storage** (main memory); it is only *stored* on secondary storage (PYQ-22).
3. The OS provides a **user-friendly interface** between the user and the hardware (PYQ-21).
4. **Multitasking** means running more than one program at a time; **time-sharing** shares one machine among several users (PYQ-23).
5. **BIOS/UEFI** initialises the hardware and loads the OS; it is not the kernel and not a device driver (TRAP-24).
6. A **program** on disk is passive; once running it becomes a **process**.

Example 19.1. *An item asks where the operating system is loaded when the computer starts. Trace the startup sequence: the machine is switched on, BIOS/UEFI checks the hardware, and the OS is loaded from secondary storage into **primary storage** (main memory), where it remains while the computer runs. If the stem instead asks which component is the core of the OS, the answer is the **kernel**.*

Exercises

1. Define an operating system and state its two main roles.
2. Where is the operating system stored, and where is it resident while the computer is running?
3. List four core functions of an operating system.
4. What is the kernel, and why is it important?
5. Distinguish a GUI from a CLI.
6. What does multitasking mean, and how does it differ from time-sharing?
7. State three ways an operating system helps to keep a computer secure.

Solutions

1. An operating system is the main system software that sits between the user and the hardware. Its two main roles are to provide a user-friendly interface and to manage the computer's resources.
2. It is stored permanently on secondary storage (such as an SSD or HDD) and is resident in primary storage (main memory) while the computer is running.
3. Any four of: process management, memory management, file management, device management, security, and the user interface.
4. The kernel is the core part of the operating system that interacts directly with the hardware. It is important because it provides the fundamental services — scheduling, memory handling, and device control — on which all other software depends.
5. A GUI (Graphical User Interface) uses icons, windows, and clicks and is easier for beginners; a CLI (Command Line Interface) requires the user to type text commands and gives precise control.
6. Multitasking means the OS runs more than one program at a time by switching the CPU among them. Time-sharing means many users share one computer at the same time, each getting a slice of CPU time.
7. Three of: user accounts with usernames and passwords; authentication at login; permissions that limit access to files and settings; and regular security updates.

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LESSON 20

Windows Explorer

20.1 What Windows Explorer is

Windows Explorer is the file-management program built into the Windows operating system. Its job is to let the user see and manage everything stored on the computer: drives, folders, and files. In Windows 10 and Windows 11 the same program is presented to the user as **File Explorer**. Internally it is the same program, whose executable file is **explorer.exe**. From one Explorer window a user can browse the entire tree of storage and perform the everyday operations of copying, moving, renaming, and deleting files.

Definition — Windows Explorer (File Explorer)

The file-management program of Windows that displays drives, folders, and files in a graphical window and lets the user copy, move, rename, and delete them. It is launched from the executable `explorer.exe`.

The main parts of the Explorer window are:

Part	What it shows
Navigation pane	The tree of drives and folders on the left.
Address bar	The full path of the folder that is currently open.
File list	The files and subfolders inside the current folder.
Ribbon / command bar	The tabs (New, Cut, Copy, Paste, Delete, View) for actions.
Search box	Files found inside the current folder matching the typed text.

20.2 Drives, folders, subfolders, and files

Storage is organised as a tree. At the top is a **drive**, identified by a drive letter. The hard disk is usually **C:**, and a USB flash drive might appear as **E:** or **F:**. A drive contains **folders**. A folder (also called a **directory**) is a container that holds files and other folders. A folder placed inside another folder is a **subfolder**. A **file** is a single item of stored data — a document, an image, a song, or a program.

Definition — Drive

A storage device or partition on which folders and files are kept, named by a drive letter such as `C:` *or* `E:`.

Definition — Folder (directory)

A named container that holds files and other folders. A folder inside a folder is called a subfolder.

Because folders can hold subfolders, and those can hold more subfolders, the whole arrangement looks like an inverted tree — the drive is the root, the folders are branches, and the files are the leaves. The top-most folder of a drive, directly under the drive letter, is called the **root folder**.

20.3 The file path

Every file and folder has one exact **path**: the address that tells Windows where it lives. The parts of a path are separated by a **backslash** (\). For example:

```
C:\Users\Student\Documents\report.docx
```

Reading left to right, `C:` is the drive, `Users`, `Student`, and `Documents` are the successive folders, and `report.docx` is the file name including its extension. The path is shown in the Explorer **address bar**, and the same information is given in a file's Properties as its **location**.

Definition — Path

The complete address of a file or folder, from the drive letter through the chain of folders to the item's own name, with parts separated by a backslash.

20.4 File names and extensions

A file name normally has two parts joined by a dot: the **name** and the **extension**. In `report.docx`, `report` is the name and `.docx` is the extension. The extension tells Windows which program should open the file and what kind of data it holds. Common extensions include `.txt` for a text file, `.docx` for a Word document, `.xlsx` for an Excel workbook, `.pptx` for a PowerPoint presentation, `.pdf` for a portable document, `.jpg` for a photo, and `.exe` for a program.

Extension	Type of file / format
`.txt`	Plain text file
`.docx`	Microsoft Word document
`.xlsx`	Microsoft Excel workbook
`.pptx`	Microsoft PowerPoint presentation
`.pdf`	Portable Document Format
`.jpg`	JPEG image
`.exe`	Executable program

Definition — Extension

The letters after the last dot in a file name (for example `.docx`) that indicate the file's type and the program that opens it.

One common confusion is between an *extension* and a *format*. An extension is the label at the end of the name, such as `.png`; the format is the way the data is actually encoded, such as PNG. The two usually match, but the words name different things, so they should not be used as interchangeable (TRM-09, TRAP-34).

20.5 Hidden files and attributes

Every file and folder carries **attributes**: settings that tell Windows how to treat it. The common attributes are **Read-only**, **Hidden**, **System**, and **Archive**. A **hidden file** or folder has the **Hidden** attribute set and is therefore not displayed by default. Windows hides important system files so that a user does not move or delete them by mistake. To display hidden items, open the **View** tab, choose **Show**, and then **Hidden items** (in Windows 10 and 11). In older versions the setting is reached through **Folder Options**.

Definition — File attribute

A property that controls how Windows treats a file or folder, such as Read-only, Hidden, System, or Archive.

20.6 Properties

Properties is the dialog box that displays the details of a selected file, folder, or drive. It is opened by pressing **Alt** + **Enter** or by choosing **right-click** → **Properties**. For a file, Properties shows the type (format), the size, the date created, the date modified, the location (path), and the

attributes. For a drive, it shows the total size, the used space, and the free space. Properties is a read-and-sometimes-edit view: changes to attributes may be made there, but the file's contents are not shown.

20.7 Selecting files

Before a file can be copied, moved, renamed, or deleted it must be selected. A single file is selected by clicking it once. A **contiguous range** of files is selected by clicking the first file, holding **Shift**, and clicking the last file. **Non-contiguous** files are selected by holding **Ctrl** and clicking each file in turn. All files in a folder are selected with **Ctrl** + **A**.

Selection	Method
One file	Click once.
A block of files in a row	Click the first, hold Shift, click the last.
Separate files scattered in the list	Hold Ctrl and click each.
Every file in the folder	Ctrl + A.

20.8 Copy

Copy makes a duplicate of the selected item while leaving the original where it is. The usual method is to press **Ctrl** + **C** to copy, open the destination folder, and press **Ctrl** + **V** to paste. The same steps can be done with **right-click** → **Copy** followed by **Paste**. Holding **Ctrl** while dragging an item also copies it. The key point about copy is that after the operation there are *two* copies: the original and the new one.

Example 20.1. *A student copies* `report.docx` *from* `Documents` *to a USB drive. After pasting, the file exists in both places. The copy in* `Documents` *is unchanged, because Copy never removes the original.*

20.9 Move

Move shifts the selected item to a new location and removes it from the old one. The standard way is to **cut** with **Ctrl** + **X**, open the destination, and **paste** with **Ctrl** + **V**. Cut followed by Paste is therefore the standard move operation. Alternatively, dragging a file to a folder on the **same drive** moves it. Dragging to a folder on a **different drive** copies it instead, because Windows assumes you intend to transfer a copy between separate devices.

Definition — Move

Relocating a file or folder from one place to another so that it no longer exists at the old location. Cut + Paste performs a move.

20.10 Rename

Rename changes only the **name** of a file or folder; its contents are untouched. The quickest method is to select the item and press **F2**, type the new name, and press **Enter**. The same command is available by right-clicking the item and choosing **Rename**. A warning worth remembering: changing the text after the dot can change the file's recognised type, so the extension should be left alone unless the type is genuinely being changed.

20.11 Delete and the Recycle Bin

Pressing the **Delete** key on a selected item moves it to the **Recycle Bin**. The Recycle Bin is a special folder that temporarily holds deleted items, giving the user a chance to recover them. An item in the Recycle Bin may be **restored** to its original location; the **Restore** command puts it back. Choosing **Empty Recycle Bin** deletes everything held in the Bin permanently.

Definition — Recycle Bin

A special Windows folder that temporarily stores files and folders deleted with the Delete key, so that they can be restored or removed permanently.

20.12 Recycle Bin vs permanent deletion

An item deleted with **Delete** is recoverable, because it sits in the Recycle Bin until the Bin is emptied. Permanent deletion happens when the user presses **Shift** + **Delete**, when the Recycle Bin is emptied, or when an item is deleted from a **USB drive**, a **network drive**, or another location that does not use the Recycle Bin. Once deleted permanently, a file cannot be restored through the normal Windows interface.

Point	Recycle Bin (Delete)	Permanent deletion
Shortcut	Delete key	Shift + Delete
Recoverable	Yes, can be restored	No, cannot be restored
Stored where	Recycle Bin folder	Removed from the disk

Permanent deletion does not mean the storage space is instantly overwritten; the freed space is simply marked as available for reuse. For exam purposes the distinction is the one that matters: Delete is recoverable, Shift+Delete is not.

20.13 Keyboard shortcuts and drag-and-drop

The following shortcuts are high-yield for the exam and should be memorised exactly.

Action	Shortcut
Copy	Ctrl + C
Cut (move)	Ctrl + X
Paste	Ctrl + V
Undo	Ctrl + Z
Select all	Ctrl + A
Rename	F2
Refresh	F5
Properties	Alt + Enter
New folder	Ctrl + Shift + N
Delete to Recycle Bin	Delete
Permanent delete	Shift + Delete

Drag-and-drop follows a fixed rule. Dragging an item to a folder on the **same drive moves** it; dragging to a folder on a **different drive copies** it. Holding **Ctrl** while dragging always **copies**, and holding **Shift** while dragging always **moves**, regardless of the drives involved.

20.14 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. **Copy** leaves the original in place; **move** (Cut + Paste) removes it. Do not confuse the two.
2. The **Delete** key sends a file to the **Recycle Bin**; it is not a permanent erase. **Shift + Delete** is the permanent erase and bypasses the Bin.
3. Deleting from a **USB drive** or **network drive** bypasses the Recycle Bin even when the plain Delete key is used.

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4. An **extension** is the label after the dot (for example `.png`); it is not the same as the **format** (PNG) (TRM-09).
5. Same-drive drag **moves**; different-drive drag **copies**. **Ctrl+drag** forces a copy; **Shift+drag** forces a move.

Example 20.1. *An item asks where a file goes when the Delete key is pressed. Trace the rule: the Delete key moves the item to the Recycle Bin, from which it can still be restored. If the stem instead says the file must be removed so that it cannot be recovered, the required operation is Shift+Delete (or emptying the Recycle Bin), not the plain Delete key.*

Exercises

1. What is Windows Explorer called in Windows 11, and what is the name of its executable program?
2. Distinguish between a folder and a subfolder, and define a file path.
3. What is a file extension, and why is an extension not the same as a file format?
4. Explain the difference between Copy and Move, and give the keyboard shortcut for each.
5. A file is deleted with the Delete key. Can it be recovered? Explain, and state one way to delete a file permanently.

Solutions

1. In Windows 11 it is called **File Explorer**. Its executable program is **explorer.exe**.
2. A folder is a container that holds files and other folders; a folder placed inside another folder is a subfolder. A path is the complete address of a file or folder, from the drive letter through the chain of folders to the item's own name, with parts separated by a backslash.
3. An extension is the letters after the last dot in a file name (for example `.docx`) that indicate the file's type. It is not the same as the format, because the extension is only the label at the end of the name, while the format is the way the data is actually encoded.
4. **Copy** duplicates the item and leaves the original in place (Ctrl + C, then Ctrl + V). **Move** relocates the item and removes it from the old location (Ctrl + X, then Ctrl + V).
5. Yes, a file deleted with the Delete key can normally be recovered, because it goes to the Recycle Bin and can be restored. It can be deleted permanently with **Shift** + **Delete** (or by emptying the Recycle Bin).

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LESSON 21

Keyboard Shortcuts

21.1 How to read a keyboard shortcut

A keyboard shortcut is a key, or a combination of keys, that performs a command directly without opening a menu. Shortcuts save time and reduce the need for the mouse, and they are a favourite topic in the JKSSB Computer Awareness paper because the answer is a single exact letter or key. In a combination written with a plus sign, such as **Ctrl+C**, the plus means: hold the first key down, then press the second key. You do not type the plus sign. Where three keys are written, such as `Ctrl+Shift+N`, hold the first two and tap the third.

Definition — Keyboard shortcut

A key or key combination that runs a command directly — the same command a menu offers — so the user reaches it from the keyboard instead of with the mouse.

The keys used to build combinations are called modifier keys. They are **Ctrl** (Control), **Alt** (Alternate), **Shift**, and the **Windows (Win)** key. A modifier key does very little on its own; it changes what the next key does. The **function keys F1** through **F12** run along the top row of the keyboard and are usually pressed on their own, without a modifier.

Key	Full form	Main role
Ctrl	Control	Acts on content: copy, cut, paste, save, find.
Alt	Alternate	Acts on windows and menus: switch, close.
Shift	Shift	Changes what is typed or selected; builds ranges.
Win	Windows key	Opens or controls system features: desktop, Explorer, lock.
Fn	Function (laptop)	Gives a second job to the F-keys on many laptops.

A practical split helps memory. The **Ctrl** combinations mostly act on *content* — copying, cutting, saving, finding. The **Alt** combinations mostly act on *windows* — switching and closing. The **Windows** key opens or controls system-level features such as the desktop, File Explorer, and the lock screen. Function keys are single keys whose job depends on the application that is active.

Most shortcut letters are chosen to be easy to remember. **C** stands for Copy, and **X** looks like a cross over the text, so it stands for Cut. **V** sits next to X on the keyboard and stands for Paste. **Z** is Undo and **Y** is Redo, the two editing-history keys. **A** is All (Select All), **S** is Save, **P** is Print, **F** is Find, and **H** opens the find-and-replace box. Learning the letter is more reliable than memorising a list, because the letter usually hints at the command.

21.2 Copy, cut, and paste

The three most common shortcuts form the clipboard group. The **clipboard** is the temporary holding area in memory where copied or cut items wait before being pasted.

Shortcut	Action
Ctrl+C	Copy the selected item; the original stays where it is.
Ctrl+X	Cut the selected item; the original is removed.
Ctrl+V	Paste the copied or cut item at the current position.

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Definition — Clipboard

A temporary area in the computer's memory that holds text, images, or files after they are copied or cut, until they are pasted.

The distinction to hold is that **Copy** and **Cut** both place the selection on the clipboard, but only **Cut** removes the original. A question that offers `Ctrl+C` and `Ctrl+X` as two options is testing exactly this difference: `Ctrl+C` leaves the original in place, and `Ctrl+X` does not. `Ctrl+V` only reads the clipboard; it never decides whether the item was copied or cut.

Example 21.1. *A candidate reads, "Which shortcut copies selected text without removing it from its original place?" The answer is `Ctrl+C`. `Ctrl+X` also puts the text on the clipboard, but it deletes the selection from the document, so it does not satisfy the phrase "without removing it".*

21.3 Undo, redo, and select all

Three Ctrl shortcuts control the editing history and the current selection.

- **Ctrl+Z** — Undo the last action. It steps backward through recent changes.
- **Ctrl+Y** — Redo the action that was just undone. It steps forward again.
- **Ctrl+A** — Select All the content of the current window or document.

Undo and redo are a matched pair and must not be swapped. `Ctrl+Z` reverses a change; `Ctrl+Y` reapplies a change that was reversed. `Ctrl+A` does not select one word; it selects everything that can be selected in the current context. In MS Office the redo key is `Ctrl+Y`; a few non-Office programs use `Ctrl+Shift+Z`, but for this exam the standard answer is `Ctrl+Y`.

21.4 Save, print, find, and replace

These four shortcuts cover saving a file and working with text inside it.

Shortcut	Action
Ctrl+S	Save the current file.
Ctrl+P	Print.
Ctrl+F	Find (open the search box for text).
Ctrl+H	Replace (find a word and substitute another).

Find and **Replace** are near neighbours. `Ctrl+F` only locates the search text and highlights it; `Ctrl+H` opens the find-and-replace function, which can also change every occurrence to a new word. Similarly, `Ctrl+S` saves and `Ctrl+P` prints; the two are often offered together as distractors. `Ctrl+S` does not print, and `Ctrl+P` does not save.

Example 21.1. *A user types a report and wants to change every occurrence of the word "received" to "recd". `Ctrl+F` would only find the word and stop there. The correct shortcut is `Ctrl+H`, which opens Replace and lets the user substitute the new text, one occurrence at a time or all at once.*

21.5 Alt-key combinations

The Alt key builds the window-management shortcuts.

- **Alt+Tab** — Switch between open windows or running applications. Repeatedly pressing Tab while holding Alt cycles through the open windows.
- **Alt+F4** — Close the active window or application.

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On the desktop with no application window active, `Alt+F4` opens the shutdown or sign-out dialog. The concept to keep clear is that `Alt+Tab` *switches* between windows and does not close any of them, while `Alt+F4` *closes* the active window. A common wrong answer reverses the two. Note that `Alt+F4` closes the whole window; saving the file is still the job of `Ctrl+S`.

21.6 Function keys

A function key is a single key, `F1` to `F12`, on the top row of the keyboard. Its action depends on the application that is running. For the exam, four function keys are high-yield.

Key	Standard action
F1	Help.
F2	Rename the selected file or folder (for example in File Explorer).
F5	Refresh the window or folder; in PowerPoint, start the slide show from the beginning.
F7	Spelling and grammar check (in MS Word).

Definition — Function key

One of the keys F1 to F12 on the top row of the keyboard, whose effect is decided by the active application rather than by a fixed system command.

In **PowerPoint**, `F5` starts a slide show from the beginning and `Shift+F5` starts it from the current slide. In **Windows**, on a folder or desktop, `F5` refreshes the view. `F7` in MS Word launches the spelling and grammar check. A documented JKSSB item asked the key that starts a slide show from the beginning, and the answer was `F5` (PYQ WO 2026 Q80, provisional key).

Example 21.1. *A candidate sees four keys: F1, F2, F5, F7. If the stem asks for Help, the answer is F1; for Rename, F2; for Refresh or slide show, F5; for the spell check, F7. Read the stem's verb first, then pick the key that matches that verb.*

The table below fixes the near neighbours that JKSSB likes to mix. Each row is one function key; the middle column is its job; the last column is the trap (the job a careless candidate might wrongly attach to it).

Key	Correct job	Common wrong answer
F1	Help	Rename or Refresh
F2	Rename a selected file or folder	Help or New
F5	Refresh; start slide show	Spell check or Save
F7	Spelling and grammar check	Slide show or Help

21.7 Windows-key shortcuts

The Windows key (the key with the Windows logo) builds system-level shortcuts.

Shortcut	Action
Win+D	Show the desktop (minimise all open windows).
Win+E	Open File Explorer.
Win+L	Lock the computer.

Mnemonic: **D** for Desktop, **E** for Explorer, **L** for Lock. These three are system shortcuts, not content shortcuts, so they do not depend on which file is open. `Win+D` does not delete files, and `Win+L` does not shut the computer down; it only locks the session, and the user must sign in again.

The **Shift** key is a modifier too. It produces capital letters and the upper symbol of a key, and `Shift+Click` selects a range of items between two clicks. `Shift+F5` starts a PowerPoint slide show from the current slide. On its own with a letter, Shift does not save, close, or print; it changes what is typed or selected.

21.8 Other high-yield keys

The shortcuts above are the core of the lesson. A few adjacent keys and combinations appear often enough in Office documents that they are worth learning alongside them.

Shortcut / key	Action
Ctrl+O	Open an existing file.
Ctrl+N	Create a new file (new document or window).
Ctrl+W	Close the current document or window.
Ctrl+B / Ctrl+I / Ctrl+U	Bold / Italic / Underline the selection.
Esc	Cancel the current action or close a dialog.
Delete	Delete the selected item; in text, erase forward.
Backspace	Erase the character to the left of the cursor.
Home / End	Move to the start / end of the line.
Ctrl+Home / Ctrl+End	Move to the top / bottom of the document.
Page Up / Page Down	Move up / down one screen.

Two cautions. First, `Ctrl+O` opens and `Ctrl+N` creates; do not confuse the two. Second, `Ctrl+W` closes a *document*, while `Alt+F4` closes the *window or application*; in a browser, `Ctrl+W` closes the current tab.

21.9 Special keys that stand alone

A few keys send a command on their own, so they are never written with a plus sign. They are not combinations, but they are learned together with shortcuts.

Key	Action
Enter	Confirm a command or start a new line.
Esc	Cancel the current action or close a dialog.
Tab	Move to the next field, cell, or control.
Caps Lock	Toggle capital letters on or off.
Backspace	Erase the character to the left of the cursor.
Delete	Erase the selected item, or the character to the right.
Insert	Toggle Insert and Overtype modes in text editors.
Home / End	Move to the start or the end of the current line.

Caps Lock and **Shift** both produce capitals, and this is a common confusion. Caps Lock changes every letter until it is switched off again, while Shift changes only the key pressed at that moment. **Delete** removes the selection or the character ahead of the cursor; **Backspace** removes what is behind the cursor.

21.10 Shortcuts in the office desktop workflow

A short walk-through on the office desktop ties the keys together. A clerk opens an existing letter with `Ctrl+O`, types the body, and underlines a heading with `Ctrl+U`. The heading is copied with `Ctrl+C`

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and pasted elsewhere with `Ctrl+V`; a wrong sentence is removed with `Ctrl+X`. A mistake is reversed with `Ctrl+Z`. The clerk searches the letter for a name with `Ctrl+F` and changes an old address to a new one with `Ctrl+H`. The finished letter is saved with `Ctrl+S` and printed with `Ctrl+P`. While the letter is open, the clerk switches to the spreadsheet with `Alt+Tab`, and when the letter is done, the window is closed with `Alt+F4`. Later, `Win+D` shows the desktop, `Win+E` opens File Explorer to find last month's folder, and `Win+L` locks the machine before the clerk steps away. This single sequence uses almost every shortcut in the lesson, which is why a scenario item is easy to build from it.

21.11 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. **Ctrl+C** copies and keeps the original; **Ctrl+X** cuts and removes the original. Do not treat them as the same command.
2. **Ctrl+Z** is Undo; **Ctrl+Y** is Redo. Reversing this pair is a common wrong answer.
3. **Ctrl+F** is Find; **Ctrl+H** is Replace. Find only locates; Replace also substitutes.
4. **Ctrl+S** is Save; **Ctrl+P** is Print. Save stores the file; Print sends it to the printer.
5. **Alt+Tab** switches between windows; **Alt+F4** closes the active window.
6. **F1** is Help, **F2** is Rename, **F5** is Refresh or slide show, and **F7** is the spell check (TRAP-32). Do not move a job from one key to another.
7. **Win+D** shows the desktop, **Win+E** opens File Explorer, and **Win+L** locks the computer.

Example 21.1. *An item asks which key is used to check spelling and grammar in MS Word. Trace the function keys: F1 is Help, F2 is Rename, F5 is Refresh or slide show, F7 is the spelling and grammar check. The answer is F7. If the stem instead asked for Rename, the answer would be F2, and if it asked for Help, the answer would be F1.*

21.12 Quick-reference sheet

Keep this single table for last-minute revision. It gathers every shortcut in this lesson.

Shortcut	Action	Shortcut	Action
Ctrl+C	Copy	Ctrl+S	Save
Ctrl+X	Cut	Ctrl+P	Print
Ctrl+V	Paste	Ctrl+F	Find
Ctrl+Z	Undo	Ctrl+H	Replace
Ctrl+Y	Redo	F1	Help
Ctrl+A	Select All	F2	Rename
Alt+Tab	Switch window	F5	Refresh / slide show
Alt+F4	Close window	F7	Spell check
Win+D	Show desktop	Win+E	File Explorer
Win+L	Lock computer	Shift+F5	Slide show from current

Exercises

1. State the action of Ctrl+C, Ctrl+X, and Ctrl+V.
2. Distinguish Ctrl+Z from Ctrl+Y, and Ctrl+F from Ctrl+H.
3. What do Alt+Tab and Alt+F4 do, and how are they different?

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4. Give the standard action of F1, F2, F5, and F7.
5. What do Win+D, Win+E, and Win+L do? What does Shift+F5 do in PowerPoint?
6. Why is the plus sign in `Ctrl+C` not typed by the user?

Solutions

1. Ctrl+C copies the selection (the original stays); Ctrl+X cuts it (the original is removed); Ctrl+V pastes the copied or cut item at the current position.
2. Ctrl+Z undoes the last action, while Ctrl+Y redoes the action that was undone. Ctrl+F opens Find, which only locates text; Ctrl+H opens Replace, which can also substitute new text for the found text.
3. Alt+Tab switches between open windows or applications without closing any of them; Alt+F4 closes the active window or application. One switches, the other closes.
4. F1 is Help; F2 renames the selected file or folder; F5 refreshes the window (or, in PowerPoint, starts the slide show from the beginning); F7 runs the spelling and grammar check in MS Word.
5. Win+D shows the desktop, Win+E opens File Explorer, and Win+L locks the computer. Shift+F5 starts a PowerPoint slide show from the current slide.
6. The plus sign is only a way of writing the combination. It means "hold the first key, then press the second"; it is not a character to be typed.

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LESSON 45

Open Source

45.1 Source code and what it means

Every program begins as **source code**: the set of instructions a programmer writes in a programming language such as C, Java, or Python. Source code is *human-readable* — a person can open the file, read it, and change it. Before the computer can run the program, a **compiler** or an **interpreter** converts the source code into a machine-readable form. That machine-readable form is called **object code**, or simply the executable, and it is what the processor actually executes. Ordinary users of a finished program normally receive only this compiled form and never see the source code.

Definition — Source code

The human-readable set of instructions, written in a programming language, that defines what a program does; it is converted into machine-readable object code by a compiler or an interpreter.

The important consequence is that whoever holds the source code controls how the software can be studied, changed, and shared. The whole topic of open source versus proprietary software grows out of this single fact.

45.2 Open source versus proprietary software

Software falls into two broad camps according to what is done with its source code.

Point	Open source	Proprietary (closed source)
Source code	Publicly available	Kept secret
View or modify	Allowed, subject to the licence	Not allowed without permission
Redistribute	Allowed, subject to the licence	Not allowed
Typical user gets	Source code plus executable	Only the compiled executable

Open-source software makes its source code public. Anyone is allowed to inspect it, modify it, and redistribute it, subject to the terms of its licence. **Proprietary software** keeps its source code confidential; the user receives only the compiled executable and may not view, change, or share the source without the owner's permission. Examples of proprietary software include Microsoft Windows, Apple macOS, Microsoft Office, Adobe Photoshop, and Oracle Database. The deciding factor between the two is the source code, not the price: some open-source software is sold, and some proprietary software is given away free of cost.

Definition — Open source

A software development model in which the source code is made publicly available so that anyone may view, modify, and redistribute it under the terms of its licence.

45.3 The open-source idea: view, modify, share

The meaning that examiners most often test is the three-part permission of an open-source licence. First, a user may **view** the source code and see exactly how the software works. Second, a user may **modify** it — fix a bug, improve it, or add a new feature. Third, a user may **share** it, redistributing copies and modified versions to others. These permissions are granted by the software's

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licence, so the exact conditions differ from one project to another; but the underlying spirit is the same three freedoms in the code.

Example 45.1. *An item states, "An open-source licence allows a user to view, change, and share the source code." This is the standard, correct description and is the answer expected in the exam. An option that says only "software available free of cost" describes* freeware*, not open source.*

45.4 Freeware, free software, public domain, commercial, shareware

Several labels are often mixed up with open source. Each has a distinct meaning.

Term	What is free	Source code
Open source	Use, study, change, share	Available
Freeware	Cost only	Usually not provided
Free software	The user's freedom	Available
Shareware	A limited trial period	Not provided
Public domain	No copyright at all	No restrictions
Commercial	Nothing (it is sold)	Usually not provided

Freeware is software given away free of cost, but its source code is usually not provided. **Free software** is about freedom, not price: the user is free to study, change, and share it. **Public domain** software has no copyright owner, so anyone may use, modify, and redistribute it without any restriction. **Shareware** is distributed as a free trial, working for a limited period or with limited features; to keep using it, the user must pay. **Commercial software** is developed and sold for profit and is often, though not always, proprietary.

Definition — Freeware

Software distributed free of cost, whose source code is usually not made available; it is free in price, not in freedom.

Definition — Shareware

Software distributed as a trial version, free for a limited time or with limited features, after which payment is required to continue using it.

45.5 Free software and its four freedoms

The term **free software** comes from the **Free Software Foundation (FSF)**, founded by **Richard Stallman**. In this context, "free" refers to liberty, not price — the well-known phrase is "free as in freedom, not free as in free beer". The FSF defines four freedoms: the freedom to run the program for any purpose; the freedom to study how it works and change it; the freedom to redistribute copies to help others; and the freedom to distribute modified versions to the public. Free software and open source are closely related and frequently overlap, but they are not identical labels. Free software stresses the ethical freedom of the user; open source stresses the practical benefits of a publicly available source code.

Definition — Free software

Software that grants the user the freedom to run, study, change, and redistribute it and its modified versions; "free" here means freedom, not zero cost.

45.6 The Open Source Initiative

Alongside the Free Software Foundation stands the **Open Source Initiative (OSI)**, founded in 1998. The OSI maintains the **Open Source Definition**, the set of criteria a licence must satisfy

to be called an open-source licence, and it reviews and approves licences against those criteria. The term **open source** itself was popularised in 1998 as a more business-friendly label than "free software". The two movements overlap heavily: most licences approved by the OSI are also free-software licences. An umbrella term often seen in questions is **FOSS**, meaning Free and Open-Source Software, or **FLOSS** (Free/Libre and Open-Source Software), which covers both traditions together.

The Open Source Definition lays down the conditions a licence must meet. The software must be freely redistributable; the source code must be included or easily obtainable; derived works and modifications must be allowed; the licence must not discriminate against any person, group, or field of use; and the rights must apply to everyone who receives the software. A licence that satisfies these conditions is called an **OSI-approved** licence. This is why "open source" is a matter of the licence, not simply of whether the software is free of cost.

Example 45.1. *Two programs are both free to download. Program A ships with its source code under an OSI-approved licence that permits modification and redistribution; Program B ships only compiled files and forbids copying. A is open source; B is* freeware. *The download price is identical, but the licences differ, and the licence is what decides the category.*

45.7 Benefits and limitations of open source

Open source is chosen for several practical reasons. The licence cost is usually zero, so organisations avoid per-seat fees. The source code can be inspected, which lets many people review it for bugs and security weaknesses. Users are not locked in to a single vendor, and the software can be customised to local needs. A large community often provides documentation and support.

Open source also has limitations that a fair answer should acknowledge. A popular project may have no formal commercial support contract, so support depends on the community or on paid third-party vendors. Compatibility with proprietary formats and integration with existing systems can be difficult. The total cost of ownership is not always zero, because training, migration, and maintenance still cost money. These points are occasionally tested as "which of the following is a limitation of open source".

45.8 Full forms and key names to remember

Direct-recall items often ask for an expansion or for the name behind a term. The following are the ones most worth memorising.

Short form	Full form / meaning
GPL	GNU General Public License
FSF	Free Software Foundation
OSI	Open Source Initiative
FOSS	Free and Open-Source Software
FLOSS	Free/Libre and Open-Source Software
AOSP	Android Open Source Project
MIT	MIT License (a permissive software licence)
BSD	Berkeley Software Distribution (its licence family)

Remember too the names behind the ideas: the Free Software Foundation was founded by **Richard Stallman**, and **Git** was created by Linus Torvalds to manage the Linux kernel's source code. **GitHub**, **GitLab**, **Bitbucket**, and **SourceForge** are the best-known hosting sites for open-source repositories.

45.9 Repositories and community development

Open-source projects live in a **repository** (often shortened to "repo"). A repository is a storage location that holds a project's source code together with every change ever made to it. **Version**

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control records this history, so an earlier version can be restored if needed. **Git** is the most widely used distributed version-control system, and popular hosting sites for repositories include **GitHub**, **GitLab**, **Bitbucket**, and **SourceForge**.

Open-source software is usually built by a **community** of contributors spread around the world. A contributor may **fork** a project, that is, make a personal copy for experimenting or improving. A **pull request** is then a proposal to merge those changes back into the main project, where **maintainers** review them. An **issue tracker** records bugs and feature requests. Because many people can inspect and improve the code, community review is treated as a key strength of open-source development.

45.10 Basic licensing concepts

By default, **copyright** gives the creator of a work exclusive rights over it. A **licence** is the legal permission that states what others may do with the software. Open-source and free-software licences fall into two broad families.

Copyleft licences, such as the GNU General Public License (GPL), require that modified versions of the software be released under the *same* licence. This keeps the software and its derivatives open. **Permissive** licences, such as the MIT License, the Apache License 2.0, and the BSD License, allow the code to be reused with very few conditions, including inside proprietary projects, provided the licence notice is kept. The licence, not the price, decides whether software is truly open source.

Definition — Copyleft

A licence condition that requires any modified or extended version of the software to be distributed under the same licence, so the work cannot be closed.

45.11 Copyright, licences, and piracy

It helps to separate three ideas. **Copyright** is the creator's legal ownership of the work. A **licence** is the permission the owner grants for specific uses. **Piracy** is the use, copying, or distribution of software in a way the licence forbids — for example, installing a proprietary program without a valid licence, or copying it to machines the licence does not cover. Open-source software is not "unlicensed": it is licensed, and its licence grants broad permissions. Public-domain software, by contrast, has no owner and therefore needs no licence at all. The exam sometimes tests this by asking whether open-source software is "free of copyright"; the answer is no, because the copyright remains with the authors and the licence grants the freedoms.

Example 45.1. *A student installs the same licensed copy of a proprietary office suite on five computers when the licence permits only one. This is **piracy**. Installing an open-source office suite such as LibreOffice on all five computers, in line with its licence, is not piracy. The difference is not the price paid but whether the use matches what the licence allows.*

45.12 Common licences at a glance

Licence	Type	Main condition
GNU GPL	Copyleft	Derivative works must use the same licence
MIT License	Permissive	Reuse allowed; keep the licence notice
Apache License 2.0	Permissive	Few limits, plus an explicit patent grant
BSD License	Permissive	Very few conditions
Creative Commons	Content licence	For creative works, not usually software

Note that **Creative Commons** licences are designed for creative works such as text, images, and music; they are not typically used to license software. For software, the GPL, MIT, Apache, and BSD licences are the ones to recognise.

45.13 Well-known open-source and proprietary software

Familiar examples help fix the concepts.

Open source	Proprietary
Linux (operating system)	Microsoft Windows
Android (mobile operating system)	Apple macOS, iOS
Mozilla Firefox (browser)	Microsoft Office
Chromium (Chrome's open-source base)	Adobe Photoshop
LibreOffice (office suite)	Oracle Database
VLC media player, GIMP, Blender	
Apache HTTP Server, MySQL, Python, Git, WordPress	

Android is the open-source **mobile operating system**, built on the open-source Linux kernel. This is a directly tested fact: when a question asks for the open-source mobile OS, the answer is Android. In contrast, iOS is a proprietary mobile operating system, and Windows and macOS are proprietary desktop operating systems.

45.14 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The test of open source is the **source code**, not the price. An open-source licence lets users **view, change, and share** the source (PYQ-26).
2. **Freeware** is free of **cost**; **free software** is about **freedom**. Do not treat the two as the same (TRAP-23).
3. **Open source**, **freeware**, and **shareware** are three different things: one shares the source, one is free of cost, one is a paid trial (TRAP-23).
4. The open-source **mobile operating system** is **Android** (PYQ-24); iOS is proprietary.
5. **GPL** is copyleft (derivatives stay open); **MIT**, **Apache 2.0**, and **BSD** are permissive.

Example 45.1. *An item asks, "Which of the following is an open-source mobile operating system?" Trace the options: Android is open source (built on the Linux kernel), while iOS, Windows, and macOS are proprietary. The answer is Android. If the stem instead asks what an open-source licence permits, the answer is the right to view, change, and share the source code.*

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Revision at a glance

Term	What it is	What it is not
Open source	Source code is public and licensed for view/change/share	Not merely free of cost
Proprietary	Source kept secret; only the executable is distributed	Not necessarily paid
Freeware	Free of cost	Not necessarily open source
Free software	Freedom to run, study, change, share	Not about zero price
Public domain	No copyright owner	Not the same as open source
Shareware	Free trial, then pay	Not free forever
Commercial	Sold for profit	Not necessarily proprietary or open
Copyleft (GPL)	Derivatives must stay open	Not permissive
Permissive (MIT)	Reuse with few conditions	Not copyleft

Exercises

1. Define source code and explain how it differs from object code.
2. State the three things an open-source licence typically permits a user to do with the source code.
3. Distinguish freeware from free software.
4. Explain what public domain software is and how it differs from open-source software.
5. Name one copyleft licence and one permissive licence, and state the main condition of each.

Solutions

1. Source code is the human-readable set of instructions written by a programmer in a programming language. It is converted by a compiler or interpreter into object code, the machine-readable form the computer actually runs.
2. View the source code, modify it, and share (redistribute) it, subject to the licence terms.
3. Freeware is free of cost but its source code is usually not provided. Free software is about freedom: the user may study, change, and share the software; "free" means liberty, not price.
4. Public domain software has no copyright owner and carries no restrictions at all. Open-source software is still copyrighted, but its licence grants the freedom to view, change, and share the source.
5. Copyleft: the GNU GPL, which requires modified versions to be released under the same licence. Permissive: the MIT License (or Apache 2.0 / BSD), which allows reuse with few conditions, such as keeping the licence notice.

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LESSON 46

Firmware, BIOS, UEFI and Drivers

46.1 What firmware is

Every computer has two broad parts: the hardware, which is the physical circuitry, and the software, which is the set of instructions. Between them sits a special kind of software called **firmware**. Firmware is software that is stored in non-volatile memory on the hardware itself and that gives the device its basic, low-level instructions. It is not opened by the user like a word processor or a web browser; it is fixed in the device and runs by itself.

Firmware runs *before* the operating system. When a computer is switched on, its memory is empty and no program is running, so something must tell the hardware how to begin. That "something" is the firmware. It starts the hardware, checks it, and then hands control to the operating system.

Definition — Firmware

Software stored in non-volatile memory that is programmed into a hardware device and provides its basic, low-level startup and control instructions, running before the operating system.

Firmware is different from ordinary application software in three ways. It is stored on the device rather than installed like a program; it is tied closely to the exact hardware; and it starts the machine rather than being started by the user.

46.2 Where firmware is stored

Firmware must be available the instant the machine is switched on, so it is kept in **non-volatile** memory — memory that keeps its contents without power. If firmware were stored in RAM, it would be lost the moment the power was switched off, and the machine could never start.

Common homes for firmware are **ROM** (Read-Only Memory), **PROM** (Programmable ROM), **EPROM** (Erasable PROM), **EEPROM** (Electrically Erasable PROM), and **flash memory**. Flash memory is the modern choice because it can be erased and rewritten while still installed, which is what makes firmware updates possible.

Definition — Non-volatile memory

Memory that retains its contents when the power is removed. ROM, flash memory, an SSD, and an HDD are non-volatile. RAM is the common volatile memory.

The important contrast is with RAM. RAM is volatile and is empty at power-on; firmware is non-volatile and is present at power-on. A common wrong statement is that firmware is loaded into RAM before the machine starts — in fact the firmware resides in ROM/flash and runs from there.

It is worth naming the members of the ROM family, because exam papers often ask which memory type can be rewritten. **ROM** is programmed at the factory and cannot be changed. **PROM** (Programmable ROM) can be written once by the user. **EPROM** (Erasable PROM) can be erased with ultraviolet light and reprogrammed. **EEPROM** (Electrically Erasable PROM) can be erased and rewritten electrically. **Flash memory** is a fast, block-erasable form of EEPROM and is the type used for modern firmware, which is why firmware can be updated in place.

Example 46.1. *An item states, "The firmware of a modern PC is stored in RAM so that it can be updated.": This is wrong on the first point. Firmware is stored in non-volatile ROM or flash memory, not in volatile RAM. Flash memory is what makes the firmware updatable, not RAM.*

46.3 Hardware, software, and firmware

It helps to place firmware among the three basic layers of a computer.

Layer	What it is	Example
Hardware	The physical parts of the computer	Processor, motherboard, keyboard
Firmware	Software fixed in non-volatile memory that controls the hardware	BIOS, UEFI
Software	Programs that the user runs or that run on the OS	MS Word, a web browser

Hardware is the body, software is the instruction set, and firmware is the low-level bridge that tells the hardware how to begin. Firmware is software by nature, but it is treated separately because it is stored on the device and is not installed or removed like an ordinary program.

46.4 Firmware on everyday devices

System firmware is not unique to computers. Many devices carry firmware that gives them their basic behaviour. A printer has firmware that controls printing; a Wi-Fi router has firmware that runs its networking; a smartphone has firmware plus an operating system; a modern television has firmware for its tuner and menu. Updating such a device usually means updating its firmware, again by flashing. This is why a firmware fix can repair a device without any change to the user's applications.

46.5 System firmware: the BIOS

On a personal computer, the most familiar firmware is the **BIOS**, which stands for **Basic Input/Output System**. It is stored on a chip on the motherboard. When the computer is switched on, the BIOS runs first. It performs the power-on self-test, initialises the basic hardware, reads its settings, and then loads the operating system.

The single most important exam fact on this topic is that the firmware which initialises the hardware and loads the operating system is the BIOS. This is directly tested (PYQ-28). The BIOS is sometimes confused with a device driver or with the kernel; neither of these loads the operating system. A device driver is loaded *by* the operating system, and the kernel is part *of* the operating system.

Definition — BIOS

The Basic Input/Output System: the system firmware on a PC motherboard that runs first at power-on, initialises the hardware, and loads the operating system.

46.6 The power-on boot sequence

The machine follows a fixed chain each time it starts. Learning this chain makes the roles of firmware, the OS, and drivers easy to separate.

1. **Power on.** Electrical power reaches the motherboard.
2. **Firmware runs.** The BIOS or UEFI begins executing from ROM/flash.
3. **Hardware initialisation and POST.** The firmware tests and configures the basic hardware.
4. **Operating system loads.** The firmware finds the boot device, reads the bootloader, and hands control to the OS.
5. **Drivers load.** Once the OS is running, it loads the device drivers it needs.
6. **Applications run.** User programs start on top of the OS.

The chain can be written compactly as: power-on → firmware → initialisation → OS → driver → application. Any question that asks "which comes first?" is answered by walking this chain from left to right.

The step in which the firmware hands control to the operating system is worth a little more detail. The firmware locates the boot device according to its configured boot order, reads a small program called the **bootloader**, and runs it. The bootloader then loads the main part of the operating system into memory. Under the legacy BIOS, the bootloader is found through the MBR (Master Boot Record); under UEFI it is found through the GPT (GUID Partition Table) scheme and a dedicated EFI system partition.

Definition — Bootloader

A small program that the firmware runs to load the operating system into memory and start it.

46.7 Boot devices and boot order

The firmware does not decide by itself which device to start from. It follows the **boot order** stored in its settings: a list such as hard disk first, then optical drive, then USB. The firmware tries each device in turn until it finds one that can start the operating system. The boot order is the reason a machine can be made to start from a USB drive by moving that drive to the top of the list, and it is one of the first things a technician checks when a computer reports that no boot device is found.

Example 46.1. *A computer is switched on and shows, "No boot device found." Walking the chain helps: the firmware runs, the POST passes, and the firmware then looks through the boot order for a startable device. If the disk has failed or its entry has moved down the list, no operating system is found and the firmware stops with this message. The failing layer is the firmware's boot configuration or the boot disk, not the application software.*

46.8 POST: the power-on self-test

POST stands for **Power-On Self-Test**. It is one of the first things the BIOS does after power-on. During POST the firmware checks that the essential hardware — CPU, RAM, keyboard, disk controller, and video — is present and responding. If a component fails, the machine may beep in a code, show an error message, or refuse to continue.

POST is a step *inside* the firmware's work. It runs *before* the operating system, not after it. A frequent trap is to place the POST after the OS has started; the correct order is firmware (including POST) first, then the OS.

Definition — POST

The Power-On Self-Test: the hardware check the BIOS runs at startup before loading the operating system.

46.9 BIOS setup and CMOS

The BIOS keeps a small set of settings: the system date and time, the order in which the machine should look for a boot device, hardware options, and sometimes passwords. These settings are stored in a tiny battery-backed memory chip called **CMOS** (Complementary Metal-Oxide-Semiconductor). Because the chip is battery-backed, the clock and boot order are remembered between sessions, even when the main power is off.

The BIOS Setup utility (sometimes called the CMOS setup) is opened by pressing a key during startup. That key is commonly **Del** or **F2**, but the exact key varies by manufacturer. Inside setup, the user can change the boot order, enable or disable built-in devices, and set passwords.

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46.10 UEFI: the modern firmware

UEFI stands for **Unified Extensible Firmware Interface**. It is the modern replacement for the legacy BIOS and performs the same basic job: it starts the hardware and loads the operating system. UEFI adds several improvements.

- It boots the machine **faster**.
- It supports very large disks through the **GPT** (GUID Partition Table) scheme, instead of the older MBR scheme with its practical limit of about 2 TB.
- It provides a **graphical** setup utility, often with mouse support.
- It offers **Secure Boot**, which helps prevent unauthorised software from loading at startup.

For exam questions, remember that "the firmware that boots the OS" may be either BIOS or UEFI depending on the age of the system. Both initialise the hardware and load the operating system; they are firmware, never drivers.

46.11 BIOS versus UEFI

The two system-firmware generations can be compared on a few clear points.

Point	BIOS	UEFI
Full form	Basic Input/Output System	Unified Extensible Firmware Interface
Age	Legacy (older)	Modern
Boot speed	Slower	Faster
Disk support	MBR (about 2 TB limit)	GPT (very large disks)
Interface	Mostly text-based	Graphical
Security	No Secure Boot	Secure Boot available

46.12 Device drivers

A **device driver** is software that lets the operating system communicate with a specific hardware device. Each device — a printer, a graphics card, a network card, a sound card — needs its own driver, because a driver translates the general commands of the operating system into the particular instructions that one device understands.

Drivers are loaded **by the operating system**, after the firmware has started the machine. They are matched to the exact device and the exact OS version. Because a driver is software, it is wrong to call it hardware, and because it is loaded by the OS, it is wrong to say that a driver initialises the hardware and loads the OS.

Definition — Device driver

Software that enables the operating system to communicate with a particular hardware device by translating general OS commands into device-specific instructions.

The three layers are easy to mix up, so keep their roles separate:

Layer	Stored where	Role
Firmware (BIOS/UEFI)	On the hardware (ROM/flash)	Starts the machine; initialises hardware; loads the OS
Device driver	With the OS, on disk	Talks to one specific device
Operating system	On disk	Manages all resources and provides the user interface

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Example 46.1. *A question asks, "Which software initialises the hardware and loads the operating system?" A device driver is offered as an option. This is the classic near-neighbour trap. The correct answer is the **BIOS/UEFI** firmware. A device driver cannot do this job: it is itself loaded by the OS, and it controls one device rather than starting the whole machine.*

46.13 Firmware, driver, OS, and kernel

Four terms are often mixed up because they all describe "software close to the hardware". Their roles are different, and the exam tests the differences.

Term	What it does	When it runs
Firmware (BIOS/UEFI)	Initialises hardware and loads the OS	First, at power-on
Kernel	The core of the OS; manages memory, processes, and devices	After the OS starts loading
Device driver	Communicates with one specific device	After the OS is running
Operating system	Manages all resources and provides the interface	After firmware, before applications

The kernel is part of the operating system, not separate firmware. A driver is loaded by the OS once the kernel is running. Only firmware runs before the OS. If an item asks which layer "starts the machine and loads the OS", the answer must be firmware — BIOS or UEFI — and never a driver or the kernel.

46.14 Firmware updates and flashing

A **firmware update** replaces the firmware stored on a device with a newer version. The process is commonly called **flashing**, because the new firmware is written into flash memory. Updates are issued to fix bugs, add features, and patch security flaws.

Firmware updates are much rarer than ordinary software updates, and they carry more risk. Firmware is low-level, so a failed update can leave a device unusable — often described as "bricking" it. Interrupting the power during flashing, or loading the wrong firmware for the model, can damage the device permanently. For this reason the standard advice is to use only the manufacturer's update for the exact model and to keep a stable power supply throughout the update.

Example 46.1. *A candidate reads, "A firmware update is called formatting." This is wrong. Formatting prepares a storage volume for files; it does not rewrite firmware. The correct term for a firmware update is **flashing**.*

46.15 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. Firmware that initialises hardware and loads the OS is **BIOS/UEFI**, not a device driver or the kernel (TRAP-24, PYQ-28).
2. A **device driver** is **software**; the OS is system software (TRAP-10).
3. Firmware is stored in **ROM/flash** and is non-volatile; RAM is volatile and empty at power-on.
4. **POST** runs **before** the operating system, as part of the firmware's startup work.
5. A driver is loaded **by the operating system**, after the firmware has started the machine.
6. The startup order is power-on, firmware, initialisation, OS, driver, application.
7. A firmware update is called **flashing**; interrupting it can damage the device.

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Example 46.1. *An item asks which layer runs first when a computer is switched on. Trace the chain: firmware runs before the OS, and the OS runs before any driver, which runs before any application. The answer is the firmware (BIOS/UEFI). If the stem instead asks what tests the hardware at startup, the answer is the POST; if it asks what lets the OS talk to one device, the answer is the device driver.*

Exercises

1. Define firmware and state where it is normally stored.
2. List the steps of the power-on boot sequence from power-on to application.
3. Expand BIOS, UEFI, and POST, and give one role of each.
4. Distinguish firmware, a device driver, and an operating system.
5. What is a firmware update called, and what is the main risk during one?

Solutions

1. Firmware is software stored in non-volatile memory that gives a device its basic, low-level startup and control instructions. It is normally stored in ROM or flash memory on the hardware itself.
2. Power on → firmware (BIOS/UEFI) runs → hardware initialisation and POST → the operating system loads → device drivers load → application programs run.
3. BIOS = Basic Input/Output System: system firmware that initialises hardware and loads the OS. UEFI = Unified Extensible Firmware Interface: the modern firmware with fast boot, GPT disk support, and Secure Boot. POST = Power-On Self-Test: the hardware check the BIOS runs at startup.
4. Firmware is stored on the hardware in ROM/flash and starts the machine and loads the OS. A device driver is software loaded by the OS that talks to one specific device. The operating system is the large system-software layer that manages all resources and provides the user interface.
5. A firmware update is called flashing. The main risk is that interrupting the power during flashing, or using the wrong firmware, can leave the device unusable.

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LESSON 52

OS PYQ Traps and System Utilities

52.1 Why this lesson is a trap drill

The computer section of the JKSSB papers does not only test whether a candidate knows a term. It tests whether the candidate can tell a term apart from its *nearest neighbour*. A typical wrong option is not a random word; it is the close confusion that "sounds right" until the exact wording is checked. Device driver versus hardware, DMA versus ordinary CPU copying, system software versus application software, and the Windows Run command `winword` are all examples of this pattern.

Because of this, the present lesson is a **trap drill**. Every question in the practice set is built around a real near-neighbour or defective-option pattern catalogued in the course knowledge base. The aim is not to learn a large amount of new material, but to build the habit of reading each option as a *claim to be tested*. The rest of these notes explain the small set of facts the traps are built on, one distinction at a time.

Definition — Trap drill

A study method in which each practice option is treated as a factual claim, and the candidate decides whether the claim is exactly correct, closely correct but wrong, or plainly wrong.

52.2 The trap catalogue for this lesson

The traps this lesson drills on are collected below. Each row states the tempting wrong belief, the correct distinction that defeats it, and the knowledge-base reference where the pattern is recorded. Read the table as a checklist: if an option matches a row on the left, it is the trap, and the right-hand column tells you what to say instead.

Tempting wrong belief	Correct distinction	Ref.
A device driver is hardware.	A device driver is software (system software); the device is hardware.	TRAP-10
DMA makes the CPU copy data "faster".	DMA moves data between an I/O device and main memory, bypassing the CPU.	§52.5
The OS runs from secondary storage.	While running, the OS resides in primary memory (RAM).	PYQ-22
Utility software is application software.	Utility software maintains the system; application software does the user's task.	§52.3
MS Word is system software.	MS Word is application software.	§52.3
MS Access is an operating system.	MS Access is a database application, not an OS.	§52.7
Windows or macOS is a Linux distribution.	Ubuntu, Fedora, and Debian are Linux distributions.	§52.7
`winword` opens WordPad or Notepad.	`winword` opens MS Word.	§52.9
A component is internal or external by importance.	The classification follows position relative to the system unit.	§52.8
An RTOS is chosen for office or graphics work.	An RTOS is chosen for fixed time deadlines.	§52.6

52.3 The operating system: role and classification

The **operating system (OS)** is the master **system software** of a computer. It sits between the user (and the application programs) on one side and the hardware on the other. Its two central jobs are to provide a *user-friendly interface* and to manage the machine's resources.

The interface may be a graphical one, such as the Windows desktop, or a command line, such as the MS-DOS prompt or the Linux shell. Either way, the user does not need to drive the hardware directly. In its resource-management role, the OS schedules the processor, allocates main memory, organises storage, and controls input and output devices from a single set of instructions. It also provides services — opening a file, printing a document, sending data over a network — that application programs call instead of talking to the hardware themselves.

Definition — Operating system

The system software that controls the computer's hardware and provides a user-friendly interface and a set of services to application programs.

Common operating systems are Windows, Linux, UNIX, MS-DOS, Android, and macOS. An application program is not an operating system. MS Word, MS Excel, MS Access, a web browser, and a media player are applications, not operating systems. This is one of the exact-recognition distinctions the exam uses.

A frequently tested additional fact is *where* the OS is when the computer is running. The OS is stored permanently on secondary storage (the disk), but while the machine is running it is loaded into **primary memory (RAM)**. The CPU executes instructions only from main memory, so the running OS must be resident there. When power is removed, the RAM copy is lost and the OS is loaded again from disk at the next boot.

52.4 System, application, and utility software

Software is grouped by its purpose. **System software** runs and manages the machine: the operating system, device drivers, utility programs, and language translators (compiler, interpreter, assembler) all belong here. **Application software** performs a task the user wants done: typing a document, preparing a spreadsheet, browsing the web. **Utility software** is a small maintenance category: tools such as Disk Cleanup, Disk Defragmenter, and antivirus programs that keep the system healthy and efficient. Utility software is closely tied to system software, but it is useful to name it separately because exam options mix the three categories.

Category	Role	Examples
System software	Runs and manages the machine	OS, device drivers, translators
Application software	Does a user's task	MS Word, Excel, a browser
Utility software	Maintains and optimises the system	Disk Cleanup, Defragmenter, antivirus

Example 52.1. *An item asks, "Which of the following is application software?" with the options Windows, MS Word, Disk Defragmenter, and a device driver. Windows is an operating system (system software); Disk Defragmenter is utility software; a device driver is system software too. Only* ***MS Word*** *is application software. The trap is that all four options are well-known pieces of software, and only the* category *separates them.*

52.5 Device drivers are software

A **device driver** is a small software program that allows the operating system to communicate with a specific hardware device. Each device — a printer, a keyboard, a graphics card, a network

adapter — needs a matching driver. The driver receives general commands from the OS and translates them into the device-specific instructions that the hardware understands. Without the correct driver, the OS cannot use the device properly.

The trap here is a classification trap. A device driver *controls* hardware, but the driver itself is **software**, and because it extends the operating system's control of the machine it is counted as **system software**. The device it drives is hardware. The OS itself is system software. An option that says "device driver is hardware" has confused the software with the device.

Definition — Device driver

A software program that lets the operating system control a particular hardware device by translating OS commands into device-specific instructions.

52.6 DMA: transferring data without the CPU

Normally, moving data between an input/output device and main memory keeps the CPU busy: the CPU reads each unit from the device and writes it to memory (or the reverse), one step at a time. **DMA — Direct Memory Access** — replaces this with a dedicated transfer. A DMA controller moves data *directly* between the I/O device and main memory while the CPU is bypassed for the duration of the transfer. The CPU starts the transfer and is notified by an interrupt when it is complete.

Definition — Direct Memory Access (DMA)

A method of data transfer in which a controller moves data between an input/output device and main memory directly, without the CPU handling each individual byte.

The advantages are clear. Because the CPU is not occupied with copying each byte, it is free to run other instructions, and large block transfers finish faster. Several traps grow from this description. DMA moves data between a *device and main memory*, not between the CPU and the device. The CPU is *bypassed* — it is not made "faster". DMA does not increase the size of RAM and does not encrypt data. The CPU still initiates the transfer and handles the completion interrupt; it is simply not in the per-byte path.

52.7 Real-time operating systems

A **real-time operating system (RTOS)** is an operating system designed so that tasks complete within **fixed time deadlines**. In an ordinary desktop OS, a task may take as long as the scheduler permits as long as it finishes eventually. In an RTOS, meeting the deadline is the whole point: a late response is a failure, even if the answer is correct.

An RTOS is therefore chosen for embedded and control systems, such as industrial machinery, medical devices, and vehicle electronics, where a delayed action can be dangerous. It is not chosen for typing a document, designing a website, or editing a photograph; those are ordinary application tasks that do not carry hard deadlines.

Definition — Real-Time Operating System (RTOS)

An operating system that guarantees that critical tasks complete within a bounded, fixed time so that deadlines are always met.

52.8 DOS, UNIX, and Linux distributions

DOS stands for **Disk Operating System**. MS-DOS was a command-line operating system: the user typed commands rather than clicking icons. **UNIX** is a multi-user, multitasking operating system that influenced many later systems. **Linux** is an open-source, UNIX-like operating system; because its source can be freely used and modified, it is distributed in many packaged forms called

distributions. Well-known Linux distributions include **Ubuntu**, **Fedora**, **Debian**, and **Linux Mint**.

This gives another exact-recognition drill. Windows and macOS are operating systems, but they are *not* Linux distributions. MS-DOS is an operating system but, again, not a Linux distribution. Only Ubuntu, Fedora, Debian, and Mint belong on the Linux-distribution list. Similarly, MS Access is a database application and is **not** an operating system at all, even though it is a familiar Microsoft product.

52.9 Internal and external hardware

Hardware is classified by its *position* relative to the system unit (the cabinet that houses the main machine). **Internal** hardware sits inside the system unit: the motherboard, the CPU, RAM, the hard disk, the power supply, and expansion cards. **External** hardware, also called a **peripheral**, sits outside the system unit and is joined to it through ports and cables: the monitor, keyboard, mouse, printer, and scanner.

The trap is that a familiar word can appear in the wrong column. A monitor and a keyboard are external; a motherboard and a RAM module are internal. The classification follows the position, not the importance of the component.

Definition — Internal vs external hardware

Internal hardware is located inside the system unit; external hardware (a peripheral) is located outside it and connected through ports.

52.10 The Windows Run box and the `winword` command

The Windows **Run** box, opened with the Windows + R shortcut, starts a program when the user types that program's command. The command `winword` opens **MS Word**. In the same way, `notepad` opens Notepad, `wordpad` opens WordPad, and `cmd` opens the Command Prompt. The command must match the program exactly.

Example 52.1. *An item asks which program is opened by typing `winword` in the Run box, offering Notepad, MS Word, WordPad, and File Explorer. The answer is **MS Word**. Notepad is opened by `notepad`, WordPad by `wordpad`, and File Explorer by `explorer`. The near-neighbours are deliberately word-processor or editor names, so the candidate who only half-remembers the command can be pulled to the wrong option.*

52.11 Running example: the office desktop

It helps to hang these facts on one concrete machine. Picture the office desktop used throughout this course: a multi-core PC running Windows, with MS Word and MS Excel installed, a printer, and an internet connection.

When the machine is switched on, the Windows OS is loaded from the disk into primary memory and stays resident there while the machine runs. MS Word and MS Excel are *application* software that the user opens to do work. Disk Cleanup and Disk Defragmenter are *utility* software that the user runs occasionally to keep the machine healthy. The printer does not work until its *device driver* — a piece of system software — is installed, so that Windows can translate its print commands into the printer's own instructions. When a large print job or file copy is under way, a *DMA* transfer can move the data between the device and main memory while the CPU continues with other work.

Example 52.1. *A single scenario can therefore generate several trap options about one machine. "Word is system software" is false — Word is application software. "The printer driver is hardware" is false — the driver is software. "The OS is stored on the disk and run from there" is half-true and therefore defective: it is stored on the disk but* run *from main memory. Recognising which category each item belongs to settles all three.*

52.12 Defective options and how to read them

Some official options are not merely wrong; they are *defective* — the wording itself is inconsistent or ambiguous. The knowledge base records this as a real feature of the papers (Anti-Pattern AP-02). A defective option is teaching material, not a fact to memorise. The safe method is to quote the option exactly, mark the defect, and then apply the stable concept.

Example 52.1. *A defective option might read, "The device driver is the hardware that stores the instructions the operating system uses to control a device." The defect is the phrase "the hardware that stores the instructions". A driver is* software *that holds those instructions; it is not hardware. The stable concept to apply is TRAP-10: a device driver is software. Once the defect is spotted, the option can be rejected without any doubt.*

The same discipline applies to near-neighbour options that are not defective but simply close. When two options are both plausible, compare them word by word against the definitions in this lesson. The distinction is usually a single term — *software* versus *hardware*, *device to memory* versus *CPU to device*, *application* versus *utility*, or `winword` versus `wordpad`. Reading for that one term is what turns a guess into a correct answer under negative marking.

52.13 Terminology ladder for this lesson

Direct-recall expansion items are the single most common JKSSB format, so the exact terms from this lesson are collected here with their full forms. Each row is one fact.

Term	Full form	One-line meaning
OS	Operating System	System software that provides an interface and manages resources
DMA	Direct Memory Access	Data moves between an I/O device and main memory with the CPU bypassed
RTOS	Real-Time Operating System	An OS that meets fixed time deadlines
DOS	Disk Operating System	A command-line operating system (MS-DOS)
Driver	Device driver	Software that lets the OS control a specific device
Utility	Utility software	Maintenance software such as Disk Cleanup or Defragmenter
Peripheral	External hardware	A device outside the system unit, joined by ports

52.14 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A device driver is **software** (system software); the device it controls is hardware (TRAP-10).
2. The OS is **system software**. MS Word is application software; Disk Cleanup is utility software.
3. While running, the OS is held in **primary memory (RAM)**, not executed from secondary storage (PYQ-22).
4. DMA moves data between a **device and main memory** and **bypasses the CPU**; it is not "faster CPU copying".
5. **RTOS** = Real-Time Operating System, chosen for fixed deadlines.
6. **DOS** = Disk Operating System; Linux distributions include Ubuntu, Fedora, and Debian. `winword` opens MS Word.

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Exercises

1. Classify each as system, application, or utility software: Windows, MS Word, Disk Defragmenter, a device driver, Disk Cleanup.
2. Explain why a device driver is described as software even though it controls hardware.
3. Describe a DMA transfer: name the two locations between which data moves and state the role of the CPU.
4. State the full form of RTOS and give one reason an RTOS is used.
5. Which of these are operating systems, and which are Linux distributions: Ubuntu, Windows, Debian, MS Access, Fedora?
6. What does typing `winword` in the Windows Run box do, and how does it differ from `wordpad`?

Solutions

1. System software: Windows and the device driver. Application software: MS Word. Utility software: Disk Defragmenter and Disk Cleanup.
2. The driver is a stored set of instructions that the OS loads and runs to talk to the device. It has no physical body of its own; the hardware is the device. A program that extends OS control of hardware is therefore software, and specifically system software.
3. Data moves directly between the input/output device and main memory. A DMA controller performs the transfer; the CPU starts it and is interrupted when it finishes, but it does not handle each byte.
4. RTOS = Real-Time Operating System. It is used where tasks must complete within fixed time deadlines, as in embedded and control systems.
5. Operating systems: Ubuntu (also a Linux distribution), Windows, Debian (also a Linux distribution), Fedora (also a Linux distribution). MS Access is not an operating system; it is a database application.
6. Typing `winword` opens MS Word. Typing `wordpad` opens WordPad, a different text editor. The two commands open different programs and must not be confused.

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LESSON 60

Current OS, Processes, Linux and Utilities

60.1 Deadlock and its four necessary conditions

A **deadlock** is a situation in which a group of processes are each waiting for a resource that is held by another process in the group, so that none of them can ever proceed. A simple picture is two processes: process A holds a printer and waits for a scanner, while process B holds the scanner and waits for the printer. Each waits for the other, and both stay stuck forever.

For a deadlock to be possible, four conditions must hold *at the same time*. They are usually listed as follows.

Condition	Meaning
Mutual exclusion	A resource is held by only one process at a time.
Hold and wait	A process holds at least one resource while waiting for more.
No preemption	A resource cannot be forcibly taken away from its holder.
Circular wait	A cycle of processes exists, each waiting for a resource held by the next.

Definition — Deadlock

A state in which a set of processes are each blocked waiting for a resource held by another process in the same set, so no process can make progress.

Definition — No preemption (as a condition)

The rule that a resource already allocated to a process cannot be taken away by force; it must be released voluntarily. Its opposite, ***preemption****, is used to* recover *from a deadlock.*

60.2 Deadlock recovery by resource preemption

Once a deadlock has been detected, the operating system must break it. One recovery method is **resource preemption**: the system takes a resource away from one of the processes and gives it to another process that needs it, breaking the circular wait. The chosen process is usually rolled back to a safe earlier state, or restarted. Preemption is also used to break the *no-preemption* condition directly, since it removes the guarantee that a resource stays with its holder.

Example 60.1 (The exam contrast). *An item asks for a method of* recovering *from deadlock. The correct answer is resource preemption — the system forcibly reclaims a resource. A distractor may name "mutual exclusion" or "circular wait", but those are necessary conditions for a deadlock, not recovery methods. The other trap is to think preemption is one of the four conditions; the condition is actually* ***no preemption****. (PYQ-10)*

60.3 Processes and the scheduler

A **process** is a program in execution. When a program is launched, the operating system creates a process for it and keeps the information needed to run it — its state, its program counter, and the resources it holds. Because many processes are usually ready at once but only one can run on a given CPU core at a time, the operating system includes a **scheduler** that decides which ready process gets the CPU next.

Scheduling may be **non-preemptive**, where a running process keeps the CPU until it finishes or blocks, or **preemptive**, where the operating system can interrupt a running process and switch to another. Priorities are one of the most common ways to choose between processes, and it is the

combination of *priorities* and *preemption* that produces the problems studied in the next sections: priority inversion and starvation.

Definition — Process

A program in execution, together with the state and resources the operating system maintains for it.

Definition — Scheduler

The operating system component that selects which ready process is given the CPU next.

60.4 Priority inversion versus priority inheritance

In a priority-based system, a task with a higher priority is supposed to run before one with a lower priority. **Priority inversion** breaks that rule. It happens when a **high-priority** task is blocked because a **low-priority** task currently holds a resource the high-priority task needs. While the low-priority task holds that resource, a **medium-priority** task can preempt it and run. The result is that the high-priority task ends up waiting behind the medium-priority task, even though it should have run first — its priority has effectively been inverted.

The standard fix is **priority inheritance**. While the low-priority task holds the resource, it temporarily **inherits** the high priority of the task that is waiting for it, so it runs at that high priority, releases the resource quickly, and then returns to its original low priority. This lets the high-priority task obtain the resource without waiting behind the medium-priority task.

Point	Priority inversion	Priority inheritance
What it is	A problem in which priority order is reversed.	A solution that temporarily raises the holder's priority.
Cause	A low-priority task holds a needed resource.	Applied while a resource is held and awaited.
Effect	High-priority task waits behind a medium-priority task.	Low-priority holder finishes fast, so the high-priority task proceeds.

Definition — Priority inversion

A condition in which a high-priority task is indirectly preempted by a lower-priority task, typically because a low-priority task holds a shared resource.

Example 60.1 (What does *not* fix inversion)**.** *An item asks which method prevents priority inversion. **Round-robin** scheduling gives every ready process a turn but knows nothing about resource holding, so by itself it does not prevent inversion. The correct method is priority inheritance (or the related priority-ceiling protocol). (PYQ-11)*

60.5 Starvation and aging

Starvation (indefinite blocking) is different from deadlock. A starved process is *ready* to run and could run in principle, but it is **repeatedly passed over** by the scheduler, so it may wait for an unbounded time. A common case is a low-priority process in a strict priority system: as long as higher-priority processes keep arriving, the low-priority process never gets the CPU.

The standard remedy for starvation is **aging**: the system gradually **raises the priority** of a process the longer it waits, so that eventually it becomes the highest-priority ready process and runs.

Definition — Starvation

A scheduling condition in which a ready process is indefinitely denied the resources it needs because other processes are repeatedly preferred.

Point	Starvation	Deadlock
Can the process ever run?	Yes, in principle	No, not without outside action
Cause	Scheduling choices (long wait)	A circular resource dependency
Cure	Aging (raise priority with waiting time)	Detection + recovery (for example preemption)

60.6 Preemptive priority scheduling

In **priority scheduling**, the scheduler always picks the ready process with the **highest priority**. The scheme is **preemptive** when a process that is currently running can be **interrupted** as soon as a process with a higher priority becomes ready; the higher-priority process then runs. It is **non-preemptive** when the running process keeps the CPU until it either finishes or blocks for input/output. Preemptive priority scheduling gives quick service to important tasks, but with strict priorities it can starve low-priority tasks unless aging is added.

60.7 Real-time operating systems (RTOS)

An **RTOS** is a **Real-Time Operating System**. It is designed so that the system responds within a **bounded time**, allowing **deadlines** to be met. Correctness in a real-time system depends not only on the *result* but also on *when* the result is produced.

Definition — Real-time operating system (RTOS)

An operating system designed to process data and events within guaranteed time limits, so that deadline requirements are satisfied.

Real-time systems are classified by what happens when a deadline is missed. In a **hard real-time** system a missed deadline is a failure and can have serious or catastrophic consequences. In a **soft real-time** system a missed deadline only degrades the quality of the result; the result is still useful. The trap is to think an RTOS is simply "a very fast operating system". Speed helps, but the defining property is **meeting deadlines**.

60.8 The Linux kernel

The **kernel** is the **core of the operating system**. It is the part that runs with full control of the hardware and provides the basic services every other program relies on. In Linux, the kernel manages **processes** (through the scheduler), **memory**, **devices** (through device drivers), and **file systems**.

The kernel runs in a privileged mode called **kernel space**, while ordinary application programs run in **user space**. A user program asks the kernel for a service — reading a file, for example — through a **system call**; the kernel performs the operation and returns the result. The Linux kernel is **open-source**: its source code can be viewed, modified, and shared.

Definition — Kernel

The central component of an operating system that manages the hardware resources (processes, memory, devices, and file systems) and mediates between applications and the hardware.

The **shell** is frequently confused with the kernel. The shell is the *command interpreter* that a user types commands into; it is not the kernel. The shell passes the user's requests to the kernel, which does the actual work.

60.9 /root versus /home

Linux stores the personal files of each user in that user's **home directory**. The **superuser**, whose account name is **root**, has its home directory at `/root`. Ordinary users' home directories are normally placed under `/home`, for example `/home/alice`.

It is important not to confuse two different things with similar names:

Path	Meaning
`/`	The **root of the whole file system** — the single top directory from which all other directories branch.
`/root`	The **home directory of the root (superuser)** account.
`/home`	The parent directory holding the home directories of **ordinary users**.

The trap is to read "`/root`" as "the root of the file system". The top of the file system is `/` (a single slash). `/root` is only the superuser's own home folder.

60.10 File systems

A **file system** is the method an operating system uses to organise, store, and retrieve files and folders on a storage device. Different systems use different file systems.

File system	Full form	Typical use
NTFS	New Technology File System	Default for modern Windows installations.
FAT32	File Allocation Table, 32-bit	Older system; widely compatible with small drives and memory cards.
ext4	Fourth Extended file system	Native and common default file system for Linux.

The exam cue is the pairing: **NTFS = Windows**, **ext4 = Linux**, and **FAT32 = broad compatibility**. Full forms are commonly asked directly.

60.11 Windows utilities: Store, WordPad/Notepad, Disk Cleanup

The **Windows Store** (now the **Microsoft Store**) is the built-in **online marketplace** in Windows. Users **search, download, install, and update** apps from it, so applications come from a trusted catalogue rather than from arbitrary websites.

Windows includes two simple text editors that are easy to confuse.

Point	Notepad	WordPad
Text handling	**Plain text** only; no formatting	**Basic formatting**: bold, italic, fonts, colour
Typical save	`.txt`	`.rtf` (rich text format)
Capability	The smallest, fastest text editor	Can insert pictures and objects

Notepad is a plain-text editor: it displays and saves characters without any formatting. **WordPad** supports *basic* rich-text formatting and can include pictures, but it is still lighter than a full word processor such as MS Word.

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Disk Cleanup is a built-in Windows utility that **frees disk space**. It deletes files that are not needed, such as temporary files, thumbnail caches, downloaded program files, and the contents of the **Recycle Bin**. It is the tool to reach for when a drive is running short of free space.

60.12 Defragmenter and the SSD caution

On a traditional **hard disk drive (HDD)**, a file can become **fragmented**: its pieces are scattered in different places on the rotating disk, which makes reading slower. The **Disk Defragmenter** rearranges the pieces so that each file is stored **contiguously** (in one run), which improves access speed.

SSD caution. A **solid-state drive (SSD)** has **no moving parts**. Defragmenting it gives no speed benefit and only adds **unnecessary writes**, which reduce the drive's working life. SSDs should not be defragmented; Windows uses the **TRIM** command for SSDs instead.

Definition — Defragmentation

The process of rearranging the fragments of files on a hard disk so that each file occupies contiguous space, improving read performance.

60.13 Imaging Fax and the `xcopy /s` command

Windows Fax and Scan is the built-in **imaging and fax** utility in Windows. With a scanner it can **scan** a document and save it as an image; with a fax modem or fax service it can **send and receive faxes**. This is the tool the exam means by "Imaging Fax".

xcopy is a Windows **command-line** command that copies files and whole directory trees. Its switch `/s` copies the named directory *including its subdirectories*, but it does not copy empty ones. Adding `/e` copies empty subdirectories as well. The exact form to remember is `xcopy /s`: "copy with subdirectories".

Command / tool	What it does
`xcopy /s`	Copies files and subdirectories (empty ones excluded).
`xcopy /s /e`	Also copies empty subdirectories.
Windows Fax and Scan	Scans documents and sends/receives faxes (Imaging Fax).

60.14 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. Deadlock's four conditions are mutual exclusion, hold and wait, **no preemption**, and circular wait. **Preemption** is a **recovery** method, not a condition (PYQ-10).
2. **Priority inversion** is the problem; **priority inheritance** is the cure. Round-robin by itself does not prevent inversion (PYQ-11).
3. **Starvation** is indefinite *scheduling* delay, not deadlock; **aging** fixes starvation.
4. An **RTOS** is defined by **meeting deadlines** (hard vs soft), not merely by being fast.
5. The Linux **kernel** is the core; the **shell** is the command interpreter. `/root` is the superuser's home; `/` is the top of the file system; `/home` holds ordinary users' homes.
6. **Notepad** = plain text; **WordPad** = basic rich text. Disk Cleanup frees space; **never defragment an SSD** (use TRIM); `xcopy /s` copies subdirectories.

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Example 60.1 (Tracing a mixed item). *An item lists statements and asks which are true: (I) Preemption is one of the four necessary conditions for deadlock. (II) Priority inheritance reduces priority inversion. (III) Aging helps prevent starvation. Statement I is false: preemption is a recovery method; the condition is* no *preemption. Statements II and III are true. The answer is "II and III only".*

Exercises

1. List the four necessary conditions for deadlock, and name one method of recovering from a deadlock.
2. Explain priority inversion and state which technique resolves it.
3. Distinguish starvation from deadlock, and name the standard remedy for starvation.
4. What is an RTOS? Give the difference between a hard and a soft real-time system.
5. State the meaning of `/root` and `/home` in Linux, and name the default file system of Windows and of Linux.
6. Explain why an SSD should not be defragmented, and state what `xcopy /s` does.

Solutions

1. Mutual exclusion, hold and wait, no preemption, and circular wait. A deadlock can be recovered from by resource preemption (taking a resource away from one process and giving it to another).
2. Priority inversion is when a high-priority task waits because a low-priority task holds a needed resource, while a medium-priority task runs. **Priority inheritance** resolves it: the low-priority holder temporarily inherits the high priority and releases the resource quickly.
3. A starved process is ready but repeatedly passed over, so it may wait indefinitely; a deadlocked process can never proceed because of a circular resource dependency. **Aging** — gradually raising the waiting process's priority — prevents starvation.
4. An RTOS (Real-Time Operating System) guarantees a response within a bounded time so deadlines are met. In a **hard** real-time system a missed deadline is a failure; in a **soft** real-time system a missed deadline only degrades quality.
5. `/root` is the home directory of the superuser (root); `/home` holds the home directories of ordinary users. Windows uses **NTFS**; Linux uses **ext4**.
6. An SSD has no moving parts, so defragmenting gives no speed benefit and only adds wear; SSDs use **TRIM** instead. `xcopy /s` copies a directory including its subdirectories, excluding empty ones.

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UNIT 6

Programming & Data Structures

LESSON 51

Programming Foundations, Compiler and Data Structures

51.1 Programming languages and their levels

A computer can act only on instructions given in its own **machine language**, a pattern of binary 0s and 1s. Writing whole programs in that form is slow, error-prone, and tied to one processor. Programmers therefore write in languages that are easier for humans, and a translator converts the resulting program into machine language. Languages are placed on a **level** scale: low-level languages (machine and assembly) are close to the hardware, while high-level languages are closer to human language. The lower the level, the more closely the code reflects the actual machine instructions.

Definition — Level of a programming language

How close a language is to the underlying machine. Machine and assembly languages are low-level; C, C++, Java, and Python are high-level.

Language	Generation	Main feature
Machine language	First	Binary 0s and 1s; only language the CPU runs directly
Assembly language	Second	Short mnemonics such as ADD and MOV; machine-dependent
High-level language	Third	English-like and mathematical; machine-independent

51.2 Machine language

Machine language is the only language a processor executes directly. It is written entirely in binary, and it is *machine-dependent*: each processor family has its own instruction set, so a program written for one machine does not run on another. Machine language needs no translation, so it is fast to run, but it is very difficult to write, read, and debug by hand. It is also called the *first-generation language*.

51.3 Assembly language

Assembly language replaces binary instruction codes with short *mnemonics* such as ADD, SUB, and MOV. It is still *machine-dependent*, because each processor family has its own assembly language, and it maps almost one-to-one onto machine instructions. Assembly code cannot run directly; it must be translated into machine language by an **assembler**. Assembly language is called the *second-generation language* and is a low-level language.

A common exam trap is to call assembly language a high-level language. It is not: assembly language is a low-level, second-generation language because its statements correspond closely to machine instructions. High-level languages, by contrast, are machine-independent and use English-like statements.

51.4 High-level languages

High-level languages use English-like words and mathematical notation. Examples are C, C++, Java, Python, COBOL, and FORTRAN. They are *machine-independent*: after translation, the same source program can run on different machines. They are easier to learn, write, read, and maintain than low-level languages, but they must be translated by a **compiler** or an **interpreter** before the CPU can run them.

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51.5 Generations of programming languages

Languages are also grouped into generations, which run parallel to the level scale above. The first three generations match the three levels exactly.

Generation	Example	Main feature
First (1GL)	Machine language	Binary 0s and 1s; run directly
Second (2GL)	Assembly language	Mnemonics; needs an assembler
Third (3GL)	C, C++, Java	High-level; needs a compiler or interpreter
Fourth (4GL)	SQL	Non-procedural; states what, not how
Fifth (5GL)	Logic / AI languages	Problem-solving and constraints

For this exam, the high-yield rows are the first three: machine language is first generation, assembly language is second generation, and a high-level language such as C or Java is third generation.

51.6 Language translators: assembler, compiler, interpreter

A **translator** is a program that converts source code written in one language into another form, usually machine language. There are three common translators, each matched to a different input language and method.

Definition — Assembler

A translator that converts an assembly-language *program into machine code.*

Definition — Compiler

A translator that converts a high-level *program into machine code (object code) as a whole, translating the entire program before it runs.*

Definition — Interpreter

A translator that converts and executes a high-level program line by line*, without producing a separate object file.*

Point	Compiler	Interpreter
Input	Whole program at once	One statement at a time
Output	Object code file	No separate object file
Execution speed	Faster	Slower
Errors	Reported after full translation	Reported line by line
Debugging	Harder	Easier
Typical examples	C, C++	Python, JavaScript

The compiler translates the whole program in one pass and produces object code, so execution is fast, but a single error can delay the whole build and debugging is harder. The interpreter takes one statement at a time, translates it, and executes it immediately; execution is slower, but errors are shown line by line, which makes debugging easier.

51.7 Source code, object code, and machine code

These three terms describe a program at different stages of translation.

Definition — Source code

The program as written by the programmer in a high-level or assembly language; it is human-readable text.

Definition — Object code

The machine-readable, relocatable code produced by a compiler or assembler from source code; it is not yet linked into a final runnable program.

Definition — Machine code

The final binary instructions, in the processor's own language, that the CPU executes directly.

The path is: the programmer writes **source code**; the translator converts it into **object code**; after linking, the result is **machine code** that the processor runs. Source code is human-readable; object code and machine code are machine-readable.

A common exam trap is to ask which of these is written by the programmer. The answer is **source code**. Object code and machine code are produced by the translator, not written by hand, and they are stored as binary patterns rather than readable text.

How a program is translated

The translation of a high-level program follows a fixed sequence:

1. The programmer writes the program as **source code**.
2. A **compiler** translates the whole source file and produces an **object file**.
3. A **linker** joins the object file with any library code needed.
4. The result is a **machine-code** executable that the CPU runs.

An interpreter shortens this path: it takes each statement of the source program, translates it, and executes it immediately, so no separate object file is produced.

51.8 Algorithm

An **algorithm** is a finite, step-by-step procedure for solving a problem. It has a clear start and end, and every step is unambiguous so that anyone following it reaches the same result. An algorithm is language-independent: it can be written in plain English or in pseudocode, without committing to a particular programming language. A program is the algorithm expressed in a programming language.

Definition — Algorithm

A finite, ordered set of unambiguous steps that solves a problem, with a defined start and end.

Example 51.1. *Adding two numbers is a small algorithm: (1) start, (2) read values a and b, (3) compute $sum = a + b$, (4) display sum, (5) stop. The same steps can later be written in any programming language.*

51.9 Flowchart

A **flowchart** is a pictorial, diagrammatic representation of an algorithm. It uses standard symbols joined by arrows to show the order in which steps are carried out, which makes the flow of control easy to see at a glance.

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Symbol	Meaning
Oval (terminator)	Start and end of the process
Parallelogram	Input or output
Rectangle	Process (a computation or action)
Diamond (rhombus)	Decision (yes/no branch)
Arrow	Flow line showing direction

A flowchart for the add-two-numbers algorithm uses these symbols in order: an oval for start, a parallelogram to read the two inputs, a rectangle to compute the sum, a parallelogram to display the result, and an oval for stop. When a step involves a choice, a diamond is used, and two arrows leave it, one for each outcome.

51.10 Debugging and types of errors

Debugging is the process of finding and correcting errors, called *bugs*, in a program. Three kinds of error are commonly distinguished.

Definition — Syntax error

A violation of the rules of the language, such as a missing semicolon or a misspelled keyword. It is reported by the compiler.

Definition — Logical error

An error in the logic of the program: it compiles and runs but produces a wrong result. It is the hardest kind to find.

Definition — Runtime error

An error that occurs while the program is running, such as dividing by zero or reading a file that does not exist.

A compiler catches syntax errors but not logical errors. A program with a logical error compiles and runs, and only the output reveals that something is wrong.

51.11 Object-oriented programming basics

OOP stands for **Object-Oriented Programming**. It organises a program around objects rather than around a sequence of actions. Two central terms are the class and the object.

Definition — Class and object

A class *is a blueprint or template that defines the data and methods; an* object *is a single instance created from that class.*

Four features define the object-oriented approach.

- **Encapsulation**: binding the data and the methods that operate on it into one unit, with the data hidden from outside access.
- **Inheritance**: a new class acquires the properties and methods of an existing class.
- **Polymorphism**: "one name, many forms" — the same operation behaves differently for different objects.
- **Abstraction**: hiding the implementation details and showing only the essential features.

As an illustration, *Student* can be a class that defines the data (roll number, name, marks) and the methods (enrol, show result). A particular student such as one with roll number 12 is then an *object* of that class. The class describes the shape; each object is one concrete example of it.

51.12 Stack and LIFO

A **stack** is a linear data structure in which insertion and deletion take place at only one end, called the *top*. It follows the order **LIFO**: Last In, First Out. The last element pushed onto the stack is the first element popped from it.

Definition — Stack (LIFO)

A linear data structure that works on the Last In, First Out rule, with insertion and deletion performed at the top.

Insertion into a stack is called **PUSH** and deletion is called **POP**. The other end of the stack is the *bottom*, and no insertion or deletion happens there.

Example 51.1. *A stack of files on a desk behaves in exactly this way. The file placed on top last is the first one picked up next. That is the LIFO rule in everyday form.*

51.13 Basic data structures

A data structure is a way of organising data so that it can be used efficiently. The common structures tested at this level are the array, stack, queue, linked list, and tree.

Structure	Defining property
Array	Fixed-size elements of the same type, stored in contiguous memory and accessed by an index
Stack	Last In, First Out (LIFO); push and pop take place at the top
Queue	First In, First Out (FIFO); insertion at the rear, deletion at the front
Linked list	Nodes that hold data and a pointer to the next node; not contiguous and can grow dynamically
Tree	A hierarchical structure with a root node and child nodes

To identify a structure in an exam item, look for its defining rule: same type, fixed size, and index access means an **array**; Last In, First Out means a **stack**; First In, First Out (like people in a queue) means a **queue**; data-plus-pointer nodes that grow dynamically mean a **linked list**; and a root with branches means a **tree**.

The two most commonly confused structures are the stack and the queue. They differ only in the order in which elements leave the structure.

Point	Stack	Queue
Order	LIFO (Last In, First Out)	FIFO (First In, First Out)
Insertion	Push, at the top	Enqueue, at the rear
Deletion	Pop, at the top	Dequeue, at the front
Both ends used?	No, one end only	Yes, front and rear

51.14 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. A **compiler** translates the **whole program at once**; an **interpreter** works **line by line**.

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2. An **assembler** translates **assembly language**; a **compiler** and an **interpreter** translate **high-level** languages.
3. **Source code** is human-readable; **object code** and **machine code** are machine-readable.
4. An **algorithm** is a step-by-step procedure; a **flowchart** is its pictorial representation.
5. A **stack** is **LIFO**; a **queue** is **FIFO**. Do not swap them.
6. A **logical error** is not caught by the compiler; the program runs but the result is wrong.
7. **OOP** stands for Object-Oriented Programming; a class is a blueprint and an object is an instance of it.

Example 51.1. *An item asks which translator produces a separate object file before execution. Trace the definitions: the compiler translates the whole high-level program and produces object code, so the answer is the compiler. If the stem instead says "translates and executes line by line", the answer is the interpreter, and if it says "translates assembly language", the answer is the assembler.*

Exercises

1. Distinguish machine language, assembly language, and a high-level language by level and by dependence on the machine.
2. State the difference between an assembler, a compiler, and an interpreter.
3. Differentiate source code, object code, and machine code.
4. Define an algorithm and explain how a flowchart represents one.
5. Distinguish a syntax error, a logical error, and a runtime error.
6. Explain LIFO and name the two operations of a stack.
7. Identify the data structure: (a) elements of the same type in contiguous memory, (b) Last In First Out, (c) a root with child nodes.

Solutions

1. Machine language is the first-generation, binary, machine-dependent language the CPU runs directly. Assembly language is the second generation, uses mnemonics, and is also machine-dependent. A high-level language is third generation, uses English-like statements, and is machine-independent.
2. An assembler translates assembly language into machine code. A compiler translates a high-level program as a whole and produces object code. An interpreter translates and executes a high-level program line by line.
3. Source code is the program written by the programmer in a high-level or assembly language and is human-readable. Object code is the machine-readable code produced by the translator and is not yet linked. Machine code is the final binary code the CPU executes directly.
4. An algorithm is a finite, step-by-step, unambiguous procedure to solve a problem. A flowchart represents it pictorially using standard symbols (oval, parallelogram, rectangle, diamond, arrow) joined by flow lines.
5. A syntax error breaks the rules of the language and is reported by the compiler. A logical error lets the program run but gives a wrong result. A runtime error occurs while the program is running.
6. LIFO means Last In, First Out: the last element pushed is the first element popped. The two operations are push (insertion) and pop (deletion), both performed at the top.
7. (a) Array, (b) Stack, (c) Tree.

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UNIT 7

Office Applications & File Formats

LESSON 22

MS Word Fundamentals

22.1 What MS Word is

MS Word (Microsoft Word) is a **word processor**: a program used to create, edit, format, store, and print text documents. It is one of the main applications in the **Microsoft Office** suite. A document written in Word is saved as a file, and the modern default file format uses the **.docx** extension. Word shows the document in a rectangular window that contains the document area together with several command areas around it.

Definition — Word processor

An application that lets a user type text, change its appearance (formatting), arrange it on pages, and print it. MS Word is the common example.

A word processor separates the *text* from its *appearance*. The same words can be displayed in many different fonts, sizes, colours, and layouts, because the formatting is stored alongside the text and can be changed at any time. This separation is the central idea of the whole lesson.

Part of the window	What it is
Document area	The white page where the text is typed and shown.
Ribbon	The strip of command tabs across the top of the window.
Quick Access Toolbar	The small row of always-visible buttons at the top-left.
Status bar	The strip at the bottom showing page, word count, and view buttons.

22.2 The Ribbon and its tabs

The **Ribbon** is the broad band of commands across the top of the Word window. It replaced the older style of drop-down menus and toolbars. The Ribbon is organised in three levels:

1. **Tabs** across the top: **Home, Insert, Design, Layout, References, Mailings, Review,** and **View**.
2. **Groups** inside each tab. For example, the Home tab contains the **Font**, **Paragraph**, and **Styles** groups.
3. **Commands** inside each group, such as the Bold button or the alignment buttons.

Some tabs are **contextual**. They are not shown all the time; they appear only when the right object is selected. When a table is selected, the **Table Tools** tabs appear; when a picture is selected, the **Picture Format** tab appears. The **File** tab is special: it does not show groups like the other tabs but opens **Backstage view**, which holds document-level actions such as New, Open, Save, Save As, Print, and Close.

Definition — Ribbon

The command area of the Word window, organised into tabs, groups, and buttons, which replaced the older menu-and-toolbar layout.

For exam purposes, the key point is that the **Home** tab carries the most-used formatting commands: the *Font* tools that change how characters look, and the *Paragraph* tools that change alignment, indentation, spacing, bullets, and numbering.

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22.3 The Quick Access Toolbar

The **Quick Access Toolbar (QAT)** is the small row of buttons in the top-left corner of the window, just above the Ribbon. It is designed to hold the few commands a user needs most often, so they stay visible on every tab. By default it shows **Save**, **Undo**, and **Redo**. A user can **customise** it by adding or removing commands, and it can be moved to sit below the Ribbon if preferred.

Definition — Quick Access Toolbar

A small, customisable toolbar at the top-left of the Word window that keeps frequently used commands (by default Save, Undo, Redo) visible on every tab.

22.4 Creating, opening and saving documents

The **File** tab opens **Backstage view**, the part of Word where you work with the document as a whole rather than its contents. From here a user can create a new document, open an existing one, save, print, or close. The common keyboard shortcuts are worth memorising because direct-recall items ask for them.

Shortcut	Action
Ctrl+N	Create a new blank document.
Ctrl+O	Open an existing document.
Ctrl+S	Save the current document.
Ctrl+P	Open the Print screen.

Save writes the current document back to its file. **Save As** writes a *copy* under a new name, in a new location, or in a different file format, leaving the original as it was. The default Word format is **.docx**; documents made with much older versions use the **.doc** extension.

Example 22.1. *A student types a letter and presses Ctrl+S, saving it as* `letter.docx`*. To keep the original and also produce a version for printing, the student uses Save As and saves a second file with a new name. The first file is unchanged.*

22.5 Entering and editing text

Text is typed at the **insertion point**, which is shown by a blinking cursor. **Word wrap** moves text automatically to the next line when the current line is full; pressing **Enter** starts a new **paragraph**. The **Backspace** key deletes the character to the left of the cursor, while the **Delete** key deletes the character to the right. **Ctrl+Z** undoes the last action and **Ctrl+Y** redoes it.

Text must be selected before it can be reformatted. A **word** is selected by double-clicking it, a **paragraph** by triple-clicking, and the **whole document** with **Ctrl+A**.

22.6 Font formatting

The **Font** group on the Home tab changes how individual characters look. It sets the **font face** (the typeface, such as Arial or Times New Roman) and the **font size**. It provides the style buttons **Bold (Ctrl+B)**, **Italic (Ctrl+I)**, and **Underline (Ctrl+U)**, and it sets the **font colour** and **highlight**. It also applies effects such as subscript and superscript. Deeper options are available in the Font dialog box.

Definition — Font

The typeface and its settings (face, size, style, colour) that determine how characters are displayed. Font formatting changes the appearance of characters, not their arrangement on the page.

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22.7 Paragraph alignment

Alignment decides where each line of a paragraph sits between the left and right margins.

Alignment	Shortcut	Effect
Left	Ctrl+L	Text lines up evenly on the left; ragged on the right. This is the default.
Centre	Ctrl+E	Each line is placed in the middle of the page.
Right	Ctrl+R	Text lines up evenly on the right.
Justify	Ctrl+J	Text is flush with both left and right margins.

Justify stretches the spaces between words so that both edges of the paragraph are even, which is why books and newspapers use it. **Centre** does not stretch anything; it simply moves each line to the middle.

22.8 Indentation

Indentation pushes paragraph lines inward, away from the margin. The **left indent** and **right indent** move the whole paragraph inward from the respective side. A **first-line indent** pushes only the first line inward, as in a typical book paragraph. A **hanging indent** pushes every line *except* the first inward, which is the standard layout for a references or bibliography list.

Definition — Indentation

The inward offset of paragraph lines from the page margins. First-line indent affects only the first line; hanging indent affects every line except the first.

Indentation must not be confused with **spacing**. Indentation shifts lines *horizontally*; spacing changes the *vertical* gaps.

22.9 Line and paragraph spacing

Line spacing sets the vertical gap between lines *inside* a paragraph; common settings are single, 1.5, and double. **Paragraph spacing** sets the gap *before* and *after* a paragraph. Both are controlled from the Paragraph group on the Home tab, or from the Paragraph dialog box. Using spacing is better than pressing Enter repeatedly to make gaps, because spacing stays correct when text is edited.

22.10 Bullets and numbering

Bullets mark list items with a symbol and are used when the order of the items does not matter. **Numbering** marks items with numbers or letters and is used when the order or sequence matters. **Multilevel lists** combine levels (for example 1, then 1.1, then 1.1.1) to build nested outlines. All three are applied from the **Paragraph** group on the Home tab.

Example 22.1. *A shopping list uses **bullets** because the order is irrelevant. A set of instructions to install software uses **numbering** because the steps must be followed in order.*

22.11 Tables

A **table** arranges information in **rows** and **columns**; each intersection of a row and a column is a **cell**. A table is inserted from **Insert** → **Table**. Cells can be **merged** into a single larger cell or **split** back into separate cells, and rows and columns can be inserted, deleted, and resized. When the table is selected, the **Table Tools** contextual tabs appear.

Definition — Table

A grid of rows and columns used to arrange text in cells, inserted in Word from Insert → Table.

22.12 Images and objects

An image is placed in the document from **Insert** → **Pictures**. **Handles** around the image let the user resize and rotate it. **Text wrapping** decides how the surrounding text flows around the image; options include In Line with Text, Square, Tight, and Behind Text. When the image is selected, the **Picture Format** contextual tab appears. Charts, shapes, and **SmartArt** are inserted from the same Insert tab.

Definition — Text wrapping

The setting that controls how text flows around a picture or object. "In Line with Text" treats the image like a character; the other styles let text flow around the image.

22.13 Headers and footers

A **header** is the area at the *top* of a page and a **footer** is the area at the *bottom*. Both repeat on every page of a section, which makes them ideal for a document title (in the header) and **page numbers** (in the footer). They are inserted from **Insert** → **Header** and **Insert** → **Footer**.

22.14 Page setup

Page setup is found on the **Layout** tab. It controls the overall page geometry: **margins** (the blank space between the text and the page edges), **orientation** (**Portrait**, which is tall, or **Landscape**, which is wide), and **page size** (for example A4 or Letter). The same tab sets **columns** and the page **breaks** described next.

22.15 Page break vs section break

A **Page Break** (**Ctrl+Enter**) forces the following content to start on a *new page*, but the page formatting stays the same. A **Section Break** starts a new **section** of the document, which can carry its own page setup. The most important consequence for the exam is that only a **Section Break** allows a part of the document to have **different headers and footers**. Both are inserted from **Layout** → **Breaks**.

Example 22.1. *An item claims, "A Page Break can be used to give the first page a different header from the rest of the document." This is wrong. A Page Break only moves content to a new page; to change the header or footer part way through the document, a **Section Break** is required. The wrong option has confused the two breaks (TRAP-13).*

22.16 Printing a document

Ctrl+P opens the **Print** screen in Backstage view. The **print preview** shows exactly how the pages will appear on paper. On this screen the user selects the **printer**, the number of **copies**, and which **pages** to print, then clicks **Print** to send the job to the printer. Checking the preview first catches wrong margins, stray blank pages, and cut-off text before paper is used.

22.17 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. **Save** overwrites the current file; **Save As** stores a copy under a new name, location, or format.
2. **Ctrl+S** saves; **Ctrl+P** prints; **Ctrl+Enter** inserts a page break. Do not swap the shortcuts.
3. **Bullets** are for unordered lists; **numbering** is for ordered lists.
4. **Indentation** shifts lines horizontally; **spacing** changes vertical gaps between lines and paragraphs.
5. A **Page Break** does *not* change headers and footers; only a **Section Break** does (TRAP-13).

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Example 22.1. *An item asks which shortcut opens the Print screen. Trace the File-level actions: New is Ctrl+N, Open is Ctrl+O, Save is Ctrl+S, and Print is Ctrl+P. The answer is* ***Ctrl+P****. If the stem instead asks for Undo, the answer is Ctrl+Z, and if it asks for a page break, the answer is Ctrl+Enter.*

Exercises

1. Name the main parts of the MS Word window.
2. What is the Quick Access Toolbar, and what does it show by default?
3. Distinguish Save from Save As.
4. List the four paragraph alignments with their shortcuts.
5. Explain the difference between a Page Break and a Section Break.

Solutions

1. The document area, the Ribbon (with its tabs and groups), the Quick Access Toolbar, and the status bar.
2. The Quick Access Toolbar is the small customisable row of buttons at the top-left of the window. By default it shows Save, Undo, and Redo, and it stays visible on every tab.
3. Save writes the current document back to its existing file. Save As writes a copy under a new name, location, or format and leaves the original file unchanged.
4. Left (Ctrl+L), Centre (Ctrl+E), Right (Ctrl+R), and Justify (Ctrl+J).
5. A Page Break (Ctrl+Enter) starts the following content on a new page while keeping the same formatting. A Section Break starts a new section that can have its own page setup — including different headers and footers. Only a Section Break can change headers and footers part way through a document.

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LESSON 23

MS Word High-Yield

23.1 Why these Word tools are high-yield

MS Word is a large program, but JKSSB does not test it evenly. A small set of review and editing tools is asked again and again, usually as an exact shortcut or as a near-neighbour distinction. This lesson covers that set: Find and Replace, spell check, hyperlinks and bookmarks, styles, comments, Track Changes, Page Break versus Section Break, mail merge, and saving, printing and exporting to PDF. The aim is not to learn every menu; it is to know each tool's shortcut and the one thing it is *not*.

Definition — The high-yield set

The Word features that competitive-exam items concentrate on: Find/Replace, spell check and grammar tools, hyperlinks and bookmarks, styles, comments, Track Changes, Section Break, mail merge, and PDF export.

Term	Meaning in this lesson
Find/Replace	The tool to locate text and swap it for other text.
Spell check	Word's proofing tool for spelling and grammar.
Bookmark	A named location or selection inside a document.
Style	A named set of formatting applied to text or a paragraph.
Track Changes	A mode that records every edit with its author.
Section Break	A break that starts a new section with its own layout.
Mail merge	Combining a standard letter with a list of records.
PDF	Portable Document Format — a fixed-layout file.

23.2 Finding and replacing text

Word has two related tools. **Find** locates text in the document and is opened with **Ctrl+F**. **Find and Replace** both locates text and substitutes new text for it, and is opened with **Ctrl+H**. Inside the Find and Replace dialog, *Replace* changes the current match while *Replace All* changes every match in one step. The same dialog can find formatting, special characters (such as a paragraph mark) and text that uses a particular style.

The common confusion is between the two shortcuts. Ctrl+F only finds; it does not change anything. Ctrl+H is the one that replaces. An item that asks how to "find and replace" text expects Ctrl+H.

Example 23.1. *A student has typed the word "recieve" in ten places and wants "receive" everywhere. Press Ctrl+H, type* `recieve` *in the Find box and* `receive` *in the Replace box, then choose Replace All. Using Ctrl+F alone would only highlight the mistakes one by one, without correcting them.*

23.3 Spelling and grammar tools

Press **F7** to start the **spelling and grammar** check. Word flags a possible misspelling with a red squiggle and a possible grammar problem with another underline. Modern Word gathers these checks in the **Editor** pane, which groups Spelling, Grammar and Refinements. From the check you can *Ignore* a flagged word once, *Ignore All* every occurrence, or *Add* the word to the dictionary so it is accepted from then on.

Two automatic tools support proofing. **AutoCorrect** fixes common typing errors as you type (for example, capitalising the first letter of a sentence), and the **thesaurus** (Shift+F7) offers synonyms.

The exam usually tests only the main shortcut, F7. Do not confuse F7 with the PowerPoint slide-show keys: F5 starts a slide show and F9 has other roles in PowerPoint, but in Word F7 is proofing.

Definition — Spell check

The Word tool, opened with F7, that checks spelling and grammar and offers corrections, ignoring or adding words to the dictionary.

23.4 Hyperlinks and bookmarks

A **hyperlink** is text or an object that jumps to another place when clicked. Press **Ctrl+K** to insert one. A hyperlink can point to a web address, to a file, or to a location inside the same document. When it points inside the document, it usually points to a **bookmark**.

A **bookmark** gives a location or a selection a name, so that text, a hyperlink or a cross-reference can jump to it. Bookmarks are not printed and are not visible to the reader by default; they are an internal target. The clean distinction for the exam is that a hyperlink is the *clickable link*, while a bookmark is the *named target* it points to.

23.5 Styles

A **style** is a named collection of formatting settings: font, size, colour, spacing, alignment and indent. Instead of formatting each paragraph by hand, you apply a style. Applying a **Heading** style does more than change the look: it gives the paragraph an outline level. That is why the **Navigation Pane** can list the document's headings and why a **Table of Contents** can be built and refreshed automatically.

Style-based formatting is consistent and easy to update. If you edit a style, every paragraph that uses that style changes at once. This is the main advantage of styles over direct formatting.

Example 23.1. *A report uses the Heading 1 style for its chapter titles. Later the author wants all chapter titles in a larger font. Instead of selecting every title, the author edits the Heading 1 style once, and every chapter title updates automatically. The Navigation Pane and the table of contents also stay correct.*

23.6 Comments and Track Changes

A **comment** is a note attached to a point in the document. It lets a reviewer ask a question or suggest a change *without editing the text itself*. Comments can be replied to and marked resolved, and they are stored as metadata shown in the margin or the Reviewing Pane.

Track Changes is different: it records the actual edits. When it is on, each insertion, deletion and formatting change is marked, together with the **author's name** and the time. This is called revision metadata. The reviewer then **Accepts** or **Rejects** each change individually or all at once. Accepting a change keeps it permanently and, with it, removes the revision metadata for that change. Rejecting a change removes the edit and keeps the original text.

Definition — Track Changes

A Word mode that records every insertion, deletion and formatting edit along with the author and time, so a reviewer can accept or reject each change.

23.7 Page Break versus Section Break

A **Page Break** pushes the following text onto a new page. It does not start a new section, so the page keeps the same layout, margins, headers, footers and page numbering as the rest of the document. A quick way to insert a page break is **Ctrl+Enter**.

A **Section Break** divides the document into independent **sections**. Because each section has its own layout settings, a Section Break is what allows *different headers and footers*, different page

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numbering, different margins or a different page orientation within one document. Section Breaks are inserted from the **Layout** tab, under **Breaks**.

This is one of the most tested Word traps. A Page Break changes only the page; a Section Break changes the section. Only a Section Break can give a document two different headers or footers. (TRAP-13)

Example 23.1. *An item states, "Using Page Break allows you to have different headers and footers in the same document." This is wrong. A Page Break keeps the same section formatting. It is a* ***Section Break*** *that allows different headers and footers in one document. (PYQ-13)*

23.8 Mail merge

Mail merge is the process that combines a **main document** (a standard letter or template) with a **data source** (a list of records such as names and addresses). The main document contains **merge fields** as placeholders, and each record fills those fields to produce one personalised copy. A common data source is a table of names and addresses held in Excel or in Outlook contacts.

The typical steps are: create or select the main document; select recipients (the data source); insert merge fields; preview the results; and finish the merge by printing the letters or sending them by email. The exam definition to remember is that mail merge *combines letters with a database of addresses*. (PYQ-37, PYQ-47)

Definition — Mail merge

The Word process that combines a standard document with a data source of records, producing one personalised copy of the document for each record.

23.9 Saving, printing and exporting to PDF

Press **Ctrl+S** to save the document; Word normally saves it in the **DOCX** format. Press **Ctrl+P** to open the print options. **Save As** can change the file name, the location and the file format. Using **Export** or **Save As PDF** creates a **PDF** file.

PDF stands for **Portable Document Format**. A PDF is a *fixed-layout* file: it looks the same on every device and is good for sharing and printing, but it is not meant for easy editing. This pairs with the separate lesson on file formats, where the extension (for example, `.pdf`) and the format (PDF) are kept distinct.

23.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. **Ctrl+F** finds; **Ctrl+H** finds and replaces. Do not swap them.
2. **F7** is spelling and grammar in Word; F5 starts a PowerPoint slide show. (TRAP-32)
3. A **Section Break**, not a Page Break, enables different headers/footers in one document. (TRAP-13)
4. A **bookmark** is a named target; a **hyperlink** (**Ctrl+K**) is the clickable link.
5. **Mail Merge** combines a letter with an address database; it is not Track Changes or AutoCorrect. (TRAP-29)

Example 23.1. *An item asks which feature combines letters with a database of addresses. Trace the meaning: a standard letter plus a list of records is mail merge. If the item instead asks what records every edit with its author, the answer is Track Changes, and if it asks what fixes typing errors as you type, the answer is AutoCorrect. (PYQ-37, PYQ-47)*

Exercises

1. State the shortcuts for Find, Find and Replace, spelling and grammar, and inserting a hyperlink.
2. Explain the difference between a Page Break and a Section Break, and say which one allows different headers and footers.
3. Distinguish a bookmark from a hyperlink.
4. What does Track Changes record, and what happens when a change is accepted?
5. Define mail merge and name its two parts.

Solutions

1. Find: Ctrl+F. Find and Replace: Ctrl+H. Spelling and grammar: F7. Insert a hyperlink: Ctrl+K.
2. A Page Break moves text to a new page within the same section. A Section Break starts a new section with its own layout. Only a Section Break allows different headers and footers.
3. A bookmark is a named location or selection inside the document; it is a target and is not printed. A hyperlink is clickable text or an object that jumps to a target, which may be a bookmark, a file or a web address.
4. Track Changes records every insertion, deletion and formatting edit, together with the author's name and the time (revision metadata). Accepting a change keeps it permanently and removes that change's revision metadata.
5. Mail merge combines a main document (the standard letter or template) with a data source (a list of records such as names and addresses) to produce one personalised copy per record.

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LESSON 24

Excel Structure

24.1 What Excel is and what a workbook is

Microsoft Excel is a spreadsheet program. A spreadsheet is a grid of boxes in which you can store text, numbers, dates, and formulas, and in which the program can calculate results automatically. Excel is one of the standard office applications, and its saved document is called a **workbook**.

A **workbook** is the Excel *file* — the single item that you open, edit, and save with a file name. When you start Excel and see a grid, you are looking at one **worksheet** that belongs to a workbook. A workbook can contain many worksheets, and each worksheet is a separate page of the same file.

Definition — Workbook

The Excel file itself. A single workbook can hold one or more worksheets and is saved under one file name.

Definition — Worksheet

A single sheet of rows and columns inside a workbook. It is also called a sheet, and it has a tab at the bottom of the Excel window.

At the bottom of the Excel window you will see one or more **sheet tabs** (for example, Sheet1, Sheet2, Sheet3). A new classic workbook opens with three worksheets, and you can add or remove sheets. Clicking a tab brings that worksheet to the front. The key point for the exam is the relationship: the workbook is the container (the file), and the worksheet is the content (one page inside it).

24.2 Workbook vs worksheet

These two words are a classic near-neighbour pair, so keep the contrast sharp. The workbook is the file; the worksheet is a part of the file. One workbook can contain many worksheets, but a worksheet never contains a workbook. A worksheet is not a separate saved file — it lives inside the workbook and is saved as part of it.

Point	Workbook	Worksheet
What it is	The Excel file	One sheet inside a workbook
How many	One file	Many per workbook
Saved as	A single file name	Part of that file
Tab	—	A tab at the bottom

Example 24.1. *An item states, "A worksheet is the Excel file, and a workbook is a part of it." This is wrong. The relation is the other way round: the workbook is the file, and the worksheet is one page inside it. The wrong option has swapped the two terms.*

24.3 The worksheet grid: rows and columns

Every worksheet is a large grid made of **rows** and **columns**. Rows run horizontally across the sheet and are identified by **numbers**, which appear down the left edge (1, 2, 3, ...). Columns run vertically down the sheet and are identified by **letters**, which appear across the top (A, B, C, ...).

The column labels do not stop at Z. After A–Z, the labels continue as two-letter names: AA, AB, ..., AZ, then BA, BB, ..., BZ, and so on. This scheme continues with three-letter names until the final column. In Excel 2007 and later, a worksheet has:

Property	Value in Excel 2007 and later
Total rows	1,048,576 (numbered 1 to 1,048,576)
Total columns	16,384 (labelled A to XFD)
First column	A
Last column	XFD

These two figures are worth memorising because they are common direct-recall items. In older Excel versions (Excel 97–2003, the `.xls` format) the limits were much smaller: 65,536 rows and 256 columns, with the last column labelled IV. Modern Excel uses the larger grid ending at column XFD.

24.4 The cell and the active cell

A **cell** is the small rectangular box formed where a row and a column meet. It is the basic unit of a worksheet, and it is where you actually type your data. The whole sheet is just a very large collection of cells.

At any moment, exactly one cell is the **active cell**. The active cell is the cell that is currently selected: it has a thick border around it, and anything you type is entered into it. The active cell's address is shown in the Name Box, and its contents are shown in the Formula Bar. Even when you select a group of cells, only one of them — the active cell — is the one the next typed entry goes into.

Definition — Cell

The rectangular box formed by the intersection of a row and a column, in which data is entered.

Definition — Active cell

The cell that is currently selected. It has a thick border, receives the next typed entry, and has its address shown in the Name Box.

24.5 Cell addresses and references

Every cell has a unique **cell address** (also called a cell reference). The address is written with the **column letter first**, followed by the **row number**. For example:

- **B5** means column B, row 5.
- **C7** means column C, row 7.
- **D12** means column D, row 12.

The order is fixed: letter then number. Writing 12D, B:5, or B_5 is not a valid Excel address. Because column labels can have more than one letter, an address such as AB100 is also valid: it means column AB, row 100.

24.6 Ranges

A **range** is a group of two or more cells treated as one block. It is written by typing the addresses of two diagonally opposite corners and separating them with a **colon** (:). For example, **A1:C10** means all the cells from column A to column C and from row 1 to row 10. A range is normally rectangular, and the cells in it are contiguous.

The number of cells in a range is the number of columns multiplied by the number of rows. For example, the range B2:D5 covers three columns (B, C, D) and four rows (2, 3, 4, 5), so it contains $3 \times 4 = 12$ cells.

Definition — Range

A rectangular group of two or more selected cells, written as two corner addresses joined by a colon, such as A1:C10.

Example 24.1. *A question lists* `B5` *and calls it a range. This is wrong:* `B5` *is a* single *cell, because it has only one address and no colon. A range must have two corner addresses joined by a colon, such as* `B5:D9`*. The colon is the signal that a range, not a single cell, is meant.*

24.7 The formula bar

The **Formula Bar** is the long white bar that sits below the Ribbon and above the worksheet grid. It always shows the *contents* of the active cell. If the active cell holds plain text or a number, the Formula Bar shows that text or number. If the active cell holds a formula, the Formula Bar shows the *formula itself* (for example `=A1+A2`), while the cell itself shows the calculated result.

You can also edit the active cell in the Formula Bar by clicking in it. This is useful when a formula is long. Remember the division of labour: the cell displays the result, and the Formula Bar displays the underlying formula or data.

Definition — Formula Bar

The bar below the Ribbon that shows, and allows editing of, the contents of the active cell — including the formula behind a calculated result.

24.8 The name box

The **Name Box** is the small box at the *left end of the Formula Bar*. It normally shows the **address of the active cell** (for example, B5). If a range of cells is selected, it shows the name or the address range of that selection (for example, A1:C10). You can also type an address into the Name Box and press Enter to jump directly to that cell.

Definition — Name Box

The box at the left of the Formula Bar that shows the address of the active cell or the name of the selected range.

Keep the two bars apart. The Name Box (on the left) answers "which cell am I in?" The Formula Bar (the long part) answers "what is in that cell?"

24.9 Basic data types

Excel accepts several basic kinds of entry in a cell:

- **Text** (also called a label): letters, words, names, and headings. Text is not used in arithmetic.
- **Number**: a numeric value that can be used in calculations.
- **Date and time**: entered and displayed as a date or time, but stored internally by Excel as a number.
- **Formula**: an entry that begins with the = sign and calculates a result from other cells.

Excel also recognises **logical values** (`TRUE` and `FALSE`) and displays **error values** such as `#VALUE!` when a calculation cannot be completed. By default, Excel aligns these data types in a helpful way:

Data type	Default alignment	Example
Text	Left	Roll No
Number	Right	4500
Date	Right	15-08-2026

A quick visual check therefore tells you what Excel thinks the entry is: a value sitting on the *left* is treated as text, and a value sitting on the *right* is treated as a number (or a date, which is stored as a number). This default alignment is a common exam point.

24.10 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A **workbook** is the file; a **worksheet** is one page inside it. Do not swap them.
2. A **cell** is one box; a **range** is two or more cells joined by a colon.
3. A cell address is **column letter first, then row number** (B5), never the reverse.
4. A worksheet in Excel 2007+ has **1,048,576 rows** and **16,384 columns**; the last column is **XFD**, not IV (IV was the old `.xls` limit).
5. The **Name Box** shows the address; the **Formula Bar** shows the contents.
6. Only **one cell is active** at a time, even when a range is selected.

Example 24.1. *An item asks, "What is the total number of rows in a modern Excel worksheet?" Recall the pair: rows are the larger count, 1,048,576, and columns are 16,384. If the stem instead asks for the last column, the answer is XFD; if it asks for the number of cells in a range, multiply columns by rows.*

Exercises

1. Distinguish a workbook from a worksheet.
2. What is a cell? How is a cell address written, and give one example.
3. How many rows and how many columns does a worksheet have in Excel 2007 and later, and what is the last column label?
4. What is a range? Write the range that covers columns A to C and rows 1 to 10, and state how many cells it contains.
5. State what the Name Box and the Formula Bar each display.
6. List four basic data types that can be entered in an Excel cell and give the default alignment of text and of numbers.

Solutions

1. A workbook is the Excel file itself, saved under one file name; a worksheet is a single sheet of rows and columns inside a workbook. A workbook can contain many worksheets.
2. A cell is the box formed where a row and a column intersect. A cell address is written as the column letter followed by the row number, for example B5 (column B, row 5).
3. A worksheet has 1,048,576 rows and 16,384 columns; the last column is labelled XFD.
4. A range is a rectangular group of two or more cells written as two corner addresses joined by a colon. The range is A1:C10, and it contains $3 \times 10 = 30$ cells.

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5. The Name Box shows the address of the active cell (or the name of the selected range). The Formula Bar shows the contents of the active cell, including the formula behind a calculated result.
6. Four basic data types are text, number, date/time, and formula. Text is aligned to the left by default, and numbers to the right by default.

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LESSON 25

Excel Formulas

25.1 What a formula is

A **formula** is an instruction that tells Microsoft Excel how to calculate the value of a cell. Instead of holding a fixed number, the cell holds a rule, and Excel works out the result of that rule from the other cells, from constants, or from functions the rule refers to. The formula is typed into the formula bar; the result is shown in the cell. When any cell that the formula uses changes, Excel recalculates and shows the new result automatically.

Definition — Formula

An instruction, written in a cell or the formula bar, that calculates a value from operands such as cell references, constants, and functions. The cell stores the formula, while the sheet displays the result.

This lesson covers the four building blocks of a formula: the leading `=`, the operators, the cell references, and the way references behave when a formula is copied. It closes with the standard error codes and the distinctions that examiners test most often.

25.2 The = rule and formula syntax

Every Excel formula must begin with the equal sign `=`. This is the single fact most often tested on this topic. Typing `=A1+A2` into a cell is treated as a formula and calculated; typing `A1+A2` without the leading `=` is treated as ordinary text and shown exactly as written.

The same `=` symbol also serves as a comparison operator that tests whether two values are equal. The position tells you which role it plays: at the very start of an entry it begins a formula, while inside a formula it compares two operands.

Definition — The = rule

A formula must start with `=`. A cell entry that does not start with `=` is not a formula.

A simple formula has three parts: the leading `=`, the operands, and the operators. In the formula `=A1*B1+100`, the operands are the references `A1` and `B1` and the constant `100`, while the operators are `*` (multiply) and `+` (add).

25.3 Formula, function, and constant

It helps to separate three ideas that often appear together. A **constant** is a fixed value written directly into a formula, such as the number `100` or a piece of text in quotation marks. A **function** is a built-in calculation with a name, such as `SUM` or `AVERAGE`, which is called from inside a formula. A **formula** is the whole instruction, and it may combine constants, references, operators, and functions in one expression.

Definition — Function

A named, built-in calculation such as `SUM(A1:A10)` or `AVERAGE(B1:B5)` that is called from within a formula and returns a single value.

The distinction matters for exams because a question may ask what a formula contains rather than what it produces. The formula `=A1+A2` contains two relative references and one arithmetic operator. The formula `=SUM(A1:A10)` calls a function on a range. Both begin with `=`, and both return a value to the cell that holds them. The rest of this lesson concentrates on the references and operators that every formula uses.

25.4 Operators

An operator tells Excel what to do with the values on either side of it. Excel has four families of operators: arithmetic, comparison, text, and reference.

Arithmetic operators produce numbers.

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponent (raise to a power)
%	Percent

Two symbols are easy to get wrong. Multiplication uses the asterisk *, not the letter x or a dot; division uses the forward slash /. The caret ^ raises a number to a power, so =2^3 is 8. The percent sign % divides by 100, so =50% is 0.5.

Comparison operators compare two values and return TRUE or FALSE: = (equal to), > (greater than), < (less than), >= (greater than or equal to), <= (less than or equal to), and <> (not equal to). The operator <> is the Excel way of writing "not equal"; it is not written !=.

A comparison gives a useful result even on its own: if A1 holds 50, the formula =A1>100 shows FALSE. The operators that candidates confuse most are =, which tests equality when it appears inside a formula, and <>, which tests inequality. Note that the leading = that starts every formula and the = comparison operator are the same character used in two different roles.

Text operator. The ampersand & joins, or concatenates, two pieces of text into one. ="Total: "&A1 shows the words Total: followed by the contents of cell A1.

Reference operators build references to groups of cells. The colon : is the range operator, so A1:A10 means every cell from A1 down to A10. The comma , is the union operator, which joins two ranges into one list. A single space between two references is the intersection operator, which returns the cells common to both ranges.

The four families can be summarised in one place. Arithmetic operators (+ - * / ^%) produce numbers. Comparison operators (= > < >= <= <>) produce TRUE or FALSE. The text operator & produces a joined piece of text. The reference operators :, ,, and the space build ranges and lists of cells. Knowing which family an operator belongs to tells you what kind of result to expect.

25.5 Order of evaluation (precedence)

When a formula has several operators, Excel does not simply work from left to right. It follows a fixed order of precedence. Parentheses are evaluated first. Then the percent sign, then the exponent, then multiplication and division, then addition and subtraction. After the arithmetic comes the concatenation operator &, and finally the comparison operators.

Because multiplication and division are evaluated before addition and subtraction, =2+3*4 is 14, not 20. The multiplication 3*4 is done first to give 12, and then 2 is added. In the same way =2+3*4 differs from =(2+3)*4, which is 20. Parentheses are the reliable way to control the order of evaluation.

Example 25.1. *In =10+2*5 the multiplication is done first, so the result is 20. If the intention was to add first, the formula must be written =(10+2)*5, which gives 60. The exponent binds even tighter than multiplication, so =2*3^2 is 18, because 3^2 is evaluated before the multiplication by 2.*

25.6 Cell references

A **cell reference** names a cell by its column letter and row number. In the default A1 reference style, A1 is column A, row 1, and B5 is column B, row 5. A reference lets a formula use the current value of another cell, so the formula stays correct when that value changes.

The crucial property of a reference is not what it points at now but how it behaves when the formula is *copied*. There are three kinds of reference: relative, absolute, and mixed.

25.7 Relative references

A reference such as `A1` is **relative**. Excel stores it, in effect, as an offset from the cell that contains the formula. When the formula is copied to another cell, the reference shifts by the same distance, so it keeps pointing at the cell that lies in the same relative position.

Example 25.1. *If `=A1` is entered in cell B1 and then copied to B2, the formula becomes `=A2`. The reference moved down one row because the copied formula moved down one row. Copy the same formula from B1 to C1, and it becomes `=B1`: the reference moved one column to the right because the formula did.*

Relative references are the default and are what you want for a column of repeated calculations, such as adding each row's two numbers.

25.8 Absolute references

A reference such as `$A$1` is **absolute**. The dollar sign locks the part of the reference that follows it: the first `$` locks the column, and the second `$` locks the row. An absolute reference does not change when the formula is copied; it always points at the same cell.

Example 25.1. *If `=$A$1` is entered in B1 and copied to B5, it remains `=$A$1`. Neither the column nor the row moved. A fixed value such as a tax rate or a constant kept in one cell should be referenced absolutely, so that copying the formula does not disturb it.*

25.9 Mixed references

A **mixed** reference locks only one part. In `$A1` the column is locked and the row is relative; copying the formula to another row changes the row number but never the column. In `A$1` the row is locked and the column is relative; copying the formula to another column changes the column letter but never the row.

A useful way to remember this is that the `$` locks whatever immediately follows it. A `$` before a letter locks the column, and a `$` before a number locks the row.

25.10 Copying and filling formulas

The behaviour of a reference under copying is the heart of the topic. When a formula is copied, only the unlocked parts of each reference shift; the locked parts stay put. The table below compares the four reference types when a formula is copied one column right and one row down, from cell B1 to cell C2.

Reference	What is locked	After copying B1 to C2
`A1`	nothing	`B2`
`$A$1`	column and row	`$A$1`
`$A1`	column	`$A2`
`A$1`	row	`B$1`

Reading the table: the relative reference `A1` moved on both axes; the absolute reference `$A$1` moved on neither; `$A1` kept its column but followed the row; and `A$1` kept its row but followed the column.

A second, very common direction is copying a formula straight down a column. If `=A1*B1` sits in C1 and is copied down to C2, the formula becomes `=A2*B2`: both relative references followed the copy down one row. If the value in B1 must stay fixed, the formula should instead read `=A1*$B$1`, so that only A1 follows the copy while `$B$1` stays anchored.

The same four references behave as follows when copied one column to the right, and when copied one row down.

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Reference	Copied right (B1→C1)	Copied down (B1→B2)
A1	B1	A2
$A$1	$A$1	$A$1
$A1	$A1	$A2
A$1	B$1	A$1

Reading either column, the rule is the same: a reference moves only along the axis that is not locked. The relative reference moves on both axes, the absolute reference on neither, and each mixed reference on exactly one.

25.11 Worked examples

Example 25.1. *A price list has prices in column A and a fixed tax rate stored in cell `$B$1`. The formula in cell C2 is `=A2*$B$1`. When it is copied down, C3 becomes `=A3*$B$1` and C4 becomes `=A4*$B$1`: the price reference moves down with each row, while the rate reference stays locked on `$B$1`. If the formula had been written `=A2*B1`, copying it down would have moved the rate to B2 and then B3, giving wrong results.*

Example 25.2. *In cell B1 the formula is `=$A1`. It is copied to C2. The column lock holds A, and the relative row follows the copy down one row, so C2 contains `=$A2`. If instead the original formula were `=A$1`, copying from B1 to C2 would keep the row 1 and move the column one step from A to B, giving `=B$1`.*

Example 25.3. *A sheet lists marks in column B. To flag each student as passing, the formula `=B2>=40` is entered in C2 and copied down. It returns `TRUE` for a mark of 40 or more and `FALSE` otherwise. Because B2 is relative, the reference follows the copy down the column, so C3 tests B3 and C4 tests B4. The comparison operator produces the `TRUE`/`FALSE` result, not a number.*

Example 25.4. *A score sheet holds marks in B2:B11 and the class total in cell `$B$12`. To show each student's share as a percentage, the formula in C2 is `=B2/$B$12*100`. The relative `B2` follows the copy down the column, while `$B$12` stays anchored on the total. The operators here are division and multiplication, evaluated from left to right at the same precedence, and the result is a plain number. If the total were written as a relative reference, copying the formula down would corrupt the denominator.*

25.12 Common formula errors

When a formula cannot be evaluated, Excel shows an error value in the cell. The codes tested in exams are:

Error	Meaning
`#DIV/0!`	Division by zero, or by an empty cell.
`#VALUE!`	The wrong type of operand, such as text where a number is needed.
`#REF!`	A referenced cell was deleted, so the reference is no longer valid.
`#NAME?`	An unknown function, or a misspelled name or text entry.
`#N/A`	The value is not available, as when a lookup finds no match.
`#NULL!`	The two ranges given do not intersect.

Remember the two that examiners use most: `#DIV/0!` for division by zero and `#REF!` for a reference that has been broken by deleting a cell. A related warning is the **circular reference**, which appears when a formula refers, directly or through a chain of other formulas, back to its own cell, so Excel cannot work out a value. Deleting the referenced row or column is the usual cause of `#REF!`, while supplying the wrong type of operand, such as text where a number is expected, produces `#VALUE!`.

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25.13 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A formula must start with `=` (TRAP-30, PYQ-38). The options `+`, `%`, and `@` are distractors.
2. `$` before a letter locks the column; `$` before a number locks the row. Do not swap these.
3. **Relative** references change on copy; **absolute** references do not. A **mixed** reference changes on one axis only.
4. `#REF!` means a broken reference; `#DIV/0!` means division by zero. Do not swap the error codes.
5. `<>` is the Excel operator for "not equal"; multiplication is `*` and division is `/`.
6. A **function** such as `SUM` is called from inside a formula, and a **constant** is a fixed value written directly; both are operands, not operators.

Example 25.1. *An item asks what `=$A$1` becomes when copied from B1 to B5. Because both the column and the row are locked by the two dollar signs, the reference does not move: the answer is `=$A$1`. If the question instead gives `=$A1`, the column is locked but the row is relative, so the answer is `=$A5`. The dollar signs, not the row numbers, decide the answer.*

Exercises

1. Which character must every Excel formula begin with, and what happens if it is omitted?
2. List the arithmetic operators for addition, subtraction, multiplication, division, and exponent.
3. Explain the difference between a relative reference `A1` and an absolute reference `$A$1`.
4. In the mixed reference `$A1`, which part is locked and which part changes when the formula is copied?
5. The formula `=A1` is in cell B1. What does B2 contain after the formula is copied to it? What would B2 contain if the formula were `=$A$1`?
6. State the meaning of the errors `#DIV/0!` and `#REF!`.
7. What is the result of the formula `=2+3*4`, and why?
8. Name the Excel operator that joins two pieces of text, and the operator that means "not equal to".

Solutions

1. A formula must begin with `=`. Without it, the entry is stored and shown as ordinary text rather than calculated.
2. Addition `+`, subtraction `-`, multiplication `*`, division `/`, and exponent `^`.
3. `A1` is relative: on copying, both its column and its row shift by the same offset as the formula. `$A$1` is absolute: the dollar signs lock the column and the row, so it does not change on copying.
4. In `$A1` the column A is locked; the row is relative, so the row number changes when the formula is copied to a different row.
5. Copying `=A1` from B1 to B2 gives `=A2`, because the relative reference follows the copy down one row. Copying `=$A$1` to B2 leaves it as `=$A$1`, because both parts are locked.
6. `#DIV/0!` means division by zero or by an empty cell; `#REF!` means a reference is invalid because a referenced cell was deleted.
7. The result is 14. Multiplication is evaluated before addition, so `3*4` is 12, and `2+12` is 14.
8. The ampersand `&` joins (concatenates) text, and `<>` means "not equal to".

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LESSON 26

Excel Functions

26.1 From formulas to functions

MS Excel is built around the idea that a value can be *calculated* instead of typed. A cell that begins with the equal sign `=` holds a calculation, and Excel works out the result. Such a calculation is called a **formula**. A simple formula is `=A1+A2`, which adds the values of two cells. A formula can also use operators such as `+`, `-`, `*`, and `/` to add, subtract, multiply, and divide.

Typing every cell into a formula becomes slow and error-prone when there are many cells. Excel therefore provides **functions**: ready-made, named operations that do a common job for you. `SUM` adds a range of numbers, `AVERAGE` finds their mean, and `MAX` finds the largest. A function is always written *inside* a formula. The function name is followed by parentheses that contain the *arguments* — the cells or values the function should work on.

Definition — Formula

An expression that begins with the equal sign `=` and calculates a value, for example `=A1+A2` or `=SUM(A1:A10)`.

Definition — Function

A predefined, named built-in operation, such as `SUM`, `COUNT`, or `IF`, that is used inside a formula to perform a particular calculation on its arguments.

The distinction to hold is simple. A **formula** is the whole calculation. A **function** is a built-in tool that performs one part of that calculation. Every function is written as a formula, but not every formula contains a function: `=A1+A2` is a formula with no function in it, while `=SUM(A1:A10)` is a formula that uses one function. A formula must begin with `=`; if the equal sign is missing, Excel treats the entry as plain text and shows it as typed.

26.2 Function syntax and arguments

Every function follows the same shape: a name, then a pair of parentheses, and inside the parentheses the arguments separated by commas. This is written as `=FUNCTION(argument1, argument2)`. The **name** tells Excel which operation to perform. The **arguments** tell it which cells or values to use. For example, in `=SUM(A1:A10)` the name is `SUM` and the single argument is the range `A1:A10`.

A **range** is a block of cells written with a colon between the first and last cell. `A1:A10` means the cells A1, A2, A3, and so on down to A10. A function can take more than one argument: `=SUM(A1:A5, C1:C5)` adds the two ranges together. The comma separates one argument from the next. Knowing the name, the parentheses, and the colon is enough to read any of the basic functions in this lesson.

Function	Example	What it returns
`SUM`	`=SUM(A1:A10)`	The total of the numbers in the range.
`AVERAGE`	`=AVERAGE(A1:A10)`	The arithmetic mean of the numbers.
`COUNT`	`=COUNT(A1:A10)`	The number of cells holding numbers.
`COUNTA`	`=COUNTA(A1:A10)`	The number of non-empty cells.
`MAX`	`=MAX(A1:A10)`	The largest number in the range.
`MIN`	`=MIN(A1:A10)`	The smallest number in the range.
`IF`	`=IF(A1>50,"Y","N")`	One of two values, chosen by the test.

26.3 SUM and AVERAGE

SUM returns the total of the numbers in its arguments. Its most common form is `=SUM(A1:A10)`, which adds all the numbers in the range A1:A10. Cells that hold text and cells that are blank are ignored; only numbers are added. SUM accepts several ranges, so `=SUM(A1:A5,B1:B5)` adds both blocks in one step.

AVERAGE returns the arithmetic mean of the numbers in its arguments. `=AVERAGE(A1:A10)` adds the numbers and divides by how many numbers there are. The important detail is what counts in the divisor: blank cells and cells holding text are *not* counted. AVERAGE divides by the number of *numeric* cells, not by the total number of cells in the range. This is the source of many wrong answers.

Example 26.1. *Suppose A1:A5 holds the values 10, 20, and 30, while A4 and A5 are blank.* `=SUM(A1:A5)` *returns* 60. `=AVERAGE(A1:A5)` *returns* $(10 + 20 + 30)/3 = 20$, *not* $60/5 = 12$. *Excel divides by the three cells that actually hold numbers, because the two blanks are ignored.*

26.4 COUNT and COUNTA: the near-neighbour pair

COUNT and COUNTA look almost the same but count different things, and the difference is a favourite exam trap. `COUNT(counts)` only the cells in a range that contain *numbers*. COUNTA (the "A" stands for *all*) counts every cell that is *not empty*, whether it holds a number or text. Both functions ignore blank cells.

Definition — COUNT

A function that returns the number of cells in a range that contain numeric values. Text, blanks, and logical values are not counted.

Definition — COUNTA

A function that returns the number of cells in a range that are not empty. It counts numbers, text, and error values alike; only truly blank cells are ignored.

The contrast is easiest to see with a mixed range. Suppose A1:A10 holds six numbers, two text labels, and two blank cells. `=COUNT(A1:A10)` returns 6, because only the numeric cells are counted. `=COUNTA(A1:A10)` returns 8, because the six numbers and the two text labels are all non-empty. Both functions ignore the two blanks. A related function, COUNTBLANK, does the opposite and counts the empty cells, returning 2 for the same range. For the exam, the pair to memorise is: **COUNT = numbers only; COUNTA = all non-empty cells**.

26.5 MAX and MIN

MAX returns the largest number in a range, and MIN returns the smallest. Their forms are `=MAX(C1:C10)` and, for the smallest, `=MIN(C1:C10)`. Like SUM, AVERAGE, and COUNT, they work on numbers and ignore text and blank cells. A useful way to remember them is that MAX is the maximum and MIN is the minimum; do not confuse MIN with the word "minimum cell", which is not a function at all.

Example 26.1. *A range holds the marks 42, 67, 55, 91, and 38.* `=MAX(C1:C5)` *returns* 91, *the top mark.* `=MIN(C1:C5)` *returns* 38, *the lowest mark.* `=AVERAGE(C1:C5)` *returns* 58.6, *the mean of the five numbers.*

26.6 The IF function and nested IF

IF lets a cell show one of two results depending on whether a condition is true or false. Its syntax is `=IF(logical_test, value_if_true, value_if_false)`. The first argument is a *logical test* that can be answered only with true or false, such as `A1>50`. The second argument is the value or text to return when the test is true, and the third is the value to return when it is false. Thus `=IF(A1>50,"Pass","Fail")` shows `Pass` when A1 is more than 50 and `Fail` otherwise.

A **nested IF** places a second IF inside the value-if-false part of the first, so more than one condition can be tested. A typical form is `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))`. Excel checks the conditions in the order they are written, and the first test that is true decides the result.

Example 26.1. *Take the formula `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))` and let A1 be blank. In a numeric comparison Excel reads a blank cell as 0. Since 0 is neither greater than 50 nor greater than 25, both tests are false and the formula returns the final value, `Low`. This is the behaviour the Junior Assistant 2026 paper examined (KB PYQ-14, JA 2026 Q74).*

When the input is *non-numeric* text, the behaviour depends on what the formula does with it. A comparison of text against a number is still answered by Excel, but if the formula tries to *calculate* with text — for example, `=A1*2` where A1 holds a word — Excel cannot do the arithmetic and returns the error value `#VALUE!`. The safe exam reading is: a blank is treated as 0 in a numeric test, and text that is arithmetic-only produces `#VALUE!`.

26.7 A worked sheet, function by function

A small sheet makes the behaviour of every function concrete. Column A holds five student names — Anita, Bilal, Chandan, Divya, and Esha — and column B holds the marks 40, 55, 70, and 60, with Esha's mark (B5) left blank because she has not been marked yet. Reading the sheet function by function shows what each one does with the blank.

Formula	Result	Why
`=SUM(B1:B5)`	225	Adds 40, 55, 70, 60; the blank is ignored.
`=AVERAGE(B1:B5)`	56.25	Divides 225 by 4 numeric cells, not by 5.
`=COUNT(B1:B5)`	4	Counts the four cells that hold numbers.
`=COUNTA(B1:B5)`	4	Counts the four non-empty marks; B5 is blank.
`=MAX(B1:B5)`	70	The largest of the four marks.
`=MIN(B1:B5)`	40	The smallest of the four marks.
`=IF(B5>=40,"Pass","Fail")`	Fail	B5 is blank, read as 0, so the test fails.

The last row links the sheet back to the nested-IF behaviour: a blank cell is read as zero in a numeric test, so Esha is shown as not yet passing. Note also that `COUNTA` on the names column, `=COUNTA(A1:A5)`, would return 5, because all five names are present even though only four marks are. This is exactly how the two count functions are used together to spot missing data.

26.8 Other basic functions worth knowing

Beyond the core set, a few basic functions appear in Office-suite questions. `COUNTIF` counts the cells that meet one condition, and `SUMIF` adds the cells that meet one condition; both extend COUNT and SUM. On the statistics side, `MEDIAN` returns the middle value of the numbers, while `MAX` and `MIN` return the ends of the range. `ROUND` rounds a number to a stated number of digits. `LEN` counts the characters in a text value. `TODAY` returns the current date, and `NOW` returns the current date and time.

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Function	What it returns
COUNTIF	The number of cells meeting one condition.
SUMIF	The total of the cells meeting one condition.
MEDIAN	The middle value of the numbers.
ROUND	A number rounded to a set number of digits.
LEN	The number of characters in a text value.
TODAY	The current date.

At JKSSB depth, the exam is satisfied by knowing what each function returns and that every one of them is written inside a formula that starts with =.

26.9 AutoSum

AutoSum is a button, not a function, that quickly inserts a SUM formula. When the active cell is placed below or beside a column or row of numbers, AutoSum looks at the adjacent numbers and writes a =SUM(...) formula over the range it selects. Pressing the button therefore saves the user from typing the range by hand. The keyboard shortcut for AutoSum is Alt + =. Its drop-down arrow also offers Average, Count Numbers, Max, and Min, so the same button can insert any of the core functions, not only SUM.

Definition — AutoSum

An Excel button that automatically inserts a SUM formula for the range next to the active cell; the shortcut is Alt + =.

26.10 Combining functions in one formula

Functions can be nested inside one another, so the result of one function becomes the argument of another. For example, =ROUND(AVERAGE(A1:A10),1) first finds the mean of A1:A10 and then rounds it to one decimal place. In the same way, =IF(COUNT(B1:B5)=0,0,AVERAGE(B1:B5)) shows 0 when the range contains no numbers at all, instead of producing an error, and otherwise shows the mean of the numbers present. Nesting is read from the inside out: the innermost function is calculated first, and its result is passed outwards to the next function.

Example 26.1. *Consider =IF(COUNT(C1:C6)=0,"No data",AVERAGE(C1:C6)). If C1:C6 holds only text or blanks, COUNT returns 0, the test is true, and the cell shows No data. If the range holds numbers, the test is false and the cell shows their average. This pattern is common on real marks sheets, where a column may be empty early in the term.*

The arguments of a function can also be single cells, several separate cells, whole ranges, or even constant values such as =SUM(A1, 5, B3:B7). A reference such as A1 can be relative or absolute ($A$1); that choice does not change what the function does, only how the reference behaves when the formula is copied.

26.11 Error values

When a formula cannot be calculated, Excel displays an *error value* beginning with the hash sign #. Each error points to a different kind of problem, and recognising them is a common direct-recall question.

Error value	What it means
`#DIV/0!`	A number is divided by zero or by a blank cell.
`#VALUE!`	The wrong type of argument is used — for example, text where a number is needed.
`#NAME?`	Excel does not recognise a name in the formula, often a misspelt function name.
`#REF!`	A cell reference is not valid, usually because the referenced cell was deleted.
`#N/A`	The value is not available to the formula, as when a lookup finds no match.
`#NUM!`	A number in the formula is not valid for the operation.
`#NULL!`	A space (intersection) between two ranges that do not intersect.

Two clarifications prevent confusion. First, the string `#####` that sometimes appears in a cell is *not* an error value; it only means the column is too narrow to display the number, and it disappears when the column is widened. Second, `#DIV/0!` also appears when the divisor is a blank cell, because a blank is read as zero in the division. Reading each error as a clue to what Excel could not do is the fastest way to answer these items.

Example 26.1. *The error value points at the mistake that produced it. A formula `=A1/0`, or `=A1/B1` where B1 is blank, shows `#DIV/0!`. A formula `=A1+B1` where A1 holds text shows `#VALUE!`. A misspelt function such as `=SUMM(A1:A5)` shows `#NAME?`. Deleting a cell that a formula refers to turns the reference into `#REF!`. A lookup that finds no match leaves the result `#N/A`, and a mathematically invalid argument such as `=SQRT(-1)` gives `#NUM!`.*

26.12 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. A **formula** starts with `=`; a **function** is a named built-in operation used inside a formula. Without the equal sign, the entry is text (TRAP-30).
2. **COUNT** counts numbers only; **COUNTA** counts all non-empty cells. Do not swap this near-neighbour pair.
3. **AVERAGE** divides by the number of numeric cells, not by the total number of cells.
4. A **blank** cell is read as zero in a numeric test, so a blank fails a "greater than" condition; text used in arithmetic gives `#VALUE!` (KB PYQ-14).
5. **AutoSum** inserts `=SUM(...)` and its shortcut is `Alt + =`, not a Ctrl combination.

Example 26.1. *An item asks which function counts a range that mixes numbers and labels. Read the function name carefully: `COUNT` returns only the numeric cells, while `COUNTA` returns every non-empty cell. If the stem asks for the count of entries of any kind, the answer is `COUNTA`; if it asks for the count of numbers, the answer is `COUNT`. A second item may ask which error appears when a cell is divided by a blank; the blank counts as zero, so the error is `#DIV/0!`.*

Exercises

1. Explain the difference between a formula and a function in MS Excel.
2. A range A1:A8 contains 5 numbers, 2 text labels, and 1 blank cell. State the values returned by `=COUNT(A1:A8)` and `=COUNTA(A1:A8)`.

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3. Write the general syntax of the IF function and explain each of its three arguments.
4. A cell holds `=IF(A1>10,"Yes",IF(A1>0,"Small","No"))` and A1 is blank. What is shown, and why?
5. Match each error value with its cause: `#DIV/0!`, `#VALUE!`, `#NAME?`, `#REF!`.

Solutions

1. A formula is any calculation that begins with `=`, such as `=A1+A2`. A function is a predefined, named operation such as `SUM` or `IF` that is used inside a formula. Every function is written as a formula, but a formula need not contain a function.
2. `=COUNT(A1:A8)` returns 5, because only the numeric cells are counted. `=COUNTA(A1:A8)` returns 7, because the 5 numbers and the 2 text labels are all non-empty; both ignore the blank.
3. `=IF(logical_test, value_if_true, value_if_false)`. The first argument is a test answered true or false; the second is the value returned when it is true; the third is the value returned when it is false.
4. The formula shows `No`. A blank is read as zero in a numeric test, so A1 is neither greater than 10 nor greater than 0, both tests are false, and the final value `No` is returned.
5. `#DIV/0!` — division by zero or by a blank cell; `#VALUE!` — the wrong type of argument, such as text where a number is needed; `#NAME?` — an unrecognised name, often a misspelt function; `#REF!` — an invalid cell reference after the referred cell was deleted.

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LESSON 27

Excel Data Tools

27.1 What Excel data tools do

Once data has been entered into a worksheet, it usually has to be organised, narrowed, summarised, and presented. Excel groups the features that do this under what we can call its data tools: **sort**, **filter**, **tables**, **charts**, **PivotTables and PivotCharts**, **printing**, **protection**, and **what-if analysis**. These are the features a JKSSB candidate is expected to recognise by name and by one-line purpose.

Definition — Data tools

The worksheet features that arrange, narrow, summarise, present, print, and safeguard data without changing the underlying values.

The single idea that holds the whole lesson together is this: most of these tools change the *view* of the data. Sorting, filtering, and a PivotTable reorganise what you see on the screen; the original values in the cells stay exactly where they were. The exceptions are deliberate edits — a sort does physically move rows, and a user can type new data — but the summary tools themselves are read-only with respect to the source.

Tool	One-line purpose
Sort	Arrange rows in a chosen order.
Filter	Show only the rows that meet a condition.
Excel table	A named, auto-expanding, formatted range of data.
Chart	A picture of the data, linked to its source range.
PivotTable	A fast summary of a large table of data.
PivotChart	A chart built on a PivotTable.
Protection	Restrict editing of cells or workbook structure.
What-if analysis	Test how the result changes with different inputs.

27.2 Sorting

Sorting rearranges rows into an order chosen by the user. **Ascending** order goes from A to Z, or from the smallest number to the largest. **Descending** order goes from Z to A, or from the largest number to the smallest. A sort is performed on a *column*, but the whole row moves with that column so that each record stays intact.

The **Sort** dialog (**Data > Sort**) can order data by **more than one level**. For example, a payroll sheet can be sorted by Department first, and then by Salary within each department. The dialog's **Add Level** button adds these secondary and tertiary keys.

A **custom list** sorts by a defined order rather than alphabetically: days of the week and months are the standard examples. Excel can also sort by cell colour, font colour, or cell icon. When the range has titles, the option **My data has headers** must be ticked so that the header row is not sorted along with the data.

Example 27.1. *A teacher's marksheet has the columns Roll No., Name, and Marks. Clicking anywhere in the Marks column and choosing Sort Largest to Smallest reorders the rows so the highest marks appear first. If only the Marks column is selected and the rest of the row does not move, the records are scrambled — the names no longer match the marks.*

Trap watch. Sorting a **single column** on its own scrambles the records. Always sort the whole table, or convert the data to an Excel table first.

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27.3 Filtering

Filtering displays only the rows that satisfy a condition and hides the rest. The hidden rows are **not deleted**; they are temporarily out of sight. AutoFilter is switched on and off with **Data > Filter**, whose keyboard shortcut is **Ctrl+Shift+L**. Once it is on, drop-down arrows appear in the header row of each column.

Definition — Filter

A tool that shows only the rows matching a condition while hiding — not removing — the other rows.

Clicking a drop-down arrow allows two broad kinds of narrowing:

- a **value filter**, where one or more ticks are kept from a list of the column's values;
- a **condition filter**, where a rule is applied. Text Filters, Number Filters, and Date Filters offer common rules such as *Equals*, *Greater Than*, *Begins With*, and *Between*.

Top 10 and **Above Average** are ready-made condition filters. **Advanced Filter** is used for more complex criteria and can copy the matching rows to another location on the sheet.

Clearing the filter with **Clear Filter** restores every row. The key contrast to remember is that filtering *hides* rows; it does not remove them and does not change the data.

Example 27.1. *A sales sheet has a Region column. Applying a value filter and ticking only "North" hides all other regions. The rows for South, East, and West are still present in the worksheet; clearing the filter brings them back unchanged.*

27.4 Excel tables

An **Excel table** is a range of data that has been converted with **Insert > Table** (shortcut **Ctrl+T**) and given a table style. A table has a **name** (by default *Table1*), and it behaves as a single object rather than as loose cells.

Definition — Excel table

A named, formatted range of data that automatically expands to include new rows and columns and whose formulas use structured references.

The practical advantages of a table over a plain range are:

- it **auto-expands** when a new row is typed immediately below it;
- it shows **banded rows** that improve readability;
- it adds **filter buttons** to the header row automatically;
- it can display an optional **Total Row**; and
- its formulas use **structured references** such as **Table1[Sales]** instead of cell addresses such as **B2**.

Because a structured reference points at the whole column, it automatically covers new rows that are added to the table. A formula written with cell addresses would have to be extended by hand.

Point	Plain range	Excel table
Name	None	Assigned name (Table1)
Grows	Manually	Auto-expands
Formulas	Cell addresses	Structured references
Filters	Manual AutoFilter	Built-in header buttons

A table is *not* a PivotTable. A table is the **source data**; a PivotTable is one of the summaries that can be built on top of it.

27.5 Charts

A **chart** is a graphical picture of worksheet data. Because the chart is **linked** to its source range, editing the numbers in the cells updates the chart automatically. The common chart types and their uses are:

Chart type	Best used for
Column / bar	Comparing values across categories.
Line	Showing a trend, usually over time.
Pie / doughnut	Showing the parts of a single whole.
Scatter (XY)	Showing the relationship between two numeric variables.

The parts of a chart carry their own names, and exam items often test them. The **chart area** is the whole chart, including the frame. Inside it, the **plot area** is the region where the data is actually drawn. Each set of values is a **data series**, and the **legend** names each series. The **category axis** (usually the X-axis) carries the labels, while the **value axis** (the Y-axis) carries the numbers. **Data labels** display the exact value at each data point, and **gridlines** help read the values.

A chart normally floats on the same worksheet as its data, but it can be moved to a separate **chart sheet** that contains only the chart.

Example 27.1. *A school records the number of students in each class. A column chart compares the classes at a glance. The same data as a line chart would suggest a trend across the classes, which may not be meaningful; for parts of one total, such as the share of boys and girls in the school, a pie chart is the right choice.*

27.6 PivotTables and PivotCharts

A **PivotTable** summarises a large table of data in seconds. Instead of writing formulas, the user **drags** fields into four areas:

Area	What goes there
Rows	Field values listed down the left side of the summary.
Columns	Field values spread across the top of the summary.
Values	The numbers that are summarised (Sum, Count, Average, and more).
Filters	Fields used to filter the whole PivotTable.

The **Values** area chooses the aggregation: by default a numeric field is summed and a text field is counted, but Count, Average, Max, Min, and other functions can be chosen. Rearranging the field layout re-computes the summary instantly — this quick re-orientation is what gives the PivotTable its name.

Definition — PivotTable

An interactive summary that aggregates a large range of data by dragging fields into Rows, Columns, Values, and Filters areas.

A **PivotChart** is a chart built on a PivotTable and linked to it. When the PivotTable layout changes, the PivotChart changes with it.

The most important contrast is that a PivotTable **summarises** the source data; it does not replace it and does not change it. If the source data is edited, the PivotTable must be **refreshed** to

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show the new result. Deleting the source range breaks the PivotTable, and a PivotTable is never the same thing as an Excel table.

27.7 Printing worksheets

A worksheet is printed from **File > Print**, and its layout is controlled from the **Page Layout** tab. The key settings are:

- **Orientation** — **Portrait** for a tall page or **Landscape** for a wide one;
- **Paper size**, **margins**, and **headers/footers**;
- **Print Area** — select a range and choose **Set Print Area** to print only that part of the sheet;
- **Print Titles** — repeats chosen header rows or columns at the top or left of every printed page, so a long table keeps its headings; and
- **Scaling** — options such as "Fit Sheet on One Page" shrink the printout to a chosen width or number of pages.

Gridlines are the faint cell borders seen on screen. They are **not printed by default**; the **Print** gridlines option under Sheet Options must be enabled to show them on paper. **Page breaks** can be inserted to control where a new printed page begins.

Example 27.1. *A 200-row price list is being printed. Without Print Titles, only the first page shows the column headings and later pages are unreadable. Setting the heading row as a print title repeats it on every page. If the list is a little too wide, landscape orientation or "Fit All Columns on One Page" keeps it tidy.*

27.8 Protecting worksheets and workbooks

Protection restricts changes that can be made to a workbook. **Protect Sheet** (**Review > Protect Sheet**) prevents cells on that sheet from being edited, while allowing selected actions such as selecting locked cells or formatting. **Protect Workbook** protects the **structure** of the workbook — preventing sheets from being added, deleted, renamed, or moved — and can also protect its windows.

Definition — Protection

A restriction placed on a worksheet or workbook that prevents specified changes, optionally guarded by a password.

Cell locking is the detail most candidates miss. Every cell is **Locked** by default in the Format Cells dialog, but this lock has **no effect until the sheet is protected**. To let a user edit some cells and not others, the chosen cells must first be **unlocked**, and then the sheet must be protected.

Protection is not the same as encryption. Protection restricts editing inside Excel; it is not the password that encrypts the whole file (that is a separate file-level password).

27.9 What-if analysis

What-if analysis answers the question "what would the result be if an input changed?". Excel provides three tools for this at the depth a JKSSB candidate needs:

Tool	What it does
Goal Seek	Finds the input value needed to reach a target result.
Scenario Manager	Stores and compares several sets of input values.
Data Table	Shows how an output changes as one or two inputs change.

All three are found under **Data > What-If Analysis**. Goal Seek works backwards from a target: for example, it finds the selling price that makes the profit exactly a desired figure. Scenario Manager keeps named alternatives such as "Best case" and "Worst case". A Data Table is a grid that shows a range of results for one or two changing inputs.

27.10 Near-neighbour contrast: COUNTIFS vs SUMPRODUCT

To **count rows that satisfy several conditions at the same time**, Excel offers two tools that students often confuse.

COUNTIFS is the direct multi-criterion counting function. Its arguments are written as **paired** ranges and criteria:

```
COUNTIFS(range1, crit1, range2, crit2, …)
```

It applies **AND** logic: a row is counted only if it meets *every* range-and-criteria pair. For example, `COUNTIFS(A2:A100,"North",B2:B100,">500")` counts the rows that are in the North region *and* have a value greater than 500.

SUMPRODUCT is a general array function. It can also count multi-criterion rows by multiplying boolean results:

```
SUMPRODUCT((A2:A100="North")*(B2:B100>500))
```

Each TRUE/FALSE comparison becomes 1 or 0 when multiplied, and SUMPRODUCT adds the products, giving the same count. Because the multiplication already coerces the boolean values to numbers, SUMPRODUCT does not need the `--` (double unary) that is sometimes seen when boolean arrays are fed to other functions.

For **OR** logic — a row that meets either of two conditions — the two matches are *added*, for example `COUNTIF(range,"North")+COUNTIF(range,"South")`.

Example 27.1. *A marksheet has a Stream column and a Marks column. To count students who are in the Science stream* and *scored more than 80, write*

```
COUNTIFS(Stream,"Science",Marks,">80").
```

The same count can be obtained with

```
SUMPRODUCT((Stream="Science")*(Marks>80)).
```

Confusing the paired arguments of COUNTIFS, or thinking COUNTIFS cannot handle two conditions at once, is the classic error (TRAP-14, PYQ-15).

27.11 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **Sort** rearranges rows into an order; **filter** shows only matching rows and **hides** the rest — it does not delete them.
2. A **PivotTable** summarises the source data but is **not** the source table; it must be **refreshed** after the data changes.
3. A PivotTable is **not** an Excel table. The table is the source; the PivotTable is a summary built on it.
4. **COUNTIFS** counts with AND across **paired** range-and-criteria arguments; **SUMPRODUCT** does the same with boolean arrays (TRAP-14).
5. **Locked** cells are protected only **after** the sheet is protected; the default lock has no effect on its own.
6. **Gridlines** are **not printed** unless Print gridlines is enabled.

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Example 27.1. *An item states, "A PivotTable changes the source data and updates on its own." Both halves are wrong. A PivotTable never changes the source; and after the source changes it does not update by itself — the user must refresh it. The stable concept is: PivotTable = summary, source = untouched, refresh = manual.*

Exercises

1. Distinguish between sorting and filtering, and state what happens to the rows that a filter does not show.
2. Name two features that an Excel table provides but a plain range does not.
3. Match each chart to its best use: column, line, pie, scatter.
4. List the four areas of a PivotTable and say what goes into each.
5. Explain why a PivotTable must be refreshed after the source data changes.
6. What is the difference between Protect Sheet and Protect Workbook? When does a locked cell actually become protected?
7. Write the COUNTIFS formula that counts rows in the North region with a value greater than 500, and give the equivalent SUMPRODUCT formula.

Solutions

1. Sorting rearranges rows into a chosen order; filtering shows only the rows matching a condition. A filter *hides* the other rows temporarily; it does not delete them, and clearing the filter restores them.
2. An Excel table has a name and auto-expands when new rows are typed; it also provides structured references, banded rows, and built-in filter buttons. A plain range has none of these by default.
3. A column chart compares values across categories; a line chart shows a trend over time; a pie chart shows parts of one whole; a scatter chart shows the relationship between two numeric variables.
4. Rows — field values down the left; Columns — field values across the top; Values — the numbers that are summarised; Filters — fields used to filter the whole PivotTable.
5. A PivotTable stores only a summary view; it does not track the source automatically. After the source data changes, the user must Refresh the PivotTable so the summary is recalculated.
6. Protect Sheet stops cells on a sheet from being edited; Protect Workbook protects the workbook structure (adding, deleting, renaming, or moving sheets). A cell is Locked by default, but the lock takes effect only after the sheet is protected.
7. `COUNTIFS(A2:A100,"North",B2:B100,">500")`; the equivalent is `SUMPRODUCT((A2:A100="North")*(B2:B100>500))`.

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LESSON 28

Access Database Basics

28.1 What a database is

A **database** is an organised collection of related data, stored so that it can be searched, updated, and retrieved easily. The data is not kept as a loose pile of separate files; it is arranged in a planned structure. A simple example is a school's information: the list of students, their classes, and their marks all belong together as related information, and a database keeps them that way.

A useful comparison is a spreadsheet. A spreadsheet (such as Excel) is built for calculation on a simple grid. A database is built for storing and linking large amounts of related data, and it can hold many tables that are related to one another. When a record system grows large, a database keeps the data consistent and avoids unnecessary duplication.

Definition — Database

An organised collection of related data, stored so that it can be searched, updated, and retrieved easily.

28.2 Database vs DBMS

A database and a DBMS are not the same thing, and the difference is a common exam point. The **database** is the collection of data itself. A **DBMS** (Database Management System) is the software that manages that data. The database is data; the DBMS is a program.

Point	Database	DBMS
What it is	The organised collection of data	The software that manages the data
Nature	Data	Program
Example	A school's records	MS Access, Oracle, MySQL

Definition — DBMS (Database Management System)

The software that stores a database and lets users insert, update, delete, and retrieve data while controlling access and keeping the data consistent.

A **relational database management system** (RDBMS) is a DBMS that keeps data in related tables. Microsoft Access is an example of an RDBMS.

28.3 What a DBMS does

A DBMS performs several jobs. It *stores* the data and controls how it is kept on disk. It lets users *insert*, *update*, *delete*, and *retrieve* records. It controls *access*, so that only permitted users can read or change the data. It also keeps the data *consistent* — different parts of the database are not allowed to contradict each other — and protects it from corruption.

Common DBMS software includes MS Access, Oracle, MySQL, and SQL Server. For this course, the important point is that Access is a DBMS: it is the program that manages a database, not the data itself.

28.4 Microsoft Access in the Office suite

Microsoft Access is the database program of the Microsoft Office suite. In the suite, Word handles text documents, Excel handles spreadsheets, PowerPoint handles presentations, and Access handles databases. Access is a relational DBMS: it stores its data in tables, and it also holds

the queries, forms, and reports that work on that data. All of these objects are kept inside a single database file for each database.

Example 28.1. *A student is asked, "Which MS Office program is used to store and manage databases?" The answer is* ***Access****. Excel is a spreadsheet program, Word is a word processor, and PowerPoint is for presentations; none of them is a DBMS.*

28.5 The objects in an Access database

An Access database contains several kinds of object, each with one job.

Object	Job
Table	Stores the actual data in rows and columns
Query	Asks a question and retrieves selected data
Form	Provides a screen to enter and view data
Report	Presents data in a printable layout

The data itself always lives in **tables**. The other objects work on that data: a query retrieves it, a form is used to enter and view it, and a report presents it for printing.

28.6 Tables, records, and fields

A **table** is the basic object where data is stored. Inside a table, data is arranged in rows and columns. Each **column** is one **field** — one kind of information, such as Name or Marks. Each **row** is one **record** — one complete entry, such as the full details of one student. A table therefore holds many records, and every record has the same set of fields.

Term	Meaning	Position in a table
Field	One kind of information	A column
Record	One complete entry	A row
Table	A collection of related records	The whole grid

Definition — Record

One complete entry in a table, held in a single row and made up of one value for each field.

Definition — Field

One kind of information in a table, held in a single column; each field has a name and a data type.

Example 28.1. *In a Students table the columns are RollNo, Name, Marks, and Passed. One row — for example, "1, Ayaan, 78, Yes" — is one record. The Name column is the Name field. The whole set of rows together is the table.*

28.7 Data types in Access

Every field has a **data type**, which is the kind of value it can hold. Common Access data types are:

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Data type	Used for
Short Text	Names, codes, and other text
Number	Values used in calculations
Date/Time	Dates and times
Yes/No	Only True or False (a Boolean value)
AutoNumber	A unique number added automatically for each new record

Because a field accepts only values that match its data type, the data stays clean. A field set to Number will not accept ordinary text, and a Yes/No field accepts only a true-or-false value.

28.8 Field properties and validation

A **field property** is a setting that describes how a field behaves. Some important properties are **Field Size**, which limits how much a field can hold; **Default Value**, which fills in a value automatically; **Required**, which, when set to Yes, stops the field from being left blank; and **Input Mask**, which fixes the pattern of entry, such as the format of a phone number.

Validation means checking that entered data is sensible and allowed. A **Validation Rule** states the condition a value must meet. For example, the rule `Between 0 And 100` accepts only values in that range. A **Validation Text** is the message shown when the rule is violated.

Definition — Validation

The checking of entered data against a rule so that only sensible, allowed values are stored in a field.

Example 28.1. *On a Marks field, the Validation Rule `Between 0 And 100` is set. A user types 150. Access rejects the entry because it fails the rule, and shows the Validation Text as a message. A valid value such as 78 is accepted normally.*

28.9 Primary key and index

A **primary key** is a field that uniquely identifies each record. Its value is different for every record and is never left blank. In the Students table, RollNo can serve as the primary key because no two students share a roll number. An AutoNumber field is often used as a ready-made primary key, because Access fills it with a fresh unique number for every new record.

An **index** is a special structure that makes searching and sorting a table faster. It works like the index of a book: instead of reading every page, it points straight to the right place. A primary key is automatically indexed. An index takes extra space and can slightly slow down the work of adding data, so it is created only where it is genuinely useful.

Definition — Primary key

A field, or set of fields, whose value uniquely identifies every record in a table; it cannot contain duplicates and cannot be left blank.

28.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. A **database** is the data; a **DBMS** is the software that manages it. Do not swap them.
2. A **field** is a column; a **record** is a row; a **table** is the whole collection of related records.
3. **Validation** checks the value entered; an **index** speeds up searching. They do different jobs.
4. A **primary key** is unique and cannot be blank; it cannot contain duplicate values.

5. Microsoft **Access** is a DBMS and part of the MS Office suite; it is not a spreadsheet.

Example 28.1. *An item asks which object stores the actual data in an Access database. Trace the objects: a query retrieves data, a form is for entry, and a report is for printing, but only the* ***table*** *stores the data. If the stem instead asks which field uniquely identifies a record, the answer is the* ***primary key****; if it asks which field setting stops a blank entry, the answer is* ***Required****.*

Exercises

1. Define a database and explain how it differs from a DBMS.
2. Distinguish a field, a record, and a table, stating the position of each in a table.
3. Name any four Access data types and give one use of each.
4. Explain the difference between a validation rule and an index.
5. What is a primary key, and why can it not contain duplicate values?

Solutions

1. A database is an organised collection of related data. A DBMS is the software that manages that data — it stores it, allows insert, update, delete, and retrieve operations, and controls access. The database is the data; the DBMS is the program.
2. A field is one kind of information and is a column in a table. A record is one complete entry and is a row in a table. A table is the whole collection of related records, arranged in rows and columns.
3. Any four of: Short Text (names and codes), Number (values used in calculations), Date/Time (dates and times), Yes/No (True or False), AutoNumber (a unique number for each new record).
4. A validation rule checks that entered data meets a condition, such as `Between 0 And 100`, and rejects values that fail it. An index is a structure that speeds up searching and sorting; it does not check values.
5. A primary key is a field whose value uniquely identifies each record. It cannot contain duplicates because two records with the same key value could not be told apart, and uniqueness is the whole purpose of the key.

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LESSON 29

Access Keys and Relationships

29.1 Why keys and relationships matter

A database keeps data in **tables**. A table is a grid: each **record** is one row, and each **field** is one column that holds a single kind of value. When a table holds many records, values such as a name, a city, or a date can repeat, so a row cannot be recognised by those values alone. A database therefore gives each row a reliable **identifier**, called a key, and it links rows that belong together using **relationships**. Keys provide identity; relationships join tables that share data; referential integrity keeps those joins correct.

Definition — Table, record, field

*A **table** stores data about one subject as rows and columns. A **record** is one complete row. A **field** is one column, holding one attribute for every record.*

Term	Meaning in this lesson
Primary key (PK)	A field or set of fields that uniquely identifies each record.
Foreign key (FK)	A field that refers to the primary key of another table.
Composite key	A primary key made of two or more fields together.
Candidate key	Any field that could serve as the primary key.
Alternate key	A candidate key that was not chosen as the primary key.
Junction table	A table that links two tables in a many-to-many relationship.
Referential integrity	The rule that a foreign-key value must match an existing primary-key value.

The rest of the lesson uses these terms in one consistent sense. A **primary key** identifies a record inside its own table. A **foreign key** points from one table to another table's primary key. A **composite key** is simply a primary key built from more than one field. A **junction table** is the third table that implements a many-to-many relationship.

29.2 The primary key

A **primary key (PK)** is a field, or a set of fields, that **uniquely identifies** each record in a table. It has three firm rules. First, its value must be **unique**: no two records may share the same primary-key value. Second, it **cannot be left blank** (it cannot contain a null value), because a blank value cannot identify anything. Third, each table has **at most one** primary key, although that one key may be built from several fields.

Definition — Primary key

A field (or set of fields) whose value is unique for every record and is never blank, so it can identify each record in the table.

Example 29.1. *In an Employee table, the field `EmployeeID` is a good primary key: every employee receives a different number, and no employee is stored without one. The employee's name would be a poor choice, because two employees can share a name.*

Access can generate a primary key automatically with the **AutoNumber** data type, which inserts a unique number for each new record. A primary key is what makes a table the **"one" side** of a one-to-many relationship.

29.3 Candidate keys, alternate keys and composite keys

Any field that could serve as the primary key is called a **candidate key**. When a table has more than one candidate key, only one is selected as the primary key; the remaining candidates are called **alternate keys**. For example, a Student table might have both `RollNumber` and `Email` as candidate keys; if `RollNumber` is chosen, then `Email` becomes an alternate key.

A **composite key** (also called a **compound key**) is a primary key made of **two or more fields used together**. The combination is unique even when each field on its own is not. Composite keys appear most often in junction tables, where two foreign keys together form the primary key.

Definition — Composite key

A primary key that consists of two or more fields, whose combined value is unique for every record.

Example 29.1. *An OrderDetails table records each product line within an order. Neither `OrderID` nor `ProductID` is unique by itself, but the pair (`OrderID` + `ProductID`) is unique for every line item, so the pair can be the primary key. This is a composite key.*

A good primary key is also **stable** and **meaningless**: it should not need to change later. This is why Access often uses an AutoNumber field rather than a value such as a name or a phone number, which a person may edit.

Key	What it does
Primary key	Uniquely identifies a record; unique and not blank; at most one per table.
Candidate key	Any field eligible to become the primary key.
Alternate key	A candidate key that was not chosen as the primary key.
Composite key	A primary key made of two or more fields together.
Foreign key	A field that refers to another table's primary key; may repeat.

29.4 The foreign key

A **foreign key (FK)** is a field in one table that refers to the **primary key** of another table. The foreign key is the link that creates a relationship between the two tables. Unlike a primary key, a foreign key **may repeat** and **may be blank**; it is not unique. The table that holds the foreign key is the **child** (or related) table, and the table whose primary key is referenced is the **parent** table.

Definition — Foreign key

A field in one table whose values must match the primary-key values of another table, used to link the two tables.

Example 29.1. *In an Employee table, the field `DepartmentID` is a foreign key that refers to the `DepartmentID` primary key in the Department table. Many employees can share the same `DepartmentID`, so the foreign key repeats; that is expected and correct.*

Point	Primary key	Foreign key
Purpose	Uniquely identifies a record	Links to another table
Uniqueness	Must be unique	May repeat
Blank allowed	No	Yes
Count per table	At most one	Can be more than one

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29.5 Types of relationships

A **relationship** works between two tables that share a common field — usually a primary key in one table and a foreign key in the other. Access supports three relationship types: **one-to-one**, **one-to-many**, and **many-to-many**. Relationships are created and edited in the **Relationships** window.

Type	Meaning	How it is built
One-to-one	Each record in one table matches at most one in the other	A shared field that is unique in both tables
One-to-many	One record matches many records in the other table	Primary key on the "one" side, foreign key on the "many" side (most common)
Many-to-many	Many records match many records	Two one-to-many relationships through a junction table

The rule to hold on to is simple: the **"one" side** always contributes the **primary key**, and the **"many" side** always carries the **foreign key**. A one-to-one relationship is just a special case in which both sides are unique.

29.5.1 One-to-one

In a **one-to-one** relationship, each record in the first table matches at most one record in the second table, and vice versa. It is used to split a table with many fields into smaller parts, or to keep sensitive data in a separate table that can be secured on its own. It is the **least common** of the three types.

Example 29.1. *An Employee table and an EmployeeConfidential table are linked one-to-one: each employee has exactly one confidential record, and each confidential record belongs to exactly one employee.*

29.5.2 One-to-many

In a **one-to-many** relationship, one record in the first table matches **many** records in the second table. The "one" side holds the **primary key**, and the "many" side holds the **foreign key**. This is the **most common** relationship type used in Access.

Example 29.2. *One Department has many Employees, but each Employee belongs to only one Department. The Department table is the "one" side and the Employee table is the "many" side, linked by* `DepartmentID`*.*

29.5.3 Many-to-many and the junction table

In a **many-to-many** relationship, many records in the first table match many records in the second. Access **cannot** store a many-to-many relationship directly. Instead it is split into **two one-to-many** relationships by adding a third table called a **junction table** (also known as an **associative** or **linking** table). The junction table contains a foreign key for each of the two other tables.

Definition — Junction table

A table that connects two tables in a many-to-many relationship, holding a foreign key for each of them.

A junction table's own primary key is usually a **composite key** formed from the two foreign keys. It may also carry extra fields that describe the link itself, such as a quantity, a grade, or a date.

Example 29.3. *Students and Courses have a many-to-many relationship: a student takes many courses, and a course has many students. Access implements this with an Enrollments junction table that holds* `StudentID` *and* `CourseID` *as its composite primary key, together with a* `Grade` *field.*

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29.6 Creating a relationship in Access

In the Access interface, relationships are built and edited in a small number of steps. On the **Database Tools** tab, the **Relationships** button opens the **Relationships** window. The tables to be linked are added to this window, and then the **primary-key field** of one table is **dragged onto** the matching foreign-key field of the other table.

1. Open the **Relationships** window from the **Database Tools** tab.
2. Add the two tables that will be linked.
3. Drag the primary key of the "one" table onto the foreign key of the "many" table.
4. In the **Edit Relationships** dialog box, choose **Enforce Referential Integrity**.
5. Optionally tick **Cascade Update** and **Cascade Delete**, then confirm.

Example 29.1. *To link Departments and Employees, drag `DepartmentID` from the Department (parent) table onto `DepartmentID` in the Employee (child) table, then enforce referential integrity. Access now refuses to store an employee whose `DepartmentID` does not exist in the Department table.*

29.7 Referential integrity

Referential integrity is the rule that a foreign-key value must **match an existing primary-key value** in the related table (or be blank). It prevents **orphan records**, which are child rows that point to a parent row that does not exist. In Access, referential integrity is switched on in the **Edit Relationships** dialog box (opened from the **Relationships** window).

Definition — Referential integrity

A rule that keeps the foreign key in a child table consistent with the primary key in the parent table: a child value must match an existing parent value or be null.

When referential integrity is enforced, Access offers two related options. **Cascade Update** means that if the parent primary-key value changes, Access automatically updates the matching foreign-key values in the child table. **Cascade Delete** means that if a parent record is deleted, Access automatically deletes the matching child records as well.

Example 29.1. *If a Department record is deleted and Cascade Delete is on, all Employees that belong to that Department are deleted too. Without referential integrity, Access would refuse to delete the Department while those Employees still point to it.*

29.8 Access file formats: `.mdb` vs `.accdb`

The file **extension** shows which Access format a database uses. `.mdb` (*Microsoft Database*) is the **older** Access file format, used by Access 2003 and earlier versions. `.accdb` (*Access Database*) is the **newer, default** format introduced with Access 2007 and used since. Access 2007 and later can still open an old `.mdb` file, but new databases are saved as `.accdb` by default.

Extension	Access version	Notes
`.mdb`	Access 2003 and earlier	The older format; supports user-level security.
`.accdb`	Access 2007 and later	The newer default; supports attachment, multi-valued and calculated fields.

The newer `.accdb` format added **attachment** fields, **multi-valued** fields, and **calculated** fields, and it integrates better with SharePoint and Outlook. Keep the pairing straight: `.mdb` is the *old* format and `.accdb` is the *new* format. Reversing "old" and "new" is a common exam trap.

29.9 Exam retrieval and common confusions

Six distinctions clear up most errors on this topic.

1. A primary key is **unique** and **not blank**; a foreign key **may repeat** and **may be blank**.
2. A table has **at most one** primary key, but it can hold **several** foreign keys.
3. A foreign key refers to the **primary key of another table**, not to its own table's primary key.
4. **One-to-many** is the most common relationship; a **many-to-many** relationship is never stored directly but is built with a **junction table**.
5. A junction table commonly has a **composite primary key** made from the two foreign keys.
6. `.mdb` is the **older** Access format; `.accdb` is the **newer** format.

Example 29.1. *An item asks which key may contain duplicate values. Walk through the rules: a primary key must be unique and not blank, so it cannot repeat; a foreign key may repeat and may be blank; a composite key is still a primary key and so is unique as a combination. The answer is the foreign key.*

Example 29.2. *An item states that "a many-to-many relationship is created by adding a foreign key to one of the two tables." This is wrong. A many-to-many relationship cannot be stored in either table directly; a **junction table** is added, and two one-to-many relationships are built through it. The wrong option has skipped the junction table.*

Quick self-check

Before the exercise set, confirm you can state each of these in one line.

- A primary key is unique, never blank, and there is at most one per table.
- A foreign key links to another table's primary key and may repeat.
- A composite key is a primary key of two or more fields.
- One-to-many is the most common relationship; the "one" side supplies the primary key and the "many" side the foreign key.
- A many-to-many relationship is always built with a junction table.
- Referential integrity keeps foreign-key values matching primary-key values and offers Cascade Update and Cascade Delete.
- `.mdb` is the older format; `.accdb` is the newer one.

Exercises

1. Define a primary key and state its two main rules about uniqueness and blank values.
2. Explain the difference between a primary key and a foreign key.
3. What is a composite key? Give one example where it is used.
4. Name the three relationship types in Access and say which is the most common.
5. Why does a many-to-many relationship need a junction table?
6. State the rule of referential integrity and name its two cascade options.
7. Distinguish the Access file formats `.mdb` and `.accdb`.

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Solutions

1. A primary key is a field (or set of fields) that uniquely identifies each record. Its value must be unique — no two records may share it — and it cannot be left blank (no null value). A table has at most one primary key.
2. A primary key uniquely identifies records inside its own table and cannot repeat or be blank. A foreign key is a field that refers to another table's primary key; it may repeat and may be blank, and it creates the relationship between the two tables.
3. A composite key is a primary key made of two or more fields used together, where the combination is unique. It is used in a junction table — for example, an Enrollments table whose primary key is (`StudentID` + `CourseID`).
4. The three types are one-to-one, one-to-many, and many-to-many. The **one-to-many** relationship is the most common.
5. Access cannot store a many-to-many relationship directly. It is split into two one-to-many relationships by a junction table that holds a foreign key for each of the two tables.
6. Referential integrity means a foreign-key value must match an existing primary-key value in the related table (or be blank), which prevents orphan records. Its two options are **Cascade Update** and **Cascade Delete**.
7. `.mdb` is the older Access format used by Access 2003 and earlier. `.accdb` is the newer, default format used from Access 2007 onward, adding attachment, multi-valued and calculated fields.

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LESSON 30

Access Queries, Forms and Reports

30.1 The Access objects and the object sequence

An Access database is built from a small set of **objects**. The **table** stores the actual data. The **query** asks a question and retrieves selected data. The **form** provides a screen for entering and viewing data. The **report** presents data in a formatted, printable layout. These four objects are used together in a natural order: data is stored in a table, selected by a query, entered through a form, and printed by a report. That order — **Table** → **Query** → **Form** → **Report** — is the object sequence this lesson is built around.

Definition — Database object

A named component of an Access database that performs one job. The main objects are the table, query, form and report.

Object	Job
Table	Stores the actual data in rows and columns.
Query	Asks a question and retrieves selected data.
Form	Provides an on-screen interface to enter and view data.
Report	Presents data in a formatted, printable layout.

The rest of this lesson uses these terms in one consistent sense. A **table** is the store. A **query** is the question. A **form** is the on-screen entry screen. A **report** is the printed output.

30.2 The table is the source of data

A table stores data in **fields** (columns) and **records** (rows). Every other object draws its data from a table or from a query. A form or report built on a table shows that table's data; built on a query, it shows only the rows the query selected. No data is stored in a query, a form or a report — those objects only *present* what the table holds. This is why the table is called the foundation of the database.

30.3 What a query is

A **query** is a question asked of the database. It retrieves data from one or more tables, usually by a condition. A query does not normally store new data of its own; it shows a *result set* drawn from the tables each time it is opened. Queries can also be used to *change* data; those are called **action queries**, and they are discussed below.

Definition — Query

An Access object that selects, filters, or acts on records according to the rules the user sets, drawing its data from one or more tables.

30.4 The Select query: retrieving on a condition

A **Select query** retrieves the records that satisfy a **condition** (also called a *criterion*). It is the most common query type in Access. The condition decides *which rows appear*, and the chosen fields decide *which columns appear*. For example, a Select query on a Students table could show only those students whose `Marks` are greater than 60.

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Example 30.1. *An item asks: "What is a query that retrieves data on the basis of a condition called?" The query is only* retrieving *data — it is not changing it — so it is a* ***Select query****. This is the concept tested by PYQ-39 (Website Operator 2026, query retrieving data on a condition).*

30.5 Building a query: Design view and criteria

Queries are usually built in **Query Design** view, or typed directly in **SQL** view. The design grid is called **QBE — Query By Example**. Each column of the grid holds one **field**, and the **Criteria** row holds the condition for that field. A criterion of `>60` on the `Marks` field, for instance, keeps only the rows in which marks are above 60.

Definition — QBE (Query By Example)

The grid-based design view in Access in which a query is built by choosing fields and typing criteria, without writing the SQL by hand.

Example 30.1. *For a Students table, place* `Marks` *in the field row and* `>60` *in the Criteria row. When the query runs, Access returns only the students who scored more than 60.*

30.6 Action queries: Update, Append and Delete

An **action query** changes the stored data. The three common types are:

- An **Update query** changes the existing values in records.
- An **Append query** adds records from one table to another.
- A **Delete query** removes the records that match a condition.

Because these queries alter the table itself, they are treated with care in Access. The distinction between retrieving data and changing data is the key one: a Select query only reads, while Update, Append and Delete queries write.

Query	What it does
Select	Retrieves records that satisfy a condition; does not change data.
Update	Changes existing values in records.
Append	Adds records from one table to another.
Delete	Removes records that match a condition.

30.7 Select versus action queries

The nearest confusion in this topic is between a Select query and the action queries. A **Select query retrieves** data on a condition and leaves the tables unchanged. An **action query** (Update, Append, Delete) **modifies** the stored data. If a question asks which query *changes* records, the answer is an action query; if it asks which query *shows* records that meet a condition, the answer is a Select query. This is the distinction recorded as TRAP-31.

Example 30.1. *An item lists Select, Update, Append and Delete queries and asks which one does* ***not*** *modify the data. Select is the only one that leaves the tables unchanged, so Select is the answer.*

30.8 Lookup fields

A **lookup** field lets the user *pick a value from a list* instead of typing it. The list usually comes from a table or a query — for example, a list of department names from a Departments table. A lookup is often used for a **foreign key**, so that the value entered matches an existing value in the related table. The benefit is fewer typing errors and more consistent data. A lookup field only *chooses* a value from a list; it does not create a new table or a new query.

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Definition — Lookup field

A field that displays a list of values, drawn from a table or query, so the user can select one instead of typing it.

30.9 Basic SQL awareness

SQL stands for **Structured Query Language**. It is the standard language for working with relational databases, and Access runs SQL behind the scenes whenever a query is run. A retrieval statement has the general shape `SELECT fields FROM table WHERE condition;`. The `SELECT` keyword names the fields to show, `FROM` names the table, and `WHERE` gives the condition. To retrieve only the students with marks above 60, one would write `SELECT Name FROM Students WHERE Marks > 60;`.

Definition — SQL

Structured Query Language — the standard language used to define and retrieve data in a relational database.

30.10 Forms: entering and viewing data

A **form** is an object that provides an on-screen interface for entering, viewing and editing data. A form is usually based on a table or a query. It shows one record at a time (or a few), with clearly labelled fields, and makes data entry easier and safer than typing directly into a table grid. Where the table is the raw store, the form is the friendly screen the user actually works with.

30.11 Reports: printable output

A **report** presents data in a formatted, printable layout. It can group records and show totals and summaries. A report is **read-only**: it is meant for printing and sharing, not for editing data. Like a form, a report can be based on a table or a query. The report is the output object of the database.

Example 30.1. *A school can keep all student marks in a table, use a Select query to pick the students who passed, enter new students through a form, and print the pass list as a report. The same data moves, in order, through the Table → Query → Form → Report sequence.*

30.12 Forms versus reports

The nearest confusion between the two presentation objects is worth stating plainly. A **form** is used *on screen* to enter and view data, and it allows editing. A **report** is *read-only* and is used to *print* data and present summaries. A form answers the question "how do I put data in and look at it?"; a report answers "how do I print it?"

Point	Form	Report
Purpose	Enter and view data on screen	Print and present data
Data change	Allows editing	Read-only
Typical use	Data-entry screen	Printed summary

30.13 Exam retrieval and common confusions

Six distinctions clear up most errors on this topic.

1. A **Select** query **retrieves** data on a condition; **Update**, **Append** and **Delete** queries **modify** data (TRAP-31).
2. Data is stored in a **table**, **not** in a query, form or report.
3. A **form** is for on-screen entry and viewing; a **report** is a printable, read-only output.

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4. The object sequence is **Table** → **Query** → **Form** → **Report**.
5. **SQL** stands for **Structured Query Language**; `SELECT ... FROM ... WHERE` retrieves data on a condition.
6. A **lookup** field picks a value from a list supplied by a table or query.

Example 30.1. *An item states, "A report is used to enter new records into the database." This is wrong. Entering records is the job of a **form**; a report is **read-only** and is used for printing. The wrong option has swapped the form and the report.*

Exercises

1. Name the four main objects of an Access database and state the job of each.
2. Give the correct object sequence for storing, selecting, entering and printing data.
3. What is a Select query, and how does it differ from an Update query?
4. Name the three common action queries and say what each one does.
5. What is a lookup field, and why is it useful?
6. Expand SQL and write the general form of a retrieval statement.
7. Distinguish a form from a report.

Solutions

1. **Table** stores the data; **Query** asks a question and retrieves selected data; **Form** provides an on-screen interface to enter and view data; **Report** presents data in a formatted, printable layout.
2. **Table** → **Query** → **Form** → **Report**. The table stores the data, the query selects it, the form enters it, and the report prints it.
3. A Select query **retrieves** records that satisfy a condition and leaves the tables unchanged. An Update query **changes** the existing values in records, so it modifies the stored data.
4. **Update** changes existing values in records; **Append** adds records from one table to another; **Delete** removes records that match a condition.
5. A lookup field displays a list of values, drawn from a table or query, so the user selects one instead of typing it. It reduces typing errors and keeps values consistent, and it is often used for a foreign key.
6. SQL is **Structured Query Language**. A retrieval statement has the form `SELECT fields FROM table WHERE condition;`, where `SELECT` names the fields, `FROM` names the table, and `WHERE` gives the condition.
7. A **form** is an on-screen interface used to enter and view data, and it allows editing. A **report** is a read-only, formatted output used for printing and summarising data.

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LESSON 31

PowerPoint Basics

31.1 What PowerPoint is

MS PowerPoint is the presentation program of the MS Office suite. A **presentation** is a file made of one or more pages, and each page is called a **slide**. The slides are shown one after another as a **slide show** on a screen or projector. The official syllabus writes this unit as "Powerpoint Presentation", and the usual short form is **PPT**. A single slide can hold text, pictures, charts, tables, audio, and video.

Definition — Presentation, slide, slide show

A presentation is the whole file; a slide is one page of it; a slide show is the full-screen sequential display of those slides to an audience.

Term	Meaning in this lesson
Presentation	The whole PowerPoint file, made of slides.
Slide	One page of the presentation.
Placeholder	The dotted-line box on a slide that holds content.
Layout	The arrangement of placeholders on a slide.
Theme	A coordinated set of colours, fonts, and effects.
Template	A ready-made starter file with a theme, layouts, and sample content.
Slide Master	The top-level slide that controls all linked layouts and slides.
Notes	The speaker's reminders below a slide, unseen by the audience.
Handout	A printed sheet carrying several slides for the audience.

The rest of this lesson uses these terms in one consistent sense. A **slide** is the page, a **placeholder** is the box that holds content on that page, and a **layout** decides where those boxes sit. A **theme** restyles the file, a **template** starts a new file from a ready design, and the **Slide Master** controls the design of every linked slide at once.

31.2 Full forms and file extensions

Three full forms carry direct-recall marks. **PPT** stands for PowerPoint Presentation. **PPTX** stands for PowerPoint XML Presentation and is the default format since Office 2007. **PPSX** stands for PowerPoint Slide Show. Two more extensions complete the family: **POTX** is the PowerPoint Template, and **PPTM** is the macro-enabled presentation.

Definition — PowerPoint file extensions

The editable working file uses `.pptx`; the template uses `.potx`; the show file that opens directly as a slide show uses `.ppsx`; the macro-enabled file uses `.pptm`. Presentations can also be saved or exported as `.pdf`.

The working distinction is simple. Double-clicking a `.pptx` file opens it for editing, while double-clicking a `.ppsx` file launches the slide show directly. A `.potx` file is never the finished talk; it is the reusable pattern from which new talks are started. Confusing these three extensions is one of the most common wrong answers in this unit.

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31.3 Slides, placeholders, and layouts

A slide is a single page of a presentation. Content is not typed freely onto the page background; it goes into **placeholders**, the dotted-line boxes marked Title, Text, Picture, Chart, or Table. Clicking inside a placeholder allows typing, and the small icons inside a content placeholder insert a picture, chart, table, or SmartArt graphic directly.

Definition — Placeholder

A dotted-line container on a slide that holds one kind of content (title, text, picture, chart, table, or media).

A **layout** is the arrangement of those placeholders. The common layouts are Title Slide, Title and Content, Section Header, Two Content, Comparison, Title Only, and Blank. A new slide is added from the Home tab through New Slide, which also offers the layout gallery; the layout of an existing slide is changed through Home tab to Layout. The Blank layout carries no placeholders at all, so every object on it must be inserted by hand.

Example 31.1. *A student builds a five-slide class talk on the office desktop. The first slide uses the Title Slide layout (one title plus one subtitle placeholder); each following point uses Title and Content; the closing slide reuses Section Header. Changing the layout of slide 3 from Title and Content to Two Content rearranges only slide 3; nothing else in the file moves.*

31.4 Themes, templates, and the Slide Master

A **theme** is a coordinated set of colours, fonts, and effects applied from the Design tab. One click restyles every slide in the file, because the theme is a design rule, not content. A **template** is a ready-made pattern file (extension `.potx`) that bundles a theme with a set of layouts and sometimes sample text. Starting a new presentation from a template reuses that whole design instead of building it again.

Definition — Theme and template

A theme is a coordinated set of colours, fonts, and effects. A template is a reusable starter file (`.potx`) carrying a theme plus layouts and sample content.

The **Slide Master** is the top-level slide that controls the design of all layouts and slides linked to it. Editing the font, the logo position, or the footer on the Slide Master changes every linked slide at once, which is far faster than editing forty slides one by one. It is opened from the View tab through Slide Master and left through Close Master View. PowerPoint has three masters in total: the Slide Master for on-screen slides, the **Handout Master** for printed handout layouts, and the **Notes Master** for printed notes pages.

Example 31.1. *An item states, "The Handout Master controls the design of all on-screen slides." This is wrong. The Slide Master controls the on-screen slides; the Handout Master controls only the printed handout layouts. The wrong option has swapped two of the three masters.*

31.5 Views: Normal, Outline, Sorter, Reading, Show, Notes Page

PowerPoint offers several views, each with one job. **Normal view** is the default editing view, with three areas: the large slide pane in the centre, the slides or outline pane with thumbnails on the left, and the notes pane below. All typing, formatting, and inserting happen here. **Outline View** shows only the text outline of every slide, which suits restructuring long talks.

Slide Sorter view shows thumbnails of all slides together, so slides can be reordered by drag and drop and checked, deleted, or duplicated at a glance. **Reading View** shows the show almost full-screen for reading on the author's own screen, with simple navigation but without editing tools. **Slide Show view** is the real full-screen presentation to an audience. **Notes Page view** shows one

slide above its speaker-notes text, one page per slide, exactly as it will print.

Definition — Normal view

The default editing view of PowerPoint, combining the slide pane, the slides or outline pane, and the notes pane.

Example 31.1. *A student has twelve slides in the wrong order. The fastest fix is Slide Sorter view: drag the thumbnails into the right sequence. Opening each slide in Normal view and cutting and pasting would work but wastes time; choosing Reading View would not help at all, because Reading View cannot reorder anything.*

31.6 Notes, handouts, and printing

Notes (speaker notes) are reminders typed in the notes pane below a slide. The audience never sees them during the show; they exist so the speaker does not forget a figure, a definition, or the next point. During the show the speaker reads them only in Presenter View on the speaker's own screen. Notes print through File to Print to Notes Pages.

Definition — Speaker notes and handouts

Speaker notes help the speaker and stay hidden from the audience. Handouts help the audience and are printed sheets carrying several slides per page.

Handouts are those printed sheets: 1, 2, 3, 4, 6, or 9 slides per page. The most examined choice is 6 slides per page, and the 3-per-page layout adds ruled lines beside each slide so the audience can write. Their design is controlled by the Handout Master. The full print path is File to Print, with choices for Full Page Slides, Handouts, Notes Pages, and Outline View.

31.7 Running and navigating the slide show

Two shortcuts start the show. **F5** begins from the first slide, and **Shift+F5** begins from the current slide. This pair is a confirmed past item (WO 2026 Q80 tested F5 for "from the beginning"). During the show, N, Space, Right arrow, or Enter moves forward; P or Left arrow moves back; Esc ends the show. Pressing B blanks the screen to black and pressing W blanks it to white, which is useful while the speaker talks without a slide. The Slide Show tab holds timing, rehearsal, and monitor settings.

A **transition** is the movement between one slide and the next (Fade, Push, Wipe, Morph), set on the Transitions tab with one transition per slide. An **animation** is the movement of an object on a slide, set on the Animations tab: Entrance, Emphasis, Exit, and Motion Paths, managed in the Animation Pane. SmartArt conversion behaviour (which content cannot convert without re-entry) has also been examined (JA 2026 Q77), so candidates should practise the Insert to SmartArt path on text and picture lists rather than assume every object converts.

31.8 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The **slide** is the page; the **placeholder** is the dotted box holding content; the **layout** arranges those boxes.
2. **.pptx** is the editable file, **.ppsx** is the show file, and **.potx** is the template. Do not swap them.
3. A **theme** restyles (colours, fonts, effects); a **template** is the reusable starter file.
4. The **Slide Master** controls on-screen slides; the Handout and Notes Masters control print layouts.

5. **Normal** is the default editing view; **Slide Sorter** reorders; Reading reviews; **notes** serve the speaker and **handouts** serve the audience.

6. **F5** starts from the beginning and **Shift+F5** from the current slide; a **transition** moves between slides while an **animation** moves objects on a slide.

Example 31.1. *An item asks which view is the default editing view and offers Normal, Slide Sorter, Reading, and Slide Show. Trace the definitions: Normal is the editing view with the slide, thumbnail, and notes panes; Slide Sorter manages order; Reading reviews; Slide Show presents. The answer is Normal view. If the stem instead asks which view reorders slides by drag and drop, the answer changes to Slide Sorter.*

Exercises

1. Distinguish a slide, a placeholder, and a layout, giving one example of each.

2. Expand PPT, PPTX, and PPSX, and state the file extension of a PowerPoint template.

3. Distinguish a theme from a template.

4. What does the Slide Master control, and how do the Handout and Notes Masters differ from it?

5. List four PowerPoint views and state the job of each.

6. Distinguish speaker notes from handouts, and state which viewers each serves.

7. Which keys start a show from the beginning and from the current slide, and which key ends the show?

8. Distinguish a transition from an animation.

Solutions

1. A slide is one page of the presentation (for example, slide 2 of a class talk). A placeholder is the dotted-line box holding content on that page (for example, the Title placeholder). A layout is the arrangement of those boxes (for example, Title and Content).

2. PPT is PowerPoint Presentation; PPTX is PowerPoint XML Presentation; PPSX is PowerPoint Slide Show. A PowerPoint template uses `.potx`.

3. A theme is a coordinated set of colours, fonts, and effects that restyles a file; a template is a reusable starter file (`.potx`) that bundles a theme with layouts and sample content.

4. The Slide Master controls the design of all linked on-screen slides and layouts. The Handout Master controls printed handout layouts and the Notes Master controls printed notes pages; neither restyles the on-screen slides.

5. Normal is the default editing view; Slide Sorter shows thumbnails for reordering; Reading View shows the show for on-screen review; Slide Show presents full-screen; Notes Page shows one slide above its notes text. Any four with correct jobs earn full credit.

6. Speaker notes are the speaker's hidden reminders below a slide; they serve the speaker. Handouts are printed multi-slide sheets; they serve the audience.

7. F5 starts from the beginning; Shift+F5 starts from the current slide; Esc ends the show.

8. A transition is the movement between one slide and the next; an animation is the movement of an object on a slide (Entrance, Emphasis, Exit, Motion Paths).

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LESSON 32

PowerPoint Animation and Transition

32.1 Animation and transition: the core distinction

PowerPoint can make a presentation move, but it uses two different tools for two different kinds of movement. An **animation** is an effect applied to one object or to text *inside* a single slide. A **transition** is an effect applied to the movement *from one slide to the next*. The simplest way to keep them apart is to ask where the movement happens: movement inside a slide is animation; movement between slides is a transition.

Both kinds of effect are added from the PowerPoint ribbon. Animation is managed on the **Animations** tab, and transitions are managed on the **Transitions** tab. Neither tool changes the content of the slides; they change only how that content appears and how the deck moves from slide to slide.

Definition — Animation

An effect applied to a single object or to text on a slide — for example making a picture appear, or making a title grow. Animation acts within a slide.

Definition — Transition

An effect applied to the change from one slide to the next, such as Fade, Push, or Morph. A transition acts between slides.

Point	Animation	Transition
Where it applies	An object or text on a slide	The change between two slides
Ribbon tab	Animations	Transitions
Nearest confusion	Transitions	Entrance/Exit effects

32.2 The four animation effects

When you select an object and open the Animations tab, PowerPoint groups the effects into four categories. Each category answers a different question about the object.

Effect	What it does
Entrance	Makes an object **appear** on the slide.
Exit	Makes an object **leave** the slide.
Emphasis	Draws attention to an object that is **already visible**, staying on the slide.
Motion Path	Moves an object **along a defined path**, such as a line or curve.

Entrance effects are used to reveal content step by step: the title or a bullet point appears when the presenter is ready. **Exit** effects are the opposite — they take an object off the slide once it is no longer needed. An entrance and an exit are not the same as an emphasis: an emphasis effect does not add or remove the object, it only makes the object stand out while it remains on the slide. A **motion path** is different again: it changes the object's position, moving it from one place to another along a drawn path.

Example 32.1. *A presenter wants a logo to travel from the left edge to the corner of the slide. That is a **Motion Path**, because the object's position changes during the show. If instead the logo simply spins once and stops in the same place, that is an **Emphasis** effect. If the logo is not on the slide at all until clicked, that is an **Entrance** effect.*

Animations can be set to start when the slide appears or when the presenter clicks, and each effect has a duration. The order and timing of the effects on one slide are controlled from the **Animation Pane**.

32.3 Transitions and Morph

A transition plays as the presentation moves from the current slide to the next. Common transitions include **Fade**, **Push**, **Wipe**, and **Dissolve**. Each transition can be given a duration and an optional sound. From the Transitions tab, the **Apply To All** button copies the chosen transition to every slide in the deck.

Transitions are often grouped by the strength of their effect. **Subtle** transitions, such as Fade and Wipe, are gentle and professional. **Exciting** transitions, such as Push and Split, use stronger movement. **Dynamic Content** transitions respond to the slide content itself.

Definition — Morph

A modern transition that smoothly moves and animates objects as one slide changes into the next. Morph works best when two slides share similar objects, so the objects appear to glide from their old position to their new one.

Example 32.1. *Slide 1 shows a circle on the left side; slide 2 shows the same circle on the right side. With **Morph** applied to slide 2, the circle appears to slide smoothly across the screen instead of jumping. Because Morph is a transition, it is applied from the **Transitions** tab, not the Animations tab.*

32.4 Starting and controlling the slide show

The slide show has a small set of shortcut keys that JKSSB papers ask about directly. The most important is **F5**, which starts the slide show from the beginning — that is, from the first slide. **Shift+F5** starts the slide show from the current slide — the one selected. **Alt+F5** starts the presentation in **Presenter View**. **Esc** ends a running slide show.

Key	Action
F5	Start the slide show from the **beginning**.
Shift+F5	Start the slide show from the **current** slide.
Alt+F5	Start the presentation in **Presenter View**.
Esc	**End** the slide show.

Once the show is running, more controls are available. The keys **N**, **Enter**, **Space**, or the right arrow advance to the next slide or animation; **P** or the left arrow go back to the previous slide. Pressing **B** turns the screen black and **W** turns it white, which is useful for pausing discussion. Typing a slide number and pressing **Enter** jumps straight to that slide.

Example 32.1. *An item asks, "Which key starts the slide show from the beginning?" Trace the key set: F5 is the full show from the first slide, Shift+F5 is from the current slide, Alt+F5 opens Presenter View, and F7 is the spelling and grammar check. The answer is **F5**. A distractor that offers F7 is testing whether you confuse the show shortcut with spell check.*

32.5 Speaker notes, views and handouts

The **Notes pane** sits below the slide in Normal view and holds **speaker notes** — the reminders the presenter reads while presenting. Notes are not shown on the projected slide during a normal slide show; in **Presenter View** the notes are visible only to the presenter, along with the current slide, the next slide, and a timer.

PowerPoint provides several views for different jobs. **Normal view** is used to edit one slide, with an outline on the side and the notes pane below. **Outline view** works with the text of all slides as an

outline. **Slide Sorter view** shows small **thumbnails** of every slide and is the view for rearranging slides. **Notes Page view** shows a slide together with its speaker notes. **Reading View** reviews the show in a window without going full screen, and **Slide Show view** presents the deck full screen.

A **handout** is the printed version of the presentation that is given to the **audience**. Handouts usually place several slides on one page — two, three, four, six, or nine per page — and some handout layouts include lined space for the audience to write. The key contrast is simple: the notes pane is for the presenter, and a handout is for the audience.

Definition — Handout

A printed or exported layout of the presentation for the audience, normally showing several slides on a single page, often with room for notes.

32.6 SmartArt conversion and its limits

SmartArt is a graphic that arranges text into a designed layout such as a list, process, or hierarchy. A SmartArt graphic can be converted into **shapes** or into **text**. Converting to **text** turns the graphic into a plain bulleted list, so the graphical layout is lost. Converting to **shapes** keeps individual shapes but removes the SmartArt arrangement, and once the conversion is done the graphic cannot be turned back into SmartArt — the content that depended on the layout must be **re-entered** or re-applied by hand.

Example 32.1. *A student has a SmartArt process diagram and converts it to text to reuse the wording in a document. The words survive, but the arrows, boxes, and arranged layout do not. To get the diagram back, the SmartArt must be created again, because a text conversion has dropped the design. The JA 2026 paper tested this kind of limit — which content cannot convert cleanly without re-entry — as its SmartArt item (PYQ-17).*

32.7 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. Animation applies to an **object inside a slide**; a transition applies between **slides**. Movement inside a slide is animation; movement between slides is a transition.
2. **Entrance**, **Exit**, **Emphasis**, and **Motion Path** are animation effects, not transitions.
3. **F5** starts from the beginning; **Shift+F5** starts from the current slide; **Alt+F5** opens Presenter View; **Esc** ends the show. Do not confuse F5 with F7 (spell check) (TRAP-32).
4. The **Notes pane** is for the presenter; a **handout** is for the audience; **Slide Sorter** is for reordering slides.
5. **SmartArt** converted to text loses its graphical layout and may need the content re-entered.

Example 32.1. *An item lists four effects and asks which one is a transition. Read each option against the definition: Entrance, Exit, and Emphasis all act on an object within a slide, while Fade acts on the move between slides. Fade is the transition. The same logic applies when an option offers Morph: Morph is a transition, so it belongs with Fade, not with the animation effects.*

Exercises

1. Distinguish an animation from a transition, giving one example of each.
2. Name the four animation effect categories and state what each does.
3. State the action of F5, Shift+F5, Alt+F5, and Esc during a slide show.
4. Explain what the Notes pane is for and how a handout differs from it.
5. What happens to a SmartArt graphic when it is converted to text?

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Solutions

1. An animation is an effect on one object or text inside a slide (for example, a picture entering). A transition is an effect on the change between two slides (for example, a Fade).
2. Entrance makes an object appear; Exit makes an object leave; Emphasis draws attention to an object that stays on the slide; Motion Path moves an object along a defined path.
3. F5 starts the show from the beginning; Shift+F5 starts from the current slide; Alt+F5 starts Presenter View; Esc ends the show.
4. The Notes pane holds speaker notes for the presenter and is not shown to the audience during a normal show. A handout is the printed layout given to the audience, usually with several slides per page and room to write.
5. It becomes a plain bulleted list; the graphical layout is lost, and the design must be re-applied or the SmartArt re-created.

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LESSON 33

PDF and File Formats

33.1 What a file format is

Every file on a computer is stored as a long series of bits. On its own, a string of bits says nothing about whether it is a letter, a photograph, a spreadsheet, or a slide show. The **file format** is the rule that tells the software how those bits are arranged — where the text begins, where the image data sits, and how the page should be drawn. When a program opens a file, it reads the format to know how to interpret the contents.

The **extension** is the short suffix at the end of a file name, such as `.pdf`, `.docx`, or `.png`. The extension is a label that helps the operating system choose which program should open the file. It is convenient but only a name: the real identity of the file is its format.

Definition — File format

The defined structure or rule by which data is stored in a file, so that software can correctly read, display, and save its contents.

Definition — Extension

The suffix at the end of a file name, usually written with a dot, that indicates the file's intended type (for example `.pdf`, `.docx`, `.png`).

Term	Meaning in this lesson
Format	The actual structure of the data inside the file.
Extension	The dot-suffix in the file name, such as `.docx`.
Fixed layout	A format that stores exact positions, so the page looks the same everywhere.
Plain text	A file of ordinary characters with no formatting.

33.2 Extension versus format (the core trap)

The single most repeated confusion in this topic is treating the extension and the format as the same thing. They are not. In the file name `report.docx`, the base name is `report`, the extension is `docx` (written `.docx`), and the format is DOCX, the Office Open XML word-processing format. Likewise, `.png` is an extension, while PNG (Portable Network Graphics) is the format.

The extension can also be changed easily, and doing so does not convert the file. Renaming a photograph from `pic.jpg` to `pic.png` does not turn a JPEG image into a PNG image; it only changes the label. This is why the exam stresses the distinction: the extension is a name, the format is the content.

Example 33.1. *An item states, "`.png` is an image file format." The statement mixes the two ideas. `.png` is the **extension**; PNG is the **format**. A correct statement would be, "PNG is an image file format, and `.png` is its usual extension."*

33.3 Common document formats

The Office applications each save their work in a standard format. These are the Office Open XML formats, and each has its own extension.

Extension	Format	Typical use
`.docx`	Word document	Text, formatting, and images (MS Word)
`.xlsx`	Excel workbook	Worksheets, formulas, and charts (MS Excel)
`.pptx`	PowerPoint presentation	Slides and slide shows (PowerPoint)
`.csv`	Comma-Separated Values	Plain-text rows with values separated by commas
`.txt`	Plain text	Unformatted characters only

A **word-processing document** (DOCX) stores formatted text. A **workbook** (XLSX) stores worksheets with formulas and numbers. A **presentation** (PPTX) stores slides. A **CSV** file is plain text in which each line is a record and commas separate the fields; it is often used to move data between programs. A **TXT** file holds plain characters and no formatting at all.

33.4 Common image extensions

Images also come in many formats, each suited to a different purpose. **PNG** (Portable Network Graphics) uses lossless compression and supports transparency, so it is good for logos and screenshots. **JPG/JPEG** uses lossy compression, so it is good for photographs where a small loss of quality is acceptable. **GIF** (Graphics Interchange Format) handles simple images and short animations. **BMP** (Bitmap) stores a raw image and is usually larger. **TIFF** (Tagged Image File Format) is used for high-quality scanning and printing. **SVG** (Scalable Vector Graphics) stores a drawing as lines and shapes, so it scales without blurring.

Extension	Format	Character
`.png`	Portable Network Graphics	Lossless, transparency
`.jpg`	JPEG image	Lossy, photos
`.gif`	Graphics Interchange Format	Simple images, animation
`.bmp`	Bitmap	Raw, larger
`.tiff`	Tagged Image File Format	High quality, scanning/print
`.svg`	Scalable Vector Graphics	Vector, scales without blur

33.5 PDF: the Portable Document Format

PDF stands for **Portable Document Format**. It is a **fixed-layout** format: every item on the page is placed at a fixed position. Because the file also carries the fonts and images it needs, a PDF looks the **same on every device and printer**. This is why PDF is used for e-books, forms, official letters, admit cards, and result sheets, where the exact appearance matters.

Contrast this with a word-processing document, which is a *re-flowable* format: text re-flows to fit the screen, so the layout can shift when the font or device changes. A PDF, by design, does not re-flow; it is a picture of a finished page expressed as instructions.

Definition — PDF

Portable Document Format — a fixed-layout file format that preserves the appearance of a document so it renders identically across devices and printers.

33.6 Text and scanned PDFs

Not every PDF contains real text. It is important to separate two kinds.

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A **text PDF** (sometimes called an editable-source PDF) is generated from a source document such as a Word file. Its pages contain genuine text and vector graphics. The text can be selected, copied, and searched, and the file is usually smaller.

A **scanned PDF** (image-only PDF) is produced by scanning printed pages or photographing them. Each page is stored as an *image*, much like a photograph. The text is readable to the human eye but not to the computer: it cannot be selected, copied, or searched. Scanned files are also usually larger.

Example 33.1. *An item says, "In a scanned PDF, you can search for a word using* `Ctrl+F`*." This is wrong. A scanned PDF is only a picture of the page, so its text is not searchable until OCR is applied. In a text PDF, the same search works immediately.*

33.7 OCR: Optical Character Recognition

OCR stands for **Optical Character Recognition**. It is the technique that reads the *picture of the text* and converts it into **machine-readable text**. Once OCR has been run, the text in a scanned document can be selected, searched, copied, and edited just like the text in a text PDF.

OCR is used on scanned pages and on photographs of documents. It is the bridge that turns an image-only PDF into a searchable one. The recognition is not always perfect, so the result should be checked.

Definition — OCR (Optical Character Recognition)

The process of converting an image of text into machine-readable, editable, and searchable text.

33.8 Encryption and password protection

A PDF can be protected so that its contents are confidential. **Encryption** scrambles the file so that it cannot be read without the correct key or password. Two kinds of password are common. A **password to open** (user password) is required before the file will open at all. A **permissions password** (owner password) restricts what a reader may do — for example, printing, copying, or editing.

Encryption provides **confidentiality**: only the authorised reader can see the contents. It does not prove who created the file, and it is not the same as a digital signature.

33.9 Digital signature

A **digital signature** is attached to a file to prove **who signed it**. It also proves **integrity**: if the file is altered after it has been signed, the signature no longer matches and the change is detected. A digital signature uses a **digital certificate** issued by a trusted authority.

A digital signature provides **authenticity** (the signer is verified) and **integrity** (the file has not changed). It does *not* hide the contents. This is the distinction the exam most often tests: a digital signature is about *who* and *whether it changed*, while encryption is about *who can read it*.

Definition — Digital signature

An electronic signature that verifies the identity of the signer and detects any change made to the file after signing, providing authenticity and integrity.

Point	Encryption	Digital signature
Main purpose	Confidentiality	Authenticity and integrity
Result	Contents unreadable	Signer verified, tampering detected
Needs	Key / password	Digital certificate

Example 33.1. *An item states, "A digital signature encrypts the PDF so that only the recipient can read it." The statement is wrong. The encryption that keeps others from reading the file is* ***encryption****. A digital signature does not hide the content; it confirms the signer and shows whether the file has been changed.*

33.10 Redaction and PDF/A

Redaction means **permanently removing** sensitive content before a file is shared. Names, account numbers, addresses, or photographs may be redacted. A proper redaction deletes the underlying text or image; it is not enough to draw a black rectangle over it, because hidden text can often still be selected and copied. A careful reader should always check that the removed text is truly gone.

PDF/A is a special ISO-standard version of PDF designed for **long-term archiving**. It embeds the fonts and colour information inside the file so that the document will render exactly the same years later, even on future software. The "A" in PDF/A stands for *archival*. It is used by libraries, courts, and government records, and is meant for preservation rather than everyday editing.

33.11 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The **extension** is the file-name suffix; the **format** is the structure of the data. `.png` is an extension; PNG is the format (TRAP-34).
2. A **scanned PDF** is an image; its text cannot be selected or searched until **OCR** is applied. A text PDF is searchable at once.
3. **Encryption** gives confidentiality; a **digital signature** gives authenticity and integrity. They are not the same (TRM-11, TRAP-15).
4. **Redaction** removes sensitive content permanently; a black box drawn over text is not sufficient.
5. **PDF/A** is for long-term **archiving**; ordinary PDF is the everyday fixed-layout format.

Example 33.1. *An item asks which format keeps a fixed layout and looks the same on every device. The answer is PDF. If the stem instead asks which format stores spreadsheet data, the answer is XLSX; if it asks which is a plain-text data format, the answer is CSV; and if it asks which format is meant for long-term archiving, the answer is PDF/A.*

Exercises

1. Distinguish a file extension from a file format, with one example.
2. State the full form of PDF and explain why PDF is called a fixed-layout format.
3. Explain the difference between a text PDF and a scanned PDF, and where OCR fits in.
4. Distinguish encryption from a digital signature in terms of the security goal each provides.
5. What is redaction, and why is drawing a black box over text not a safe redaction? What is PDF/A used for?

Solutions

1. The extension is the dot-suffix in the file name, such as `.docx`; it is only a label. The format is the actual structure of the data inside the file, such as the DOCX word-processing format. The same extension can be renamed without changing the format.
2. PDF stands for Portable Document Format. It is called a fixed-layout format because every item is placed at a fixed position on the page, and the fonts and images are carried in the file, so the document looks the same on every device and printer.

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3. A text PDF contains real, machine-readable text, so it can be selected, copied, and searched. A scanned PDF stores each page as an image, so its text cannot be selected or searched. OCR (Optical Character Recognition) converts the image of the text into machine-readable text, after which the scanned PDF becomes searchable.
4. Encryption provides confidentiality: it scrambles the file so only someone with the correct key or password can read it. A digital signature provides authenticity and integrity: it proves who signed the file and detects any change made after signing. It does not hide the contents.
5. Redaction is the permanent removal of sensitive content before a file is shared. Drawing a black box over text is not safe because the hidden text may still be selectable and copyable underneath. PDF/A is a standardised PDF version for long-term archiving that embeds fonts and colour information so the document renders the same in the future.

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LESSON 54

Office Shortcut and Product Trap Lab

54.1 Why this lesson is a trap drill

The computer section of JKSSB papers does not reward vague recognition. It tests the **exact key** and the **exact product category**. A wrong option is usually not random; it is the nearest neighbour to the right answer — the key that differs by one letter, the same function key in a different program, or a familiar software name placed in the wrong category. The safe method is to read every option as a claim to be tested and to keep the exact English term: `F5`, `F7`, `Ctrl+H`, `$A$1`. This lesson gathers the Word, PowerPoint, Excel, and product-category traps that JKSSB reuses.

54.2 MS Word: exact shortcut keys

Word has a small set of shortcuts that JKSSB asks again and again. Learn each key with its full action.

Shortcut	Action in MS Word
F7	Spelling & Grammar (spell check)
Shift+F7	Thesaurus
Ctrl+H	Find and Replace
Ctrl+F	Find only (opens the Navigation pane)
Ctrl+K	Insert Hyperlink
Ctrl+X	Cut the selection
Ctrl+C	Copy the selection
Ctrl+V	Paste the Clipboard contents

Definition — Find and Replace

*A Word tool, opened with **Ctrl+H**, that searches for a word or phrase and replaces it with another across the document.*

The one distinction to fix first is **Ctrl+H** versus **Ctrl+F**. `Ctrl+H` opens Find and Replace, so it can change text; `Ctrl+F` only finds text. The second is **F7** versus **Shift+F7**: `F7` checks spelling and grammar, while `Shift+F7` opens the Thesaurus. A third is **Ctrl+K**, which inserts a hyperlink, not a comment or a caption.

54.3 Word views and review tools

Word offers several views of the same document. **Print Layout** shows the document exactly as it will appear on the printed page, with margins, headers, and footers in place. The other common views are Web Layout, Outline, and Draft. For exam purposes, the key phrase "as it will appear when printed" points to Print Layout.

Track Changes is Word's review tool. When it is switched on, every insertion, deletion, and formatting change is recorded rather than applied silently. Each marked change can later be **accepted** or **rejected** by a reviewer. Track Changes also stores the author's name and the time of the edit as document metadata. The important trap is that Track Changes does *not* delete the original text permanently: the original remains available until a reviewer accepts or rejects the change (PYQ-12).

Example 54.1. *A candidate reads the option "Track Changes permanently deletes the original text, so it can never be restored." This is false. Track Changes preserves the original text and marks the edit; the reviewer decides whether to accept or reject it. The trap works because "records changes" sounds close to "deletes the old version."*

54.4 Word: Section Break versus Page Break

A **Page Break** starts a new page but keeps the document in the same section. A **Section Break** starts a new section. The distinction matters because only a Section Break lets the pages that follow use a *different* header or footer. This is a repeated trap: the wrong option claims that a Page Break changes the header or footer, which it does not (PYQ-13; TRAP-13).

Definition — Section Break

A break that divides a document into sections, so each section can have its own page setup, including a different header and footer.

54.5 Word feature trap: Mail Merge

Mail Merge combines a **main document** (the common letter or label) with a **data source** such as an address database, and produces many personalised copies. It is a Word feature. The trap is to confuse it with Track Changes or AutoCorrect. The reliable clue in the question is the pair "combines letters with an address database", which points to Mail Merge and not to any other Word tool (PYQ-37; TRAP-29).

54.6 PowerPoint: the show keys

PowerPoint's presentation shortcuts are a favourite near-neighbour set.

Shortcut	Action in PowerPoint
F5	Start the slide show from the beginning
Shift+F5	Start the slide show from the current slide
Alt+F5	Start the presentation in Presenter View
Esc	End the running slide show

The classic trap is to call `F5` the spell-check key. In PowerPoint, `F5` starts the show from the first slide; `F7` checks spelling. `Shift+F5` begins from the slide you are on, and `Alt+F5` opens Presenter View, which shows the speaker notes on the presenter's screen while the audience sees only the slides. `Esc` ends the show but does not jump to the first slide (PYQ-40; TRAP-32).

54.7 Handouts, Slide Master, and orientation

Handouts are a print layout that places several slides on one page for the audience, usually with lined space for notes. The **Handout Master** controls the layout of handouts and notes pages.

The **Slide Master** is the design template behind the whole presentation. It controls the fonts, colours, background, and placeholder positions of every slide that uses it. Editing the Slide Master changes all dependent slides at once, which is exactly why it is used for a consistent look.

Orientation is the direction of the page. **Portrait** is taller than it is wide; **Landscape** is wider than it is tall. Slide orientation is set in PowerPoint, and notes and handouts have their own orientation setting.

54.8 PowerPoint: converting to SmartArt

SmartArt turns text into a visual diagram such as a process, cycle, or hierarchy. A **bulleted list**, a **block of text**, and **indented text** can all be converted directly to SmartArt. A **table of data** cannot be converted directly, because a table is a structured object rather than a plain text string; the Convert to SmartArt command is disabled for it. To use table content, the text must first be copied into a text box (PYQ-17).

54.9 Excel: the formula palette

Every Excel formula begins with the = sign (PYQ-38; TRAP-30). The **formula palette** — the Insert Function dialog — is the helper that lets you build a formula. It lists the function and shows

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each **argument** in its own box, so you can fill in the values one at a time. It is opened from the **fx** button next to the formula bar. The palette guides formula entry; it does not resize cells or change data.

Definition — Formula palette

The Excel dialog box that lets a user select a function and enter its arguments while building a formula.

54.10 Excel: cell references and the $ sign

A cell reference tells Excel which cell to use. The reference type decides what happens when the formula is copied to another cell.

Reference	Type	Behaviour when copied
A1	Relative	Both column and row shift
$A$1	Absolute	Neither column nor row changes
$A1	Mixed	Column fixed, row shifts
A$1	Mixed	Row fixed, column shifts

The **$** sign locks the part immediately after it. In `$A$1` both the column and the row are locked, so the reference is **absolute**: it always points to the same cell no matter where the formula is copied. `$A1` locks the column only, and `A$1` locks the row only; both are **mixed** references. While editing a reference, pressing **F4** cycles through relative, absolute, and mixed forms.

Example 54.1. *A formula in `D4` reads `=B4*C4`. Copied down to `D5`, it becomes `=B5*C5`, because the references are relative. If instead the formula is `=$B$4*$C$4`, it stays `=$B$4*$C$4` in every cell, because the `$` signs make both references absolute.*

54.11 Product-category traps: Word, Excel, Access, MS-DOS

JKSSB often places four familiar names together and asks which one belongs to a given category. Match each name to its exact category.

Program	Correct category
MS Word	Word processor (text documents)
MS Excel	Spreadsheet (cells and formulas)
MS Access	Database management system (tables, queries, reports)
MS-DOS	Disk Operating System (a command-line operating system)

The trap is that all four are familiar computing names. **MS Access** is a database, *not* a spreadsheet; **MS Excel** is a spreadsheet, *not* a database; and none of Word, Excel, or Access is an operating system. **MS-DOS** is the only operating system in the group.

54.12 Exam retrieval and common confusions

1. Word **F7** is spell check; **Ctrl+H** is Find and Replace; **Ctrl+K** is Insert Hyperlink.
2. **Section Break**, not Page Break, allows a different header/footer on later pages (TRAP-13).
3. PowerPoint **F5** starts from the beginning, **Shift+F5** from the current slide, **Alt+F5** is Presenter View, and **Esc** ends the show (TRAP-32).
4. A **table** cannot be converted directly to SmartArt.

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5. Excel **$A$1** is an absolute reference; every formula starts with =.
6. **MS Access** is a database; **MS-DOS** is an operating system.

Example 54.1. *A question asks which key starts a PowerPoint slide show from the beginning. Trace the set:* `F5` *begins the show,* `Shift+F5` *starts from the current slide,* `Alt+F5` *opens Presenter View, and* `Esc` *ends the show. The answer is* `F5`*. If the stem instead says "from the current slide", the answer is* `Shift+F5`*.*

Exercises

1. State the action of F7, Ctrl+H, and Ctrl+K in MS Word.
2. Distinguish Print Layout from the other Word views.
3. Explain what Track Changes records and why it does not permanently delete the original text.
4. Distinguish a Page Break from a Section Break, and say which one allows a different header.
5. Give the PowerPoint keys for starting a show from the beginning, from the current slide, for Presenter View, and for ending the show.
6. Explain the difference between the relative reference A1 and the absolute reference $A$1 in Excel.
7. Classify MS Word, MS Excel, MS Access, and MS-DOS by category.

Solutions

1. `F7` opens Spelling & Grammar; `Ctrl+H` opens Find and Replace; `Ctrl+K` inserts a hyperlink.
2. Print Layout shows the document as it will appear on the printed page, with margins and headers in place. Web Layout, Outline, and Draft are the other common views and are not print-accurate.
3. Track Changes records insertions, deletions, and formatting edits, along with author and time metadata. It marks the edits instead of applying them silently, so the original text is preserved until a reviewer accepts or rejects each change.
4. A Page Break starts a new page within the same section; a Section Break starts a new section. Only a Section Break allows a different header or footer on the following pages.
5. `F5` from the beginning; `Shift+F5` from the current slide; `Alt+F5` for Presenter View; `Esc` to end the show.
6. A1 is relative, so both its column and row shift when the formula is copied. $A$1 is absolute, so neither the column nor the row changes; the reference always points to the same cell.
7. MS Word is a word processor; MS Excel is a spreadsheet; MS Access is a database management system; MS-DOS is an operating system.

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LESSON 55

File Formats and Named Technology Associations

55.1 Extension and format are two different things

Every file stored on a computer follows some fixed plan for arranging its data. That plan is the **format** of the file. The format decides which program can open the file and what the file is able to store: a picture, a document, a table, or a database. An **image format** stores a picture; a **document format** stores a printable page; a **plain-text format** stores only characters.

A file name usually ends with an **extension** — the letters written after the last dot. In the name `report.png`, the extension is `png`. The extension is only a label; it tells the reader which format is expected inside. The extension and the format are therefore related but not the same thing. The exam tests this distinction directly, so it is stated here in its own section.

Definition — File format

A fixed way of arranging the data in a file so that a particular program can read it, write it, and display it correctly.

Definition — File extension

The letters after the last dot in a file name, such as `png` in `report.png`. The extension signals the file's format.

Term	Meaning in this lesson
Extension	The letters after the last dot in a file name.
Format	The actual arrangement of data inside the file.
PNG	Portable Network Graphics, an image format.
PDF	Portable Document Format, a fixed-layout document format.
CSV	Comma-Separated Values, a table stored as plain text.
TXT	A plain text file with no formatting.
`.mdb` / `.accdb`	Older and newer Microsoft Access database formats.

55.2 Image formats

The best-known image format for this course is **PNG**, which stands for **Portable Network Graphics**. PNG is a raster image format: it stores a picture as a grid of pixels. It uses *lossless* compression, meaning the picture is saved without permanently throwing away detail, and it supports transparency.

Other common image formats are **JPEG** (written `.jpg` or `.jpeg`), a *lossy* format often used for photographs, and **GIF** (`.gif`), which also supports simple animation. Typical image extensions are `.png`, `.jpg`, `.jpeg`, `.gif`, `.bmp`, and `.tiff`. The key exam point is simple: PNG, JPEG, and GIF are all **image formats**, and none of them is a document, spreadsheet, or database format.

Definition — Raster image

An image stored as a grid of pixels. PNG and JPEG are raster formats.

55.3 PDF: the Portable Document Format

PDF stands for **Portable Document Format**, a format created by Adobe. Its defining property is a **fixed layout**: a PDF keeps the exact page layout, fonts, and formatting on every device, so the

document looks the same whether it is opened on a desktop, a phone, or a printer. This is why PDF is the standard format for sharing a finished document that must not shift or reflow.

A PDF opens in a **PDF reader**, the best-known being **Adobe Acrobat Reader**. A **web browser** can also display a PDF, but reading a PDF is not the same as editing an editable word-processing document. A PDF may contain real, selectable text, or it may be a **scanned image** of each page. A scanned PDF is only a picture; to make its text searchable and selectable, **OCR** (Optical Character Recognition) must be applied. **PDF/A** is an archival PDF standard meant for long-term preservation.

Example 55.1. *A notice is scanned on a photocopier and saved as a PDF. The PDF now contains pictures of the pages, not typed text. Selecting a sentence with the mouse does nothing, and searching for a word finds nothing. Running OCR on the file creates a text layer, after which the same words can be searched and copied. The format is still PDF; only the internal content has changed from an image to searchable text.*

55.4 The Office document family

Microsoft Office files use a matching family of extensions that are easy to remember as a group.

Extension	Format	Program
`.docx`	Word document	Microsoft Word
`.xlsx`	Excel workbook	Microsoft Excel
`.pptx`	PowerPoint presentation	Microsoft PowerPoint

These are the modern **Office Open XML** formats. The older, pre-2007 extensions were `.doc` (Word), `.xls` (Excel), and `.ppt` (PowerPoint). For an exam, the pairing to hold is: **.docx = Word**, **.xlsx = Excel**, **.pptx = PowerPoint**; each extension always belongs to the same program, in the modern and the older naming alike.

55.5 Plain-data formats: CSV and TXT

CSV stands for **Comma-Separated Values**. A CSV file stores a table as plain text, with the values in each row separated by commas. Because it is only text, a CSV file can be opened by a spreadsheet program such as Excel, by a database, or by a simple text editor. The commas are the field separators; the file itself carries no formatting such as bold text or colours.

TXT denotes a **plain text** file. It stores only characters, with no formatting and no table structure. Both CSV and TXT are simple, vendor-neutral formats: unlike a `.docx` or `.xlsx` file, they do not belong to one program. The distinction to keep is that **CSV holds a table written as text separated by commas**, while **TXT holds plain characters with no inherent structure**.

55.6 Microsoft Access databases: `.mdb` and `.accdb`

Microsoft Access is the database program of the Office family, and it uses two extensions that candidates often confuse. `.mdb` is the **older** Access database format, used by Access 2003 and earlier. `.accdb` is the **newer** Access database format, used from Access 2007 onward. Both files can hold the standard database objects: tables, queries, forms, reports, macros, and modules.

The important point is that **both** `.mdb` and `.accdb` are **Microsoft Access database** files. Neither is a Word document, an Excel workbook, or a PowerPoint presentation.

55.7 Named technology associations

This section is the high-yield part of the lesson: matching a named technology to the organisation that produced it or the purpose it serves.

Firefox is a **web browser** developed by **Mozilla**. Mozilla is the organisation, and Firefox is its browser product; Mozilla also produces **Thunderbird**, an email client. Firefox must not be confused with Microsoft's browser, **Edge** (or the older Internet Explorer), nor with a PDF reader.

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C-DAC stands for the **Centre for Development of Advanced Computing**. It is an Indian research and development organisation under the **Ministry of Electronics and Information Technology (MeitY)**. **PARAM** is a family of **supercomputers** developed by C-DAC; **PARAM 8000** was India's first indigenous supercomputer. The pairing to remember is **PARAM = Indian supercomputer, C-DAC = the organisation that developed it**.

55.8 Software roles: browser, PDF reader, and remote support

Software is often tested by the role it performs rather than by its maker.

Role	Example
Web browser	Firefox, Google Chrome, Microsoft Edge
PDF reader	Adobe Acrobat Reader
Remote-support tool	TeamViewer, AnyDesk, Chrome Remote Desktop

A **web browser** is used to visit web pages; a **PDF reader** is used to open PDF documents; a **remote-support tool** lets one computer control or view another over a network. Firefox is a browser and Adobe Acrobat Reader is a PDF reader; neither is a remote-support tool. Match the software to its *job*, not to a similar-sounding name.

Example 55.1. *A help-desk officer needs to see a distant user's screen to fix a problem. The right tool is a remote-support program such as TeamViewer or AnyDesk,* not *a browser and* not *a PDF reader. If the user then sends a manual as a fixed-layout file, it will arrive as a PDF and open in a PDF reader. If the user sends a spreadsheet of data, it will be an XLSX (Excel) or a CSV. Each task has its own format and its own tool.*

55.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. An **extension** is the letters after the dot; a **format** is how the data is stored. "Extension = format" is wrong (TRM-09, TRAP-34).
2. **PNG, JPEG, and GIF are image formats**. CSV is **not** an image format; it is a plain-text table format.
3. **PDF** is a fixed-layout document format. A scanned PDF is a picture and needs **OCR** to become searchable.
4. **.docx = Word**, **.xlsx = Excel**, **.pptx = PowerPoint**; the extension always matches the program.
5. **.mdb** and **.accdb** are **both Access database** files — older and newer respectively.
6. **Firefox is a browser by Mozilla**; **PARAM is a supercomputer by C-DAC**. Do not swap the makers.

Exercises

1. Distinguish a file *extension* from a file *format*, using `report.png` as the example.
2. Name any three image formats and state one property of PNG.
3. Explain what is meant by a fixed-layout format, and name the format created by Adobe that behaves this way.
4. Match each extension to its program: `.docx`, `.xlsx`, `.pptx`, `.accdb`.
5. What does CSV stand for, and how is its data arranged?
6. (a) Who develops Firefox? (b) What is PARAM, and which organisation develops it?

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Solutions

1. The extension is the suffix after the last dot — here `png`. The format is the way the data inside the file is actually stored — here PNG, the Portable Network Graphics image format. The extension signals the format but is not the format itself.
2. PNG (Portable Network Graphics), JPEG (`.jpg/.jpeg`), and GIF (`.gif`), among others. PNG uses lossless compression and supports transparency.
3. A fixed-layout format keeps the exact page layout, fonts, and formatting on every device, so the document looks the same wherever it is opened. PDF (Portable Document Format) is the format created by Adobe that behaves this way.
4. `.docx` → Microsoft Word; `.xlsx` → Microsoft Excel; `.pptx` → Microsoft PowerPoint; `.accdb` → Microsoft Access (newer database format).
5. CSV stands for Comma-Separated Values. Its data is stored as a table in plain text, with the values in each row separated by commas.
6. (a) Firefox is developed by Mozilla. (b) PARAM is a family of supercomputers, developed by C-DAC (the Centre for Development of Advanced Computing); PARAM 8000 was India's first indigenous supercomputer.

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LESSON 61

Current Office, Email and Statement Evaluation

61.1 Word: structured content and content controls

Most Word documents are free-form, but many office documents — forms, templates, and records — need *structure*: a fixed place for a name, a date, a choice from a list, or a check box. Word provides **content controls** for exactly this task. A content control is an individual element that you insert at a chosen point in a document or template; the user can type or choose only what the control allows.

Definition — Content control

A single, structured element inserted into a Word document or template to capture one piece of information in a controlled way.

Common content controls include plain text, rich text, picture, drop-down list, combo box, date picker, and check box. They are inserted from the **Developer** tab, which is hidden by default and is enabled through Word Options → Customize Ribbon. A content control can restrict editing at that spot and can enforce limited **data validation**: a drop-down list allows only the listed items, and a date picker allows only a date. This is the fact behind the JA 2026 word item (PYQ-12): a content control *can* enforce data validation, so the statement that it cannot is the incorrect one.

Content controls should not be confused with *placeholders*, which belong to PowerPoint slide layouts, or with form *fields* from older Word versions. In this course, "content control" means the Word Developer-tab element that structures a document.

61.2 First-line and hanging indents

Indentation decides how far a paragraph's text sits from the left (or right) margin. Word distinguishes two special paragraph indents.

Indent	Meaning
First-line indent	Only the **first line** of the paragraph is indented; the remaining lines start at the left margin.
Hanging indent	Every line **except the first** is indented; the first line hangs out to the left. Used for references, bibliographies, and bulleted lists.

Definition — Hanging indent

A paragraph indent in which all lines except the first are pushed inward, so the first line "hangs" at the left margin.

These are paragraph-level formatting settings. Two paragraphs can carry different indents even when they use the *same style*, because a paragraph style supplies a default that individual paragraphs can override. The usual wrong statement claims that a hanging indent shifts the first line; in fact the first line is the one that stays out, and all the others move inward.

61.3 The Navigation Pane

The **Navigation Pane** is a Word panel opened from the **View** tab. It is a browsing aid rather than a formatting tool. It presents three views or tabs.

1. **Headings**: an outline of the document built from paragraphs that carry a *Heading style* (Heading 1, Heading 2, and so on).

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2. **Pages**: thumbnail pictures of each page.
3. **Results**: the matches for text typed into the search box.

Because the Headings view is driven by heading styles, applying a Heading style does two things at once: it gives the paragraph an outline level and it makes the paragraph appear in the Navigation Pane (and in an automatic Table of Contents).

Example 61.1. *A student writes a long report. Applying* Heading 1 *to each chapter title makes every chapter appear in the Navigation Pane, so the reader can jump straight to any chapter. A reviewer's* comment *on a sentence does* not *appear in the Navigation Pane; comments are read in the markup/comments pane. That distinction is the trap in JA 2026 (PYQ-12).*

61.4 Track Changes and revision metadata

Track Changes is Word's review tool. When it is switched on, every edit is recorded as a *revision* instead of being applied silently. Three kinds of change are tracked: insertions, deletions, and formatting changes. Each recorded change stores **metadata** — the **author's name** and the **date and time** of the edit.

Definition — Revision metadata with Track Changes

The information Word stores alongside a tracked change: who made it and when.

A tracked change can be **accepted** or **rejected**. Accepting a change keeps the new text as final and **removes the revision metadata** attached to that change; rejecting a change restores the original text. Before sharing a document, Word's *Inspect Document* feature can be used to strip hidden personal information such as author names.

61.5 Section Break versus Page Break

Word documents are divided into **sections**, and page-level settings belong to a section. The two breaks behave quite differently.

Break	What it does	What it changes
Page Break	Moves the following text to the next page.	Nothing else; the document stays in the same section with the same headers, footers, and margins.
Section Break	Starts a **new section**.	Each section may have its own headers and footers, margins, page orientation, columns, and page numbering.

Definition — Section Break

A break that ends the current section and begins a new one, so that page-level formatting can differ from that point onward.

Example 61.1. *A report needs no header on the title page but a running header on every later page. Inserting a single Page Break will not do it, because both pages remain in one section. The correct tool is a* ***Section Break*** *after the title page, after which the new section can carry its own header. This is the exact distinction tested in JA 2026 (PYQ-13): Section Break, not Page Break, allows different headers and footers in one document.*

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61.6 Excel: nested IF, blank and non-numeric input

An **IF** function chooses between two values: `=IF(condition, value_if_true, value_if_false)`. A **nested IF** places a second IF inside one of those two positions, which lets one formula test several conditions in turn:

```
=IF(A1>50,"High",IF(A1>25,"Medium","Low")).
```

Excel evaluates the conditions from left to right and returns the value of the **first** condition that is TRUE; once a condition is satisfied, the remaining tests are skipped.

Example 61.1. *With A1 = 70 the first condition (70 > 50) is TRUE, so the result is* High. *With A1 = 30 the first condition is FALSE but the second (30 > 25) is TRUE, so the result is* Medium. *With A1 = 10 both are FALSE, so the result is* Low.

The behaviour on **blank** and **non-numeric** input is the trap (PYQ-14). In a numeric comparison Excel treats a **blank cell as zero**. Therefore, if A1 is empty, `A1>50` is FALSE and `A1>25` is also FALSE, so the formula falls through to the final value — *Low*. A blank cell does not make the formula stop or produce an error; it is simply read as 0.

Non-numeric **text** is different again. Text is not a number, so:

- arithmetic on it fails — for example `=A1+5` with text in A1 returns the error `#VALUE!`;
- `ISNUMBER` returns FALSE for a text cell and for a blank cell.

For this reason, robust formulas guard the input, for instance with `ISBLANK`, `ISNUMBER`, or `IFERROR`, before using the value in a calculation.

61.7 COUNTIFS and SUMPRODUCT

Counting rows that meet *several* conditions at once is a frequent task. Excel offers two routes.

Definition — COUNTIFS

A function that counts the rows satisfying all *of the given range-and-criteria pairs (AND logic):* `COUNTIFS(range1, criteria1, range2, criteria2, …)`.

The **COUNTIFS** function takes *paired* arguments: a range followed by the criterion that applies to it, repeated for each condition. All the ranges must be the same size. The singular **COUNTIF** handles one condition; the plural **COUNTIFS** handles several. Text criteria can use the wildcards `*` (any sequence of characters) and `?` (any one character).

Definition — SUMPRODUCT

A function that multiplies corresponding elements of the given arrays and returns the sum of those products.

On its own, `SUMPRODUCT(A1:A3,B1:B3)` returns `A1*B1 + A2*B2 + A3*B3`. It can also be used to *count*, by building boolean arrays and multiplying them:

```
=SUMPRODUCT((A1:A10>50)*(B1:B10="Yes"))
```

counts the rows in which the first condition AND the second condition are both true, because the product is 1 only when both parts are 1. The double-unary form `=SUMPRODUCT(--(A1:A10>50))` is a common way to coerce TRUE/FALSE into 1/0.

Example 61.1. *To count how many students scored above 50* and *belong to the Science stream, either* `=COUNTIFS(Marks,">50",Stream,"Science")` *or* `=SUMPRODUCT((Marks>50)*(Stream="Science"))` *will work. The stable distinction tested in JA 2026 (PYQ-15, TRAP-14) is that COUNTIFS takes several criteria directly, while SUMPRODUCT does arithmetic on arrays and can therefore both* add *and* count *— it does not "only add."*

61.8 Email: filters, signatures, POP3 and privacy

Filters and rules

An email **filter**, usually called a **rule** in desktop mail clients, acts automatically on messages. A rule has two parts: a **condition** (sender, recipient, subject, keywords, presence of an attachment) and an **action** (move to a folder, flag, forward, delete, mark as read). A typical rule sends all mail from one mailing list into a dedicated folder, keeping the inbox tidy.

Digital signature versus encryption

These two security features are constantly confused (PYQ-16, TRAP-15).

Point	Digital signature	Encryption
Purpose	Authenticity and integrity	Confidentiality
What it establishes	Who sent the message, and that it was not altered in transit	That only the intended reader can read the contents
Does it hide the content?	No	Yes

A **digital signature** does *not* encrypt the message body. It proves the sender's identity and detects tampering. It is **encryption** that scrambles the content so that only the holder of the correct key can read it.

POP3 and IMAP

POP3 stands for **Post Office Protocol version 3**. It is a mail *retrieval* protocol: it **downloads** messages from the mail server to the local device. By default it typically **removes** the messages from the server, so once downloaded the mail lives only on that machine and is not kept in sync across devices. **IMAP** works differently: it leaves the mail on the server and keeps all devices synchronised. The stable distinction (PYQ-16, TRAP-17) is that POP3 downloads to the local machine, whereas IMAP syncs. **SMTP**, in contrast, is the protocol for *sending* mail.

BCC and recipient privacy

An email has three recipient fields.

Field	Full form	Visibility
To	—	Primary recipients; visible to everyone.
Cc	Carbon Copy	Copied recipients; visible to everyone.
Bcc	Blind Carbon Copy	Addresses hidden from the *other* recipients.

The key point (PYQ-16, TRAP-16) is what Bcc hides. Bcc hides the blind recipients from *each other* and from the To/Cc recipients; it does *not* hide them from the **sender**, who always sees the complete recipient list. A common wrong statement says that Bcc hides the sender and recipients from one another — the sender is never hidden.

61.9 PowerPoint SmartArt conversion

SmartArt turns a plain list into a diagram. PowerPoint can convert content in both directions, with one firm limit.

- **Text to SmartArt**: select a bulleted list and choose **Home** → Convert to SmartArt Graphic; choose a layout. The text is preserved as the diagram's content.
- **SmartArt to text**: with the graphic selected, use SmartArt Tools Design → Convert → Convert to Text. You get a bulleted list back; the shapes and any pictures in the graphic are *lost*.

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- **SmartArt to shapes**: Convert to Shapes breaks the graphic into ordinary shapes. Once that is done, the shapes **cannot** be converted back into SmartArt.

Example 61.1. *A presenter has six bullet points and wants a cycle diagram. She selects the list and uses* Convert to SmartArt. *Later she needs the plain list again, so she uses* Convert to Text *and the six points return as bullets. What cannot be undone is the step to individual shapes: after* Convert to Shapes*, rebuilding the SmartArt would mean starting over (the "re-entry" point behind PYQ-17).*

The direct source of Convert to SmartArt is *text*. A picture or a chart is not transformed into SmartArt in one step; you would instead insert a picture-type SmartArt layout and place the images by hand.

61.10 Exam retrieval and common confusions

The following distinctions prevent most errors in this unit.

1. Different headers and footers require a **Section Break**, not a Page Break (TRAP-13).
2. A **hanging indent** shifts every line except the first; a first-line indent shifts only the first line.
3. A **blank cell** is treated as 0 in a comparison; it does not cause an error (PYQ-14).
4. **COUNTIFS** counts with AND logic over paired arguments; **SUMPRODUCT** multiplies arrays and can both add and count (TRAP-14).
5. A **digital signature** gives authenticity and integrity, not confidentiality; **encryption** gives confidentiality (TRAP-15).
6. **POP3 downloads** to the local machine; **IMAP** syncs (TRAP-17).
7. **Bcc** hides recipients from each other, never from the sender (TRAP-16).
8. **Navigation Pane** shows headings, pages, and search results, not comments.

Example 61.1. *An item states, "A digital signature encrypts the message so that only the recipient can read it." This is wrong. The signature establishes the sender and integrity; encryption is the feature that hides the content. The wrong option has merged two distinct security services.*

Exercises

1. Distinguish a first-line indent from a hanging indent, with one use for each.
2. State why a Section Break, and not a Page Break, is needed for different headers and footers.
3. Explain what a nested IF returns when several conditions are true at once.
4. What does `=IF(A1>50,"High","Low")` display when A1 is blank, and why?
5. Give the general form of COUNTIFS, and state one thing SUMPRODUCT can do that COUNTIFS cannot.
6. Distinguish a digital signature from encryption in one sentence.
7. What does POP3 do with mail, and how does that differ from IMAP?
8. In a message, who can see the addresses placed in the Bcc field?
9. Describe two directions of SmartArt conversion and one that cannot be reversed.

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Solutions

1. A first-line indent indents only the first line (normal paragraph openings); a hanging indent indents every line except the first (references and bulleted lists).
2. Page-level settings such as headers and footers belong to a section. A Page Break keeps the same section, so its header/footer cannot differ; a Section Break starts a new section that can carry its own header and footer.
3. A nested IF returns the value of the *first* condition that is TRUE; once a condition is satisfied the remaining tests are skipped.
4. It displays *Low*, because a blank cell is treated as 0, and 0 is not greater than 50, so the FALSE branch is taken.
5. `COUNTIFS(range1, criteria1, range2, criteria2, …)`. SUMPRODUCT can multiply corresponding array elements and return their sum (arithmetic), so it can add as well as count; COUNTIFS only counts.
6. A digital signature proves the sender's identity and that the message was not altered, whereas encryption hides the content so only the intended reader can read it.
7. POP3 downloads mail from the server to the local device and by default removes it from the server; IMAP leaves mail on the server and keeps devices synchronised.
8. The sender can see all Bcc addresses, and each Bcc recipient is hidden from the other recipients (To, Cc, and the other Bcc recipients).
9. Text converts to SmartArt (Convert to SmartArt), and SmartArt converts back to text (Convert to Text). Converting SmartArt to shapes is *not* reversible — the shapes cannot be turned back into SmartArt.

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UNIT 8

Internet, Web & Email

LESSON 34

Internet, WWW and Browser

34.1 The Internet: the network itself

The Internet is a global network of connected computers and networks. It is best described as a **network of networks**: smaller networks joined together so that any connected device can reach any other. It carries many services — the World Wide Web, e-mail, file transfer, and more. The Internet is the physical infrastructure underneath all of these: the cables, routers, servers, and the devices connected to them.

Definition — Internet

The global network of interconnected computers and networks over which services such as the Web and e-mail run.

Term	Meaning in this lesson
Internet	The global network of networks; the infrastructure.
WWW (World Wide Web)	A service on the Internet: linked web pages and sites.
Website	A collection of related webpages under one address.
Webpage	A single document on the Web, shown in a browser.
Homepage	The main or starting page of a website.
Browser	Software that opens and displays websites and webpages.
Search engine	A website that finds other pages matching typed keywords.
Client	The device or program that requests a service or resource.
Server	The computer or program that supplies services to clients.
ISP (Internet Service Provider)	The company or organisation that provides the Internet connection.
Hyperlink	Clickable text or image that jumps to another page or section.
URL (Uniform Resource Locator)	The address of a webpage or resource on the Web.

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The rest of this lesson uses these terms in one consistent sense. The **Internet** is the network. The **WWW** is the linked-pages service running on that network. A **webpage** is one document, a **website** is the collection, and the **homepage** is the collection's front door. A **browser** opens pages; a **search engine** finds pages. A **client** requests and a **server** supplies. An **ISP** provides the connection itself. A **hyperlink** points to another page, and a **URL** is a page's address.

34.2 Internet vs WWW

The single most tested confusion in this lesson is Internet versus WWW. The Internet is the whole global network; the WWW — World Wide Web — is one service that runs on the Internet, namely the system of linked web pages and sites. E-mail is another service on the same Internet, and it is not part of the Web. Access follows the same layering: a device reaches the Internet through an ISP connection, and it reaches the Web through a browser running over that Internet connection.

Definition — World Wide Web (WWW)

The system of linked web pages and websites accessed over the Internet through a browser; one service among several that the Internet carries.

Example 34.1. *An item states, "The Internet and the World Wide Web are the same thing." This is wrong. The Internet is the global network of networks, while the WWW is one service running on it. The wrong option has collapsed the infrastructure and one of its services into a single term.*

34.3 Website, webpage, and homepage

A **webpage** is a single document on the Web, shown inside a browser. A **website** is a collection of related webpages grouped under one address (domain). The **homepage** is the main or starting page of a website — the page that loads when the site's base address is typed. In order of size: the homepage is one page, any single document is a webpage, and the whole collection is the website.

Definition — Website, webpage, homepage

A webpage is one document on the Web; a website is the collection of related webpages under one address; the homepage is the website's main or starting page.

Each webpage of a website has its own URL, and the homepage URL is the site's base address. Typing that address in the browser's address bar loads the homepage first, and hyperlinks on it lead to the site's remaining pages.

34.4 Browser

A **browser** is software used to open and view websites and webpages. It sends a request for a page and displays the page that comes back. Common browsers are Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari. A browser is software installed on the user's device; it is not the Internet itself, and it is not a search engine.

Definition — Browser

Software that requests webpages and displays them; examples are Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.

34.5 Search engine

A **search engine** is a website (a service) that finds other webpages matching keywords typed by the user. Examples are Google, Bing, and Yahoo. It is reached *through* a browser: the browser is the tool open on the device, and the search-engine site is loaded inside it. Searching therefore needs both layers — a browser open on the device, and a search-engine site loaded in that browser.

Definition — Search engine

A website that returns the pages matching a user's keywords; examples are Google, Bing, and Yahoo.

The browser-versus-search-engine table is worth memorising as a whole. A browser is software (an application) whose role is to open and display pages; a search engine is a website whose role is to find pages for keywords. Chrome, Firefox, Edge, and Safari belong in the first column; Google, Bing, and Yahoo belong in the second. The classic trap swaps them: it calls Chrome a search engine or Google a browser.

34.6 Client and server

A **client** is the device or program that requests a service or resource. The office PC's browser asking for a webpage is acting as a client: it asks, and it does not supply the page itself. A **server** is the computer (or program) that supplies services or resources to clients. A web server stores websites and sends pages to the browsers that ask for them. One server can serve many clients at the same time, and servers stay switched on and connected so that clients can reach them at any time.

Definition — Client and server

The client requests a service or resource; the server supplies it. On the Web, the browser is the client and the web server holds the pages.

The two always appear as a pair, in three steps. First, the client (the browser) sends a request for a page. Second, the server receives the request and sends the page back. Third, the browser displays the received page. This request-and-response pair is the client-server model of the Web: typing a site address and pressing Enter sends a request, the site's server returns the homepage, and the browser shows it.

34.7 ISP: the connection provider

ISP stands for Internet Service Provider. It is the company or organisation that provides the Internet connection to homes and offices. Without an ISP connection, the device cannot reach the Internet at all, so the ISP link comes before everything else: no ISP means no browsing, no search, and no e-mail. This is the tested fact of WO 2026 Q74 (PYQ-34): ISP expands to Internet Service Provider, and no other expansion is accepted.

Definition — ISP (Internet Service Provider)

The company or organisation that provides the Internet connection to homes and offices.

34.8 Hyperlink and URL

A **hyperlink** (link) is clickable text or an image that jumps to another page or section. Clicking it loads the linked webpage, which may sit on the same site or on a different one. Links are what make the Web a "web": pages joined to pages. Note the division of labour: a hyperlink points the way, and the browser follows it when clicked.

A **URL** — Uniform Resource Locator — is the address of a webpage or resource on the Web. Typing the site's URL in the browser's address bar loads its homepage. The URL names the page; the hyperlink jumps to it.

Definition — Hyperlink and URL

A hyperlink is clickable text or an image that leads to another page; a URL (Uniform Resource Locator) is the address of a webpage or resource.

34.9 Full-form ladder of this lesson

Direct-recall full-form items are among the most common JKSSB computer formats, so learn each expansion exactly and give the full form on first mention. **ISP** — Internet Service Provider — gives the Internet connection. **WWW** — World Wide Web — is the linked system of pages and sites. **URL** — Uniform Resource Locator — is the address of a page or resource. **HTTP** — HyperText Transfer Protocol — is the set of rules browsers and servers follow when requesting and delivering pages. **HTML** — HyperText Markup Language — is the language in which webpages are written.

34.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The Internet is **not** the Web; the WWW is **one service** running on the Internet.
2. A browser is **not** a search engine. Chrome is a **browser**; Google is a **search engine**.
3. A website is the **collection**; a webpage is **one document**; the homepage is the **first page**.
4. The client **requests** and the server **supplies**. The ISP — Internet Service Provider — gives the connection (PYQ-34).

5. A hyperlink **points**; the browser **opens**. A URL is the page's address.

Example 34.1. *An item asks which software is used to view websites. Trace the roles: the browser opens and displays pages, while the search engine finds pages and the ISP provides the connection. The answer is the browser (for example Chrome). If the stem instead asks which site finds pages for keywords, the answer is the search engine (for example Google), and if it asks who provides the connection, the answer is the ISP.*

Exercises

1. Distinguish the Internet from the World Wide Web.
2. Distinguish a website from a webpage and from a homepage.
3. Distinguish a browser from a search engine, giving two examples of each.
4. Explain the roles of client, server, and ISP when a page is opened.
5. Define hyperlink and URL, and explain how the two work together.

Solutions

1. The Internet is the global network of networks — the infrastructure. The WWW (World Wide Web) is one service running on that infrastructure: the system of linked web pages and sites.
2. A webpage is one document on the Web; a website is the collection of related webpages under one address; the homepage is the website's main or starting page.
3. A browser is software that opens and displays pages (Chrome, Firefox, Edge, Safari). A search engine is a website that finds pages for keywords (Google, Bing, Yahoo); it is reached through a browser.
4. The client (browser) requests the page, the server (web server) supplies it, and the ISP (Internet Service Provider) provides the Internet connection over which the request and response travel.
5. A hyperlink is clickable text or an image that jumps to another page; a URL (Uniform Resource Locator) is a page's address. Clicking the hyperlink tells the browser to load the page named by that URL.

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LESSON 35

URL, DNS, HTTP and HTTPS

35.1 What a URL is

A URL (Uniform Resource Locator) is the full address of one resource on the Internet: one page, file, image, or service. Typing a URL tells the browser both how to connect and exactly what to fetch.

Definition — URL

The complete address of a single Internet resource, naming the protocol (scheme), the host, and the location of the resource on that host, including the optional port, path, and query.

Term	Meaning in this lesson
URL	The full address of one resource, such as a page or file.
Scheme	The protocol word before the colon, such as https or http.
Domain name	The readable host name, such as www.example.in.
TLD	Top-Level Domain: the last label, such as .in or .com.
Subdomain	A prefix section of the name, such as www or mail.
Port	The door number on the server (80 for HTTP, 443 for HTTPS).
Path	The location of the page inside the site.
Query	Parameters after a ?, such as ?topic=dns.
IP address	The numeric address of a device (IPv4 is 32-bit, IPv6 is 128-bit).
DNS	Domain Name System: translates domain names into IP addresses.
HTTP	Hypertext Transfer Protocol: carries web pages.
HTTPS	Hypertext Transfer Protocol Secure: HTTP protected by TLS.
TLS	Transport Layer Security: encrypts the exchange (successor of SSL).

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The rest of this lesson uses these terms in one consistent sense. A **URL** is the whole address; the **domain name** is only the host part inside it. **DNS** translates that host name into an **IP address**. **HTTP** moves the page, and **HTTPS** is HTTP carried inside a **TLS**-encrypted channel.

35.2 Anatomy of a URL

Consider the example used throughout this lesson:

```
https://www.example.in:443/study/computer?topic=dns&lang=en
```

Read it left to right. The **scheme** is `https`: the protocol the browser must speak. The `://` is only a separator between the scheme and the host. The host is `www.example.in`, in which `www` is the **subdomain**, `example` is the registered domain, and `.in` is the **TLD** (Top-Level Domain). Common TLDs include `.com`, `.org`, `.edu`, `.gov`, and country labels such as `.in`.

After the host comes the optional **port**, `:443`: the door number on the server. The **path**, `/study/computer`, locates the page inside the site. The **query**, `?topic=dns&lang=en`, passes extra parameters to the server. Port and query are optional: when the port is missing, the browser fills in the default for the scheme.

Definition — TLD (Top-Level Domain)

The last label of a domain name, such as .com, .org, .edu, .gov, or a country label such as .in.

Example 35.1. *Split `mail.example.com`. The subdomain is `mail`, the registered domain is `example`, and the TLD is `.com`. One registered domain can carry many such subdomains (`www`, `mail`, `blog`), each pointing to a different service, while the domain itself stays the same.*

35.3 Ports, paths, and queries

The **port** selects which service on the server receives the connection. Port **80** is the default for HTTP and port **443** is the default for HTTPS, which is why both numbers are usually hidden in the address bar. The **path** names the page or file within the site, and the **query** carries name–value pairs after a `?` mark. Neither the port nor the query changes which host is contacted; the host is fixed by the domain name alone.

35.4 IP addresses: the real destination

An IP (Internet Protocol) address is the numeric address every device uses to send and receive data on the network. People type names; machines route by numbers.

Definition — IP address

The numeric address of a device on a TCP/IP network, used for routing data to that device.

IPv4 (Internet Protocol version 4) is **32-bit** and is written as four decimal groups, for example `192.0.2.44`. **IPv6** (Internet Protocol version 6) is **128-bit** and is written in hexadecimal groups. It is never 256-bit; that figure is a repeated wrong option. IPv6 exists because the IPv4 space is too small for the number of connected devices.

35.5 DNS: the phone book of the Internet

DNS (Domain Name System) is the distributed naming system that translates domain names into IP addresses. It does not carry page content and does not assign addresses to devices; it only answers the question "which IP address belongs to this name?"

Definition — DNS

The hierarchical, distributed system that maps each domain name to the IP address of its server, with answers cached to speed up repeat visits.

Resolution runs as a short chain. The browser first checks its own cache for the name. If the name is unknown, the **resolver** (usually run by the ISP — Internet Service Provider) asks the DNS hierarchy: the root servers, then the TLD servers for labels such as `.in`, then the domain's own authoritative server. That authoritative server returns the IP address, the resolver caches the answer for a fixed time, and the browser receives the number it can connect to.

Example 35.1. *On the office desktop, typing `www.example.in` first triggers a DNS lookup. The resolver walks the hierarchy, learns the address (say `192.0.2.44`), caches it, and only then does the browser open a connection to that number. Typing the IP address directly would skip DNS, but nobody memorises numbers for every site.*

35.6 HTTP: carrying web pages

HTTP (Hypertext Transfer Protocol) is the request–response protocol of the web. The browser sends a request naming the path (and query); the server returns the page. HTTP runs by default over port 80. Plain HTTP provides no encryption: the request and the page travel in readable form, so anyone on the path between browser and server can read them.

Definition — HTTP

The protocol by which a web client requests a resource and the server returns it. Plain HTTP does not encrypt the exchange.

35.7 HTTPS and TLS: the secure form

HTTPS (Hypertext Transfer Protocol Secure) is HTTP protected by TLS. TLS (Transport Layer Security) is the successor of the older SSL (Secure Sockets Layer); the phrase "SSL/TLS" survives only as a name for the same family of protection. Before any page moves, the browser and server run a TLS handshake: the server proves its identity with a certificate, and the two sides agree on encryption keys. The HTTP request and response then travel inside that encrypted channel.

Definition — HTTPS

HTTP carried inside a TLS-encrypted channel, by default over port 443 and shown with a padlock in the address bar.

TLS provides three things at once: **confidentiality** through encryption (only the two ends can read the data), **integrity** (tampering is detected), and server **authentication** (the certificate ties the site to its claimed name). HTTPS therefore protects the trip; it is not a different page language.

35.8 The full flow: from typed URL to secure page

Put the pieces together for `https://www.example.in/study/computer`. The browser parses the URL: scheme `https`, host `www.example.in`, path `/study/computer`. DNS resolves the host to an IP address. The browser opens a connection to that IP on port 443, the HTTPS default. The TLS handshake secures the channel, and the HTTP request for the path travels inside it. The server returns the page, and the browser renders it with the padlock shown.

35.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A **URL** addresses one resource; the **domain name** is only the host part inside it.
2. **DNS** translates names into IP addresses. It does not assign addresses and does not carry page content.
3. **IPv4** is 32-bit; **IPv6** is 128-bit, never 256-bit.
4. **HTTP** uses port 80; **HTTPS** uses port 443. Do not reverse them.
5. **HTTPS** is HTTP over **TLS**. HTTP alone gives no encryption.
6. The **scheme** is the protocol word (`https`), not the domain name that follows it.

Example 35.1. *An item asks which part of `https://mail.example.com/inbox` is the subdomain. Trace the host `mail.example.com` from the right: the TLD is `.com`, the domain is `example`, and the remaining prefix `mail` is the subdomain. If the stem instead asks for the scheme, the answer is `https`; if it asks for the default port of that scheme, the answer is 443.*

Exercises

1. Expand URL, DNS, HTTP, HTTPS, TLS, IP, TLD, and ISP.
2. Label every part of `https://blog.example.org:443/notes/net?ch=1`.
3. Distinguish a URL from a domain name.
4. Describe the steps of DNS resolution when a name is not cached.
5. Contrast HTTP with HTTPS on security, default port, and address-bar form.

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Solutions

1. URL — Uniform Resource Locator; DNS — Domain Name System; HTTP — Hypertext Transfer Protocol; HTTPS — Hypertext Transfer Protocol Secure; TLS — Transport Layer Security; IP — Internet Protocol; TLD — Top-Level Domain; ISP — Internet Service Provider.
2. Scheme `https`; subdomain `blog`; domain `example`; TLD `.org`; port `443`; path `/notes/net`; query `?ch=1`.
3. A URL is the full address of one resource (scheme, host, port, path, query). A domain name is only the readable host part inside it.
4. The browser checks its cache; then the resolver asks the DNS hierarchy (root, then TLD, then the authoritative server), which returns the IP address; the resolver caches it and hands it to the browser.
5. HTTP carries pages without encryption over port 80 (`http://`); HTTPS is HTTP protected by TLS over port 443 (`https://` with a padlock).

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LESSON 36

Web Browsing, Search, Upload and Download

36.1 The browser, tabs, bookmarks, and history

A **web browser** is the software used to open and view web pages on the internet. Common examples are Google Chrome, Mozilla Firefox, and Microsoft Edge. The browser sends a request for a page and then displays the returned text, images, and links. The address typed to reach a page is the **URL** — Uniform Resource Locator — and it is typed in the **address bar** at the top of the browser window.

Definition — Web browser

Software that requests web pages from servers and displays them to the user.

Term	Meaning in this lesson
Web browser	Software (Chrome, Firefox, Edge) that opens and displays web pages.
Tab	One open page inside the browser window; many tabs can stay open together.
Bookmark	A saved shortcut to a page for quick return without retyping the address.
Browsing history	The date-wise list of pages visited earlier, kept by the browser.
URL	Uniform Resource Locator: the address of a page, typed in the address bar.
Search engine	A website (Google, Bing) that finds pages matching typed words.

Each open page lives in its own **tab**, so several pages can stay open in one window. A new tab is opened with **Ctrl+T**. A **bookmark** is a saved shortcut to a page: instead of typing the address again, the saved bookmark is clicked. It is created with **Ctrl+D**. **Browsing history** is the list of pages visited earlier, stored by date, and it is opened with **Ctrl+H**. The distinction is simple: history answers "where was I", while a bookmark answers "where do I want to return".

Example 36.1. *While preparing a report on the office desktop, the clerk keeps the office portal open in one tab and the search results in a second tab, and bookmarks the portal page so it can be opened directly the next morning. The history list that evening shows both pages in order.*

36.2 How a search engine works

A **search engine** is a website, such as Google or Bing, that finds pages matching the words typed by the user. It works in three ordered stages: crawling, indexing, and ranking.

Definition — Crawling

The collection step: automatic programs called crawlers or spiders visit pages and follow their links to discover new and changed pages.

Definition — Indexing

The storage step: the crawled pages are kept in a large index, like the index at the back of a book, so matches can be found quickly.

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Definition — Ranking

The ordering step: when words are typed, matching pages are picked from the index and ordered by relevance on the search results page.

Crawling comes first because without it the engine would never learn that a new page exists. Indexing comes second: the collected pages are stored in the index before any search is typed. Ranking comes last: only after the user types words does the engine select matches from that ready index and order them. The anchor to remember is the order itself: **crawl, then index, then rank**.

36.3 Sponsored results and organic results

The matches chosen by relevance are called **organic results**. In addition, the results page shows **sponsored results**: paid placements that appear because an advertiser paid for them, not because they ranked highest on relevance. A sponsored result always carries a visible **Ad** or **Sponsored** label. The practical consequence is direct: being listed first on the page does not always mean being the most relevant match, since the top position may be a paid placement.

Example 36.1. *A search for a printer shows two blocks: the first carries a small "Ad" label and links to a seller who paid for the position, while the entries below it are the organic results ordered by relevance. The topmost entry is therefore not necessarily the best match.*

36.4 Cookies, cache, and history

These three stores are the most confused triple in this lesson. Each stores a different thing, and each is stored by a different party.

Definition — Cookie

A small file a website stores on the computer to remember that visitor: login state, cart items, and preferences such as language.

Definition — Browser cache

Copies of page parts (images, styles, scripts) kept by the browser on the computer so repeat visits load faster.

Definition — Browsing history

The list of visited page addresses, kept by the browser by date.

Item	What it stores	Who stores it
Cookie	Visitor data: login, cart, choices	The website
Cache	Copies of page files: images, styles	The browser
History	List of visited page addresses	The browser

Cookies are set by websites and read back on the next visit, which is why a user stays logged in. The cache is managed by the browser: on the next visit the saved copies load from the disk instead of being fetched again, so the page opens faster. Clearing them has different effects. Deleting **cookies** signs the user out of sites and clears saved preferences. Clearing the **cache** does not sign the user out; it only makes the next visit slightly slower. Clearing **history** removes the record of visited addresses but changes neither logins nor saved page copies.

36.5 Downloading and uploading

Downloading means copying a file from the internet to the user's computer: saving a form PDF from the office portal, or saving a photo from a web page. **Uploading** means sending a file from the user's computer to the internet: attaching a document to an email, or submitting a scanned form on a portal. The direction is always judged from the user's machine: download moves data **down** to the machine, upload moves data **up** from the machine to a server.

Definition — Downloading

Copying a file from the internet (a server) to the user's computer.

Definition — Uploading

Sending a file from the user's computer to the internet (a server).

Point	Download	Upload
Direction	Server to computer	Computer to server
Typical use	Opening pages, saving files	Mail attachments, form submission
Typical speed	Faster	Slower

Home and office connections usually provide a faster download speed than upload speed, because receiving pages and files is the more common need. A downloaded file, once saved, opens without needing the internet; an upload needs the connection at the moment of sending.

36.6 File conversion

File conversion means changing a file from one format to another while keeping the same content. The common office need is converting a Word document (**.docx**) to **PDF** before submitting it on a portal. Conversion changes the **format**, not the transfer direction: a converted file must still be uploaded or downloaded separately. The **extension** (such as .pdf, .docx, .jpg) is the name tag that says which format the file is stored in, while the **format** (such as PDF, PNG) is the storage method itself. The two words must not be conflated: the extension names the format; it is not the format.

Example 36.1. *The clerk finishes the report in Word, converts the .docx file to PDF, and then uploads the PDF on the portal. The conversion changed the format; the upload changed the location. If the portal instead needed a photograph, a .JPG image would be converted to the needed picture format the same way.*

36.7 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The search order is **crawl**, then **index**, then **rank**. The index is built before the search is typed.
2. **Sponsored** means a paid placement with an Ad label; **organic** means ranked by relevance. The top of the page is not always the best match.
3. **Cookie** (website; visitor data) vs **cache** (browser; page copies) vs **history** (browser; visited list). Do not swap the three.
4. **Download** means internet to computer; **upload** means computer to internet. Judge the direction from the user's machine.
5. **Conversion** changes the format; **transfer** changes the location. The extension (.png) names the format (PNG); it is not the format itself.

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Exercises

1. List the three stages of a search engine in order and say what each stage does.
2. Distinguish a sponsored result from an organic result.
3. Distinguish a cookie from the browser cache and from browsing history.
4. Explain the difference between downloading and uploading, giving one office example of each.
5. What does file conversion change, and how is the extension related to the format?

Solutions

1. Crawling (crawlers visit pages and follow links to collect them), then indexing (pages stored in a large index), then ranking (matches picked from the index and ordered by relevance after words are typed).
2. A sponsored result is a paid placement carrying an Ad or Sponsored label; an organic result is a normal match chosen and ordered by relevance.
3. A cookie is visitor data (login, cart, choices) stored by the website; the cache is copies of page files (images, styles) stored by the browser for faster repeat visits; history is the browser's date-wise list of visited addresses.
4. Downloading copies a file from the internet to the computer, such as saving a form PDF from the portal; uploading sends a file from the computer to the internet, such as attaching a document to an email.
5. Conversion changes the format while keeping the content (for example .docx to PDF). The extension (.pdf, .docx) is the name tag for the format; the format is the storage method itself.

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LESSON 37

Email Basics

37.1 What email is

Electronic mail (email) is a message sent from one electronic mailbox to another over a network such as the Internet. Each user owns a mailbox identified by a unique email address. The sender composes a message, directs it with the recipient fields, and sends it; the message then waits in each recipient's mailbox until it is opened. Because delivery does not require both parties to be present at the same time, email is called a store-and-forward service.

Definition — Email address

A unique identifier of a mailbox, written as a user name, then the @ symbol, then the domain name, for example `assistant@gmail.com`.

Term	Meaning in this lesson
User name	The part of the address left of the @; the mailbox on the server.
Domain name	The part of the address right of the @; the mail server holding the mailbox.
To / Cc / Bcc	Recipient fields: primary recipient, visible copy, hidden copy.
Subject / body	The topic line and the main written content.
Attachment	A file sent along with the message.
Inbox / Sent / Drafts	Folders for received, sent, and unfinished mail.
Spam / Trash	Folders for filtered junk and for deleted messages.
Webmail / email client	Mail used in a browser vs mail used in an installed app.

The rest of this lesson uses these terms in one consistent sense. The **user name** is whatever stands left of the @ and the **domain name** is whatever stands right of it. **To** is the main recipient, **Cc** (Carbon Copy) is a visible additional recipient, and **Bcc** (Blind Carbon Copy) is a hidden additional recipient. **Subject** is the topic line, **body** is the message text, and an **attachment** is a separate file carried with the message.

37.2 The email address

Every email address contains exactly one @ symbol, read as "at". The @ separates the user name on the left from the domain name on the right. In `assistant@gmail.com`, `assistant` is the user name and `gmail.com` is the domain name. The domain name identifies which mail server holds the mailbox; the user name identifies which mailbox on that server. This is the point tested when an item asks what the @ separates: the answer is the user name and the domain name (PYQ-35).

Definition — Domain name

The part of an email address right of the @ that identifies the mail server or organisation holding the mailbox, such as `gmail.com`.

Addresses are written without spaces, and the @ appears exactly once. Anything before it belongs to the user name; anything after it belongs to the domain. Options that say the @ separates the domain and the host, or two parts of the domain, are therefore wrong.

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37.3 Recipient fields: To, Cc and Bcc

The three recipient fields decide who receives the message and who can see whom. **To** names the primary recipient, the person the message is mainly for. **Cc** (Carbon Copy) names additional recipients who receive a visible copy; every recipient can see the To and Cc lists. **Bcc** (Blind Carbon Copy) names additional recipients whose addresses are hidden from the other recipients.

Definition — Bcc

A recipient field whose addresses are hidden from the other recipients of the same message; each Bcc recipient is concealed from the To, Cc, and fellow Bcc recipients.

The word "blind" refers to this concealment. A Bcc recipient still receives the message normally. The other recipients simply cannot see that address. The sender, who composed the message, always knows the full list. To and Cc addresses, by contrast, are visible to everyone who receives the message. This is the distinction behind the common trap: Bcc hides recipients from each other, not from the sender, and it does not hide the To or Cc lists from anyone (TRAP-16).

Example 37.1. *The officer sends a notice To the assistant and Cc the clerk, with the head office on Bcc. The assistant and the clerk see each other's addresses but cannot see the head office address. The head office receives the same notice. The sender (the officer) knows all three addresses.*

For the examination, read every Bcc option word by word. An option that says Bcc hides the sender is wrong; an option that says Bcc hides the To list from the recipients is wrong; the correct statement is that Bcc recipients are hidden from each other while To and Cc remain visible to all.

37.4 Subject, body and attachments

A message has two content parts and an optional file part. The **subject** is a single short line stating the topic, such as "Leave application for 12 October". It is displayed in the message list before the message is opened, so a clear subject helps the recipient identify the message. The **body** is the main written content: the greeting, the message itself, and the closing.

An **attachment** is a file sent along with the message, usually shown with a paperclip icon. Common attachments are documents, spreadsheets, presentations, PDFs, and images. Attaching a file sends a copy to the recipient; the sender keeps the original. Two practical points matter for the examination. First, mail servers impose a size limit, so a very large file is rejected or must be shared as a link instead. Second, an attachment can carry a virus, so an unexpected attachment should not be opened blindly.

37.5 Folders: Inbox, Sent, Drafts, Spam, Trash

Received mail arrives in the **Inbox**. Every account has exactly one Inbox; it is the default view when the mailbox is opened. A copy of every message that is sent is kept in the **Sent** folder (sometimes labelled Sent Items or Sent Mail). A message that has been started but not yet sent can be saved in **Drafts**; it waits there until it is completed and sent.

Definition — Drafts

The folder holding composed messages that have been saved but not yet sent.

Unwanted or bulk mail is separated into the **Spam** folder (also called Junk). The mail service filters suspected junk out of the Inbox automatically; a wanted message found there should be moved back or marked "not spam". Messages the user deletes go to the **Trash** folder (also called Deleted Items). They remain recoverable until the folder is emptied; emptying Trash or Spam permanently removes those messages.

The five folders are most easily confused in pairs. Inbox receives while Sent keeps sent copies; Drafts holds work not yet sent while Sent holds work already sent; Spam holds filtered junk while

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Trash holds what the user deleted. No other folder does another's job: drafts never send themselves, sent copies never return to drafts, and spam is not the delete bin.

37.6 Reply, Reply All and Forward

Three actions dispose of a received message. **Reply** composes an answer addressed only to the sender. **Reply All** composes an answer addressed to the sender plus all the To and Cc recipients of the original message. **Forward** sends the received message onward to new recipients chosen by the forwarder, usually with a short added note on top.

Definition — Reply All

A response addressed to the sender and to every visible (To and Cc) recipient of the original message; Bcc recipients of the original are never re-added.

Two consequences follow. First, Reply All can only reach the addresses visible on the original message, so hidden Bcc addresses are never included. Second, a forwarded message carries the original content (and often its attachments) inside the new message; the forwarder becomes the new sender. For the examination, match the verb to the action: "answers only the sender" is Reply; "answers everyone visible" is Reply All; "sends it onward to new recipients" is Forward.

37.7 Webmail versus an email client

Email can be used in two ways. **Webmail** means using mail through a web browser, with the mailbox kept on the provider's server; Gmail, Yahoo Mail, and Outlook.com are webmail services. An **email client** is an installed application used for mail, such as Microsoft Outlook or Mozilla Thunderbird; it can keep or download copies on the local machine.

Closely tied to the client are the incoming-mail protocols. **POP3** (Post Office Protocol version 3) downloads mail to the local machine, usually removing it from the server. **IMAP** (Internet Message Access Protocol) keeps mail synced on the server, so a phone, an office computer, and a laptop all show the same folders. Outgoing mail in both arrangements is normally sent with **SMTP** (Simple Mail Transfer Protocol). The one-line contrast to remember is: POP3 downloads, IMAP syncs, SMTP sends.

Example 37.1. *The assistant reads office mail in two places: through Gmail in the browser (webmail) and through the Outlook app on the office computer (email client) configured with IMAP, so a message read on one device shows as read on the other. Had the client used POP3, the downloaded message would sit on that one computer instead of staying synced on the server.*

37.8 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The @ separates the **user name** (left) from the **domain name** (right). It does not separate domain and host (PYQ-35, TRAP-28).
2. **Bcc** hides recipients from **each other**; the sender always knows, and To/Cc stay visible to all (TRAP-16).
3. **Cc** is Carbon Copy (visible); **Bcc** is Blind Carbon Copy (hidden). Do not swap "blind" and "visible".
4. **Reply** answers the sender; **Reply All** answers all visible recipients; **Forward** sends the message onward.
5. **Drafts** holds unsent work, **Sent** holds sent copies, **Inbox** holds received mail, **Spam** holds filtered junk, and **Trash** holds deleted mail.
6. **Webmail** runs in a browser; an **email client** is installed software. **POP3** downloads mail to the machine while **IMAP** keeps it synced on the server.

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Exercises

1. In the address `assistant@gmail.com`, identify the user name and the domain name. What does the @ separate?
2. Distinguish To, Cc, and Bcc by stating who can see whom.
3. Distinguish the subject, the body, and an attachment.
4. State which folder holds each of the following: an unfinished message, a copy of a sent message, a newly arrived message, filtered junk, a deleted message.
5. Distinguish Reply, Reply All, and Forward.
6. Distinguish webmail from an email client, and POP3 from IMAP.

Solutions

1. The user name is `assistant` and the domain name is `gmail.com`. The @ separates the user name (left) from the domain name (right).
2. To is the primary recipient and Cc is a visible copy; both lists are seen by all recipients. Bcc is a hidden copy: each Bcc address is concealed from the other recipients, while the sender knows the full list.
3. The subject is the short topic line shown in the message list; the body is the main written content; an attachment is a separate file carried with the message (shown with a paperclip icon).
4. Unfinished: Drafts. Sent copy: Sent. Newly arrived: Inbox. Filtered junk: Spam (junk). Deleted: Trash (deleted items), recoverable until emptied.
5. Reply answers only the sender; Reply All answers the sender plus all visible To and Cc recipients (never Bcc); Forward sends the received message onward to new recipients.
6. Webmail is mail used through a browser with the mailbox on the server (Gmail, Yahoo Mail, Outlook.com); an email client is installed software (Outlook, Thunderbird). POP3 downloads mail to the local machine, usually removing it from the server; IMAP keeps mail synced on the server so all devices show the same folders.

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LESSON 38

SMTP, POP and IMAP

38.1 Why email needs protocols

Sending and reading mail involve different programs and different servers. The sender writes a message in a mail program, that program hands it to a mail server, the message travels between servers, waits in the recipient's mailbox, and is finally collected by the recipient's program. For all these independent programs to work together, they follow fixed rules called **protocols**. One protocol sends mail out, and two other protocols bring it in, each in its own way. Secure transmission and protection against mail threats sit on top of these three protocols.

Definition — Email protocol

A fixed set of rules that a mail program and a mail server follow when sending, forwarding, or collecting mail, so any standard program can work with any standard server.

Term	Meaning in this lesson
SMTP	Simple Mail Transfer Protocol; the sending protocol.
POP3	Post Office Protocol version 3; a receiving protocol that downloads mail to one machine.
IMAP	Internet Message Access Protocol; a receiving protocol that syncs mail across devices.
SSL/TLS	Secure Sockets Layer / Transport Layer Security; the secure channel for mail transmission.
Spam	Bulk unwanted mail, usually junk or advertising.
Phishing	A fake message impersonating a trusted sender to steal secrets.
Spoofing	Forging the sender address so mail looks as if it came from someone trusted.

The rest of this lesson uses these terms in one consistent sense. **SMTP** always means the sending step. **POP3** always means downloading to one local machine. **IMAP** always means keeping every device in step with the server copy. **SSL/TLS** always means the secure transmission channel. **Spam**, **phishing**, and **spoofing** are three distinct threats defined above.

38.2 SMTP: the sending protocol

SMTP (Simple Mail Transfer Protocol) is the protocol used to send mail. When the sender presses send, the mail program hands the message to the sender's mail server using SMTP, and the servers forward it toward the recipient's mail server using SMTP. SMTP moves the message out of the sender and along the path between servers until it reaches the recipient's mailbox.

Definition — SMTP

The protocol that sends mail from the sender's program to the mail server and forwards it between mail servers toward the recipient's mailbox.

Two limits matter for the exam. First, SMTP only *sends*; it does not deliver mail into the recipient's hands for reading. Second, the receiving step that follows is the job of POP3 or IMAP, described next. Any option that says SMTP downloads mail to the inbox, or that POP3 sends mail, reverses the direction and is wrong.

38.3 POP3: download to one machine

POP3 (Post Office Protocol version 3) is a receiving protocol. It pulls mail from the mail server down to one local machine. After the download, the message is typically removed from the server, so the only copy lives on that one computer. Mail downloaded this way can be read offline, but a second device will not see the same mailbox, because the server copy is gone.

Definition — POP3

A receiving protocol that downloads mail from the server to one local machine, usually removing it from the server.

POP3 suits a single computer where mail is collected and read in one place. Its weakness is exactly its method: because the mail leaves the server, bookmarks such as read marks, folders, and deletions do not travel to other devices.

Example 38.1. *A clerk collects office mail only on the office desktop. With POP3, each morning's mail downloads to that desktop and is removed from the server. The clerk can read it offline at the desk, but the same messages do not appear on the clerk's phone, because the server copy is gone.*

38.4 IMAP: sync across devices

IMAP (Internet Message Access Protocol) is also a receiving protocol, but it works differently from POP3. The mail stays on the server, and the mail program shows a synced view of it: folders, read and unread marks, deletions, and moves are all mirrored between the device and the server. When the reader acts on one device, every other device shows the same change, because each of them reflects the same server copy.

Definition — IMAP

A receiving protocol that keeps mail on the server and syncs each device's view — folders, read marks, deletions — with that server copy.

IMAP suits a reader with several devices, such as a phone and a desktop, who must see the same inbox everywhere. Its cost is that the mailbox lives on the server, so reading new mail needs a connection, and the server mailbox must have room.

38.5 POP3 vs IMAP and the full mail flow

Both POP3 and IMAP receive mail; neither sends it. The difference is the method. POP3 downloads to one machine and usually clears the server; IMAP keeps the server copy and syncs every device with it.

Point	POP3	IMAP
Job	Receives mail	Receives mail
Method	Downloads to one machine	Syncs a view of the server copy
Server copy	Usually removed	Kept on the server
Many devices	Poor fit	Designed for it
Offline reading	Natural (copy is local)	Needs a connection for new mail

The full journey of one message combines the sending protocol with one of the receiving protocols. The sender's program sends the message by SMTP to the sender's mail server; that server forwards it by SMTP to the recipient's mail server, where it waits in the mailbox; the recipient's program then collects it with POP3 (download) or IMAP (sync). In one line: sender, then SMTP, then the recipient server, then POP3 or IMAP, then the reader. Sending is always SMTP; the receiving side is POP3 or IMAP.

38.6 To, Cc, BCC, filters, and signatures

The **To** field names the main recipients of a message, and **Cc** (carbon copy) names others kept in the loop. Everyone who receives the message can see both lists. **BCC** (blind carbon copy) names recipients who are hidden from each other: nobody on the To or Cc list sees who is on BCC, and BCC recipients do not see each other. The sender always knows the full list; BCC hides recipients from each other, not from the sender.

Filters, also called **rules**, sort incoming mail automatically. A filter examines the headers and content of each message, such as the sender, the subject, or words in the body, and then files, flags, or deletes matching mail — for example, moving all newsletters to one folder. A mail **signature** is different: it is a fixed block of sender details, such as name, post, and phone number, added at the end of outgoing messages. It identifies the sender but does not encrypt or otherwise protect the message.

38.7 Digital signature vs encryption, and secure transmission

A **digital signature** proves who sent a message and that it was not changed on the way: it gives authenticity and integrity. **Encryption** does something different: it scrambles the message so that only the intended recipient can read it, giving confidentiality. A signed message can still be read by anyone — a signature is not secrecy. An encrypted message can still be forwarded — encryption hides the content but does not by itself prove who sent it. Confusing the two is a tested trap (TRAP-15): a signature never encrypts the body.

Definition — Digital signature vs encryption

A digital signature provides authenticity (who sent it) and integrity (it was not altered); encryption provides confidentiality (only the intended recipient can read it).

When mail travels between programs and servers, the channel itself can be protected by **SSL** (Secure Sockets Layer) and its successor **TLS** (Transport Layer Security). Together **SSL/TLS** provide the secure channel over which mail is transmitted, so that others on the path cannot read it. For the exam, the answer is direct: the secure email transmission protocol is SSL/TLS (PYQ-36). The official Junior Assistant item behind PYQ-16 separately confirms the POP3, BCC, filter, and signature-versus-encryption facts used in this lesson.

Example 38.1. *An item states, "A digitally signed email can be read only by its recipient." This is wrong. The signature proves the sender and detects tampering, but anyone can still read the body. Only encryption makes the body readable solely by the intended recipient. The wrong option has assigned encryption's job to the signature.*

38.8 Spam, phishing, spoofing, and malicious content

Spam is bulk unwanted mail, usually advertising or junk sent to many addresses at once. Its defining mark is that it is bulk and unwanted; it clutters the inbox and may also carry fraud or malware. Spam filters move suspected junk to a separate folder using the sender's reputation and content rules.

Phishing is a fake message that pretends to be a trusted sender — a bank, an office, or a colleague — in order to steal passwords or money. It typically creates urgency ("verify now", "account blocked", "claim within 24 hours") and asks the reader to click a link or open an attachment and hand over secrets. The defence is to check the real sender address and never enter passwords through a link reached from mail.

Spoofing forges the sender address so that a message looks as if it came from someone trusted. Phishing messages often use spoofing, but the two terms are distinct: phishing is the fraud attempt, and spoofing is the forged-sender trick it may use. A **malicious link** leads to a fake login page or a malware download; hovering over it shows an address that does not match the claimed sender. A **malicious attachment** carries malware behind an innocent-looking file name. The safe rule is

to verify an unexpected message through another channel before clicking any link or opening any attachment.

38.9 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. SMTP **sends**; POP3 and IMAP **receive**. Reversing the direction is a common wrong answer.
2. POP3 **downloads** mail to one local machine (usually clearing the server); IMAP **syncs** every device with the server copy. Do not swap them (TRAP-17).
3. A digital signature proves the sender and detects tampering; it does **not** encrypt the body. Encryption alone provides secrecy (TRAP-15).
4. BCC hides recipients **from each other**; the sender always knows the full list, and To/Cc are visible to all (PYQ-16).
5. Secure transmission of email is **SSL/TLS** (PYQ-36). A signature is not a transmission protocol.
6. Spam is **bulk unwanted** mail; phishing is **fake trusted mail that steals secrets**; spoofing is the **forged sender address** either may use.

Example 38.1. *An item asks which protocol suits reading the same office mailbox from a phone and a desktop. Trace the methods: POP3 downloads to one machine and clears the server, so the second device misses the mail; IMAP keeps the server copy and syncs every device with it. The answer is IMAP. If the stem instead asks which protocol sends the message out, the answer is SMTP, and if it asks which channel makes transmission secure, the answer is SSL/TLS.*

Exercises

1. State the full form of SMTP, POP3, IMAP, SSL, and TLS, and give the job of each of the first three.
2. Explain the difference between POP3 and IMAP when the same mailbox is read from two devices.
3. Trace one message from the sender to the reader, naming the protocol used at each step.
4. Distinguish a digital signature from encryption, and state which protocol secures email transmission.
5. Define spam, phishing, and spoofing, and state the safe rule for unexpected links and attachments.

Solutions

1. SMTP is Simple Mail Transfer Protocol and sends mail; POP3 is Post Office Protocol version 3 and downloads received mail to one machine; IMAP is Internet Message Access Protocol and syncs received mail across devices; SSL is Secure Sockets Layer and TLS is Transport Layer Security, together the secure transmission channel.
2. With POP3 the mail downloads to the first machine and usually leaves the server, so the second device does not see it; with IMAP the mail stays on the server and both devices show the same synced view of folders, read marks, and deletions.
3. The sender's program sends the message by SMTP to the sender's server; the servers forward it by SMTP to the recipient's server, where it waits; the recipient collects it with POP3 (download) or IMAP (sync).
4. A digital signature provides authenticity and integrity (who sent it, unaltered); encryption provides confidentiality (only the recipient can read it). Secure transmission uses SSL/TLS.
5. Spam is bulk unwanted mail; phishing is a fake trusted message that steals secrets through urgency plus a link or attachment; spoofing forges the sender address. Verify an unexpected message by another channel before clicking any link or opening any attachment.

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LESSON 39

E-Banking

39.1 What e-banking covers

Electronic banking, or e-banking, means using banking services without visiting the branch. The account itself stays in the bank; only the channel used to reach it changes. **Net banking** is banking through the bank's website in a browser. **Mobile banking** is banking through the bank's app on the phone. **UPI** (Unified Payments Interface) apps move money directly between bank accounts from the phone. An ATM, the net-banking site, the mobile app, and a UPI app are therefore four doors into the same account.

Definition — E-banking

The use of banking services through electronic channels — net banking, mobile banking, UPI, and ATMs — instead of visiting the branch in person.

Term	Meaning in this lesson
Net banking	Banking through the bank's website in a browser.
Mobile banking	Banking through the bank's official app.
UPI	Unified Payments Interface; instant phone-based payments between accounts.
VPA	Virtual Payment Address (for example name@bank); identifies the account without the account number.
ATM PIN	Personal Identification Number used with the physical card.
UPI PIN	Personal Identification Number authorising UPI payments.
CVV	Card Verification Value printed on the card.
OTP	One-Time Password sent by SMS, valid for one transaction only.
MFA	Multi-Factor Authentication; checking two or more factors together.

The rest of this lesson uses these terms in one consistent sense. A **PIN** is a secret typed by the holder. **CVV** is printed on the card. **OTP** arrives by SMS and expires. **MFA** is the combination of two or more different proofs of identity.

39.2 UPI in detail

UPI stands for Unified Payments Interface. Each user is identified by a **VPA** (Virtual Payment Address), such as name@bank, so money can be sent without sharing the account number or IFSC. Transfers are **instant** and work **24** $\times$ **7**, including Sundays and holidays. Every payment is authorised on the payer's own phone by entering the **UPI PIN**, which is set separately for each bank account inside the UPI app. The central rule of UPI safety is short: entering the UPI PIN always sends money, and receiving money never needs any PIN.

Other electronic transfer modes appear in nearby options and should be kept in one line each. **IMPS** (Immediate Payment Service) is an instant bank-to-bank transfer available 24 $\times$ 7. **NEFT** (National Electronic Funds Transfer) settles in batches and has no minimum amount. **RTGS** (Real Time Gross Settlement) settles large-value transfers instantly and carries a minimum amount. UPI and IMPS are the instant, phone-friendly modes; NEFT is batched; RTGS is for large values.

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Definition — UPI

Unified Payments Interface: an instant, round-the-clock system for moving money between bank accounts from a phone, identified by VPA and authorised by a UPI PIN.

39.3 The four secret codes

Four codes look alike and are deliberately mixed in distractors. Each lives in a different place and does a different job.

ATM PIN. The Personal Identification Number used with the physical card at an ATM or shop terminal. It is memorised, typed on the machine, and authorises card transactions on that channel. It never authorises a UPI payment.

UPI PIN. The Personal Identification Number typed in the phone to authorise a UPI payment. It is set per bank account in the UPI app and can differ from the ATM PIN. Approving any screen with the UPI PIN sends money.

CVV. The Card Verification Value: the 3-digit number printed on the back of the card (4 digits on the front for some cards). It is entered with the card number and expiry date for online card payments and proves the card is in hand. It is fixed and reusable for the life of the card.

OTP. The One-Time Password: a short numeric code sent by SMS that is valid for one transaction only and for a few minutes. It is generated fresh each time and expires after use or after the time window.

Code	Where it lives	How long it lasts	Typical use
ATM PIN	In memory	Until changed	Card at ATM or shop terminal
UPI PIN	In memory	Until changed	Authorising a UPI payment
CVV	Printed on card	Life of the card	Online card payment with card number
OTP	Sent by SMS	One transaction, minutes	Second approval for a login or payment

Example 39.1. *A question states, "The 3-digit number on the back of the card was sent by SMS and expired after one use." This mixes two codes. The number printed on the back of the card is the **CVV**, which is fixed and reusable; the code that arrives by SMS and expires after one use is the **OTP**. The stem has attached OTP behaviour to the CVV name.*

39.4 Authentication and multi-factor authentication

Authentication means proving "I am the account holder" before the bank acts. It uses three kinds of factor. **Something you know** is a password or PIN. **Something you have** is the phone, the card, or the OTP that arrives on the registered mobile number. **Something you are** is a fingerprint or face scan.

Definition — Multi-Factor Authentication (MFA)

Authentication that checks two or more different factors together, for example a password (know) plus an SMS OTP (have), or a PIN plus a fingerprint.

A login password plus an SMS OTP is therefore **two-factor** authentication, not one: the password proves knowledge and the OTP proves possession of the registered phone. Asking for the OTP a second time does not add a new factor; a fingerprint or card would. The exam trap is to call a password-plus-OTP login "single-factor" or to call two passwords "multi-factor"; factors must be of *different* kinds.

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39.5 QR codes and collect requests

A **QR** (Quick Response) code is the square barcode the merchant displays. Scanning it with the UPI app fills in the merchant's payment address; the payer then checks the **merchant name and amount** on their own screen and approves with the UPI PIN. A QR never asks for a PIN or OTP by itself. A common fraud is a fake QR sticker pasted over the real one, so the displayed name on the payer's screen is the check that matters: if the name does not match the shop, stop.

A **collect request** is a message asking the receiver to pay: it appears as "pay Rs. X to Y, approve with UPI PIN". Approving it and entering the UPI PIN sends the receiver's money to the requester. No refund, cashback, prize, or buyer ever needs the seller to enter a UPI PIN; entering the PIN always means paying. Unexpected collect requests, especially ones with remarks like "refund" or "cashback", must be declined or ignored. To receive money, the receiver only shares a VPA or QR and enters nothing.

Example 39.1. *A buyer sends a Rs. 500 collect request to a seller with the remark "advance, approve to receive". The seller's screen shows "pay Rs. 500", and approving with the UPI PIN would send Rs. 500 to the buyer. The remark is false: receiving needs no PIN. The safe action is to decline and ask the buyer to send the payment instead.*

39.6 Safe-banking rules

The rules below are tested almost verbatim, so they are stated exactly.

1. The bank **never** asks for OTP, CVV, ATM PIN, or UPI PIN by call, SMS, email, or link. Any such request is fraud.
2. **Never share** an OTP for any reason — to "cancel", to "verify", or to "receive" a payment. Sharing it completes the transaction the fraudster started.
3. Do not open **KYC-update or account-blocked** links from SMS. Open the bank's site or app by typing the address or using the official app.
4. **Never install** a screen-sharing or remote-access app at a caller's request; it exposes OTPs, PINs, and balances on the screen.
5. Use **typed addresses and official apps**, and check the **padlock (https)** on the banking site before logging in.
6. Avoid **public Wi-Fi** for banking; prefer mobile data or a trusted network.
7. **Log out** after net banking, lock the UPI app with the device lock, and keep **SMS and email alerts on** so an unknown debit is noticed at once.

39.7 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. ATM PIN is for the **card** channel; UPI PIN is for the **UPI** channel. They are set and used separately.
2. CVV is **printed on the card** and reusable; OTP is **sent by SMS** for one transaction and expires.
3. Entering the UPI PIN **always pays**; receiving money needs **no PIN**.
4. A password plus an SMS OTP is **two-factor** authentication (know + have); two passwords are still one factor.
5. The bank **never asks** for OTP, CVV, or any PIN. Any caller who asks is not the bank.

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6. A QR is verified by the **name and amount on your own screen**; a pasted-over sticker is defeated by that check.

Example 39.1. *An item asks what is needed to receive a UPI refund. Trace the rule: receiving needs a VPA or QR and no PIN; any screen asking for the UPI PIN is a payment screen. The answer is that no PIN or OTP is entered to receive; the "approve to receive" request is a collect-request fraud.*

Exercises

1. Expand UPI, OTP, CVV, PIN, QR, and MFA.
2. Distinguish the ATM PIN from the UPI PIN: where is each used?
3. Distinguish CVV from OTP: where does each live, and how long does each last?
4. Explain why a login password plus an SMS OTP counts as two-factor authentication.
5. A seller receives a collect request marked "refund, approve to receive". What should the seller do, and why?

Solutions

1. UPI — Unified Payments Interface; OTP — One-Time Password; CVV — Card Verification Value; PIN — Personal Identification Number; QR — Quick Response code; MFA — Multi-Factor Authentication.
2. The ATM PIN is typed on a machine with the physical card at an ATM or shop terminal. The UPI PIN is typed in the phone to authorise a UPI payment, set per bank account in the UPI app. Neither works on the other's channel.
3. CVV is printed on the card and is reusable for the life of the card; it is entered with the card number for online card payments. OTP is sent by SMS, is valid for one transaction only and a few minutes, and then expires.
4. The password is something the user knows and the OTP on the registered phone is something the user has. Two different factor kinds are checked, so the login is two-factor (a form of MFA).
5. Decline or ignore it. A collect request approved with the UPI PIN sends the seller's money; receiving a refund needs no PIN. The remark is a fraud pattern.

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LESSON 53

Networking Legacy Services and Web Standards

53.1 How to read this lesson

This is a trap-drill lesson. Almost every concept here has a near-neighbour: another term that sounds similar or sits next to it in a list but does a different job. The exam builds its wrong options out of exactly those neighbours, and it often asks the question with a negative stem (NOT, EXCEPT, or an incorrect statement). The defence is always the same: state what the term is, what it is not, and which neighbour it is confused with.

Term	Meaning in this lesson
Circuit switching	A communication method that reserves a dedicated path for the whole session before data moves.
Packet switching	A communication method that breaks data into packets, each travelling independently with no reserved path.
Telnet	TELecommunication NETwork; a remote-login service for running commands on a distant computer.
FTP	File Transfer Protocol; the service for uploading and downloading files.
SMTP	Simple Mail Transfer Protocol; sends mail from the sender toward the receiver's server.
POP3	Post Office Protocol version 3; downloads mail to the local machine.
IMAP	Internet Message Access Protocol; keeps mail on the server and syncs it across devices.
W3C	World Wide Web Consortium; the body that develops open web standards.
DHTML	Dynamic HyperText Markup Language; HTML combined with styling and scripting for dynamic pages.
URL	Uniform Resource Locator; the full address of a web resource.
DNS	Domain Name System; translates domain names into IP addresses.
IP address	The numeric address of a device on the network.
OSI model	The seven-layer networking reference model.
Physical port	A hardware socket on the machine.
Logical port	A number identifying a service or process in networking software.

53.2 Packet switching vs circuit switching

Circuit switching establishes a dedicated communication path between the two ends before any data flows, and that path stays reserved for the whole session. The classic example is the old telephone call: the line is yours until you hang up, whether or not anyone is speaking.

Definition — Circuit switching

A switching method in which a dedicated path is set up and reserved for the entire communication session before data transfer begins.

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Packet switching works the opposite way. Data is broken into small units called packets, and each packet travels independently through the network. No path is reserved in advance; packets from many senders share the same links. The Internet runs on packet switching.

Definition — Packet switching

A switching method in which data is divided into packets that travel independently over shared links, with no dedicated path reserved in advance.

Example 53.1. *An item states, "The Internet reserves a dedicated circuit for each browsing session." This is wrong. The Internet uses packet switching: pages arrive as independent packets over shared links, and no path is reserved for the session. The wrong option has described circuit switching instead.*

53.3 Telnet and FTP

Telnet (TELecommunication NETwork) is a remote-login service. It lets a user connect to another computer over the network and run commands there as if sitting at its keyboard. Telnet sends its traffic, including passwords, in plain text. It is not a file-transfer service and not a mail service.

FTP (File Transfer Protocol) is the file-transfer service. It moves files between computers: uploading from the local machine to a server and downloading back. It is not a remote-login shell and not a mail protocol.

The exam tests the triple together: Telnet logs in remotely, FTP moves files, and SMTP sends mail. Any option that swaps these roles — "Telnet transfers files" or "FTP sends mail" — is a planted near-neighbour distractor.

Definition — Telnet

A network service that provides remote login to another computer so the user can run commands on it.

Definition — FTP

A network protocol used to transfer files between computers on a network.

53.4 SMTP, POP3, and IMAP

Mail handling is split across three protocols, and the exam tests the split. **SMTP** (Simple Mail Transfer Protocol) *sends* mail: it carries a message from the sender's machine toward the receiver's mail server. **POP3** (Post Office Protocol version 3) *receives by download*: it pulls mail down to the local machine, typically removing it from the server. **IMAP** (Internet Message Access Protocol) *receives by sync*: it keeps mail on the server so the same mailbox can be read from several devices.

Definition — SMTP / POP3 / IMAP roles

SMTP sends mail; POP3 downloads mail to the local machine; IMAP keeps mail on the server and synchronises it across devices.

Three attached traps come from real papers. First, an option saying "POP3 keeps mail synced on the server" is wrong; that synced behaviour is IMAP (TRAP-17). Second, "a digital signature encrypts the body so only the recipient can read it" is wrong: a digital signature proves authenticity and integrity, while encryption provides confidentiality (TRAP-15). Third, "BCC hides the sender and the recipients from each other" is wrong: BCC hides recipients from each other, the sender always knows everyone, and To and Cc names remain visible to all recipients (TRAP-16). Finally, in any mail address the @ sign separates the user name from the domain name (TRAP-28).

Example 53.1. *An item asks which protocol downloads mail to the local machine. The answer is POP3. If the stem instead asks which keeps the mailbox synced on the server across a phone and a desktop, the answer is IMAP, and if it asks which sends the message in the first place, the answer is SMTP.*

53.5 W3C and DHTML

The **W3C** (World Wide Web Consortium) develops the open standards of the World Wide Web, such as HTML and CSS. It is a standards body — not a browser, not a search engine, and not an Internet service provider. The near-neighbour is the **ISP** (Internet Service Provider), which sells Internet access; W3C writes the standards that the access carries.

Definition — W3C

The World Wide Web Consortium: the international body that develops open standards for the World Wide Web.

HTML (HyperText Markup Language) marks up the static structure of a web page. **DHTML** (Dynamic HyperText Markup Language) is not a separate replacement language; it is the combination of HTML with styling and scripting that makes pages respond and change. The trap option calls DHTML a brand-new language that replaces HTML. The correct statement is that DHTML builds on HTML to add dynamic behaviour.

53.6 URL, DNS, and IP addresses

A **URL** (Uniform Resource Locator) is the full address of a resource on the web. A **domain name** is the human-readable part inside it. The **DNS** (Domain Name System) translates that human-readable name into a numeric address. The **IP address** is that numeric address: it identifies a device on the network.

The chain in one line is: type a URL, DNS resolves its domain name, and the browser reaches the IP address. The exam swaps the links: an option saying "DNS is the numeric address" confuses DNS with the IP address, and one saying "the URL is the numeric address" confuses the URL with the IP address.

53.7 IPv4 vs IPv6

IPv4 uses 32-bit addresses, written as four decimal groups. The address space proved too small, so **IPv6** was introduced with 128-bit addresses and a fixed header of 40 bytes (PYQ-18).

Definition — IPv4 / IPv6 address size

IPv4 addresses are 32 bits long; IPv6 addresses are 128 bits long.

The single most repeated distractor on this topic claims IPv6 is 256-bit (TRAP-18, from JA 2026 Q78). It is always wrong: IPv6 is 128-bit, never 256-bit. A second attached trap from the same paper concerns companion protocols: **ICMP** is a control and diagnostic protocol that carries no application data and uses no ports (TRAP-19); **BGP** routes between autonomous systems (inter-domain) and does not route inside one local network (TRAP-20); and **HTTP/3** runs over QUIC over UDP, not over TCP (TRAP-21, from JA 2026 Q79).

53.8 The seven OSI layers

The **OSI model** (Open Systems Interconnection) is the seven-layer reference model for networking. From top to bottom the layers are Application, Presentation, Session, Transport, Network, Data Link, and Physical. A memory hook is "All People Seem To Need Data Processing".

Definition — OSI model

A seven-layer reference model for network communication: Application, Presentation, Session, Transport, Network, Data Link, Physical.

The near-neighbour trap names four layers. Four is the layer count of the TCP/IP model, not the OSI model. When the stem says OSI, the number is always seven; an option offering four has imported the wrong model.

53.9 Physical vs logical ports

A **physical port** is a hardware socket on the machine: the place where a cable or device plugs in. A **logical port** is a number in networking software that identifies a service or process. The trap swaps them: a logical port number is not a hardware socket, and a physical socket is not a service number.

Definition — Physical vs logical port

A physical port is a hardware socket for cables or devices; a logical port is a number identifying a network service or process.

53.10 Exam retrieval and common confusions

Nine distinctions prevent most errors on this topic.

1. Circuit switching reserves a dedicated path; packet switching sends independent packets with no reserved path.
2. Telnet gives remote login; FTP transfers files; SMTP sends mail. Do not swap the three roles.
3. POP3 downloads mail to the local machine; IMAP keeps mail synced on the server (TRAP-17).
4. A digital signature proves authenticity and integrity; encryption provides confidentiality (TRAP-15).
5. BCC hides recipients from each other; the sender always knows all recipients (TRAP-16).
6. IPv6 is 128-bit with a 40-byte header, never 256-bit (TRAP-18).
7. ICMP carries no application data; BGP is inter-domain; HTTP/3 uses QUIC over UDP (TRAP-19, TRAP-20, TRAP-21).
8. The OSI model has seven layers; four layers is the TCP/IP model.
9. A physical port is a hardware socket; a logical port is a service number.

Example 53.1. *An item asks which statement is incorrect. Work each option against its near-neighbour: does the switching option reserve or share the path, does the protocol option match the service to its role, does the address option use 128-bit for IPv6, does the layer option say seven for OSI? The option that breaks its pair is the incorrect statement the stem asks for.*

Exercises

1. Distinguish circuit switching from packet switching in one sentence each.
2. Match Telnet, FTP, and SMTP to their roles.
3. State the roles of SMTP, POP3, and IMAP, and say which one keeps mail synced on the server.
4. Expand W3C, DHTML, URL, DNS, FTP, and SMTP.
5. State the bit lengths of IPv4 and IPv6, and name the OSI layers from top to bottom.

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Solutions

1. Circuit switching reserves a dedicated path for the whole session before data moves; packet switching breaks data into independently travelling packets with no reserved path.
2. Telnet gives remote login to another computer; FTP transfers files between computers; SMTP sends mail toward the receiver's server.
3. SMTP sends mail, POP3 downloads it to the local machine, and IMAP keeps it on the server and syncs it; the synced-on-server protocol is IMAP.
4. W3C — World Wide Web Consortium; DHTML — Dynamic HyperText Markup Language; URL — Uniform Resource Locator; DNS — Domain Name System; FTP — File Transfer Protocol; SMTP — Simple Mail Transfer Protocol.
5. IPv4 is 32-bit and IPv6 is 128-bit; the OSI layers from the top are Application, Presentation, Session, Transport, Network, Data Link, Physical.

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LESSON 63

Current Web, HTML, JavaScript, VBScript and ASP.NET

63.1 Static pages, markup, scripts, and the server

A web page can be static or dynamic. A **static** page shows the same content to every visitor. A **dynamic** page builds its content, or changes it, at view time. Three layers combine to produce the modern page. **Markup** describes structure: HTML lays out headings, paragraphs, and links. **Scripting** adds behaviour: small programs in JavaScript or VBScript run inside the page and react to the visitor. **Server technology** such as ASP.NET builds the page on the server before the browser ever sees it.

Definition — Markup language

A language that tags text to say what each part is (heading, paragraph, link), rather than giving computational instructions.

Term	Meaning in this lesson
HTML	HyperText Markup Language; structures the page; stored as ASCII text.
Tag	A named element in angle brackets, e.g. `<p>`.
Attribute	Detail inside the opening tag as `name="value"`.
DHTML	Dynamic HTML: HTML plus styles plus scripts for interactive pages.
XML	eXtensible Markup Language: carries data with user-defined tags.
JavaScript	Netscape browser scripting language; `+` joins strings.
VBScript	Microsoft scripting language; variables declared with `Dim`.
ASP.NET	Microsoft server framework; `Session` stores one user's data.
Meta-search engine	Queries other search engines; Dogpile is the classic example.

The rest of this lesson uses these terms in one consistent sense. A **tag** names an element; an **attribute** refines it. **DHTML** is not a new language but a combination. **XML** carries data while HTML displays it. **Java** and **JavaScript** are different languages despite the similar names.

63.2 HTML: a markup language in an ASCII-text document

HTML (HyperText Markup Language) is a markup language, not a programming language. It does not compute; it labels. Each piece of text is wrapped in tags that say what it is: a heading, a paragraph, a link. The browser reads these tags and renders the page, and the tags themselves are not displayed.

An HTML document is an **ASCII-text document**. It is plain text and can be written and read in any text editor. No special software is needed to create one, and nothing about it is compiled to binary. This is a favourite direct-recall fact: HTML files are plain ASCII text.

Definition — HTML

HyperText Markup Language: the markup language that structures web pages, stored as plain ASCII text and rendered by the browser.

Example 63.1. *A student opens a page source and sees `<html>`, `<body>`, and `<p>Hello</p>`. These are tags, readable as text in the source, but the visitor only sees the rendered words. The document*

on disk is ordinary text, which is why the lesson calls HTML an ASCII-text document.

63.3 Tags, attributes, and validity

A **tag** names an element and is written in angle brackets. Most elements have three parts: an **opening tag** such as `<p>`, the content, and a **closing tag** with a slash such as `</p>`. An **attribute** adds detail and always sits inside the opening tag in the form `name="value"`. In `<body bgcolor="yellow">`, `body` is the tag and `bgcolor="yellow"` is the attribute. In `<font size="4">`, `font` is the tag and `size` is the attribute.

A tag or attribute is **valid** when the language defines it. An invented name is simply not recognised by the browser. Exam items test this by mixing real names (`h1`, `p`, `marquee`, `font`, `body`, `html` as tags; `size`, `bgcolor` as attributes) with invented ones. The method is always the same: match the name against the defined set, and never confuse a tag with an attribute.

63.4 The `h1` heading and the `marquee` tag

Headings run from `h1` through `h6`. **`h1`** is the largest, top-level heading, and the numbers grow as the headings shrink, so `h6` is the smallest. It is written as `<h1>Title</h1>`. The single most common error is reversing this order, so fix the anchor now: **`h1` is biggest**.

The **`marquee`** tag makes text scroll across the page, written as `<marquee>Welcome</marquee>`. It is a presentation tag: it moves text but adds no structure. It belongs to older, browser-specific HTML rather than strict modern structure, which is exactly why exam setters like it as a recall item.

Definition — Heading levels

Six levels `h1` to `h6`; `h1` renders largest and `h6` smallest.

63.5 DHTML: the dynamic combination

DHTML (Dynamic HyperText Markup Language) is not a separate language. It is the combination of HTML with style sheets and scripts working together, so that a page can change after it has loaded — for example, text that reacts when the mouse passes over it. Plain HTML is static; a DHTML page is interactive without needing a reload.

Definition — DHTML

Dynamic HyperText Markup Language: HTML combined with style sheets and scripts so a page can change after loading.

The exam contrast is always the same. If the stem says the page is fixed once sent, the answer is HTML. If it says the page reacts in the browser without reloading, the answer is DHTML.

63.6 XML: data, not display

XML (eXtensible Markup Language) carries and stores data; it does not lay out a page the way HTML does. Its tags are **user-defined**: the author invents tags such as `<city>` to label the data being carried. HTML uses a fixed tag set for display; XML uses an open tag set for data. That is the whole classification: HTML displays, XML transports.

Definition — XML

eXtensible Markup Language: a markup language for storing and transporting data with author-defined tags.

Example 63.1. *An item asks which language is used to transport structured data between systems with custom tags. HTML is wrong because its tags are fixed and presentational. The answer is XML, whose tags the author defines for the data at hand.*

63.7 Java and JavaScript are different languages

Java is a full programming language: code is compiled to bytecode and runs on the Java platform. **JavaScript** is a lightweight scripting language: code is interpreted in the browser to make pages interactive. They share part of a name and nothing else. Every exam cycle reuses this trap, so treat any option that calls JavaScript a version or subset of Java as wrong on sight.

JavaScript was developed at **Netscape** for the Navigator browser in the mid-1990s. It first shipped under other names (Mocha, then LiveScript) before the JavaScript name was settled. For exam purposes the anchor is one line: JavaScript means Netscape, just as VBScript means Microsoft.

63.8 JavaScript string concatenation with +

In JavaScript the `+` operator joins (concatenates) strings. Joining `"Good"` and `"Morning"` gives `"GoodMorning"`. Mixing text and a number with `+` also joins as text. For example, joining `"Total: "` with `5` gives `"Total: 5"`. The same symbol adds numbers, but wherever a string is involved it concatenates. Items test this by asking what an expression evaluates to; the method is mechanical: join the pieces as text.

Example 63.1. *`"Web" + "Page"` evaluates to `"WebPage"` — joined, not added. An option showing a numeric sum for two quoted strings has confused the two roles of `+`.*

63.9 VBScript and the `Dim` statement

VBScript (Visual Basic Scripting Edition) is Microsoft's scripting language in the Basic family, used for page scripting and small Windows tasks. Variables are declared with the **`Dim`** statement, short for Dimension: `Dim city` declares a variable named `city`. The exam anchor is direct: see `Dim` and think VBScript variable declaration. No other language in this lesson declares variables that way.

63.10 ASP.NET and the `Session` object

ASP.NET is Microsoft's framework for building dynamic web pages on the server. The server runs the code, builds the page, and sends plain HTML to the browser. The **`Session`** object stores data for **one user across many pages**, such as a login name kept while the visitor moves from page to page. Its nearest neighbour is the **Application** object, which is shared by all users. The distinction is per-user versus shared: `Session` keeps one visitor's data; it is never shared across visitors.

Definition — Session object

The ASP.NET object that holds data for a single user across multiple page requests.

63.11 Meta-search engines: Dogpile

A **search engine** such as Google or Bing keeps its own index of pages. A **meta-search engine** keeps no index of its own; it sends the query to other search engines and combines their results. **Dogpile** is the classic exam example of a meta-search engine. The memory hook is in the name: Dogpile piles up other engines' results and shows them together.

63.12 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. HTML is an **ASCII-text** document. An option calling it a compiled binary or a database file is wrong.
2. **`h1`** is the largest heading; **`marquee`** scrolls text. Do not reverse the heading order.
3. A **tag** names the element; an **attribute** sits inside the opening tag as `name="value"`. `font` is a tag; `size` is its attribute.

4. **Java** is a full compiled language; **JavaScript** is a Netscape browser scripting language. Similar names, different languages.
5. In JavaScript `+` concatenates strings; in VBScript `Dim` declares a variable.
6. The ASP.NET `Session` object is per-user, not shared; **Dogpile** is a meta-search engine, not a search engine with its own index.

Example 63.1. *An item asks what `<h1>` produces and offers "smallest heading" as an option. Trace the heading ladder: `h1` is the top level and renders largest, with `h6` smallest. The option has reversed the ladder. If the stem instead asks which tag scrolls text, scan for `marquee`; if it asks which object keeps one user's login across pages, scan for `Session`.*

Exercises

1. Expand HTML, DHTML, XML, and ASP.NET, and state in one line what each does.
2. Explain why an HTML document is called an ASCII-text document.
3. Distinguish a tag from an attribute, giving one example of each.
4. Which heading tag is largest, and which tag scrolls text across the page?
5. Distinguish Java from JavaScript, and state where JavaScript originated.
6. What does the `+` operator do to two strings in JavaScript? What does `Dim` do in VBScript?
7. What does the ASP.NET `Session` object store, and how does it differ from Application state?
8. What is a meta-search engine, and why is Dogpile an example?

Solutions

1. HTML — HyperText Markup Language: structures web pages. DHTML — Dynamic HyperText Markup Language: HTML plus styles plus scripts for interactive pages. XML — eXtensible Markup Language: carries data with user-defined tags. ASP.NET: Microsoft's server framework for building dynamic pages.
2. It is plain text: it can be written and read in any text editor, with no compilation step. The browser renders the tags at view time.
3. A tag names an element in angle brackets (e.g. `<p>`); an attribute sits inside the opening tag as `name="value"` (e.g. `size="4"` in `<font size="4">`).
4. `h1` is the largest heading (`h6` smallest); `marquee` scrolls text.
5. Java is a full compiled programming language running on the Java platform; JavaScript is a lightweight browser scripting language developed at Netscape.
6. `+` joins (concatenates) the strings as text. `Dim` declares a variable in VBScript.
7. `Session` stores data for one user across many pages (e.g. a login name). Application state is shared by all users; `Session` is per-user.
8. A meta-search engine keeps no index and combines results from other search engines. Dogpile does exactly this, so it is the classic example.

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UNIT 9

Networking

LESSON 40

Networking Basics

40.1 What a network is: nodes, hosts, clients, servers

A computer network is formed when two or more computers and related devices are connected so that they can share data and resources. The shared resources include files, printers, an internet connection, and application software. On the office desktop, opening a shared folder, printing to a shared printer, and browsing the web all use the same underlying network.

Definition — Computer network

A set of two or more connected computing devices (nodes) that can exchange data and share resources such as files, printers, and an internet connection.

Every participant in a network is called a **node**. A node can be a computer, a printer, a switch, or any device with a network address. A **host** is a node that can send or receive data over the network, such as a desktop computer, a laptop, or a network printer. In practice every host is a node, but the word host is used when the device is an end point that produces or consumes data.

Term	Meaning in this lesson
Network	Two or more connected devices that share data and resources.
Node	Any participant device in the network.
Host	A node (computer, printer) that sends or receives data.
Client	The device or program that requests a service.
Server	The computer that provides the service and answers requests.

The client–server pair is the most examined relationship in this block. The **client** is the device (or program) that requests a service — a web page, a file, or a print job. The **server** is the computer that provides that service: it stores the shared files or pages and answers the requests of many clients. When the office desktop opens a file stored on the department's shared machine, the desktop is the client and the shared machine is the server.

Definition — Client and server

The client requests a service; the server stores the shared resource and provides it in response to client requests.

40.2 Networks by area: PAN, LAN, MAN, WAN

Networks are classified by the geographical area they cover. Four names must be learned together with their full forms, because the exam tests both the expansion and the coverage in the same item.

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Type	Full form	Coverage and example
PAN	Personal Area Network	Around one person: phone, laptop, and earphones linked by Bluetooth.
LAN	Local Area Network	One building or campus: an office, school, or laboratory network.
MAN	Metropolitan Area Network	One city: a bank's branches linked across a town.
WAN	Wide Area Network	A country or the world: the Internet is the largest WAN.

A **PAN** (Personal Area Network) covers the smallest area — the devices around a single person, such as a phone connected to a laptop and wireless earphones. A **LAN** (Local Area Network) covers one building or campus, such as all the computers of one office floor connected through a switch. A **MAN** (Metropolitan Area Network) spans a city, for example the branches of one organisation linked across a town. A **WAN** (Wide Area Network) spans a country or the whole world; the Internet is the largest example of a WAN.

Definition — PAN, LAN, MAN, WAN

PAN covers one person; LAN covers one building or campus; MAN covers one city; WAN covers a country or the world.

Example 40.1. *A question gives four coverages — one person, one building, one city, and the whole world — and asks for the matching names. Read the area first: person means PAN, building means LAN, city means MAN, and worldwide means WAN. The area decides the answer before the full form is even expanded.*

40.3 WLAN, intranet, Internet, and ISP

Three further names complete the vocabulary. A **WLAN** (Wireless Local Area Network) is a LAN built with wireless signals instead of cables: the office Wi-Fi through which laptops connect without a cable is a WLAN. An **intranet** is a private network inside one organisation, usable only by its own staff — for example the company's internal portal. The **Internet** is the worldwide public network of connected networks, open to anyone with access. The two words differ by one prefix and are a standing trap: intranet is private to one organisation, while the Internet is public and worldwide.

An **ISP** (Internet Service Provider) is the company that provides a customer with access to the Internet. The home or office connects through its ISP — over a phone line, cable, fibre, or mobile link — and the ISP carries the traffic onward to the Internet.

Definition — WLAN, intranet, Internet, ISP

A WLAN is a wireless LAN; an intranet is a private organisational network; the Internet is the worldwide public network; an ISP (Internet Service Provider) gives customers access to the Internet.

40.4 Topology: bus, star, ring, mesh, tree

The word **topology** means the arrangement by which nodes are connected in a network. Five topologies are examined: bus, star, ring, mesh, and tree. Each one decides how data travels and what happens when one cable or device fails, and the exam repeatedly asks exactly that: which topology has a central device, and which one fails fully when the main cable breaks.

In a **bus** topology, all nodes share one single main cable called the bus. It is simple and cheap for a small network. Data sent by one node travels along the whole bus and is accepted by the addressed node. Its weakness follows from its shape: if the main cable breaks, the whole network fails.

In a **star** topology, every node connects to one central device (a hub or a switch), and all data passes through that centre. If one spoke cable fails, only that node is affected and the rest keep working. But if the central device itself fails, the whole network stops. The ordinary office LAN — one switch with many desktops plugged into it — is a star, and it is the most common topology in exam examples.

In a **ring** topology, each node connects to exactly two neighbours, forming a closed circle. Data travels around the ring, usually in one direction, passed from node to node until it reaches its destination. Each intermediate node repeats the data it is not addressed to. A break at one point disturbs the whole ring, unless a second (dual) ring provides a backup path.

In a **mesh** topology, every node has a direct link to many (in a full mesh, all) other nodes. It is the most reliable topology: if one path fails, another path carries the data. It is also the costliest, because it needs the most cable and the most ports.

In a **tree** topology, several star networks are joined through a central backbone, like the branches of a tree. It suits a large campus: each department is a star, and the stars join at the centre. Its weakness is the backbone: if the backbone fails, whole branches lose connection.

Topology	Shape	What one failure does
Bus	One shared cable	Main cable breaks: the whole network fails.
Star	Centre with spokes	One spoke fails: one node affected; centre fails: all fail.
Ring	Closed circle	One break disturbs the ring.
Mesh	Many direct links	One path fails: traffic uses another path.
Tree	Stars joined on a backbone	Backbone fails: whole branches fail.

Definition — Topology

The physical or logical arrangement of nodes in a network: bus (one cable), star (central device), ring (closed circle), mesh (many direct links), and tree (stars joined on a backbone).

Example 40.1. *An item states, "In which topology does the failure of the central device bring down the whole network?" Trace the table: only the star routes every message through one centre, so the answer is star. If the stem instead says the main cable broke and everything stopped, the shape is the single shared cable, so the answer is bus.*

40.5 Transmission media: guided and unguided

Transmission media are the paths that carry data signals from one device to another. They divide into two families. **Guided (wired) media** confine the signal inside a physical cable. **Unguided (wireless) media** carry the signal through air or space without a cable.

There are three guided media. **Twisted-pair cable** consists of pairs of copper wires twisted together; it is the ordinary telephone and office-LAN cable — cheap and suitable for short ranges. **Coaxial cable** has a central copper wire with insulation and a metal shield around it; it is used for cable TV and older LANs and has better shielding than twisted pair. **Fibre-optic cable** uses thin glass or plastic strands carrying light signals; it is the fastest medium with the longest range and is immune to electrical noise, but it is also the costliest. Ranked by speed and range, fibre-optic comes first, then coaxial, then twisted pair.

There are four unguided media at this level. **Radio waves** travel in all directions and are used for FM radio and for short-range WLAN and Bluetooth links. **Microwaves** travel in a straight line and need line of sight; they are used for point-to-point links between towers. **Satellite links** are microwave signals relayed through a satellite; they cover very long distances but introduce delay. **Infrared** works over a very short range, is blocked by walls, and is used for TV remotes.

Definition — Guided and unguided media

Guided media carry signals inside a cable (twisted pair, coaxial, fibre-optic); unguided media carry signals through air or space (radio waves, microwaves, satellite links, infrared).

40.6 Networking devices and their roles

Ten devices are examined, and device–role matching is the highest-yield item type of this lesson. Each device has one role to memorise.

A **NIC** (Network Interface Card) is the card — or the built-in circuit on the motherboard — that lets a computer physically connect to a network. Without it the machine has no network port to plug into.

A **modem** (modulator-demodulator) converts digital computer signals into analogue signals for a telephone line and back again. It is the device that connects a home or office to the ISP over a line that cannot carry digital signals directly.

A **hub** connects several nodes in a star and broadcasts every incoming signal to all of its ports; the addressed node accepts the data and the rest ignore it. A **switch** also connects nodes in a star, but it forwards each message only to the addressed port. The switch is therefore smarter and faster than the hub, because it does not flood every port with every message.

A **repeater** regenerates a weak signal so that it can travel further along a long cable. It works at the signal level only: it does not choose a path and does not filter traffic. A **bridge** connects two LAN segments and filters traffic, passing across only the data meant for the other side.

A **router** connects different networks — for example a home LAN to the ISP's network — and chooses the best path for each packet of data. A **gateway** connects networks that use different protocols or data formats, translating between them so the two sides can understand each other. An **access point** is the wireless base station that lets Wi-Fi devices join a wired network. A **firewall** blocks unauthorised access while letting legitimate traffic pass; it is a security device that can be hardware, software, or both.

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Device	Role (one line)
NIC (Network Interface Card)	Connects a computer to the network.
Modem	Converts digital to analogue and back; links the site to the ISP.
Hub	Broadcasts every incoming signal to all ports.
Switch	Forwards each message only to the addressed port.
Repeater	Regenerates a weak signal for a longer run.
Bridge	Joins two LAN segments and filters traffic between them.
Router	Joins different networks and chooses the best path.
Gateway	Translates between networks using different protocols.
Access point	Lets wireless (Wi-Fi) devices join the wired network.
Firewall	Blocks unauthorised access; a security device.

Definition — The four most confused pairs

Hub sends to all, switch sends to the addressed port; repeater regenerates the signal, router chooses the path; bridge joins two similar LAN segments, gateway translates between different protocols; modem connects to the line, firewall guards the entry.

40.7 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The client **requests** and the server **provides**. Do not swap the roles.
2. Read the area first: one person is PAN, one building is LAN, one city is MAN, worldwide is WAN.
3. **Intranet** is private to one organisation; the **Internet** is public and worldwide.
4. A hub broadcasts to **all** ports; a switch forwards only to the **addressed** port.
5. A repeater only **regenerates** the signal; the router **chooses the path**. A bridge joins two LAN segments; a gateway translates between **different** protocols.
6. Fibre-optic is the **fastest** guided medium; twisted pair is the slowest of the three cables.

Example 40.1. *An item asks which device "connects different networks and selects the best path for each packet." Walk the device table: the hub and switch stay inside one network, the repeater only strengthens the signal, and the gateway translates protocols — only the router joins different networks and chooses paths, so the answer is router.*

Exercises

1. Distinguish a node, a host, a client, and a server.
2. Expand PAN, LAN, MAN, WAN, WLAN, NIC, and ISP, and state the coverage of each network type.
3. Distinguish an intranet from the Internet.
4. Describe the five topologies and state, for each, what a single failure does.
5. Distinguish guided from unguided transmission media, giving the three cables and the four wireless media of this lesson.
6. Match each of the ten devices of this lesson to its one-line role.

Solutions

1. A node is any participant device in the network. A host is a node (computer, printer) that sends or receives data. A client is the device or program that requests a service; a server is the computer that provides the service and answers client requests.
2. PAN — Personal Area Network (around one person); LAN — Local Area Network (one building or campus); MAN — Metropolitan Area Network (one city); WAN — Wide Area Network (a country or the world); WLAN — Wireless Local Area Network (a LAN built with wireless signals); NIC — Network Interface Card (lets a computer connect to a network); ISP — Internet Service Provider (gives customers Internet access).
3. An intranet is a private network inside one organisation, usable only by its staff. The Internet is the worldwide public network of connected networks, open to anyone with access.
4. Bus: one shared cable; a break in the main cable fails the whole network. Star: every node joins one centre; a spoke failure affects one node, a centre failure stops all. Ring: closed circle of neighbours; one break disturbs the ring. Mesh: many direct links; one failed path leaves others. Tree: stars joined on a backbone; a backbone failure drops whole branches.
5. Guided media confine the signal in a cable: twisted-pair (cheap, short range), coaxial (shielded, e.g. cable TV), and fibre-optic (light signals; fastest, longest range, costliest). Unguided media use air or space: radio waves (all directions, short range), microwaves (straight line, towers), satellite links (long distance, with delay), and infrared (very short range, blocked by walls).
6. NIC connects the computer; modem converts digital–analogue for the line to the ISP; hub broadcasts to all ports; switch forwards to the addressed port; repeater regenerates a weak signal; bridge joins two LAN segments with filtering; router joins different networks and chooses the path; gateway translates between different protocols; access point admits Wi-Fi devices; firewall blocks unauthorised access.

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LESSON 41

Networking Protocols

41.1 Why protocols and addresses matter

Computers on a network can understand each other only because they follow agreed rules. A **protocol** is such a set of rules: it fixes what messages look like and in which order the machines exchange them. Different jobs need different protocols. Opening a web page, moving a file, sending an email, translating a name into an address, and handing out addresses to new devices are five different jobs, so they use five different protocols. The exam tests each protocol by its **exact English name** together with the single role it performs.

Definition — Protocol

A fixed set of rules governing communication between devices on a network, covering message format and exchange order. TCP, IP, DNS, DHCP, HTTP, HTTPS, FTP, Telnet, and SMTP are the protocols of this lesson.

Term	Meaning in this lesson
TCP	Transmission Control Protocol: reliable, ordered, connection-oriented delivery.
UDP	User Datagram Protocol: fast, connectionless, best-effort delivery.
IP	Internet Protocol: addressed, best-effort carriage of packets across networks.
ICMP	Internet Control Message Protocol: control and diagnostic messages only.
DNS	Domain Name System: translates names into IP addresses.
DHCP	Dynamic Host Configuration Protocol: automatically assigns addresses.
HTTP / HTTPS	HyperText Transfer Protocol (plain) and its Secure version for the web.
FTP	File Transfer Protocol: moves files between computers.
Telnet	Plain-text remote login protocol.
SMTP	Simple Mail Transfer Protocol: sends outgoing mail.
OSI model	The seven-layer reference model from Physical to Application.
Port	A number naming the service on a machine (HTTP 80, HTTPS 443).

The rest of this lesson uses these terms in one consistent sense. **TCP** and **UDP** are the two transport protocols. **IP** names the addressing protocol with its two versions, IPv4 and IPv6. **ICMP** is the helper that reports errors, never the carrier of user data. **DNS** and **DHCP** are the two address services: one resolves names, the other assigns addresses. **HTTP**, **HTTPS**, **FTP**, **Telnet**, and **SMTP** are application protocols, each with one job and one well-known port.

41.2 TCP and UDP

The transport layer offers two opposite bargains. **TCP** (Transmission Control Protocol) is connection-oriented: it sets up the connection, numbers every segment, retransmits whatever is lost, and delivers the bytes complete and in order. What TCP guarantees is *ordering and correctness*; it does not promise when the data will arrive. **UDP** (User Datagram Protocol) is connectionless: it sends each

datagram independently, with no setup, no ordering, and no retransmission. What UDP offers in return is speed and low overhead.

Definition — TCP

A connection-oriented transport protocol that delivers data reliably and in order through sequencing, acknowledgement, and retransmission.

Definition — UDP

A connectionless transport protocol that sends independent datagrams quickly with no delivery, ordering, or error-recovery guarantee.

Choose between them by what the task values most. A file, a web page, or an email must arrive complete and correct, so it travels over TCP. Live voice, a video stream, or a quick DNS lookup values speed over perfection, so it travels over UDP, where a lost fragment is simply skipped.

Example 41.1. *An item states, "TCP guarantees fast delivery of every packet." This is wrong. TCP guarantees that the bytes arrive complete and in order (PYQ-18); speed is exactly what TCP sacrifices. The fast, best-effort protocol is UDP.*

41.3 IPv4 and IPv6

The **Internet Protocol (IP)** carries addressed packets from the sender to the receiver across networks. Each device holds an IP address, and routers read that address to forward the packet hop by hop. IP by itself is unreliable: delivery is best-effort, and it is TCP above it that adds reliability.

IPv4 uses 32-bit addresses written as four decimal octets, for example 192.168.1.10. Its header is variable in length (20 to 60 bytes). **IPv6** uses 128-bit addresses written as eight hexadecimal groups, for example 2001:0db8::1, and its header is fixed at 40 bytes (PYQ-18, TRM-10). The longer address space is the whole point of IPv6: it supplies vastly more addresses than IPv4.

Definition — IPv4

The 32-bit version of the Internet Protocol, with dotted-decimal addresses and a variable-length header.

Definition — IPv6

The 128-bit version of the Internet Protocol, with hexadecimal addresses and a fixed 40-byte header.

41.4 ICMP: the control channel

ICMP (Internet Control Message Protocol) carries control and diagnostic messages about the network itself: error reports such as "destination unreachable" and reachability probes. Its two familiar tools are **ping**, which asks whether a host is reachable, and **traceroute**, which shows the path packets take. ICMP transfers no application data and uses no port numbers (TRAP-19). That last point is the exam's favourite trap: any option giving ICMP a port, a file transfer, or a mail delivery is wrong.

41.5 DNS and DHCP

DNS (Domain Name System) translates a human name such as `jkssb.nic.in` into the IP address the network needs. Without it every user would have to memorise numeric addresses. DNS queries travel mainly over UDP on port 53, falling back to TCP for large zone transfers.

DHCP (Dynamic Host Configuration Protocol) automatically assigns an IP address, subnet mask, default gateway, and DNS server address to each device that joins the network. Without it every ad-

dress would have to be typed in by hand. The home router and the office network each run a DHCP server; the joining laptop or phone acts as the DHCP client, using UDP ports 67 and 68.

Definition — DNS

The distributed naming service that resolves a domain name into its IP address. It answers "what number goes with this name?"

Definition — DHCP

The service that automatically assigns IP configuration (address, mask, gateway, DNS server) to devices joining a network. It answers "what address may this newcomer use?"

Example 41.1. *An item asks which service gives a newly connected office desktop its IP address. Trace the pair: DNS resolves names the user already has, while DHCP hands out addresses to newcomers. The answer is DHCP. If the stem instead asks what turns* `jkssb.nic.in` *into an address, the answer is DNS.*

41.6 Application protocols: web, files, remote login, mail

HTTP (HyperText Transfer Protocol, port 80) fetches web pages in plain text. **HTTPS** (HyperText Transfer Protocol Secure, port 443) is the same protocol with an SSL/TLS encryption layer, shown by `https://` and the padlock. The S stands for Secure (WO 2026 Q76 territory).

FTP (File Transfer Protocol, port 21) moves files between computers; it is neither email nor web browsing. **Telnet** (port 23) opens a plain-text remote login session and is today replaced by the encrypted SSH. **SMTP** (Simple Mail Transfer Protocol, port 25) sends outgoing mail to the server. Receiving is a separate job handled by **POP3** (Post Office Protocol version 3, port 110), which downloads mail to the local machine (PYQ-16, TRAP-17), and **IMAP** (Internet Message Access Protocol, port 143), which keeps mail synced on the server.

41.7 The seven OSI layers

The **OSI model** (Open Systems Interconnection) is a seven-layer reference frame for describing network communication:

No.	Layer	Job
7	Application	Services the user sees: HTTP, FTP, SMTP, DNS, DHCP.
6	Presentation	Translation, encryption, compression.
5	Session	Opens, manages, and closes sessions between machines.
4	Transport	End-to-end delivery: TCP and UDP.
3	Network	Addressing and routing: IP (IPv4, IPv6) and ICMP.
2	Data Link	Frames on a single link: switch, MAC address.
1	Physical	Bits on the wire, cable, or radio signal.

Data moves down the layers on sending and up the layers on receiving, each layer adding (or reading) its own header. For this exam, three placements carry almost all the marks: the application names (HTTP, FTP, SMTP, DNS, DHCP) at layer 7, TCP and UDP at layer 4, and IP and ICMP at layer 3.

Example 41.1. *Opening a web page uses HTTP at the Application layer, TCP at the Transport layer, and IP at the Network layer: one protocol per layer, working together. An item asking "which layer does IP belong to" is answered by the same table: the Network layer (layer 3).*

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41.8 Port awareness

An **IP address** names the machine; a **port number** names the service on that machine. Ports 0–1023 are the well-known ports. The exam set is: FTP 21, Telnet 23, SMTP 25, DNS 53, HTTP 80, POP3 110, IMAP 143, HTTPS 443. DHCP uses UDP ports 67 and 68, and ICMP uses no ports at all. The recommended drill is to cover the service names and recall the numbers in the chain 21–23–25–53–80–443.

41.9 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. IPv6 is **128-bit**, never 256-bit; its header is fixed at 40 bytes (TRAP-18, TRM-10).
2. ICMP is **control/diagnostic** only: no application data, no ports (TRAP-19).
3. TCP promises **order**, not timing; UDP promises neither (PYQ-18).
4. **SMTP sends** mail; **POP3/IMAP receive** it (TRAP-17).
5. **DNS resolves** names to addresses; **DHCP assigns** addresses to joining devices.
6. HTTP is port **80** and plain; HTTPS is port **443** with SSL/TLS.
7. TCP and UDP live at the **Transport** layer; IP and ICMP at the **Network** layer; HTTP, FTP, SMTP, DNS, and DHCP at the **Application** layer.

Exercises

1. Distinguish TCP from UDP on reliability, connection setup, and one typical use each.
2. Contrast IPv4 with IPv6 on address length, notation, and header size.
3. Explain why ICMP is described as a control protocol and name its two common tools.
4. Distinguish DNS from DHCP with one sentence on each role.
5. List the seven OSI layers in order with the protocols of this lesson placed at their layers.

Solutions

1. TCP is connection-oriented, reliable, and ordered: it numbers segments and retransmits losses; typical uses are web pages, email, and file transfer. UDP is connectionless and best-effort: no setup, ordering, or retransmission; typical uses are streaming, voice calls, and DNS queries.
2. IPv4 is 32-bit, dotted-decimal (four octets), with a variable 20–60 byte header. IPv6 is 128-bit, hexadecimal (eight groups), with a fixed 40-byte header.
3. ICMP carries control and diagnostic messages about the network (errors, reachability) rather than user data, and it uses no ports. Its tools are ping (reachability) and traceroute (path).
4. DNS translates a domain name into its IP address. DHCP automatically assigns an IP address and related configuration to a device joining the network.
5. 7 Application (HTTP, HTTPS, FTP, Telnet, SMTP, POP3, IMAP, DNS, DHCP); 6 Presentation; 5 Session; 4 Transport (TCP, UDP); 3 Network (IP, ICMP); 2 Data Link; 1 Physical.

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LESSON 62

Current Internet, Networking and Cloud IaaS

62.1 TCP: ordered and reliable, but with no timing promise

A running Internet conversation must survive lost, duplicated, and reordered packets. **TCP** (Transmission Control Protocol) is the transport protocol that hides this mess from the application. It is *connection-oriented*: the two ends set up a connection before any data moves. Once the connection exists, TCP numbers every byte, acknowledges what arrives, and retransmits whatever is lost, so the receiving application sees the bytes in exactly the order they were sent. That is the double guarantee to memorise: **ordered** delivery and **reliable** delivery.

What TCP does not promise is timing. It never claims the data will arrive within a fixed time, because retransmissions take as long as they take. This "no timing guarantee" point is tested directly (PYQ-18): an option that claims TCP promises delivery by a deadline is wrong.

Definition — TCP

A connection-oriented transport protocol that delivers bytes in order and retransmits lost segments, with no guarantee about delivery time.

Term	Meaning in this lesson
TCP	Transmission Control Protocol; ordered and reliable transport.
UDP	User Datagram Protocol; connectionless, fast, unreliable transport.
ICMP	Internet Control Message Protocol; control and diagnostic messages only.
BGP	Border Gateway Protocol; inter-domain routing between autonomous systems.
IPv6	128-bit next-generation Internet addressing; no broadcast.
QUIC	Quick UDP Internet Connections; the UDP-based transport under HTTP/3.
DNS	Domain Name System; translates domain names into IP addresses.
PIR	Public Interest Registry; operates the .org registry.
ICANN	Internet Corporation for Assigned Names and Numbers; global name/number coordination.
ARPANET	Advanced Research Projects Agency Network (1969); ancestor of the Internet.
IaaS	Infrastructure as a Service; provider manages hardware, customer manages OS and above.

The rest of this lesson uses these terms in one consistent sense. **TCP** means ordered and reliable transport. **UDP** means the fast, connectionless alternative with neither guarantee. **ICMP** means the control-message protocol that never carries application data. **BGP** means routing between autonomous systems, never inside one LAN. **IPv6** means 128-bit addresses with no broadcast. **QUIC** means the UDP-based transport that HTTP/3 uses instead of TCP. **DNS** means the name-to-address directory. **PIR** runs the .org registry under **ICANN** coordination. **ARPANET** is the 1969 packet-switched ancestor of the Internet. **IaaS** splits responsibility so that the provider manages networking hardware while the customer manages the operating system, applications, and data.

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62.2 UDP and the TCP/UDP contrast

UDP (User Datagram Protocol) is TCP's near-neighbour and the most common distractor for it. UDP is *connectionless*: each datagram is sent independently with no setup. It provides no ordering guarantee, no reliability guarantee, and no retransmission. Its header is small, so it is faster and cheaper per packet than TCP.

Definition — UDP

A connectionless transport protocol that sends independent datagrams with no ordering, reliability, or retransmission guarantee.

Point	TCP	UDP
Connection	Connection-oriented	Connectionless
Ordering	Guaranteed	Not guaranteed
Reliability	Retransmits lost data	No retransmission
Speed	Slower (acknowledgements, state)	Faster (light header)
Typical use	Web pages, mail, file transfer	Live voice, video, games, DNS lookups

The exam tests this pair by swapping one property: an option that says UDP guarantees order, or that TCP is connectionless, is always wrong. The anchor is the direction of the trade: TCP buys correctness at the cost of speed, while UDP buys speed at the cost of correctness.

Example 62.1. *An item states, "UDP guarantees ordered delivery of datagrams." This is wrong. Ordering is a TCP guarantee; UDP provides no ordering at all. The wrong option has moved a TCP property onto UDP.*

62.3 ICMP: the network's own messenger

While TCP and UDP carry user data, **ICMP** (Internet Control Message Protocol) carries messages *about* the network. Routers and hosts use it to report errors — destination unreachable, time exceeded, parameter problems — and diagnostic tools such as **ping** (echo request and reply) and **traceroute** are built on ICMP messages.

Definition — ICMP

The control and diagnostic protocol of the Internet layer; reports errors and supports ping and traceroute, and never carries application data.

Three facts together close every ICMP trap (PYQ-18, PYQ-19, TRAP-19). First, ICMP transfers no application data: it never moves a file, a mail message, or a web page. Second, it uses no ports: ports belong to TCP and UDP, and an option that assigns ICMP a port number is wrong. Third, it is neither a routing protocol like BGP nor a mail protocol like SMTP; it only reports on conditions the other protocols encounter.

62.4 BGP: routing between autonomous systems

The Internet is not one network but a federation of large independently managed networks called **autonomous systems** (an ISP, a university backbone, a government network). **BGP** (Border Gateway Protocol) is the protocol that routes *between* them. It exchanges reachability information across autonomous-system borders and selects the paths that traffic takes over the Internet backbone.

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Definition — BGP

The inter-domain routing protocol that exchanges reachability between autonomous systems on the Internet backbone.

The tested contrast is BGP against the *interior* protocols **OSPF** (Open Shortest Path First) and **RIP** (Routing Information Protocol), which route *inside* a single domain (PYQ-19, TRAP-20). Any option that places BGP inside one small local network has swapped the two scopes. The one-word shortcut is "inter-domain": whenever the stem says inter-domain or between autonomous systems, the answer is BGP.

62.5 IPv6: address space, header, and no broadcast

IPv4 uses 32-bit addresses (about 4.3 billion), which ran out under the growth of the Internet. **IPv6** replaces them with **128-bit** addresses, written as eight groups of hexadecimal digits, giving about 3.4×10^{38} addresses. This vast space is the standard reason given for IPv6 removing the need for address-sharing workarounds such as **NAT** (Network Address Translation), which lets many private addresses share one public IPv4 address (PYQ-19).

Definition — IPv6 address

A 128-bit Internet-layer address, written as eight hexadecimal groups, with a fixed 40-byte header.

Two further IPv6 facts are tested. The fixed IPv6 header is **40 bytes** long (PYQ-18). And IPv6 has **no broadcast**: IPv4's three delivery styles (unicast to one receiver, multicast to a group, broadcast to everyone) become unicast, multicast, and anycast (nearest of several receivers) in IPv6, with a multicast to the all-nodes group replacing the broadcast. An option mentioning an "IPv6 broadcast address" is therefore self-defeating. The 256-bit figure that appears as a distractor is always wrong (TRAP-18).

Example 62.1. *An item offers "IPv6 uses 256-bit addresses" as one option. This is wrong. IPv6 uses 128-bit addresses; 256 is double the correct size and matches no Internet-layer standard. The option confuses IPv6 with an unrelated larger number.*

62.6 HTTP/3 and QUIC

Web traffic travelled over TCP for decades: HTTP/1.1 and HTTP/2 both run over TCP. **HTTP/3** breaks that pattern. It runs over **QUIC** (Quick UDP Internet Connections), which itself runs over **UDP**, not over TCP. QUIC rebuilds the needed reliability, congestion control, and security on top of UDP, gaining faster connection setup and independent streams without head-of-line blocking across streams.

Definition — QUIC

A UDP-based transport protocol carrying HTTP/3, providing its own reliability and security instead of using TCP.

The trap writes itself: any option claiming HTTP/3 relies on TCP has the protocol stack backwards (PYQ-19, TRAP-21). The correct chain is HTTP/3 over QUIC over UDP.

62.7 DNS and the .org / PIR / ICANN chain

DNS (Domain Name System) is the Internet's directory: it translates human-readable domain names such as `example.org` into numeric IP addresses that routers can use. Resolution runs hierarchically from root servers down through top-level-domain servers to the domain's own authoritative servers. DNS is a lookup service, not a routing protocol and not a mail-transfer protocol, and confusing it with either is a standard wrong answer.

The governance half of the topic names three actors. **.org** is a generic top-level domain originally meant for non-profit organisations. Its registry is operated by **PIR** (Public Interest Registry). **ICANN** (Internet Corporation for Assigned Names and Numbers) coordinates domain names and IP addresses at the global level. The exam distinction is role-based: PIR *runs* the .org registry, while ICANN *coordinates* the naming system as a whole.

62.8 ARPANET and packet switching

ARPANET (Advanced Research Projects Agency Network), live in **1969**, is the ancestor of the Internet. Its contribution was proving that **packet switching** works across long distances: splitting data into addressed packets that travel independently and are reassembled at the destination, with the network routing around failed links. Later ARPANET work produced the TCP/IP protocol family.

Definition — Packet switching

Splitting data into addressed packets that may take different routes and are reassembled at the destination.

The tested contrast is packet switching against **circuit switching**, the older telephone model that holds one dedicated path for the whole call. The Internet uses packet switching; the traditional telephone call used circuit switching. Options that assign a dedicated reserved circuit to the Internet have the two models swapped.

62.9 The router: joining networks by reading IP addresses

A **router** connects different networks and forwards packets between them. It does this by reading each packet's destination **IP address** and choosing the next hop from its **routing table**. In assertion–reason form: the assertion (a router joins networks) and the reason (it reads the destination IP and consults its table) are both true, and the reason correctly explains the assertion, because address-reading is *how* the router forwards between networks.

The near-neighbours are the **switch**, which forwards *inside* one network using MAC addresses, and the **hub**, which merely repeats everything it receives. An option that says a router forwards within one network using MAC addresses describes a switch, not a router.

62.10 IaaS: the provider manages the hardware

Cloud service models split responsibility between provider and customer. In **IaaS** (Infrastructure as a Service), the **provider** manages the physical infrastructure: servers, storage hardware, **networking hardware**, and virtualisation. The **customer** manages everything above that layer: the operating system, runtime, applications, and data. The exam anchor is the provider's side: in IaaS the provider manages the networking hardware (PYQ-20).

Definition — IaaS responsibility split

The provider manages infrastructure (networking and storage hardware, virtualisation); the customer manages the operating system, runtime, applications, and data.

Moving down the ladder, the provider manages more. In **PaaS** (Platform as a Service) the provider additionally manages the operating system and runtime, leaving the customer its applications and data. In **SaaS** (Software as a Service) the provider manages everything up to the finished application, and the customer only manages its own data and settings. The trap claims the IaaS provider manages the customer's OS or data; it never does (TRAP-22).

Example 62.1. *An item states, "In IaaS the provider manages the customer's operating system and data." This is wrong. Those belong to the customer; the provider stops at the infrastructure layer (networking and storage hardware plus virtualisation). The option describes SaaS, not IaaS.*

62.11 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. TCP guarantees **order and reliability** but no timing; UDP guarantees **neither** (PYQ-18).
2. ICMP carries **control messages only**: no application data, no ports (TRAP-19).
3. BGP is **inter-domain** (between autonomous systems); OSPF and RIP are intra-domain (TRAP-20).
4. IPv6 is **128-bit** with a **40-byte** header; 256-bit is always wrong (PYQ-18, TRAP-18).
5. IPv6 has **no broadcast**; the replacement is **multicast** to the all-nodes group (PYQ-19).
6. HTTP/3 uses **QUIC over UDP**, never TCP (TRAP-21).
7. In IaaS the **provider** manages networking hardware; the **customer** manages the OS, applications, and data (TRAP-22).

Example 62.1. *An item asks which protocol routes between autonomous systems on the Internet backbone. Trace the scope word: "between autonomous systems" is the definition of inter-domain routing, and the inter-domain protocol is BGP. OSPF and RIP are interior protocols, and ICMP is not a routing protocol at all, so all three are eliminated by scope.*

Exercises

1. State the two guarantees TCP provides and the one it does not.
2. Distinguish TCP from UDP on connection, ordering, reliability, and speed.
3. What does ICMP carry, and what two things does it never do?
4. Distinguish BGP from OSPF/RIP by scope.
5. State the IPv6 address size and header size, and explain what replaces the broadcast.
6. Explain the HTTP/3 protocol stack in one chain of three names.
7. Distinguish the roles of PIR and ICANN for the .org domain.
8. State the IaaS responsibility split in two sentences.

Solutions

1. TCP guarantees ordered delivery and reliable delivery (lost segments are retransmitted). It provides no timing guarantee.
2. TCP is connection-oriented, ordered, and reliable but slower; UDP is connectionless, unordered, and unreliable (no retransmission) but faster.
3. ICMP carries control and diagnostic messages (errors, ping, traceroute). It never transfers application data and never uses ports.
4. BGP routes between autonomous systems (inter-domain, Internet backbone); OSPF and RIP route inside a single domain.
5. IPv6 uses 128-bit addresses with a fixed 40-byte header. It has no broadcast; a multicast to the all-nodes group replaces it.
6. HTTP/3 runs over QUIC, which runs over UDP: HTTP/3 over QUIC over UDP.
7. PIR operates the .org registry; ICANN coordinates domain names and IP addresses globally.
8. The provider manages the infrastructure (networking and storage hardware, virtualisation); the customer manages the operating system, runtime, applications, and data.

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UNIT 10

Cybersecurity & IT Law

LESSON 42

Malware

42.1 What "malware" means

Malware is short for **malicious software**: any program written to harm a computer, steal from it, or misuse it. The official syllabus names **computer virus and latest anti-virus** explicitly, and the exam's computer block tests the near neighbours as well: worm, Trojan, ransomware, spyware, adware, rootkit, keylogger, bot and botnet, backdoor, logic bomb, and cryptojacking.

Definition — Malware

Any software designed to damage a computer, steal data from it, or misuse its resources without the owner's consent.

Term	Meaning in this lesson
Virus	Malware that attaches to a host file and needs human action to run.
Worm	Standalone, self-spreading malware that travels over networks.
Trojan (Trojan horse)	Malware disguised as useful software; does not replicate itself.
Ransomware	Malware that locks files and demands payment for release.
Spyware	Malware that secretly watches the user and steals data.
Adware	Malware that floods the machine with unwanted advertisements.
Keylogger	A spying tool that records every key pressed.
Rootkit	Malware that hides itself and other malware deep in the system.
Backdoor	A secret entry path into the system bypassing normal checks.
Bot / botnet	A remotely controlled machine / a network of such machines.
Logic bomb	Malicious code that waits for a trigger, then strikes.
Cryptojacking	Secret use of the victim's computer to mine cryptocurrency.
Anti-virus	Software that detects, blocks, and removes known malware.

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The rest of this lesson uses these terms in one consistent sense. A **virus** always needs a host file and a human launch. A **worm** needs neither. A **Trojan** pretends to be legitimate and does not replicate. **Ransomware** encrypts and demands money. **Spyware** hides and steals; **adware** shows ads. A **keylogger** records keystrokes. A **rootkit** conceals; a **backdoor** re-admits. A **botnet** is a controlled network (robot + network). A **logic bomb** waits for its trigger. **Cryptojacking** mines on the victim's machine.

42.2 Virus, worm, and Trojan

A **virus** attaches itself to a host program or file and stays inert until a person opens, runs, or shares that file. Common carriers are email attachments, pen drives, and downloaded files. Once launched it can damage, delete, or corrupt files. The defining limit is dependence: no host file and no human action, no spread.

Definition — Virus

Malware that attaches to a host program or file and requires human action to execute and spread.

A **worm** is standalone and self-replicating. It copies itself from machine to machine over network connections, exploiting security gaps, without any host file and without any user click. Its main harm

is consumption: networks clog, systems slow, and services jam under the load of endless self-copies.

Definition — Worm

Standalone malware that replicates and spreads across networks by itself, without a host file or human action.

A **Trojan**, or Trojan horse, takes its name from deception: it arrives looking like something useful — a free game, a handy tool, a harmless document. The user installs or opens it willingly, and the hidden malicious part runs alongside. Unlike a virus or a worm, a Trojan does not replicate itself; each new machine needs its own act of trickery. Once inside, it can steal data, install spyware or ransomware, or open a backdoor for the attacker's later return.

Definition — Trojan (Trojan horse)

Malware disguised as legitimate software that tricks the user into installing it; it does not replicate itself.

The three are best remembered as one table: a virus needs a host file and a user action; a worm needs neither; a Trojan needs the user's trust and does not self-replicate. File damage points to a virus, network-wide slowdown with no clicks points to a worm, and a "useful" download that betrays the machine points to a Trojan.

Example 42.1. *An item describes a program that "spreads from computer to computer over the network without any user opening a file." Trace the table: needs no host and no click means worm. If instead the stem says the malware "was installed as part of a free utility the user downloaded," the deception plus no replication means Trojan.*

42.3 Ransomware

Ransomware locks the victim out of their own data, usually by encrypting files (or the whole screen), and then displays a demand: pay the ransom to receive the unlock key. The "pay to unlock" message, often with a countdown or a payment address, is its signature.

Ransomware usually arrives the Trojan way — a harmful attachment or link the user opens — and then encrypts documents, photos, and databases in minutes. The expected defences are regular **offline backups**, updated operating system and **anti-virus** software, and caution with attachments. The exam's standing line is that paying the ransom does not guarantee recovery; the backup is what defeats the demand.

Definition — Ransomware

Malware that encrypts the victim's files (or locks the screen) and demands payment for the key to restore them.

42.4 Spyware, adware, and the keylogger

Spyware is the silent watcher. It hides on the machine, observes browsing habits, logins, and personal details, and sends them to a third party. It deliberately causes no visible damage so the victim suspects nothing. It commonly arrives bundled with free downloads or deceptive installers, and its purpose is theft by observation: passwords, bank details, and identity data.

Adware is the noisy opposite. It floods the screen with unwanted advertisements — pop-ups, banners, redirected searches — to earn money from ad traffic. Heavy adware also slows the machine and may carry spyware along with it, so the exam treats persistent pop-ups as more than a nuisance.

A **keylogger** is a focused spying tool: it records every key pressed and forwards the log to the attacker, harvesting usernames, passwords, and card numbers. Keyloggers exist as software and as small hardware devices fitted between the keyboard and the computer.

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Definition — Spyware vs adware

Spyware hides and steals information quietly; adware displays unwanted advertisements openly. A keylogger is a spying tool that records keystrokes.

Example 42.1. *An item reports "constant pop-up advertisements and redirected searches." The visible ads mean adware. If instead the item reports "passwords stolen although nothing on screen looked wrong," the silent theft means spyware, and if it specifies "every keystroke was recorded," the tool is a keylogger.*

42.5 Rootkit and backdoor

A **rootkit** is malware whose job is concealment. It lodges deep in the system, often with administrator-level control, and hides files, processes, and network activity from the user and from ordinary detection tools — including hiding other malware.

A **backdoor** is the secret entry a rootkit or Trojan protects: a path into the system that bypasses normal login and security checks, left open so the attacker can return at will. The pairing to remember is that Trojans and rootkits commonly install backdoors; the rootkit hides the intruder, and the backdoor keeps the door open.

Definition — Rootkit and backdoor

A rootkit conceals an intruder and its malware deep in the system; a backdoor is a hidden entry path that bypasses normal authentication.

42.6 Bot and botnet, logic bomb, cryptojacking

A **bot** is an ordinary computer that has been compromised and is now obeyed remotely; a **botnet** is a network of such machines (the word joins robot + network) under one master. Botnets send spam, spread further malware, and launch large-scale attacks that no single machine could mount. The owner typically notices only slowness.

A **logic bomb** is malicious code already planted that waits for a trigger — a date, an event, or a specific action — and strikes only when the condition is met. It does not spread; it waits.

Cryptojacking is theft of computing power: the attacker secretly runs cryptocurrency mining on the victim's machine, so the computer runs hot and slow while the attacker collects the earnings. Like a bot, its outward sign is an otherwise unexplained slowdown.

42.7 How malware reaches a machine, and how to stop it

Malware reaches users through a short list of routes: harmful email attachments and links, infected pen drives, bundled free-software installers, downloads from untrusted sites, and unpatched system software with known gaps.

The defences form a matching list. **Anti-virus** software detects, blocks, quarantines, and removes known malware; both it and the operating system must be kept updated, since each update closes gaps and refreshes the list of known threats. A **firewall** filters incoming and outgoing network traffic and blocks suspicious connections, but it is a barrier, not a cleaner: it does not remove malware already inside. Safe habits complete the set: do not open unknown attachments or links, scan pen drives before use, download only from trusted sources, and keep regular **offline backups** so ransomware loses its leverage.

42.8 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. A virus needs a host file **and** a human action; a worm needs **neither**; a Trojan disguises itself and **does not replicate**.

2. Spyware **hides and steals**; adware **shows ads**. A keylogger records **keystrokes**.

3. Ransomware **encrypts and demands payment**; its signature is the "pay to unlock" message, and the defence is the **offline backup**.
4. A rootkit **hides** the intruder; a backdoor is the **secret re-entry**. Trojans commonly install backdoors.
5. A botnet is a **network** of remotely controlled machines (robot + network), used for spam and large attacks.
6. A logic bomb **waits for a trigger**; cryptojacking **mines cryptocurrency** on the victim's machine.
7. Anti-virus **removes** malware; a firewall **filters** traffic and does not clean an infected machine.

Example 42.1. *An item asks which malware "encrypts the user's files and demands money for the decryption key." The lock plus the demand means ransomware — not a worm (self-spreading), not adware (ads), not spyware (silent theft). If the stem instead says the program "waits until a fixed date and then deletes files," the waiting trigger means a logic bomb.*

Exercises

1. Distinguish a virus from a worm from a Trojan on three points: host file, self-spreading, and user action.
2. Contrast spyware with adware. Where does a keylogger fit?
3. Explain what ransomware does and name the defence that defeats its demand.
4. Distinguish a rootkit from a backdoor, and explain how they work together.
5. Define bot/botnet, logic bomb, and cryptojacking in one sentence each, naming each type's outward sign.

Solutions

1. A virus needs a host file and a human launch and does not spread by itself; a worm is standalone and self-spreading over networks with no host and no click; a Trojan pretends to be legitimate software, is installed willingly, and does not replicate itself.
2. Spyware hides and secretly steals user data; adware openly floods the machine with unwanted ads. A keylogger is a focused spying tool that records every keystroke to steal credentials.
3. Ransomware encrypts the victim's files (or locks the screen) and demands payment for the unlock key. Regular offline backups defeat the demand, since the data can be restored without paying.
4. A rootkit hides the intruder and its malware deep in the system; a backdoor is a secret entry bypassing normal checks. Trojans and rootkits install backdoors so the attacker can return unseen.
5. A bot is a remotely controlled compromised machine and a botnet is a network of them (sign: slowness during spam or attacks); a logic bomb waits for a trigger such as a date and then strikes (sign: a timed surprise); cryptojacking secretly mines cryptocurrency on the victim's machine (sign: hot, slow computer).

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LESSON 43

Hacking, Phishing and Spoofing

43.1 Why attacks succeed

Most attacks on ordinary computer users do not defeat the machine by force. They persuade the person in front of the screen to hand over the key: a password, a One-Time Password, a click, or a file download. The attacker needs a weakness to use and a moment in which the victim acts without verifying. The weakness may be technical, such as a short reused password or outdated software, or human, such as trust in an authoritative-sounding caller, hurry, or fear of an account being blocked. This lesson stays at a safe awareness level: it names each attack, shows how to recognise it, and teaches the correct defensive response. It does not teach how to carry out any attack.

Definition — Attack chain

*A **vulnerability** is a weakness in software, settings, or human habit that can be misused. An **exploit** is a method that takes advantage of such a weakness. An **attack** is the complete harmful act against a person or system. The **payload** is the harmful element the attack delivers, such as stolen credentials, stolen money, or a malicious file.*

Term	Meaning in this lesson
Hacking	Gaining unauthorised access to a computer, account, or network.
Phishing	A fake message, mail, or call imitating a trusted sender to steal credentials or money.
Spoofing	Faking an identity, such as a sender address, caller number, or website address.
Social engineering	Tricking a person, rather than breaking software, into giving access.
Password attack	An attempt to discover a password by guessing, a word list, exhaustive trial, or observation.
Malicious link	A web address that leads to a fake page or an unwanted download.
Malicious attachment	A file that harms the computer or steals data when opened.
OTP	One-Time Password: a single-use code; never shared with anyone.

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The rest of this lesson uses these terms in one consistent sense. **Hacking** is the broad act of unauthorised access. **Phishing** is fraud by imitation that asks the victim to act. **Spoofing** is the forged identity that makes the fraud look genuine. **Social engineering** is the human manipulation behind both. A **vulnerability** is the weakness, the **exploit** is the method, and the **payload** is the harm delivered.

43.2 Hacking

Hacking means gaining **unauthorised access** to a computer system, an online account, or a network. A **hacker** is the person who performs this act. Access without permission is unlawful even when nothing is copied or damaged; the absence of permission is what defines the act, not the amount of harm that follows.

An **ethical hacker** is a security tester who probes systems **with written permission** from the owner in order to find and fix weaknesses before criminals exploit them. The permission is the entire difference: the same technical step performed with written authorisation is a security test, and performed without it is an attack.

Definition — Hacking

Gaining access to a computer, account, or network without the owner's permission.

Example 43.1. *A student opens a colleague's mailbox by guessing the password and reads the mail. Even though nothing is deleted, this is hacking, because the access was unauthorised. A bank that hires a tester to try its login page, with a signed agreement, is instead conducting an ethical security test.*

43.3 Password attacks

Passwords remain the most common lock on an account, so attackers try to discover them first. Four methods appear repeatedly in examinations.

Guessing tries obvious choices: the user's name, birth year, phone number, or common strings such as `password123`. **Dictionary attack** tries every entry in a list of common passwords automatically, which succeeds quickly against short familiar words. **Brute-force attack** tries every possible combination of characters until one works; it always succeeds eventually, but a longer password makes the search impractically slow. **Shoulder surfing** is direct observation: watching the keyboard or screen while the victim types a password or PIN.

The defence follows from the methods. Use a **long** password that mixes letters, digits, and symbols, and never reuse one password across accounts. Never disclose a password or an **OTP (One-Time Password)** to anyone, including a caller who claims to represent the bank or the office. Lock the screen when leaving the desk, shield the keypad while typing, and switch on **two-step verification** (also called two-factor authentication) wherever it is offered, so that the password alone is not enough to enter the account.

Definition — Brute-force vs dictionary attack

*A **dictionary attack** tries each word in a ready list of common passwords. A **brute-force attack** tries every possible character combination in turn.*

43.4 Social engineering

Social engineering is the practice of tricking a person into granting access, instead of breaking the software technically. Two named forms recur. **Pretexting** invents a false identity and story — a bank officer verifying records, an IT support engineer fixing the account — to extract a password, OTP, or remote-access permission. **Baiting** offers a lure — a prize, a free recharge, free software — that triggers the victim to click a link or download a file.

The standard signal is manufactured urgency: words such as **urgent**, **blocked**, **suspended**, and **verify now**, combined with a demand to act immediately without checking. The safe response is fixed: slow down, verify the claim through an independently known channel (the published helpline or the real website), and never disclose a password or OTP under pressure.

Definition — Social engineering

Manipulating a person, rather than a machine, into revealing credentials or granting access.

43.5 Phishing

Phishing is a fake message, email, text, or call that **imitates a trusted sender** — a bank, an office, a delivery service — in order to steal passwords, OTPs, card details, or money. The message always contains a call to action: click the enclosed link, open the attached file, or call back the given number. The link leads to a look-alike login page that records whatever is typed; the attachment installs the payload when opened.

Named channels help classification. Fraudulent text messages are called **smishing** (SMS phishing) and fraudulent voice calls are called **vishing** (voice phishing), but both are phishing by the

same definition: imitation plus a request for action. The recognition signs are stable: an urgent or threatening tone, spelling and grammar errors, a sender address that does not match the claimed organisation, and a link whose real address differs from the genuine website.

Definition — Phishing

A fraudulent message that imitates a trusted sender and asks the victim to click, open, reply, or pay, in order to steal credentials or money.

Example 43.1. *A clerk receives a mail headed "Salary slip ready — download within 24 hours or the account will be blocked." The sender shows the office name, but the reply address is an unknown mailbox and the enclosed link points to a misspelled portal. The mail is phishing: imitation (the office name), pressure (blocking within 24 hours), and a call to action (the link). The safe response is to ignore the link, open the real portal by typing its known address, and report the message.*

43.6 Spoofing

Spoofing means **faking an identity** so that a message, call, or website appears to come from someone it does not. Three forms cover the examination scope. **Email spoofing** forges the "From" address so the mail displays a trusted name that never sent it. **Caller ID spoofing** forges the number shown on the phone so the call appears to come from a bank or office. **Website (URL) spoofing** registers a look-alike address — for instance `bank-verlfy.com` in place of the genuine `bank-verify.com` — and hosts a fake login page there.

Spoofing is best understood as a disguise rather than a complete fraud. A phishing message very often **uses** spoofing — a forged sender address — to look trustworthy, but spoofing can also appear without any message at all, as in a forged website waiting for visitors. The check is the same in every case: confirm the identity through a known channel instead of trusting what is displayed.

Definition — Spoofing

Presenting a forged sender address, caller number, or website address as though it were genuine.

43.7 Phishing vs spoofing vs hacking

These three terms form the mandatory near-neighbour set of the lesson, and examination options deliberately mix them. **Hacking** is the broadest: any unauthorised access. **Phishing** is fraud by imitation that asks the victim to act: click, open, reply, or pay. **Spoofing** is the forged identity itself: the faked address, number, or site. A single incident can contain all three — a forged sender address (spoofing) carries a fake bank mail asking for the OTP (phishing), and the stolen OTP is then used to enter the account (hacking) — but each word answers a different question: was access taken without permission, was a fake message asking for action, or was an identity forged?

43.8 Malicious links and attachments

A **malicious link** is a web address that leads to a fake login page or starts an unwanted download. Before clicking, hover over the link with the mouse to reveal the real address, or long-press it on a phone to preview the destination. Safer still, ignore the link entirely and type the website's known address by hand. Shortened addresses, misspelled brand names, and extra words inserted into a familiar address, arriving inside an unexpected or urgent message, are warning signs.

A **malicious attachment** is a file that harms the computer or steals data once opened. Never open an unexpected attachment, even when the displayed sender is a known name, because that name may itself be spoofed; confirm with the sender through a separate channel first. Be especially cautious with documents that ask the reader to "enable macros" or "enable content" on opening, since that step is what activates many payloads. In short: the link is the address, the attachment is the file, and both can carry the payload.

43.9 Safe habits and response after targeting

Four habits block the large majority of these attacks. First, verify every urgent request through a **known channel** — the published helpline or the manually typed website — before acting. Second, keep the operating system, browser, and **antivirus** software updated, and scan external drives before opening their files. Third, keep a regular **backup** of important files so that one incident cannot destroy the only copy. Fourth, use the mail client's **report phishing** option on a suspicious message rather than replying to it or forwarding it to colleagues.

If a suspicious page has already been opened, do not type any password, OTP, or card details into it. Close the page, delete the message, and do not call back any number printed inside it. Change the affected password by visiting the real website directly, inform the bank or office through its published helpline, and preserve a record — a screenshot and the sender's address — for the complaint. The order is fixed: stop, verify on the real channel, change credentials, report.

43.10 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. Phishing **imitates** a trusted sender to steal data or money; spoofing **forges** the identity it wears; hacking is the broader **unauthorised access**.
2. OTP stands for **One-Time Password**. No genuine bank, office, or support desk ever asks for it by call or message.
3. A **dictionary** attack tries a ready word list; a **brute-force** attack tries every combination.
4. A **vulnerability** is the weakness, an **exploit** is the method that uses it, and the **payload** is the harm delivered.
5. Ethical hacking requires **written permission**; without it, access is unauthorised.
6. The safe response to any urgent message is fixed: do not click, verify through a known channel, and report.

Example 43.1. *An item describes a call in which the displayed number matches the bank's helpline and the caller demands the OTP to "unblock" the account. Trace the definitions: the displayed number is forged (spoofing), the call imitates the bank and demands the OTP (phishing by vishing, using social engineering), and any resulting entry into the account is unauthorised access (hacking). The correct answer names phishing, and the correct action is to refuse, disconnect, and verify through the published helpline.*

Exercises

1. Define hacking, and state what separates an ethical security test from an attack.
2. Distinguish a dictionary attack from a brute-force attack.
3. Define phishing and list four recognition signs of a phishing message.
4. Distinguish phishing from spoofing in one sentence each, and explain how they combine.
5. State the safe response when an unexpected message demands an OTP urgently.

Solutions

1. Hacking is gaining unauthorised access to a computer, account, or network. An ethical security test is performed with the owner's written permission in order to find and fix weaknesses; without that permission the same act is an attack.
2. A dictionary attack tries each entry of a ready list of common passwords; a brute-force attack tries every possible character combination in turn, so longer passwords resist it.

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3. Phishing is a fraudulent message imitating a trusted sender in order to steal credentials or money. Signs: urgent or threatening tone, spelling errors, a sender address that does not match the claimed organisation, and a link whose real address differs from the genuine site.
4. Phishing is fraud by imitation that asks the victim to act; spoofing is the forged identity itself. A phishing message often wears a spoofed sender address to look trustworthy.
5. Do not click, reply, or disclose the OTP; verify through an independently known channel (published helpline or manually typed website), change credentials on the real site if exposed, and report the message.

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LESSON 44

Cybersecurity Prevention

44.1 Prevention and the CIA triad

Cleaning an infected machine is slow and uncertain, so the first goal of security is prevention: small repeated habits plus a few tools that each do one distinct job. The standard way to judge whether protection is complete is the **CIA triad**: **confidentiality**, **integrity**, and **availability**.

Definition — Confidentiality

Only authorised persons can read the data. Passwords, encryption, and MFA protect it.

Definition — Integrity

The data is accurate and has not been changed by unauthorised means. Clean backups, digital signatures, and timely patching protect it.

Definition — Availability

The data and systems are reachable when needed. Backups, antivirus software, and firewalls protect it.

Term	Meaning in this lesson
Antivirus	Software that detects, blocks, and removes malicious programs.
Signature	A stored pattern of a known virus, used for recognition.
Real-time scanning	Checking files at the moment they are opened, downloaded, or copied.
Quarantine	Holding a suspicious file in an isolated area so it cannot run.
Firewall	A filter that allows or blocks network traffic by rule.
Patch	A small vendor fix for a known security weakness.
MFA — Multi-Factor Authentication	Proving identity with two or more independent factors.
Backup	A separate clean copy of data kept for recovery.
HTTPS	The secure, encrypted form of web browsing.
WPA2/WPA3	Encryption standards that protect a Wi-Fi link.
Autorun	The feature that launches programs automatically when a drive is connected.

One attack can break more than one leg of the triad. Stealing a password over open Wi-Fi breaks confidentiality; altering marks in a result file breaks integrity; ransomware that locks every document breaks availability. The rest of this lesson maps each preventive control to the goal it serves.

44.2 Antivirus: signatures, scanning, and quarantine

Antivirus software detects, blocks, and removes malicious programs (malware). Its oldest method is the **signature**: a stored pattern of a known virus, like a fingerprint kept on record. The antivirus compares each file against its signature database and flags anything that matches. Because new malware appears every day, the signature database must be updated regularly; an outdated antivirus misses new threats.

Definition — Signature

A stored pattern identifying a known malicious program, against which files are compared.

Signatures catch known malware, but they cannot recognise a brand-new (zero-day) threat. Behaviour-based detection watches what a program *does* — for example, trying to encrypt every document — and stops it even when no signature matches.

Two scanning modes matter. **Real-time scanning** checks files at the moment they are opened, downloaded, or copied, before harm is done. A **full (on-demand) scan** checks the whole system when the user asks for it. When either mode finds something suspicious, the antivirus moves the file to **quarantine**: an isolated area where the file cannot run or spread. Quarantine is not deletion. The file is held safely until the user deletes it or restores it (for example, when a clean file was flagged by mistake).

Example 44.1. *A pen drive is plugged into the office desktop and a file on it matches a known virus signature. Real-time scanning blocks the file the moment it is opened and moves it to quarantine, so it cannot run. The user then deletes the quarantined file. If instead the file had been a clean office template flagged by mistake, quarantine would allow restoring it — deletion would not.*

44.3 Firewall: the network gatekeeper

A **firewall** monitors incoming and outgoing network traffic and blocks whatever its rules forbid. Its job is to stop unauthorised network access: a stranger on the internet reaching into the office computer, or a malicious program on the computer sending data out. A firewall can be software running on the computer or hardware placed at the network boundary; offices commonly use both.

Definition — Firewall

A hardware or software filter that allows or blocks network traffic according to a set of rules.

The firewall–antivirus boundary is a favourite exam trap. A firewall blocks bad *traffic*; it does not remove viruses already saved on the disk. Antivirus removes bad *files*; it does not, by itself, patrol the network boundary. The two complement each other and neither replaces the other.

44.4 Patching and updates

A **patch** is a small fix released by the vendor for a known security weakness, and a **software update** delivers such fixes (plus features). Attackers study published weaknesses, so any delay in installing patches leaves known doors open. The operating system, the web browser, and the antivirus itself must all be updated; each has flaws attackers can exploit. Automatic updates are the safest setting for most users.

Example 44.1. *The office desktop offers an operating-system update. Installing it closes a published weakness the same week it became known. Postponing it for months means every attacker who read the same notice knows exactly which door is still open on that machine.*

44.5 Strong passwords and MFA

A **strong password** is long and mixes uppercase letters, lowercase letters, numbers, and symbols. It must be unique per site, never shared, and never left at its factory default: routers and office devices ship with well-known default passwords that must be changed immediately. A **password manager** stores one distinct strong password per site so the user only memorises a single master secret.

MFA stands for Multi-Factor Authentication. It demands two or more independent proofs of identity: something you know (the password), plus something you have (a one-time code on the phone) or something you are (a fingerprint). Even when the password leaks, the attacker still lacks the second factor, so the account holds. Note the distinction the exam tests: a longer password is

still one factor; MFA is the extra check, not a stronger password.

Definition — MFA (Multi-Factor Authentication)

An authentication method that requires two or more independent factors — for example, a password plus a phone code or fingerprint.

44.6 Backup: the availability control

A **backup** is a separate clean copy of important data, kept so work can continue when the original is lost, damaged, or locked by ransomware. Backup is the great protector of availability. Good backups are regular (so little work is lost), kept offline or off-site (so one fire, theft, or ransomware attack cannot destroy both copies), and tested by trial restores. A backup that cannot be restored is not a backup.

The backup–antivirus boundary is another tested pair. Backup restores *lost* data; antivirus removes *malicious* data. A machine can have fresh antivirus and still lose a year's work to a failed disk if there is no backup.

44.7 Safe browsing

Most infections arrive through the browser, so browsing habits are a control in their own right. Look for **HTTPS** and the padlock symbol before entering passwords or bank details; HTTPS encrypts the session between the browser and the site. Treat unexpected links and attachments as hostile: **phishing** messages imitate trusted senders (a bank, an office, a colleague) to steal credentials. Download software only from official or trusted sources, log out on shared computers, and never save passwords on a machine others use.

44.8 Secure Wi-Fi

The wireless link needs its own protection. Home and office networks must use **WPA2 or WPA3** encryption with a strong password, and the router's default admin password must be changed. Open (password-free) public hotspots expose traffic, so sensitive logins — banking, email, office accounts — should wait for mobile data, a trusted network, or a **VPN** (Virtual Private Network), which encrypts traffic across untrusted links. Two extra habits help: turn off automatic connection to unknown hotspots, and keep software updated so known wireless flaws are patched. Note the layering the exam tests: HTTPS protects the website session, while WPA2/WPA3 protects the wireless link itself; both layers matter.

Definition — VPN (Virtual Private Network)

An encrypted tunnel that carries traffic safely across an untrusted network.

44.9 Removable-media safety

USB drives and memory cards carry malware between machines. Scan every drive with antivirus *before* opening its files, and never plug an unknown found drive into the office computer. Disable **autorun** so a newly connected drive cannot launch programs by itself. Handle drives gently at the end as well: use the safe-eject (safely remove hardware) option before pulling the device out, so files are not corrupted.

44.10 Which control does what

This table is the highest-yield page in the lesson; most questions are matching items built from its rows.

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Control	Main job
Antivirus + signatures	Detect and remove known malware files.
Real-time scanning + quarantine	Check on access; isolate suspects before harm.
Firewall	Block unauthorised network traffic.
Patching / updates	Close known software weaknesses.
Strong password + MFA	Prove identity; block account theft.
Backup	Restore data after loss or ransomware.
HTTPS + safe browsing	Protect sessions; dodge phishing and bad downloads.
WPA2/WPA3 + VPN	Protect data travelling on the network link.
Scan USB + disable autorun	Stop malware entering on removable drives.

A useful habit: read the stem's verb. "Remove" points to antivirus, "block traffic" to the firewall, "prove identity" to passwords and MFA, "restore" to backup, and "protect the link" to WPA2/WPA3 or a VPN.

44.11 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The firewall blocks **traffic**; the antivirus removes **files**. Do not swap them.
2. Quarantine **isolates**; deletion **removes**. A quarantined file can be restored; a deleted one cannot.
3. MFA adds a **second proof of identity**; a longer password is still one factor.
4. Backup restores **availability**; encryption protects **confidentiality**. A backup is not an antivirus and an antivirus is not a backup.
5. HTTPS protects the **website session**; WPA2/WPA3 protects the **Wi-Fi link**.
6. Confidentiality keeps secrets secret, integrity keeps data correct, and availability keeps systems reachable. Map each attack to the leg it breaks before choosing the control.

Example 44.1. *An item asks which control lets the office resume work after ransomware locks every document. Trace the verb: "resume work after loss" means restoring availability, so the answer is the offline backup. Antivirus (removes malware) and the firewall (blocks traffic) cannot unlock or recreate the lost files.*

Exercises

1. State the three goals of the CIA triad in one line each.
2. Explain what a virus signature is and why it must be updated.
3. Distinguish quarantine from deletion.
4. Distinguish a firewall from antivirus software.
5. Explain why a long password alone is not MFA, and what MFA adds.

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Solutions

1. Confidentiality: only authorised persons can read the data. Integrity: the data is accurate and unaltered by unauthorised means. Availability: the data and systems are reachable when needed.
2. A signature is a stored pattern of a known virus used for recognition. New malware appears daily, so the database must be updated or new threats will not match anything on record.
3. Quarantine isolates a suspicious file in a safe area where it cannot run, and the file can later be restored; deletion removes the file, and it cannot be recovered from quarantine-style holding.
4. A firewall monitors network traffic and blocks what its rules forbid; it does not clean infected files. Antivirus detects and removes malicious files on the machine; it does not by itself patrol the network boundary.
5. A long password is still a single factor (something you know). MFA (Multi-Factor Authentication) adds at least one independent factor, such as a phone code or fingerprint, so a leaked password alone is not enough.

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LESSON 56

Cybersecurity, IT Act and Security Controls

56.1 Why security, and the CIA triad

A working computer holds records worth protecting: student marks, account details, personal files, and official correspondence on the office desktop. **Cybersecurity** is the practice of protecting computers, networks, and data against theft, damage, unauthorised access, and misuse. Three goals organise the whole topic, remembered as the **CIA triad**: **confidentiality** (only authorised persons can read the data), **integrity** (the data is not altered by unauthorised means), and **availability** (the system stays usable when needed). Every threat in this lesson attacks one of these goals, and every control defends one.

Definition — Cybersecurity

The protection of computer systems, networks, and data against theft, damage, disruption, and unauthorised access, so that confidentiality, integrity, and availability are preserved.

Term	Meaning in this lesson
Malware	Umbrella term for malicious software written to harm or exploit.
Virus	Malware that attaches to a host file and needs human action to spread.
Worm	Standalone self-copying malware that spreads over a network unaided.
Trojan horse	Malware disguised as useful software; opens a backdoor; does not replicate.
Ransomware	Malware that locks or encrypts files and demands payment.
Encryption	Coding data so only authorised persons can read it.
SSL / TLS	Secure Sockets Layer / Transport Layer Security; secure transmission protocols.
Firewall	Barrier filtering network traffic against rules.
Quarantine	Isolation of a suspicious file so it cannot run.
MFA	Multi-Factor Authentication; two or more identity checks at login.

The rest of this lesson uses these terms in one consistent sense. **Malware** is always the general term, and virus, worm, Trojan, and ransomware are its specific types. **Encryption** always means hiding content (confidentiality), while **digital signature** always means proving the sender and intactness (authenticity and integrity). The exam tests exactly these pairings, so the wording here is kept fixed.

56.2 The Information Technology Act, 2000

The **Information Technology Act, 2000** is India's law for the electronic world. Its scope has three parts: it gives **legal validity to electronic records and electronic signatures**, so that online transactions and electronic governance stand on a legal footing; it enables **electronic governance** (official records, notices, and transactions in electronic form); and it defines **cyber contraventions and offences**, such as unauthorised access to a computer, damage to computer systems and data, and fraud or cheating committed through a computer.

Definition — Information Technology Act, 2000

India's legislation of the year 2000 that recognises electronic records and electronic signatures in law, supports electronic governance and electronic commerce, and provides for cyber contraventions and offences.

Two exam points follow. First, the year is **2000**; later changes to the law were amendments to the 2000 Act, not a fresh Act, so an option calling the Act itself "the IT Act of 2008" is wrong. Second, the scope words are **electronic records, electronic governance, and cybercrime**: an option that limits the Act to hardware prices, typing tests, or office furniture has left the scope entirely. No section numbers are needed for this lesson; the exam fact is the Act's year and its three branches of coverage.

56.3 Malware: the umbrella and its members

Malware, short for malicious software, is any program written to harm a computer, steal data, or disrupt work. It is the umbrella term: the question stem may say "malware" when it means the general class, and name virus, worm, Trojan, or ransomware when it means one member. Two further nuisance programs belong nearby: **spyware**, which secretly watches the user's activity and reports it, and **adware**, which displays unwanted advertisements.

Definition — Malware

Malicious software: any program designed to damage, disrupt, or gain unauthorised access to a computer system. Virus, worm, Trojan horse, and ransomware are types of malware.

A **virus** attaches itself to a host file or program and spreads when that file is run or shared. It needs human action — opening an infected attachment, running an infected file, or carrying it on a pen drive — and once it runs it can corrupt, delete, or alter files. A **worm**, by contrast, is standalone: it copies itself and spreads across a network without any host file and without waiting for a user to click. Worms consume bandwidth and system resources, so their signature effect is a slowed or crashed network. A **Trojan horse** pretends to be useful software — a free utility, a game, a handy tool — but hides a harmful purpose inside. It does not replicate by itself; instead it opens a **backdoor** through which an attacker can steal passwords, spy on the user, or control the machine. **Ransomware** encrypts the victim's files or locks the screen and demands payment (a ransom) for release. Paying does not guarantee recovery; the reliable defence is a clean **backup** kept separate from the machine.

Example 56.1. *An item describes a program that "looks like a useful download, does not copy itself, but lets an attacker read passwords." Trace the three contrasts: it needs no host file to be described (so not a virus), it does not self-spread (so not a worm), and it demands no payment (so not ransomware). The false-useful disguise plus the backdoor is the Trojan horse.*

56.4 Antivirus, quarantine, and patching

Antivirus software detects, blocks, and removes malware. It works by recognising known malware signatures and by flagging suspicious behaviour, and it must be updated regularly because new malware appears every day. When the antivirus meets a file it cannot yet judge, it places it in **quarantine**: an isolated holding area where the file cannot run or spread while the user or administrator decides its fate. Quarantine is isolation, not deletion and not a backup — the file is still present but harmless while isolated.

Definition — Quarantine

The isolation of a suspicious file by security software so that it cannot execute or spread, pending inspection, removal, or restoration.

Patching is the companion hygiene step: installing the vendor-supplied updates that fix known security flaws in the operating system and applications. An unpatched machine remains open to attacks that the vendor has already fixed, so "install updates promptly" is the exam's standard correct advice. Antivirus cleans what is already inside, quarantine holds what is suspect, and patching closes the holes through which the next attack would enter.

56.5 Phishing and anti-phishing

Phishing is not malware at all but deception: a fake message — mail, SMS, or website — that impersonates a trusted sender (a bank, an office, a delivery firm) to steal passwords, OTPs, or money. Its warning signs are urgency and threat ("your account will be blocked"), unexpected attachments, links whose real destination does not match the claimed sender, and any request for a password or OTP. The defence, called **anti-phishing** practice, is procedural: verify the sender through a separate channel, hover over or long-press links to inspect them before clicking, never share an OTP or password by message, and report the message to the service provider or office.

Definition — Phishing

A fraudulent message impersonating a trusted sender, designed to trick the recipient into revealing passwords, OTPs, or financial details.

Example 56.1. *An item shows a mail reading "Dear customer, your account is blocked; click here and enter your OTP to restore it." The correct response is to treat it as phishing: do not click, do not enter the OTP, and verify with the bank through its known number or site. A genuine bank or office never asks for a password or OTP by mail or message, so any option advising compliance is the trap.*

56.6 Encryption, digital signatures, and SSL/TLS

Encryption converts readable data (plaintext) into coded form (ciphertext) so that only authorised persons holding the key can read it. Its goal is confidentiality. A **digital signature**, by contrast, proves who sent a message and that it was not altered in transit; its goals are authenticity and integrity. The pair must never be swapped: a signature does not hide the message (anyone can still read it), and encryption by itself does not prove who sent it. This is the same distinction tested in the email lesson (TRAP-15), and it returns here because ransomware's "encryption" step uses the same idea against the victim.

SSL (Secure Sockets Layer) and its successor **TLS** (Transport Layer Security) provide a secure, encrypted channel over a network. They protect data **in transit**: mail submission, web logins, and online forms, signalled by the padlock in the browser address bar. The lesson's anchor fact, already seen in WO 2026 Q76, is that the secure email transmission protocol is **SSL/TLS**. Note the limit: SSL/TLS protect the journey, not the endpoints — they do not clean an infected file and do not replace antivirus.

Definition — Encryption

The conversion of plaintext into ciphertext with a key, so that only authorised holders of the key can recover the original content.

56.7 Firewall, MFA, and the matching map

A **firewall** is a barrier, implemented in hardware or software (or both), that filters incoming and outgoing network traffic against a set of rules. It blocks unauthorised access while letting permitted traffic pass, like a gatekeeper at the office network door checking every packet against the list. Its partner inside the machine is the antivirus: the firewall filters traffic at the boundary, while the antivirus cleans malware already inside. One does not replace the other.

MFA (Multi-Factor Authentication) protects logins by demanding two or more independent

checks — for example, a password plus an OTP, or a password plus a fingerprint. It closes the door against stolen passwords: even a leaked password is useless without the second factor. Strong, distinct passwords remain the first factor; MFA adds factors rather than replacing them.

Control	Job (the stem verb to watch)
Firewall	Filters network traffic; blocks unauthorised access.
Antivirus	Detects and removes malware on the machine.
Quarantine	Isolates a suspicious file so it cannot run.
Encryption / SSL-TLS	Hides content; protects data in transit.
Digital signature	Proves the sender and that content is intact.
Patching	Fixes known security flaws by installing updates.
MFA	Adds extra login checks beyond the password.
Backup	Recovers data after ransomware, deletion, or failure.

The matching technique is mechanical: read the verb in the stem. "Filters traffic" points to the firewall; "cleans malware" to the antivirus; "isolates" to quarantine; "hides content" to encryption; "proves sender" to digital signature; "fixes flaws" to patching; "extra login check" to MFA; "recovers after loss" to backup.

56.8 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. The IT Act is of **2000** and covers electronic records, e-governance, and cybercrime; later changes were amendments.
2. Malware is the **umbrella**; virus, worm, Trojan, and ransomware are specific types under it.
3. Virus needs a **host file plus human action**; worm is standalone and **self-spreading**; Trojan **does not replicate**; ransomware **locks files and demands payment**.
4. A digital signature **proves** (authenticity/integrity); it does not **hide** — encryption provides confidentiality (TRAP-15).
5. **SSL/TLS** protect data **in transit**; they do not clean infected files.
6. A **firewall** filters **traffic** at the boundary; antivirus cleans **malware** inside; quarantine **isolates** without deleting.
7. **MFA** adds login factors; **patching** fixes software flaws; **backup** is the recovery answer to ransomware.

Example 56.1. *An item asks which control "proves that a message came from the claimed sender and was not altered." The verb "proves" selects the digital signature, not encryption (which hides) and not the firewall (which filters). If the stem instead says "hides the content from eavesdroppers," the same map selects encryption; if it says "blocks unauthorised packets," it selects the firewall.*

Exercises

1. State the year and the three branches of scope of the Information Technology Act covered in this lesson.
2. Define malware and place virus, worm, Trojan horse, and ransomware under it in one sentence each.
3. Distinguish a virus from a worm on the two points of host file and human action.

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4. Why is a Trojan horse dangerous even though it does not replicate?
5. Explain why backup, rather than payment, is the correct response to ransomware.
6. Contrast encryption with a digital signature: which goal does each serve?
7. Using the matching map, assign one control to each verb: filters, isolates, hides, proves, fixes.

Solutions

1. The Information Technology Act is of **2000**. It gives legal validity to electronic records and electronic signatures, enables electronic governance, and defines cyber contraventions and offences.
2. Malware is malicious software written to harm or exploit. A virus attaches to a host file and needs human action; a worm self-copies across networks unaided; a Trojan disguises itself as useful software and opens a backdoor without replicating; ransomware locks or encrypts files and demands payment.
3. A virus needs a host file and a human act (opening, running, sharing) to spread; a worm is standalone, copies itself, and spreads over the network without a host file or user action.
4. The Trojan's disguise persuades the user to install it, and the hidden backdoor then lets the attacker steal passwords, spy, or control the machine — damage that needs no replication.
5. Payment does not guarantee recovery and rewards the attacker, while a clean separate backup restores the data without negotiation; hence backup is the reliable defence.
6. Encryption serves **confidentiality** (hides content); a digital signature serves **authenticity and integrity** (proves sender and intactness).
7. Filters — firewall; isolates — quarantine; hides — encryption (SSL/TLS in transit); proves — digital signature; fixes — patching (installing updates).

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LESSON 64

Current Cybersecurity, Cryptography and Cyber Law

64.1 Scope and study method

This lesson covers the current-affairs edge of the official Computer Awareness scope: computer virus and latest anti-virus, terms and abbreviations used in IT, and the role of IT in governance. It treats cybersecurity threats, cryptography basics, firewall types, and cyber law at awareness level. The exam depth is recall and classification, with at most a one-step scenario. The study method for every term is the near-neighbour contrast: what it is, what it is not, and the nearest confusion listed alongside it.

Term	Meaning in this lesson
Malware	Hostile software, including virus, worm, Trojan horse, ransomware, spyware.
Parasitic / file-infector virus	A virus that attaches itself to an executable file.
Phishing	Fraudulent messages that trick the user into revealing credentials.
Ransomware	Malicious code that locks or encrypts files and demands payment.
DDoS	Distributed Denial of Service: flood of traffic from many machines.
Hacking	Unauthorised access to a computer or account.
DES — Data Encryption Standard	Symmetric cipher working on 64-bit blocks.
Encryption	Conversion of plaintext to ciphertext for confidentiality.
Digital signature	Authenticity and integrity proof; not confidentiality.
Firewall	Barrier between a trusted and an untrusted network.
Personal data	Data identifying a person, protected by consent and purpose limits.
Copyright	Owner's exclusive rights over an original work.

64.2 Malware and the parasitic virus

Malware is the general name for hostile software. A **virus** is malware that needs a host program and some human action, such as opening an attachment, to spread. A **worm** spreads on its own across a network without a host program. A **Trojan horse** looks like useful software but hides a harmful payload, and it does not self-replicate.

Definition — Parasitic (file-infector) virus

A virus that attaches itself to an executable file. When the infected program runs, the virus code runs first and then hands control back to the host program, spreading further as infected files are shared or copied.

The parasitic virus is the file-infecting member of the family: it lives inside executable files, activates when the host program runs, and travels with shared files. An **anti-virus** detects, blocks, and removes such infections, and its database must be kept updated against the latest strains.

Example 64.1. *A student copies a free game from a friend's pen drive and runs it on the office desktop. The game file carries a file infector: the virus runs first, attaches to other executable files, and spreads when those files are shared further. Updating the anti-virus and scanning the drive before opening the files is the correct defence.*

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64.3 Phishing, ransomware, DDoS, and hacking

These four threats differ in mechanism, and the exam swaps their definitions. **Phishing** attacks the user: a fraudulent mail or message, often styled as a bank or KYC notice, tricks the user into revealing a password or OTP on a fake login page. **Ransomware** attacks the data: malicious code locks or encrypts the victim's files and demands payment for the decryption key. **DDoS** (Distributed Denial of Service) attacks availability: a flood of traffic from many machines overwhelms a server so genuine users cannot reach it. A flood from a single machine is only DoS; the distributed prefix matters. **Hacking** is the broad term for unauthorised access to a computer or account; phishing, ransomware, and DDoS are specific methods or goals within that broad idea.

Definition — Ransomware

Malicious code that locks or encrypts the victim's files and demands payment for their release.

Example 64.1. *An item describes a clerk whose files are suddenly unreadable, with a message demanding payment for the key. Tracing the mechanism — files locked, payment demanded — gives ransomware. If the stem instead described a fake bank mail asking for an OTP, the same trace gives phishing; a server slowed to a halt under flood traffic gives DDoS.*

64.4 Cryptography and DES

Cryptography protects information by transforming it so only the intended recipient can read it. **Plaintext** is the readable message, **ciphertext** is the transformed message, and a **key** controls the transformation. **Encryption** converts plaintext to ciphertext and **decryption** converts it back. In **symmetric** cryptography the same key encrypts and decrypts; in **asymmetric** cryptography a public key and a private key form a pair.

Definition — DES (Data Encryption Standard)

A symmetric cipher that processes data in 64-bit blocks. Its key is 64 bits long, of which 56 bits are effective, the remaining 8 bits being parity.

The two numbers for DES must be kept apart. The **block size** is 64 bits: that is the chunk of data processed at a time. The **effective key** is 56 bits. An option offering 128 bits confuses DES with a later cipher's block or key size.

64.5 Encryption vs digital signature

This is the TRAP-15 distinction and it recurs in papers. **Encryption** provides **confidentiality**: only the intended recipient can read the message. A **digital signature** provides **authenticity and integrity**: it proves who sent the message and that it was not altered in transit. A signature does not hide the message content, and encryption by itself does not prove the sender's identity.

Definition — Digital signature

A cryptographic proof attached to a message that establishes the sender's authenticity and the message's integrity, without by itself providing confidentiality.

Example 64.1. *An item states, "A digital signature encrypts the message body so only the recipient can read it." This is wrong. Hiding the body is encryption (confidentiality). The signature answers two different questions: who sent it and whether it was changed. The wrong option has merged the two security verbs.*

64.6 Firewalls: three types

A **firewall** is a barrier between a trusted internal network and an untrusted external network. It permits or denies traffic according to a rule set. It filters packets and connections; it does not

disinfect file-infecting viruses inside downloads that its rules allow through — that is the anti-virus's job.

A **packet-filtering** firewall examines each packet's headers — source and destination address, port, and protocol — and decides packet by packet. It is the fastest and simplest type but cannot judge a session as a whole. A **stateful-inspection** firewall examines the same headers plus the **connection state**: it remembers open sessions and drops packets that do not belong to a known session. An **application (proxy)** firewall sits between the client and the server as a go-between and inspects application-layer content and commands, such as web or mail requests.

Definition — Firewall

A network barrier that permits or denies traffic between a trusted internal network and an untrusted external network according to a rule set.

The ordering shorthand is: header only, then header plus state, then full content via proxy. Filtering checks the envelope, stateful inspection checks the conversation, and the proxy reads the letter inside.

64.7 Internet governance, privacy, and data protection

The Internet has no single owner; names and numbers are coordinated internationally, for example by ICANN (Internet Corporation for Assigned Names and Numbers). Governance concerns at awareness level are four: the **security** of networks, the **privacy** of personal data, fair **access** to the network, and which country's **law** applies when an act crosses borders. The positive side is the **role of IT in governance** (e-governance): online services, transparency, and delivery of benefits to citizens.

Personal data is any data that identifies a person, such as a name combined with a phone number, biometrics, or account details. Protection principles at awareness level are: collect data with **consent**, use it only for the stated **purpose**, keep it **secure**, and retain it no longer than needed. Privacy settings, strong passwords, and OTP discipline are the user-side defences. Misuse or leakage of personal data can attract legal consequences under cyber law.

64.8 Cyber law and copyright: awareness-level map

Cyber law at this level means two things: unauthorised access, fraud, and data misuse attract punishment, and creators hold enforceable rights over their works. The **Copyright Act** protects original literary, artistic, musical, and software works from copying without permission. The awareness-level map matches each concept to its area: the **meaning of copyright** (what exclusive rights the owner holds), **first ownership** (who owns the work first), **infringement** (what counts as violation), **fair dealing** (limited permitted uses such as private study), and **offences** (punishment for wilful infringement). Software piracy — copying licensed software without permission — is the standard infringement example. Section numbers are recall only where the question itself supplies them; otherwise the concept match is what the exam tests.

64.9 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. DES block size is **64 bits**; the effective key is 56 bits. Do not offer 128 for either (DES fact).
2. Digital signature ≠ encryption: authenticity/integrity vs confidentiality (TRAP-15/TRM-11).
3. A file infector attaches to **executable files**; it is not a worm (self-spreading) or a Trojan (non-replicating disguise).
4. Packet filter checks headers only; stateful adds **connection state**; proxy inspects **application content**.

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5. Phishing steals **credentials**; ransomware locks **files**; DDoS blocks **access**; hacking is unauthorised access generally.
6. DDoS needs **many** machines (distributed); a single-machine flood is only DoS.
7. Copyright protects the owner against **infringement**, with limited **fair-dealing** exceptions such as study — piracy is the standard violation example.

Example 64.1. *An item asks which firewall remembers open sessions and drops stray packets. Trace the triple: the packet filter decides per packet, the proxy inspects application content, and only stateful inspection tracks connection state. The answer is the stateful-inspection firewall. If the stem instead asked which device disinfects an infected executable, the answer leaves firewalls entirely and gives the anti-virus.*

Exercises

1. Define a parasitic (file-infector) virus and distinguish it from a worm and a Trojan horse.
2. State the DES block size and effective key size, and expand the abbreviation DES.
3. Distinguish packet-filtering, stateful-inspection, and application/proxy firewalls.
4. Distinguish phishing, ransomware, DDoS, and hacking in one line each.
5. Explain why a digital signature is not encryption, using TRAP-15.
6. List the four Internet-governance concerns and the four data-protection principles from this lesson.
7. Match each copyright concept (meaning, first ownership, infringement, fair dealing, offences) to what it covers.

Solutions

1. A parasitic virus attaches to an executable file, runs when the host program runs, and spreads with shared files. A worm spreads on its own across a network without a host; a Trojan horse disguises itself as useful software and does not self-replicate.
2. DES (Data Encryption Standard) is a symmetric cipher with a 64-bit block size and a 56-bit effective key (64 bits with parity).
3. Packet-filtering checks per-packet headers only; stateful inspection adds connection-state tracking; the application/proxy firewall inspects application-layer content as a go-between.
4. Phishing tricks the user into revealing credentials; ransomware locks or encrypts files for ransom; DDoS floods a service from many machines so genuine users cannot reach it; hacking is unauthorised access generally.
5. Encryption provides confidentiality (who can read it); a digital signature provides authenticity and integrity (who sent it, whether it was changed). Merging them is TRAP-15.
6. Governance concerns: security, privacy, access, jurisdiction (applicable law). Data-protection principles: consent, stated purpose, security, limited retention.
7. Meaning: the exclusive rights the owner holds; first ownership: who owns the work first; infringement: what counts as violation (for example software piracy); fair dealing: limited permitted uses such as private study; offences: punishment for wilful infringement.

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UNIT 11

Governance & Integration

LESSON 47

Humanware and IT Governance

47.1 What humanware means

A computer system is not only metal and programs. It also needs people. The human element of a computer system is called **humanware**, and it is also known as **liveware** or **peopleware**. Humanware means the people who design, build, operate, manage and use computers. It is *neither hardware nor software*; it is the people component of the system.

Definition — Humanware

The people involved with a computer system — its users, operators, administrators, programmers and support staff. Also called liveware or peopleware.

Term	Meaning in this lesson
Hardware	The physical parts of the computer.
Software	The programs and instructions the computer runs.
Humanware	The people who use and manage the hardware and software.
User	The person who uses the finished system to do a task.
Administrator	The person who manages the system, network or database.

The rest of this lesson uses these terms in one consistent sense. **Hardware** is the machinery. **Software** is the set of instructions. **Humanware** is the people.

47.2 The people in a computer system

Humanware is not a single role. Many different people take part, and each has a distinct job. The main categories are the following.

- **User (or end user)** — the person who uses the finished system to perform a task. A clerk entering data, a student reading a portal and a customer paying a bill are all users.
- **Operator** — the person who runs the computer system, feeds jobs into it and monitors routine operation.
- **Administrator** — the person who manages the system. A **system administrator** manages servers, user accounts and access rights; a **network administrator** manages the network; a **database administrator** manages the database.
- **Programmer (developer)** — the person who writes, tests and maintains the software.
- **IT support (help desk) and technician** — the people who answer user problems, troubleshoot faults, and install, repair and maintain the hardware.

The distinction to hold is simple. Administrators *manage* the system, operators *run* it, and users *use* its output.

47.3 Hardware, software and humanware

A working computer system needs all three components. Hardware is the physical machine. Software is the set of programs that tell the hardware what to do. Humanware is the people who design, run and use both. Hardware without software does nothing, and both are useless without the people who operate them. This is why humanware is taught as a full component, not as an afterthought.

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Example 47.1. *A government office installs new computers (hardware) and an accounting program (software). The system still does not work by itself. A programmer must maintain the program, a system administrator must create user accounts, an operator must run the daily jobs, and the office staff (users) must enter the data. Those people are the humanware.*

47.4 The role of IT in governance

Governance is the process of decision-making and its implementation in an organisation or a state. **IT** (Information Technology) and **ICT** (Information and Communication Technology) let government carry out that work over computers and networks. The role of IT in governance is to make government faster, cheaper, more transparent and more accessible to the people it serves.

The main benefits of using IT in governance are the following.

- **Transparency** — citizens can see the rules, the status of their applications and the decisions taken.
- **Accountability** — actions are recorded and can be traced.
- **Efficiency** — less paperwork and faster processing.
- **Convenience** — services are available from home and around the clock.
- **Lower cost** and **reduced corruption**, because manual handling and physical counters are replaced by digital processes.

The overall aim is **citizen-centric** service: the citizen is placed at the centre of every process.

47.5 IT governance versus e-governance

These two terms are a common near-neighbour pair, and the difference matters.

- **IT governance** is the process by which an organisation directs and controls its own IT resources so that they support its goals. It is *governance of IT*.
- **E-governance** is the use of IT by government to govern and deliver public services. It is *IT in governance*.

The trap is to swap them. IT governance is about managing IT well inside an organisation; e-governance is about using IT to govern the state and serve citizens.

47.6 E-governance basics

E-governance is the application of ICT by government to deliver services, share information and transact with citizens, businesses and other government agencies. It uses the internet, databases and digital records in place of paper and physical counters. It makes government services faster, wider-reaching and easier to audit.

Definition — E-governance

The use of information and communication technology to improve the way government serves and interacts with the public.

E-governance is usually described by the parties who interact. The four standard types are the following.

Type	Full form	Interaction
G2C	Government to Citizen	Services to the general public.
G2G	Government to Government	Between government departments.
G2B	Government to Business	With companies and traders.
G2E	Government to Employee	With government staff.

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This lesson keeps the treatment at awareness level. Named platforms and detailed G2x matching belong to Lesson 57.

47.7 Digital public services

A **digital public service** is a government service that a citizen can obtain online. Common examples are certificates, licences, land records, bill payment, applications and grievance redressal. Citizens reach these services through service portals and through **Common Service Centres (CSCs)**, which provide a physical access point where help is needed. Typical features are online application, status tracking, digital payment and downloadable documents. A service counts as digital when the citizen can complete the task without visiting an office in person.

47.8 Electronic records

An **electronic record** is information created, stored and retrieved in electronic (digital) form. Examples include e-documents, e-files, scanned papers, database entries and email. Electronic records are easy to store, search, retrieve, share and back up, and they replace large volumes of paper. An electronic record can have legal validity when it is authenticated, for example with a digital signature.

Definition — Electronic record

Information that is created, stored and retrieved in electronic form, such as an e-document, an e-file or a database entry.

47.9 Authentication

Authentication is the process of verifying that a user, device or system is who or what it claims to be. It answers the question: "Who are you?" Common methods are a password, a PIN, an **OTP** (One-Time Password), a biometric such as a fingerprint or iris, and a digital certificate. Authentication must succeed before the system grants access.

Definition — Authentication

The process of confirming a claimed identity before access to a system is allowed.

Authentication is often confused with **authorization**. **Authorization** is the process of granting or denying permission — it decides what a verified identity is allowed to do. Authentication proves identity; authorization grants or refuses access rights. Authentication always comes first.

Authentication relies on *factors*. There are three standard types: something you **know** (a password or PIN), something you **have** (an OTP on your phone, a token or a smart card), and something you **are** (a fingerprint, iris or face). When two different factors are combined, it is called **two-factor authentication (2FA)**; when two or more are used, it is **multi-factor authentication (MFA)**. Two passwords are not two factors, because they are the same factor used twice.

Example 47.1. *A candidate logs in to a government portal with a password and is then asked for an OTP sent to the registered mobile number. The password is something the candidate knows; the OTP is something the candidate has. Together they form two-factor authentication. The portal then decides which forms the candidate may open — that decision is authorization, not authentication.*

47.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. Humanware means **people**. It is neither hardware nor software; do not confuse it with firmware (which is software stored in hardware).
2. **IT governance** is governance *of* IT; **e-governance** is IT *in* governance. Do not swap them.

3. **G2C** is Government to **Citizen**, not Government to Company. The business interaction is G2B.
4. **Authentication** proves identity; **authorization** grants permission. Authentication comes first.
5. An **electronic record** is digital information. A printed paper file is a physical record, not an electronic one.

Example 47.1. *An item asks which component of a computer system consists of the people who operate and manage it. Hardware is the machinery and software is the set of instructions, so both are wrong. The correct answer is humanware. If the stem instead asks what improves transparency and accountability in government, the answer is the role of IT in governance, and if it asks what verifies a user's identity, the answer is authentication.*

Exercises

1. Define humanware and name four categories of people who form it.
2. Distinguish hardware, software and humanware, and explain why all three are needed.
3. Give three benefits of using IT in governance.
4. Distinguish IT governance from e-governance.
5. What are the four types of e-governance interaction, and what does each connect?
6. Define an electronic record and give two examples.
7. Distinguish authentication from authorization, and name the three standard authentication factors.

Solutions

1. Humanware (also called liveware or peopleware) is the human element of a computer system. Four categories are users (end users), operators, administrators, and programmers (with IT support and technicians).
2. Hardware is the physical machine, software is the set of programs, and humanware is the people. Hardware without software does nothing, and both need people to design, run and use them, so all three are required.
3. Any three of: transparency, accountability, efficiency, convenience, lower cost, reduced corruption, and citizen-centric service.
4. IT governance is the direction and control of an organisation's own IT resources — governance of IT. E-governance is the use of IT by government to deliver public services — IT in governance.
5. G2C (Government to Citizen) serves the public; G2G (Government to Government) links departments; G2B (Government to Business) links companies; G2E (Government to Employee) links government staff.
6. An electronic record is information created, stored and retrieved in electronic form. Examples include an e-document, an e-file, a scanned paper or a database entry.
7. Authentication verifies identity ("who are you?"); authorization grants or denies permission ("what may you do?"). The three factors are something you know, something you have, and something you are.

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LESSON 57

Digital Platforms and E-Governance

57.1 What e-Governance means

A running government delivers many services: certificates, licences, payments, benefits, and information. **e-Governance** is the use of Information and Communication Technology (ICT) by government to deliver these services. Instead of standing at a physical counter, a citizen reaches the service through an online portal. The aim is to make services accessible, efficient, transparent, and reliable.

e-Governance is about governance, not merely about technology. The technology is the tool; the purpose is better delivery of public services.

Definition — e-Governance

The use of Information and Communication Technology (ICT) by government to deliver services to citizens, businesses, and other parts of government in an accessible, efficient, transparent, and reliable way.

Term	Meaning in this lesson
e-Governance	ICT-based delivery of government services.
G2x	The interaction categories: G2C, G2G, G2B, G2E.
Digital India	The flagship Government of India programme launched on 1 July 2015.
NeGP	National e-Governance Plan, approved in May 2006.
e-Kranti	Pillar 5 of Digital India and the upgraded NeGP (NeGP 2.0).
NeGD	National e-Governance Division, created in 2009 under MeitY.
e-NAM	National Agriculture Market, the pan-India agricultural trading portal.

57.2 The G2x classification

e-Governance interactions are named by who takes part. The "G" always stands for Government, and the letter after it says who the government is interacting with.

Type	Full form	Who is served / examples
G2C	Government to Citizen	The general public. UMANG, DigiLocker, certificates, utility bills.
G2G	Government to Government	Government departments and levels. e-Office, PFMS.
G2B	Government to Business	Businesses and companies. GeM, GSTN.
G2E	Government to Employee	Government employees. An employee service book or salary portal.

The classification is the highest-yield part of this topic. An item can name a platform and ask which G2x category it belongs to, or name a category and ask which platform fits. The safe method is to ask "who is the government serving?" A citizen means G2C; another government body means G2G; a company means G2B; an employee means G2E.

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Example 57.1. *The stem asks, "GeM (Government e-Marketplace) is an example of which category?" GeM is used by government to buy goods and services from businesses, so the government is serving a business. The answer is G2B, not G2C. A student who sees only that "government uses it" may wrongly choose G2C; the deciding question is who is on the other side.*

57.3 Digital India

Digital India is a flagship programme of the Government of India. It was launched on **1 July 2015** by the Prime Minister, with the vision to transform India into a **digitally empowered society and knowledge economy**.

The programme rests on **three vision areas**:

(i) Digital Infrastructure as a Core Utility to Every Citizen;

(ii) Governance and Services on Demand;

(iii) Digital Empowerment of Citizens.

Under these vision areas the programme is built on **nine pillars**: Broadband Highways; Universal Access to Mobile Connectivity; Public Internet Access Programme; e-Governance: Reforming Government through Technology; e-Kranti: Electronic Delivery of Services; Information for All; Electronics Manufacturing; IT for Jobs; and Early Harvest Programmes. Pillar 5, e-Kranti, is the service-delivery pillar and is treated separately below.

Definition — Digital India

A flagship Government of India programme, launched on 1 July 2015, with the vision of transforming India into a digitally empowered society and knowledge economy, built on three vision areas and nine pillars.

57.4 NeGP, e-Kranti, and NeGD: three names, three things

These three names are close and are a standing source of confusion. They are kept apart here by asking what each one *is*.

The **National e-Governance Plan (NeGP)** is the older umbrella *plan*. It was approved in **May 2006**, with the vision to make all government services accessible to the common man in his locality, through common service delivery outlets, at affordable cost. It is built around **Mission Mode Projects (MMPs)** — 27 MMPs with supporting components.

e-Kranti means "electronic delivery of services". It is **Pillar 5** of the Digital India programme and is also called **NeGP 2.0**, an upgraded approach to the National e-Governance Plan. Its vision is "Transforming e-Governance for Transforming Governance", and its mission is to deliver all government services electronically through integrated and interoperable systems.

The **National e-Governance Division (NeGD)** is not a plan at all. It is an *organisation*. It was created in **2009** by the Ministry of Electronics and Information Technology (MeitY) as an independent business division under the Digital India Corporation. Its role is to support MeitY in the programme management and implementation of e-Governance projects, and to give technical and advisory support to ministries and departments.

Name	What it is	Year	One word
NeGP	Umbrella e-Governance plan with 27 MMPs	2006	The plan
e-Kranti	Pillar 5 of Digital India; NeGP 2.0	2015	The approach
NeGD	Division under MeitY supporting projects	2009	The body

Example 57.1. *An item states, "NeGD is the National e-Governance Plan approved in 2006." This is wrong on two counts. The plan approved in 2006 is **NeGP** (National e-Governance Plan). **NeGD** is the National e-Governance* Division*, an organisation created in 2009. The distractor swaps a plan for a body.*

57.5 e-NAM and agricultural marketing

e-NAM stands for **National Agriculture Market** (also written eNAM). It is a pan-India **electronic trading portal** for agricultural commodities, launched on **14 April 2016**. It is implemented by the **Small Farmers Agribusiness Consortium (SFAC)** under the Ministry of Agriculture and Farmers Welfare. The portal is available at **enam.gov.in**.

The key idea is that e-NAM **networks the existing Agriculture Produce Market Committee (APMC) mandis** to create a **unified national market** for agricultural commodities. Because trade happens online, farmers get **online competitive bidding and transparent price discovery**, and direct sale to buyers reduces the role of intermediaries.

Definition — e-NAM

The National Agriculture Market, a pan-India electronic trading portal, launched on 14 April 2016, that networks the existing APMC mandis to create a unified national market for agricultural commodities.

A near-neighbour to keep separate is **AGMARKNET**. AGMARKNET is a portal that provides agricultural marketing *information* such as market prices. e-NAM, by contrast, is the *trading* platform where the sale itself takes place. An item that asks for the agricultural marketing *trading* platform wants e-NAM.

57.6 Citizen portal, electronic records, and authentication

The **National Portal of India** is **india.gov.in**. It is a **single-window** gateway from which a citizen can reach the information and services of the Government of India. A related platform, **MyGov**, is the citizen-engagement platform for participative governance.

Electronic governance depends on giving electronic documents the same standing as paper. The **Information Technology Act, 2000** gives legal recognition to **electronic records**. An electronic record can be stored, retrieved, and used in electronic form. To be trustworthy, it must be authenticated.

Authentication proves that a message or document is genuine and comes from the claimed sender. A **digital signature** provides authenticity, integrity, and non-repudiation. **Encryption**, by contrast, provides confidentiality: it keeps the content secret from unauthorised readers. **Aadhaar** is used as a digital identity for authentication of a person.

Definition — Digital signature

A mechanism used to authenticate an electronic record. It provides authenticity, integrity, and non-repudiation, and is distinct from encryption, which provides confidentiality.

Example 57.1. *An item states, "A digital signature encrypts the document so that only the recipient can read it." This confuses two ideas. Encrypting for secrecy is **encryption**, which provides confidentiality. A **digital signature** proves who sent the document and that it was not altered; it provides authenticity and integrity. The wrong option has used the signature's name for encryption's job.*

57.7 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. e-NAM = agricultural marketing and trading; it is **not** general e-commerce.

2. GeM = G2B procurement, **not** a G2C service. e-Office and PFMS are G2G.
3. NeGP (plan, 2006), e-Kranti (pillar / NeGP 2.0, 2015), and NeGD (division, 2009) are three different things; do not swap them.
4. Digital signature = authenticity and integrity; encryption = confidentiality. They are **not** the same (TRAP-15).
5. The citizen portal of the Government of India is india.gov.in, not the Digital India or NeGD site.
6. Digital India is built on three vision areas and nine pillars; e-Kranti is Pillar 5, the service-delivery pillar.

Example 57.1. *An item asks which body supports MeitY in the programme management of e-Governance projects. Trace the three names: NeGP is a plan, e-Kranti is a pillar, and only NeGD is an organisation. The answer is NeGD.*

Exercises

1. Expand e-NAM and state its purpose in one line.
2. Distinguish G2C, G2G, G2B, and G2E, giving one example of each.
3. State the launch date and vision of the Digital India programme.
4. Explain the difference between NeGP, e-Kranti, and NeGD.
5. Define authentication and explain how a digital signature differs from encryption.

Solutions

1. e-NAM is the National Agriculture Market: a pan-India electronic trading portal that networks the existing APMC mandis to create a unified national market for agricultural commodities.
2. G2C (Government to Citizen) serves the public, e.g. UMANG; G2G (Government to Government) serves government bodies, e.g. e-Office; G2B (Government to Business) serves businesses, e.g. GeM; G2E (Government to Employee) serves government employees, e.g. an employee service book.
3. Digital India was launched on 1 July 2015, with the vision to transform India into a digitally empowered society and knowledge economy.
4. NeGP is the National e-Governance Plan, the umbrella plan approved in May 2006 with 27 MMPs. e-Kranti is Pillar 5 of Digital India and the upgraded NeGP (NeGP 2.0). NeGD is the National e-Governance Division, an organisation created in 2009 under MeitY to support e-Governance projects.
5. Authentication proves that a message or document is genuine and comes from the claimed sender. A digital signature provides authenticity, integrity, and non-repudiation; encryption provides confidentiality by keeping the content secret. They are different functions.

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